Algebraic Geometry

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July 23, 2026

Contents

Chapter 1

Basic material

This chapter is partly taken from [Hul03]. For complementary accounts of the classical affine and projective material, see [Hol12] and [Har77, Chapter I]. The scheme-theoretic viewpoint developed later is particularly well motivated in [EH00], while [Liu02] gives a more systematic treatment.

I Introduction: algebraic sets

Further reading. For additional examples and a classical treatment of affine and projective algebraic sets, see [Hul03], [Hol12], and [Har77, Chapter I, §§1–3].

I.A Ideals and affine algebraic sets

Let $\Bbbk$ be a field and consider the polynomial ring $A = \Bbbk[x_1, \ldots, x_n]$.

I.A.1 From ideals to algebraic sets

Given a subset P of A, we set

$$\mathbf{V}(P) = \{a \in \Bbbk^n \mid f(a) = 0, \forall f \in P\}.$$

Remark I.A.1. It is an easy check that, denoting by (P) the ideal generated by P, we have

$$\mathbf{V}(P) = \mathbf{V}((P)).$$

Indeed, let $a \in \mathbf{V}((P))$. Then, since any element f of P lies in (P), we have $f(a) = 0$, so $a \in \mathbf{V}(P)$. Conversely, consider $a \in \mathbf{V}(P)$. For any $f \in (P)$, there are $p_1, \ldots, p_s \in P$ and $g_1, \ldots, g_s \in A$ such that $f = g_1p_1 + \cdots + g_sp_s$. We have $p_1(a) = \cdots = p_s(a) = 0$, since $p_1, \ldots, p_s \in P$ and $a \in \mathbf{V}(P)$. Therefore, $f(a) = g_1(a)p_1(a) + \cdots + g_s(a)p_s(a) = 0$. So $a \in \mathbf{V}((P))$.

According to the previous remark, we will usually replace P by the ideal it generates and write $\mathbf{V}(P)$ without further comment.

Definition I.A.2. A subset of the form $\mathbf{V}(P)$ is called an affine algebraic set. Literally, P can be any subset of $A = \Bbbk[x_1, \ldots, x_n]$, but as we have just seen, the relevant subsets of A are the ideals of A.

Example I.A.3. In $\Bbbk^2$, the algebraic set

$$\mathbf{V}(y - x^2)$$

is the parabola with equation $y = x^2$. The algebraic set

$$\mathbf{V}(xy)$$

is the union of the two coordinate axes, since $xy = 0$ if and only if $x = 0$ or $y = 0$.

Example I.A.4. In $\Bbbk^2$, the algebraic set

$$\mathbf{V}(x^2 + y^2 - 1)$$

depends on the field $\Bbbk$. For instance, over $\mathbb{R}$ it is the usual circle, while over an algebraically closed field it should be regarded simply as the zero locus of the polynomial $x^2 + y^2 - 1$.

Lemma I.A.5. *Affine algebraic sets satisfy the following properties.*

i) We have $\mathbf{V}(0) = \Bbbk^n$ *and* $\mathbf{V}(A) = \emptyset$.

ii) For all ideals $\mathfrak{a} \subset \mathfrak{b}$*, we have* $\mathbf{V}(\mathfrak{a}) \supset \mathbf{V}(\mathfrak{b})$.

iii) For all ideals $\mathfrak{a}, \mathfrak{b}$ *of* A*, we have* $\mathbf{V}(\mathfrak{a} \cap \mathfrak{b}) = \mathbf{V}(\mathfrak{a}) \cup \mathbf{V}(\mathfrak{b})$.

iv) Given ideals $(\mathfrak{a}_i \mid i \in I)$*, we have* $\mathbf{V}(\sum_{i \in I} \mathfrak{a}_i) = \cap_{i \in I} \mathbf{V}(\mathfrak{a}_i)$.

Proof. For i), $\mathbf{V}(0) = \Bbbk^n$ is obvious, while $\mathbf{V}(A) = \emptyset$ is clear by noting that $\mathbf{V}(1) = \emptyset$. Item ii) is clear since any $b \in \mathbf{V}(\mathfrak{b})$ satisfies $f(b) = 0$, for all $f \in \mathfrak{a}$, as $\mathfrak{a} \subset \mathfrak{b}$. For iv), let us take $a \in \mathbf{V}(\sum_{i \in I} \mathfrak{a}_i)$. Then $f(a) = 0$ for all $f \in \mathfrak{a}_i$, for all $i \in I$, because $\mathfrak{a}_i \subset \sum_{i \in I} \mathfrak{a}_i$. Hence $a \in \cap_{i \in I} \mathbf{V}(\mathfrak{a}_i)$. Conversely, given $a \in \cap_{i \in I} \mathbf{V}(\mathfrak{a}_i)$, we have $f_i(a) = 0$, for all $i \in I$ and all $f_i \in \mathfrak{a}_i$. Since any $f \in \sum_{i \in I} \mathfrak{a}_i$ is a finite combination of some $(f_i \mid i \in I)$, we have $f(a) = 0$.

In conclusion, the only non-trivial item is iii). The inclusion $\mathbf{V}(\mathfrak{a}) \cup \mathbf{V}(\mathfrak{b}) \subset \mathbf{V}(\mathfrak{a} \cap \mathfrak{b})$ follows from ii), because $\mathfrak{a}$ and $\mathfrak{b}$ both contain $\mathfrak{a} \cap \mathfrak{b}$. Conversely, consider $a \in \Bbbk^n$ and assume that $a \notin \mathbf{V}(\mathfrak{a}) \cup \mathbf{V}(\mathfrak{b})$, so that $a \notin \mathbf{V}(\mathfrak{a})$ and $a \notin \mathbf{V}(\mathfrak{b})$. Then, there are $f \in \mathfrak{a}$ and $g \in \mathfrak{b}$ such that $f(a) \neq 0 \neq g(a)$. Hence $fg \in \mathfrak{a} \cap \mathfrak{b}$ gives $fg(a) \neq 0$, so $a \notin \mathbf{V}(\mathfrak{a} \cap \mathfrak{b})$. □

Exercise I.A.6. Draw the following affine algebraic sets in $\mathbb{R}^2$:

$$\mathbf{V}(xy), \qquad \mathbf{V}(y - x^2), \qquad \mathbf{V}((x-1)(x+1), y).$$

Solution. The set $\mathbf{V}(xy)$ is the union of the two coordinate axes. The set $\mathbf{V}(y - x^2)$ is the parabola $y = x^2$. Finally,

$$\mathbf{V}((x-1)(x+1), y)$$

is the set of points satisfying $y = 0$ and $(x-1)(x+1) = 0$, hence it is the two-point set $\{(-1, 0), (1, 0)\}$.

Exercise I.A.7. Show that every finite subset of $\Bbbk^n$ is an affine algebraic set.

Solution. It is enough to show that a single point is algebraic, since finite unions of algebraic sets are algebraic by Lemma iii). If $a = (a_1, \ldots, a_n) \in \Bbbk^n$, then

$$\{a\} = \mathbf{V}(x_1 - a_1, \ldots, x_n - a_n).$$

Thus any finite subset of $\Bbbk^n$ is a finite union of algebraic sets, hence is algebraic.

Exercise I.A.8 (Finite sets of points and collision of points). Let

$$A = \Bbbk[x_1, x_2]$$

and consider the finite subsets

$$\begin{aligned} X_1 &= \{(0,0)\}, \\ X_2 &= \{(0,0),(1,0)\}, \\ X_3 &= \{(0,0),(1,0),(0,1)\}, \\ X_4 &= \{(0,0),(1,0),(0,1),(1,1)\} \end{aligned}$$

of $\Bbbk^2$.

1. Compute the vanishing ideals $\mathbf{I}(X_i)$.
2. Find a basis of the $\Bbbk$-vector space $A/\mathbf{I}(X_i)$ for $1 \leq i \leq 4$.
3. Show that the ideals
$$\mathfrak{a} = (x_1, x_2) \qquad \text{and} \qquad \mathfrak{b} = (x_1, x_2)^2$$
have the same zero set, but that the quotient algebras $A/\mathfrak{a}$ and $A/\mathfrak{b}$ are not isomorphic.
4. For $a \in \Bbbk$, consider the ideals
$$\begin{aligned} \mathfrak{a}_{2,a} &= (x_2, x_1(x_1 - a)), \\ \mathfrak{a}_{3,a} &= (x_1x_2, x_1(x_1 - a), x_2(x_2 - a)), \\ \mathfrak{a}_{4,a} &= (x_1(x_1 - a), x_2(x_2 - a)). \end{aligned}$$
Determine their zero sets and find bases of the quotient algebras $A/\mathfrak{a}_{r,a}$.
5. Find generators for the relations among the displayed generators of $\mathfrak{a}_{2,a}$, $\mathfrak{a}_{3,a}$ and $\mathfrak{a}_{4,a}$.

Solution. For a point $p = (a_1, a_2) \in \Bbbk^2$, its vanishing ideal is

$$\mathbf{I}(p) = (x_1 - a_1, x_2 - a_2).$$

Moreover, the vanishing ideal of a finite union is the intersection of the vanishing ideals:

$$\mathbf{I}(Y \cup Z) = \mathbf{I}(Y) \cap \mathbf{I}(Z).$$

1. *Computation of the ideals.* For the first set, we immediately obtain
$$\mathbf{I}(X_1) = (x_1, x_2).$$
For X_2, the polynomial x_2 vanishes at both points, as does $x_1(x_1 - 1)$. We claim that
$$\mathbf{I}(X_2) = (x_2, x_1(x_1 - 1)).$$
Indeed, modulo the ideal on the right, every polynomial has a unique representative of the form
$$\alpha + \beta x_1, \qquad \alpha, \beta \in \Bbbk.$$
If such a polynomial vanishes at $(0,0)$ and $(1,0)$, then
$$\alpha = 0, \qquad \alpha + \beta = 0,$$

so $\alpha = \beta = 0$. Hence the displayed ideal contains every polynomial vanishing on X_2.

For X_3, put

$$J_3 = (x_1x_2, x_1(x_1-1), x_2(x_2-1)).$$

All three generators vanish on X_3, so

$$J_3 \subset \mathbf{I}(X_3).$$

Modulo J_3, we have

$$x_1x_2 = 0, \qquad x_1^2 = x_1, \qquad x_2^2 = x_2.$$

Thus every polynomial is congruent modulo J_3 to a linear combination of

$$1, \quad x_1, \quad x_2.$$

Suppose

$$\alpha + \beta x_1 + \gamma x_2$$

vanishes at the three points of X_3. Evaluating successively at $(0,0)$, $(1,0)$ and $(0,1)$ gives

$$\alpha = 0, \qquad \alpha + \beta = 0, \qquad \alpha + \gamma = 0.$$

Hence $\alpha = \beta = \gamma = 0$. Therefore

$$\mathbf{I}(X_3) = (x_1x_2, x_1(x_1-1), x_2(x_2-1)).$$

Finally, put

$$J_4 = (x_1(x_1-1), x_2(x_2-1)).$$

Modulo J_4, we have

$$x_1^2 = x_1, \qquad x_2^2 = x_2.$$

Consequently, every polynomial is congruent to a linear combination of

$$1, \quad x_1, \quad x_2, \quad x_1x_2.$$

If

$$\alpha + \beta x_1 + \gamma x_2 + \delta x_1x_2$$

vanishes on X_4, evaluation at the four points gives

$$\begin{aligned} \alpha &= 0, \\ \alpha + \beta &= 0, \\ \alpha + \gamma &= 0, \\ \alpha + \beta + \gamma + \delta &= 0. \end{aligned}$$

It follows that all four coefficients vanish. Hence

$$\mathbf{I}(X_4) = (x_1(x_1-1), x_2(x_2-1)).$$

We have therefore proved

$$\begin{aligned} \mathbf{I}(X_1) &= (x_1, x_2), \\ \mathbf{I}(X_2) &= (x_2, x_1(x_1-1)), \\ \mathbf{I}(X_3) &= (x_1x_2, x_1(x_1-1), x_2(x_2-1)), \\ \mathbf{I}(X_4) &= (x_1(x_1-1), x_2(x_2-1)). \end{aligned}$$

2. *Bases of the quotient algebras.* The computations above show that bases are given by

quotient algebra	basis over $\Bbbk$
$A/\mathbf{I}(X_1)$	$\{1\}$
$A/\mathbf{I}(X_2)$	$\{1, x_1\}$
$A/\mathbf{I}(X_3)$	$\{1, x_1, x_2\}$
$A/\mathbf{I}(X_4)$	$\{1, x_1, x_2, x_1x_2\}$.

In particular,

$$\dim_{\Bbbk}(A/\mathbf{I}(X_i)) = i.$$

One may also see this directly from the evaluation morphism

$$A \longrightarrow \Bbbk^{X_i}, \qquad f \longmapsto \big(f(p)\big)_{p\in X_i}.$$

Its kernel is $\mathbf{I}(X_i)$, and in the present examples it is surjective. Thus

$$A/\mathbf{I}(X_i) \simeq \Bbbk^i$$

as $\Bbbk$-algebras.

3. *Two ideals with the same zero set.* Let

$$\mathfrak{a} = (x_1, x_2), \qquad \mathfrak{b} = (x_1, x_2)^2 = (x_1^2, x_1x_2, x_2^2).$$

Since $\mathfrak{b} \subset \mathfrak{a}$, we have

$$\mathbf{V}(\mathfrak{a}) \subset \mathbf{V}(\mathfrak{b}).$$

Conversely, if $(a_1, a_2) \in \mathbf{V}(\mathfrak{b})$, then

$$a_1^2 = a_1a_2 = a_2^2 = 0.$$

Since $\Bbbk$ is a field, this implies $a_1 = a_2 = 0$. Hence

$$\mathbf{V}(\mathfrak{a}) = \mathbf{V}(\mathfrak{b}) = \{(0,0)\}.$$

On the other hand,

$$A/\mathfrak{a} \simeq \Bbbk,$$

whereas

$$A/\mathfrak{b} = \Bbbk[x_1, x_2]/(x_1^2, x_1x_2, x_2^2)$$

has basis

$$\{1, x_1, x_2\}.$$

Thus

$$\dim_{\Bbbk}(A/\mathfrak{a}) = 1, \qquad \dim_{\Bbbk}(A/\mathfrak{b}) = 3,$$

so the two quotient algebras cannot be isomorphic.

Moreover, the classes of x_1 and x_2 in $A/\mathfrak{b}$ are nonzero nilpotent elements. For example,

$$\overline{x}_1 \neq 0, \qquad \overline{x}_1^2 = 0.$$

4. *Specialisation of the ideals.* Assume first that $a \neq 0$.

   For
$$\mathfrak{a}_{2,a} = (x_2, x_1(x_1 - a)),$$
   a common zero must satisfy
$$x_2 = 0, \qquad x_1(x_1 - a) = 0.$$
   Therefore
$$\mathbf{V}(\mathfrak{a}_{2,a}) = \{(0,0), (a,0)\}.$$

   For
$$\mathfrak{a}_{3,a} = (x_1x_2, x_1(x_1 - a), x_2(x_2 - a)),$$
   we have
$$x_1 \in \{0, a\}, \qquad x_2 \in \{0, a\}, \qquad x_1x_2 = 0.$$
   The last condition excludes (a, a), and hence
$$\mathbf{V}(\mathfrak{a}_{3,a}) = \{(0,0), (a,0), (0,a)\}.$$

   Finally,
$$\mathfrak{a}_{4,a} = (x_1(x_1 - a), x_2(x_2 - a))$$
   gives independently
$$x_1 \in \{0, a\}, \qquad x_2 \in \{0, a\}.$$
   Thus
$$\mathbf{V}(\mathfrak{a}_{4,a}) = \{(0,0), (a,0), (0,a), (a,a)\}.$$

   For $a = 0$, the ideals become
$$\begin{aligned} \mathfrak{a}_{2,0} &= (x_2, x_1^2), \\ \mathfrak{a}_{3,0} &= (x_1^2, x_1x_2, x_2^2) = (x_1, x_2)^2, \\ \mathfrak{a}_{4,0} &= (x_1^2, x_2^2). \end{aligned}$$
   They all have the same zero set:
$$\mathbf{V}(\mathfrak{a}_{2,0}) = \mathbf{V}(\mathfrak{a}_{3,0}) = \mathbf{V}(\mathfrak{a}_{4,0}) = \{(0,0)\}.$$

   Thus, after the specialisation $a = 0$, respectively two, three and four points have the same underlying zero set.

   Let us now compute the quotient algebras. Modulo $\mathfrak{a}_{2,a}$, we have
$$x_2 = 0, \qquad x_1^2 = ax_1.$$
   Every polynomial therefore reduces to a linear combination of
$$1, \quad x_1.$$
   These classes are linearly independent. For $a \neq 0$, this follows by evaluating at $(0,0)$ and $(a,0)$; for $a = 0$, it is immediate from
$$A/\mathfrak{a}_{2,0} \simeq \Bbbk[x_1]/(x_1^2).$$

Hence
$$\{1, x_1\}$$
is a basis of $A/\mathfrak{a}_{2,a}$ for every $a \in \Bbbk$.

Modulo $\mathfrak{a}_{3,a}$, we have
$$x_1x_2 = 0, \qquad x_1^2 = ax_1, \qquad x_2^2 = ax_2.$$
Thus every polynomial reduces to a linear combination of
$$1, \quad x_1, \quad x_2.$$
For $a \neq 0$, their independence follows by evaluation at
$$(0,0), \quad (a,0), \quad (0,a).$$
For $a = 0$, these are the standard monomials modulo $(x_1, x_2)^2$. Consequently,
$$\{1, x_1, x_2\}$$
is a basis of $A/\mathfrak{a}_{3,a}$ for every $a \in \Bbbk$.

Finally, modulo $\mathfrak{a}_{4,a}$, we have
$$x_1^2 = ax_1, \qquad x_2^2 = ax_2.$$
Every monomial reduces to a linear combination of
$$1, \quad x_1, \quad x_2, \quad x_1x_2.$$
For $a \neq 0$, independence follows by evaluating at the four points of $\mathbf{V}(\mathfrak{a}_{4,a})$. For $a = 0$, these are the standard monomials modulo (x_1^2, x_2^2). Hence
$$\{1, x_1, x_2, x_1x_2\}$$
is a basis of $A/\mathfrak{a}_{4,a}$.

In conclusion,
$$\dim_{\Bbbk}(A/\mathfrak{a}_{r,a}) = r$$
for $r = 2, 3, 4$, independently of a.

5. *Relations among the generators.* For $\mathfrak{a}_{2,a}$, write
$$g_1 = x_2, \qquad g_2 = x_1(x_1 - a).$$
We have the relation
$$x_1(x_1 - a)g_1 - x_2g_2 = 0.$$
We claim that this relation generates all the relations between g_1 and g_2.

Indeed, suppose
$$ug_1 + vg_2 = 0.$$
Then
$$ux_2 = -vx_1(x_1 - a).$$
Since x_2 and $x_1(x_1 - a)$ are coprime, $x_1(x_1 - a)$ divides u. Thus there exists $h \in A$ such that
$$u = hx_1(x_1 - a).$$

Substituting, we obtain
$$v = -hx_2.$$
Hence
$$(u, v) = h\big(x_1(x_1 - a), -x_2\big).$$
For $\mathfrak{a}_{3,a}$, put
$$q_1 = x_1x_2, \qquad q_2 = x_1(x_1 - a), \qquad q_3 = x_2(x_2 - a).$$
We have two relations:
$$(a - x_1)q_1 + x_2q_2 = 0$$
and
$$(a - x_2)q_1 + x_1q_3 = 0.$$
We show that they generate all relations.
Suppose
$$uq_1 + vq_2 + wq_3 = 0.$$
Factoring x_1 from the first two terms gives
$$x_1\big(ux_2 + v(x_1 - a)\big) = -wx_2(x_2 - a).$$
Since x_1 is coprime to $x_2(x_2 - a)$, it follows that x_1 divides w. Write
$$w = x_1\beta$$
for some $\beta \in A$. Dividing the equality by x_1 gives
$$ux_2 + v(x_1 - a) + \beta x_2(x_2 - a) = 0,$$
and therefore
$$v(x_1 - a) = -x_2\big(u + \beta(x_2 - a)\big).$$
Since $x_1 - a$ and x_2 are coprime, x_2 divides v. Write
$$v = x_2\alpha$$
for some $\alpha \in A$. Substituting, we obtain
$$u + \alpha(x_1 - a) + \beta(x_2 - a) = 0,$$
or equivalently
$$u = \alpha(a - x_1) + \beta(a - x_2).$$
Thus
$$(u, v, w) = \alpha(a - x_1, x_2, 0) + \beta(a - x_2, 0, x_1).$$
Hence all relations are generated by
$$(a - x_1, x_2, 0) \qquad \text{and} \qquad (a - x_2, 0, x_1).$$
Finally, for $\mathfrak{a}_{4,a}$, put
$$h_1 = x_1(x_1 - a), \qquad h_2 = x_2(x_2 - a).$$

The relation

$$x_2(x_2 - a)h_1 - x_1(x_1 - a)h_2 = 0$$

generates all relations. Indeed, h_1 and h_2 are coprime, so the same argument as for $\mathfrak{a}_{2,a}$ applies.

We have therefore obtained the following generators of the modules of relations:

ideal	generating relations
$\mathfrak{a}_{2,a}$	$\big(x_1(x_1 - a), -x_2\big)$
$\mathfrak{a}_{3,a}$	$(a - x_1, x_2, 0), \quad (a - x_2, 0, x_1)$
$\mathfrak{a}_{4,a}$	$\big(x_2(x_2 - a), -x_1(x_1 - a)\big).$

In particular, the number of generators and the number of generating relations do not change when a is specialised to zero.

Exercise I.A.9. Assume that $\Bbbk$ is infinite. Show that an infinite strict subset $X \subsetneq \Bbbk$ is not an affine algebraic set.

Solution. The affine algebraic subsets of $\Bbbk$ are the subsets of the form $\mathbf{V}(f_1, \ldots, f_s)$. If all the f_i are zero, then $\mathbf{V}(f_1, \ldots, f_s) = \Bbbk$. Otherwise, one of the f_i is a non-zero polynomial, and $\mathbf{V}(f_1, \ldots, f_s)$ is contained in the finite set of roots of this polynomial. Hence an algebraic subset of $\Bbbk$ is either finite or equal to $\Bbbk$. Therefore an infinite strict subset of $\Bbbk$ is not algebraic.

I.A.2 From subsets to ideals

Given a subset $X \subset \Bbbk^n$, one defines the ideal

$$\mathbf{I}(X) = \{f \in A \mid f(a) = 0, \forall a \in X\}.$$

We call $\mathbf{I}(X)$ the ideal *associated with* X, or the ideal of *functions vanishing at* X.

Lemma I.A.10. *Ideals associated with subsets have the following properties.*

i) For all subsets $X \subset Y \subset \Bbbk^n$, we have $\mathbf{I}(X) \supset \mathbf{I}(Y)$.

ii) Given $X \subset \Bbbk^n$, we have $X \subset \mathbf{V}(\mathbf{I}(X))$. Equality holds if and only if X is an algebraic subset of $\Bbbk^n$.

iii) If $\mathfrak{a}$ is an ideal of A, then $\mathfrak{a} \subset \mathbf{I}(\mathbf{V}(\mathfrak{a}))$.

Proof. Item i) is obvious. For iii), let $f \in \mathfrak{a}$. Then, for any $a \in \mathbf{V}(\mathfrak{a})$, we have $f(a) = 0$. So $f \in \mathbf{I}(\mathbf{V}(\mathfrak{a}))$.

For ii), we let $a \in X$ and take $f \in \mathbf{I}(X)$. Then $f(a) = 0$ by definition. Since this happens for all $f \in \mathbf{I}(X)$, we have $a \in \mathbf{V}(\mathbf{I}(X))$. This shows that $X \subset \mathbf{V}(\mathbf{I}(X))$. If equality is attained, then by definition X is an algebraic set, actually an algebraic subset of $\Bbbk^n$. Conversely, if X is an algebraic subset of $\Bbbk^n$, then there is an ideal $\mathfrak{b}$ such that $X = \mathbf{V}(\mathfrak{b})$. Hence $\mathfrak{b} \subset \mathbf{I}(\mathbf{V}(\mathfrak{b})) = \mathbf{I}(X)$. So $\mathbf{V}(\mathbf{I}(X)) \subset \mathbf{V}(\mathfrak{b}) = X$. □

Example I.A.11. Let $X = \{0, 1\} \subset \Bbbk$. Then

$$\mathbf{I}(X) = (x(x - 1)).$$

Indeed, a polynomial vanishes at both 0 and 1 if and only if it is divisible by x and by $x - 1$, hence by $x(x - 1)$.

Example I.A.12. Let $X = \{(0,0)\} \subset \Bbbk^2$. Then

$$\mathbf{I}(X) = (x, y).$$

Indeed, $f \in \Bbbk[x, y]$ vanishes at the origin if and only if its constant term is zero, which is equivalent to saying that $f \in (x, y)$.

Exercise I.A.13. Compute $\mathbf{I}(X)$ for

$$X = \{(0,0), (1,0)\} \subset \Bbbk^2.$$

Solution. A polynomial $f \in \Bbbk[x, y]$ vanishes at both $(0,0)$ and $(1,0)$ if and only if its restriction $f(x, 0)$ is divisible by $x(x-1)$. Equivalently, the remainder of f modulo y lies in $(x(x-1)) \subset \Bbbk[x]$. This gives

$$\mathbf{I}(X) = (y, x(x-1)).$$

Exercise I.A.14. Let $X \subset \Bbbk^n$ and $Y \subset \Bbbk^n$. Show that

$$\mathbf{I}(X \cup Y) = \mathbf{I}(X) \cap \mathbf{I}(Y).$$

Solution. A polynomial f belongs to $\mathbf{I}(X \cup Y)$ if and only if it vanishes at every point of $X \cup Y$. This is equivalent to saying that f vanishes on X and on Y, namely that $f \in \mathbf{I}(X)$ and $f \in \mathbf{I}(Y)$. Therefore

$$\mathbf{I}(X \cup Y) = \mathbf{I}(X) \cap \mathbf{I}(Y).$$

I.B Morphisms of affine algebraic sets

Here we define morphisms of algebraic subsets as maps from one set to another, such that each coordinate of the target depends in a polynomial way on the coordinates of the source.

I.B.1 Morphisms of algebraic sets and polynomial mappings

Let us consider two affine algebraic sets

$$X \subset \Bbbk^n, \qquad Y \subset \Bbbk^m.$$

Definition I.B.1. A map $\varphi : X \longrightarrow Y$ is a *morphism of affine algebraic sets* if there exist polynomials

$$f_1, \ldots, f_m \in \Bbbk[x_1, \ldots, x_n]$$

such that

$$\varphi(a) = \big(f_1(a), \ldots, f_m(a)\big) \qquad \text{for all } a \in X.$$

In other words, a morphism is a map whose coordinate functions are given by polynomials on the ambient affine space.

Example I.B.2. Every polynomial map

$$\Bbbk^n \longrightarrow \Bbbk^m, \qquad a \longmapsto \big(f_1(a), \ldots, f_m(a)\big),$$

is a morphism. Its restriction to an affine algebraic subset is a morphism provided that its image is contained in the target algebraic set.

Example I.B.3. Let
$$X = \mathbf{V}(y - x^2) \subset \Bbbk^2.$$
The map
$$\alpha : \Bbbk \longrightarrow X, \qquad t \longmapsto (t, t^2),$$
is a morphism. The first projection
$$\pi : X \longrightarrow \Bbbk, \qquad (x, y) \longmapsto x,$$
is also a morphism, and
$$\pi \circ \alpha = \mathrm{id}_{\Bbbk}, \qquad \alpha \circ \pi = \mathrm{id}_X .$$
Thus the parabola is isomorphic, as an affine algebraic set, to the affine line.

I.B.2 Coordinate rings

Let $X \subset \Bbbk^n$ be an affine algebraic subset. The morphisms of algebraic sets $X \to \Bbbk$ are given by polynomial maps $\Bbbk^n \to \Bbbk$ of the form $(a_1, \ldots, a_n) \to f(a_1, \ldots, a_n)$. Replacing f by $f + g$, with $g \in \mathbf{I}(X)$, will not change the value of f at any point of X, so it is natural to associate with a morphism of algebraic sets $X \to \Bbbk$ an element of $\Bbbk[X] = \Bbbk[x_1, \ldots, x_n]/\mathbf{I}(X)$.

Definition I.B.4. Let $X \subset \Bbbk^n$ be an affine algebraic set. Its *coordinate ring* is
$$\Bbbk[X] := \Bbbk[x_1, \ldots, x_n]/\mathbf{I}(X).$$
An element of $\Bbbk[X]$ is called a *polynomial function* on X.

Proposition I.B.5. *Let $X \subset \Bbbk^n$ and $Y \subset \Bbbk^m$ be affine algebraic sets. Every morphism $\varphi : X \to Y$ induces a homomorphism of $\Bbbk$-algebras*
$$\varphi^* : \Bbbk[Y] \longrightarrow \Bbbk[X], \qquad h \longmapsto h \circ \varphi.$$
Moreover, the assignment $\varphi \mapsto \varphi^*$ gives a bijection
$$\mathrm{Hom}_{(\mathrm{AffAlg})_{\Bbbk}}(X, Y) \simeq \mathrm{Hom}_{(\Bbbk\text{-Alg})}\big(\Bbbk[Y], \Bbbk[X]\big).$$

Proof. Write
$$\varphi(a) = \big(f_1(a), \ldots, f_m(a)\big)$$
with $f_i \in \Bbbk[x_1, \ldots, x_n]$. If $h \in \Bbbk[y_1, \ldots, y_m]$, then
$$h \circ \varphi = h(f_1, \ldots, f_m)$$
is a polynomial function on X. If $h \in \mathbf{I}(Y)$, then h vanishes on Y, hence $h \circ \varphi$ vanishes on X. Therefore substitution induces a well-defined homomorphism
$$\varphi^* : \Bbbk[Y] \to \Bbbk[X].$$

Conversely, let
$$\psi : \Bbbk[Y] \longrightarrow \Bbbk[X]$$
be a homomorphism of $\Bbbk$-algebras. Denote by $\bar{y}_i \in \Bbbk[Y]$ the class of the coordinate y_i, and choose a polynomial $f_i \in \Bbbk[x_1, \ldots, x_n]$ representing $\psi(\bar{y}_i)$. Define
$$\varphi : X \longrightarrow \Bbbk^m, \qquad a \longmapsto \big(f_1(a), \ldots, f_m(a)\big).$$

For every $h \in \mathbf{I}(Y)$, the class of h in $\Bbbk[Y]$ is zero. Hence

$$h(f_1, \ldots, f_m) \in \mathbf{I}(X),$$

so

$$h\big(\varphi(a)\big) = 0 \qquad \text{for every } a \in X.$$

Thus $\varphi(a) \in Y$, and $\varphi : X \to Y$ is a morphism. By construction, $\varphi^* = \psi$. The two constructions are inverse to one another. $\square$

Remark I.B.6. The correspondence is contravariant: if

$$X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z$$

are morphisms, then

$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^*.$$

In particular, a morphism $\varphi : X \to Y$ is an isomorphism if and only if $\varphi^* : \Bbbk[Y] \to \Bbbk[X]$ is an isomorphism of $\Bbbk$-algebras.

Exercise I.B.7. Let

$$X = \mathbf{V}(y - x^2) \subset \Bbbk^2.$$

Compute the homomorphisms on coordinate rings induced by

$$\alpha : \Bbbk \to X, \qquad t \mapsto (t, t^2),$$

and by the projection $\pi : X \to \Bbbk$, $(x, y) \mapsto x$. Verify algebraically that they are inverse isomorphisms.

Solution. We have

$$\Bbbk[X] = \Bbbk[x, y]/(y - x^2) \simeq \Bbbk[x].$$

The pullback by α is

$$\alpha^* : \Bbbk[x, y]/(y - x^2) \longrightarrow \Bbbk[t], \qquad \bar{x} \longmapsto t, \quad \bar{y} \longmapsto t^2.$$

The pullback by π is

$$\pi^* : \Bbbk[t] \longrightarrow \Bbbk[x, y]/(y - x^2), \qquad t \longmapsto \bar{x}.$$

Since $\bar{y} = \bar{x}^2$ in $\Bbbk[X]$, these homomorphisms are inverse to one another.

Exercise I.B.8. Let $X \subset Y \subset \Bbbk^n$ be affine algebraic sets. Show that the inclusion $X \hookrightarrow Y$ is a morphism and that the induced homomorphism

$$\Bbbk[Y] \longrightarrow \Bbbk[X]$$

is surjective.

Solution. The inclusion is the restriction of the identity map of $\Bbbk^n$, hence is a morphism. Since $X \subset Y$, every polynomial vanishing on Y also vanishes on X, so

$$\mathbf{I}(Y) \subset \mathbf{I}(X).$$

Therefore the quotient map induces a surjection

$$\Bbbk[x_1, \ldots, x_n]/\mathbf{I}(Y) \longrightarrow \Bbbk[x_1, \ldots, x_n]/\mathbf{I}(X).$$

Exercise I.B.9. Consider

$$\nu : \Bbbk \longrightarrow \mathbf{V}(y^2 - x^3) \subset \Bbbk^2, \qquad t \longmapsto (t^2, t^3).$$

Show that ν is a bijective morphism, but that it is not an isomorphism of affine algebraic sets.

Solution. The map is a morphism and its image lies in the cusp because

$$(t^3)^2 = (t^2)^3.$$

It is injective: if $t^2 = u^2$ and $t^3 = u^3$, then either both are zero, or $t = t^3/t^2 = u^3/u^2 = u$. It is also surjective. Indeed, if (x, y) satisfies $y^2 = x^3$ and $x \neq 0$, then $t = y/x$ satisfies $t^2 = x$ and $t^3 = y$; if $x = 0$, then also $y = 0$.

The induced homomorphism is

$$\nu^* : \Bbbk[x, y]/(y^2 - x^3) \longrightarrow \Bbbk[t], \qquad \bar{x} \longmapsto t^2, \quad \bar{y} \longmapsto t^3.$$

Its image is $\Bbbk[t^2, t^3]$, which does not contain t. Hence ν^* is not surjective, so ν is not an isomorphism.

I.C Radicals and the Nullstellensatz

Further reading. For the classical affine theory, including the Nullstellensatz, see [Har77, Chapter I, §1]; for a more elementary presentation and further examples, see [Hul03] and [Hol12].

I.C.1 Radicals

Given a ring A, it is important to study the set of nilpotent elements, namely elements a such that $a^m = 0$ for some integer $m \geq 1$. If there is no non-zero nilpotent element, then A is said to be *reduced*.

Definition I.C.1. The *radical* of a ring A, or nilradical, is the set of nilpotent elements of A. It is denoted by $\mathrm{Rad}(A)$. More generally, given an ideal $\mathfrak{a}$ of A, we define the radical of $\mathfrak{a}$ as follows:

$$\sqrt{\mathfrak{a}} = \{a \in A \mid a^n \in \mathfrak{a} \text{ for some integer } n \geq 1\}.$$

An ideal $\mathfrak{a}$ is *radical* if $\mathfrak{a} = \sqrt{\mathfrak{a}}$.

Remark I.C.2. We have the following properties.

i) For any ideal $\mathfrak{a}$ of A, the radical $\sqrt{\mathfrak{a}}$ is an ideal of A.

ii) Any ideal $\mathfrak{a}$ of A is contained in its radical, $\mathfrak{a} \subset \sqrt{\mathfrak{a}}$.

iii) Given $X \subset \Bbbk^n$, the ideal $\mathbf{I}(X)$ is radical.

The first item follows by expanding sufficiently high powers of a sum. The second item is obvious. For the third item, given $n \in \mathbb{N}$ and $f \in A$ such that $f^n \in \mathbf{I}(X)$, we must have $f^n(a) = 0$ for all $a \in X$, so $f(a) = 0$ for all $a \in X$, hence $f \in \mathbf{I}(X)$. So $\sqrt{\mathbf{I}(X)} \subset \mathbf{I}(X)$.

Remark I.C.3. We have $\sqrt{(0)} = \mathrm{Rad}(A)$. Given an ideal $\mathfrak{a}$ of A, consider the ring $B = A/\mathfrak{a}$ and the canonical projection $\varphi : A \to B$. Then $\varphi^{-1}(\mathrm{Rad}(B)) = \sqrt{\mathfrak{a}}$.

Remark I.C.4. We note that, if $\mathfrak{a}$ is a prime ideal, then $\sqrt{\mathfrak{a}} = \mathfrak{a}$, i.e., $\mathfrak{a}$ is radical. Indeed, given $f \in A$, if $f^n \in \mathfrak{a}$ for some $n \geq 1$, then since $\mathfrak{a}$ is prime we have $f \in \mathfrak{a}$ or, when $n > 1$, $f^{n-1} \in \mathfrak{a}$, and continuing this way we finally get $f \in \mathfrak{a}$ anyway, so $\sqrt{\mathfrak{a}} \subset \mathfrak{a}$. Since the inclusion $\mathfrak{a} \subset \sqrt{\mathfrak{a}}$ holds for any ideal, we get $\mathfrak{a} = \sqrt{\mathfrak{a}}$ when $\mathfrak{a}$ is prime.

Example I.C.5. In $\Bbbk[x]$, we have

$$\sqrt{(x^m)} = (x)$$

for every $m \geq 1$. More generally, if

$$f = c\prod_{i=1}^{r}(x - a_i)^{m_i}$$

over an algebraically closed field, then

$$\sqrt{(f)} = \left(\prod_{i=1}^{r}(x - a_i)\right).$$

Example I.C.6. The so-called *ring of dual numbers* is

$$\Bbbk[x]/(x^2).$$

This ring is not reduced: the class of x is non-zero, but its square is zero. On the other hand, the ring $\Bbbk[x]/(x)$ is isomorphic to $\Bbbk$, hence is reduced.

Exercise I.C.7. Compute the radical of the ideal

$$\mathfrak{a} = (x^2, y^3) \subset \Bbbk[x, y].$$

Solution. We have $x^2 \in \mathfrak{a}$ and $y^3 \in \mathfrak{a}$, hence $x, y \in \sqrt{\mathfrak{a}}$. Therefore $(x, y) \subset \sqrt{\mathfrak{a}}$. Conversely, since $\mathfrak{a} \subset (x, y)$ and (x, y) is prime, hence radical, we have $\sqrt{\mathfrak{a}} \subset (x, y)$. Thus

$$\sqrt{(x^2, y^3)} = (x, y).$$

Exercise I.C.8. Decide whether the following rings are reduced:

$$\Bbbk[x]/(x^2), \qquad \Bbbk[x, y]/(xy), \qquad \Bbbk[x, y]/(x^2, xy).$$

Solution. The ring $\Bbbk[x]/(x^2)$ is not reduced because the class of x is nilpotent. The ring $\Bbbk[x, y]/(xy)$ is reduced if $\Bbbk[x, y]$ is reduced and (xy) is a radical ideal. Indeed,

$$(xy) = (x) \cap (y),$$

so (xy) is radical. Hence $\Bbbk[x, y]/(xy)$ is reduced. Finally, $\Bbbk[x, y]/(x^2, xy)$ is not reduced because the class of x is non-zero but satisfies $x^2 = 0$.

Exercise I.C.9. Compute $\sqrt{(x^2y^3)}$ in $\Bbbk[x, y]$.

Solution. Since $x^2y^3 \in (x^2y^3)$, we have $xy \in \sqrt{(x^2y^3)}$. Thus $(xy) \subset \sqrt{(x^2y^3)}$. Conversely, if $f \in \sqrt{(x^2y^3)}$, then $f^N \in (x^2y^3)$ for some N. Since (xy) is radical and contains (x^2y^3), we get $f \in (xy)$. Therefore

$$\sqrt{(x^2y^3)} = (xy).$$

I.C.2 Integral ring extensions

Given an inclusion of rings $A \subset B$, an element $b \in B$ is said to be *integral over* A if there is a monic polynomial $f \in A[x]$ such that $f(b) = 0$. If all elements of B are integral over A then B is said to be *integral over* A.

Lemma I.C.10. *Let* $A \subset B$ *be an inclusion of rings.*

1. *Assume that A and B are domains, with B integral over A. Then A is a field if and only if B is a field.*

2. *An element $b \in B$ is integral over A if and only if $A[b]$ is finitely generated as an A-module.*

Therefore, the A-integral elements of B are a subring of B.

Proof. We first prove the second assertion. If b is integral over A, then there is a monic relation

$$b^d + a_{d-1}b^{d-1} + \cdots + a_1 b + a_0 = 0, \qquad a_i \in A.$$

Hence every power b^m with $m \geq d$ can be rewritten as an A-linear combination of $1, b, \ldots, b^{d-1}$. Therefore $A[b]$ is generated as an A-module by $1, b, \ldots, b^{d-1}$.

Conversely, assume that $A[b]$ is finitely generated as an A-module, say by $e_1, \ldots, e_r$. For $1 \leq i \leq r$ there are coefficients $a_{i,j} \in A$, with $1 \leq j \leq r$, such that

$$be_i = \sum_{1 \leq j \leq r} a_{i,j} e_j, \qquad \text{for } 1 \leq i \leq r.$$

Thus the matrix

$$M = (b\delta_{i,j} - a_{i,j})_{1 \leq i,j \leq r}$$

annihilates the column vector $(e_1, \ldots, e_r)^{\mathrm{t}}$. Multiplying by the adjugate matrix, we see that $\det(M)$ annihilates every e_i, with $1 \leq i \leq r$, and hence all of $A[b]$. In particular, $\det(M) \cdot 1 = 0$. Since $\det(M)$ is a monic polynomial in b with coefficients in A, the element b is integral over A.

Now assume that B is integral over A. If A is a field, then every element $b \in B$ is algebraic over A. Since B is an integral domain, the inverse of any non-zero $b \in B$ belongs to the field $A(b)$ and is a polynomial in b with coefficients in A. Hence $b^{-1} \in B$, so B is a field.

Conversely, assume that B is a field and let $a \in A$ be non-zero. Then $a^{-1} \in B$. Since B is integral over A, there is a monic relation

$$(a^{-1})^d + c_{d-1}(a^{-1})^{d-1} + \cdots + c_1 a^{-1} + c_0 = 0, \qquad c_i \in A.$$

Multiplying by a^{d-1} gives

$$a^{-1} = -(c_{d-1} + c_{d-2}a + \cdots + c_0 a^{d-1}) \in A.$$

Thus every non-zero element of A is invertible, and A is a field.

Finally, if $b, c \in B$ are integral over A, then $A[b]$ is finite over A, and $A[b,c]$ is finite over $A[b]$, hence finite over A. Since $b + c$, bc and $-b$ belong to $A[b,c]$, the criterion just proved shows that they are integral over A. Therefore the integral elements form a subring of B. $\square$

Example I.C.11. The element $\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ is integral over $\mathbb{Q}$, since it satisfies the monic equation $T^2 - 2 = 0$. More generally, every element algebraic over a field $\Bbbk$ is integral over $\Bbbk$.

On the other hand, $x^{-1} \in \Bbbk[x, x^{-1}]$ is not integral over $\Bbbk[x]$. Indeed, a monic equation for x^{-1} over $\Bbbk[x]$, multiplied by a suitable power of x, would imply that $x^{-1} \in \Bbbk[x]$, which is impossible.

Exercise I.C.12. Show that if $b \in B$ is integral over A, then the image of b in $B/\mathfrak{b}$ is integral over the image of A, for every ideal $\mathfrak{b} \subset B$.

Solution. If b satisfies a monic equation

$$b^d + a_{d-1}b^{d-1} + \cdots + a_0 = 0, \qquad a_i \in A,$$

then reducing the same equation modulo $\mathfrak{b}$ gives a monic equation for the image of b in $B/\mathfrak{b}$ with coefficients in the image of A.

Lemma I.C.13. *Let κ be a field extension of a field $\Bbbk$ which is finitely generated as a $\Bbbk$-algebra. Then κ is a finite field extension of $\Bbbk$.*

Proof. We prove the statement by induction on the number of algebra generators. Write

$$\kappa = \Bbbk[\alpha_1, \dots, \alpha_n].$$

If all the α_i are algebraic over $\Bbbk$, then κ is a finite field extension of $\Bbbk$, and there is nothing to prove.

Assume, for a contradiction, that one of the generators is transcendental over $\Bbbk$; after reordering, say α_n is transcendental. Put

$$K = \Bbbk(\alpha_n).$$

Since

$$\kappa = K[\alpha_1, \dots, \alpha_{n-1}],$$

the induction hypothesis, applied over the field K, shows that $\alpha_1, \dots, \alpha_{n-1}$ are algebraic over K. Let $f_i \in K[T]$ be a monic polynomial satisfied by α_i, for $1 \le i \le n-1$. Choose a non-zero polynomial $g \in \Bbbk[\alpha_n]$ clearing all denominators of the coefficients of these polynomials. Then the α_i are integral over

$$A = \Bbbk[\alpha_n, g^{-1}].$$

Hence

$$A[\alpha_1, \dots, \alpha_{n-1}]$$

is integral over A. Moreover

$$\kappa = \Bbbk[\alpha_n][\alpha_1, \dots, \alpha_{n-1}] \subset A[\alpha_1, \dots, \alpha_{n-1}] \subset \kappa,$$

so equality holds and κ is integral over A.

Since κ is a field, the previous lemma implies that A is a field. But this is impossible: as α_n is transcendental over $\Bbbk$, the ring $\Bbbk[\alpha_n, g^{-1}]$ is a localisation of the polynomial ring $\Bbbk[\alpha_n]$ obtained by inverting only the powers of g, and it is not a field. Indeed, one can choose an irreducible polynomial in $\Bbbk[\alpha_n]$ which does not divide g, and its class is then non-invertible in $\Bbbk[\alpha_n, g^{-1}]$.

This contradiction shows that all the α_i are algebraic over $\Bbbk$. Therefore κ is finite over $\Bbbk$. □

I.C.3 Hilbert's Nullstellensatz

Theorem I.C.14. *Let $\Bbbk$ be an algebraically closed field and put $A = \Bbbk[x_1, \dots, x_n]$.*

i) Any maximal ideal of A is of the form $\mathfrak{m}_a = (x_1 - a_1, \dots, x_n - a_n)$ for some $a = (a_1, \dots, a_n) \in \Bbbk^n$.

ii) For any ideal $\mathfrak{a} \neq A$, we have $\mathbf{V}(\mathfrak{a}) \neq \emptyset$.

iii) Then, for any ideal $\mathfrak{a} \subset A$, we have $\mathbf{I}(\mathbf{V}(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

Proof. Let us prove i). First, we show that $\mathfrak{m}_a$ is maximal. To see this, note that the evaluation map

$$\varphi_a : A \to \Bbbk, \qquad f \mapsto f(a)$$

is a surjective ring homomorphism. So $\ker(\varphi_a)$ is a maximal ideal. It turns out that $\ker(\varphi_a) = \mathfrak{m}_a$. Indeed, it is clear that $\mathfrak{m}_a$ lies in $\ker(\varphi_a)$, for $\varphi_a(x_i - a_i) = 0$ for $1 \le i \le n$. Conversely, if $f \in \ker(\varphi_a)$, then $f(a) = 0$, i.e. the polynomial $f(x + a)$ has no constant term. Hence $f(x + a)$ lies in the ideal $(x_1, \dots, x_n)$, and therefore f lies in $(x_1 - a_1, \dots, x_n - a_n) = \mathfrak{m}_a$.

The main point of i) is that any maximal ideal $\mathfrak{m} \subset A$ is of the form $\mathfrak{m} = \mathfrak{m}_a$, for some $a \in \Bbbk^n$. For this, let $\kappa = A/\mathfrak{m}$, write $\pi : A \to \kappa$ for the canonical projection and $\varphi : \Bbbk \to \kappa$ for the map given by the inclusion of $\Bbbk$ in A as the set of constant polynomials, composed with π.

The morphism φ is necessarily injective, so κ is a field extension of $\Bbbk$. Also, the field κ is finitely generated as a $\Bbbk$-algebra, as A itself is generated by $x_1, \dots, x_n$ as a $\Bbbk$-algebra and κ is a quotient of A. Therefore, by Lemma I.C.13, κ is a finite field extension of $\Bbbk$. So, since $\Bbbk$ is algebraically closed, $\varphi : \Bbbk \to \kappa$ must be an isomorphism. Hence, denoting by b_i the image of x_i in κ, and $a_i = \varphi^{-1}(b_i)$, for $1 \le i \le n$, we have $\pi(x_i - a_i) = 0$ so $x_i - a_i \in \mathfrak{m}$. Hence $\mathfrak{m}_a \subset \mathfrak{m}$. But $\mathfrak{m}_a$ is maximal, and $\mathfrak{m} \neq A$, so $\mathfrak{m} = \mathfrak{m}_a$.

Let us show ii). Consider an ideal $\mathfrak{a} \subsetneq A$. Then, there is a maximal ideal containing $\mathfrak{a}$, by Krull's theorem, based on Zorn's lemma. Therefore, there is $a \in \Bbbk^n$ such that $\mathfrak{a} \subset \mathfrak{m}_a$. Hence a lies in $\mathbf{V}(\mathfrak{a})$, which is thus non-empty.

Finally, we prove iii). Let $f \in \sqrt{\mathfrak{a}}$. Then $f^n \in \mathfrak{a}$ for some $n \in \mathbb{N}$. Take $a \in \mathbf{V}(\mathfrak{a})$. Then $f^n(a) = 0$, so $f(a) = 0$. This happens for all $a \in \mathbf{V}(\mathfrak{a})$, so $f \in \mathbf{I}(\mathbf{V}(\mathfrak{a}))$.

To prove the other inclusion, we use Rabinowitsch's trick. Consider $f \in \mathbf{I}(\mathbf{V}(\mathfrak{a}))$, so $f(a) = 0$, for all $a \in \mathbf{V}(\mathfrak{a})$. Consider the ideal

$$\mathfrak{b} = \mathfrak{a} + (ft - 1) \subset A[t] = \Bbbk[x_1, \dots, x_n, t].$$

We claim that $\mathbf{V}(\mathfrak{b}) = \emptyset$. Indeed, if $(a, \lambda) \in \Bbbk^{n+1}$ were a common zero of $\mathfrak{b}$, then $a \in \mathbf{V}(\mathfrak{a})$ and $f(a)\lambda - 1 = 0$. But f vanishes on $\mathbf{V}(\mathfrak{a})$ by assumption, so $f(a) = 0$, a contradiction.

By ii), applied to the polynomial ring $A[t]$, since $\mathbf{V}(\mathfrak{b}) = \emptyset$, the ideal $\mathfrak{b}$ must be the whole ring $A[t]$. Hence there exist $h_1, \dots, h_s \in \mathfrak{a}$ and $q_1, \dots, q_s, q \in A[t]$ such that

$$1 = q_1 h_1 + \cdots + q_s h_s + q(ft - 1).$$

Passing to the quotient ring $(A/\mathfrak{a})[t]$, the first terms disappear. If we denote by $\overline{f}$ the image of f in $A/\mathfrak{a}$, we obtain

$$1 = \overline{q}(\overline{f}t - 1) \qquad \text{in } (A/\mathfrak{a})[t].$$

Write

$$\overline{q} = a_0 + a_1 t + \cdots + a_N t^N, \qquad a_i \in A/\mathfrak{a},$$

with $a_N \neq 0$ if $\overline{q} \neq 0$. Then

$$1 = (a_0 + a_1 t + \cdots + a_N t^N)(\overline{f}t - 1).$$

Comparing the coefficient of t^0, we get

$$-a_0 = 1,$$

hence $a_0 = -1$. Comparing the coefficient of t^1, we get

$$a_0 \overline{f} - a_1 = 0,$$

hence $a_1 = a_0 \overline{f} = -\overline{f}$. Comparing the coefficient of t^2, we get

$$a_1 \overline{f} - a_2 = 0,$$

hence $a_2 = -\overline{f}^2$. Continuing in the same way, we obtain

$$a_i = -\overline{f}^i \qquad \text{for } 0 \le i \le N.$$

Finally, comparing the coefficient of t^{N+1} gives

$$a_N \overline{f} = 0.$$

Since $a_N = -\overline{f}^N$, we get

$$\overline{f}^{N+1} = 0$$

in $A/\mathfrak{a}$. Therefore $f^{N+1} \in \mathfrak{a}$, and hence $f \in \sqrt{\mathfrak{a}}$. $\square$

Example I.C.15. Assume that $\Bbbk$ is algebraically closed. In $\Bbbk[x]$, the ideal

$$(x^2(x-1)^3)$$

defines the same algebraic set as its radical:

$$\mathbf{V}(x^2(x-1)^3) = \{0,1\} = \mathbf{V}(x(x-1)).$$

The Nullstellensatz says precisely that

$$\mathbf{I}(\{0,1\}) = (x(x-1)) = \sqrt{(x^2(x-1)^3)}.$$

Example I.C.16. In $\Bbbk[x,y]$, with $\Bbbk$ algebraically closed, the ideals

$$(x^2, y) \qquad \text{and} \qquad (x, y)$$

define the same algebraic set, namely the origin. The first ideal is not radical, and its radical is (x,y).

Exercise I.C.17. Let $\Bbbk$ be algebraically closed. Compute $\mathbf{I}(\mathbf{V}(\mathfrak{a}))$ for

$$\mathfrak{a} = (x^2, y^3) \subset \Bbbk[x,y].$$

Solution. We have

$$\mathbf{V}(\mathfrak{a}) = \{(0,0)\}.$$

Hence $\mathbf{I}(\mathbf{V}(\mathfrak{a})) = \mathbf{I}(\{(0,0)\}) = (x,y)$. Equivalently, by the Nullstellensatz,

$$\mathbf{I}(\mathbf{V}(\mathfrak{a})) = \sqrt{(x^2, y^3)} = (x,y).$$

Exercise I.C.18. Let $\Bbbk$ be algebraically closed and let

$$\mathfrak{a} = (xy, xz) \subset \Bbbk[x,y,z].$$

Describe $\mathbf{V}(\mathfrak{a})$ geometrically and compute $\sqrt{\mathfrak{a}}$.

Solution. The equations are $xy = 0$ and $xz = 0$, i.e. $xy = xz = 0$. Thus either $x = 0$, or both $y = 0$ and $z = 0$. Therefore

$$\mathbf{V}(\mathfrak{a}) = \mathbf{V}(x) \cup \mathbf{V}(y,z),$$

the union of the plane $x = 0$ and the x-axis. Correspondingly,

$$\sqrt{\mathfrak{a}} = (x) \cap (y,z).$$

Since $(x) \cap (y,z) = (xy, xz)$, the ideal $\mathfrak{a}$ is already radical.

I.C.4 Geometrically regular rational functions

We introduce the field of rational functions in n variables

$$K = \Bbbk(x_1, \ldots, x_n).$$

It consists of equivalence classes of fractions u/v, where $u, v \in A = \Bbbk[x_1, \ldots, x_n]$ and $v \neq 0$, with

$$\frac{u}{v} = \frac{u'}{v'} \qquad \Longleftrightarrow \qquad uv' = u'v.$$

A chosen fraction u/v defines a $\Bbbk$-valued function away from $\mathbf{V}(v)$. To indicate that it need not be defined everywhere, we sometimes write

$$\frac{u}{v} : \Bbbk^n \dashrightarrow \Bbbk.$$

We shall also consider the same fraction over an algebraic closure $\overline{\Bbbk}$.

Definition I.C.19. Let $P \subset \overline{\Bbbk}^n$. A rational function $g \in K$ is *geometrically regular on P* if it admits a representation

$$g = \frac{u}{v}, \qquad u, v \in A,$$

such that $v(a) \neq 0$ for every $a \in P$. Note that the condition concerns the rational function and not a particular fraction representing it. For instance, $x_1/x_1 = 1$ is geometrically regular everywhere, although the displayed fraction is not defined along $\mathbf{V}(x_1)$.

For $f \in A$, put

$$D(f) = \overline{\Bbbk}^n \setminus \mathbf{V}(f).$$

Every fraction of the form u/f^m, with $u \in A$ and $m \geq 0$, is geometrically regular on $D(f)$. The converse is also true.

Lemma I.C.20. *Let $f \in A$. A rational function $g \in K$ is geometrically regular on $D(f)$ if and only if*

$$g = \frac{u}{f^m}$$

for some $u \in A$ and $m \geq 0$.

Proof. One implication is immediate. Conversely, write $g = a/b$, with $a, b \in A$, $b \neq 0$, and with a and b having no common irreducible factor. Since g is geometrically regular on $D(f)$, we may choose this representation so that b has no zero on $D(f)$. Thus

$$\mathbf{V}(b) \subset \mathbf{V}(f)$$

in $\overline{\Bbbk}^n$. By the Nullstellensatz, there is an integer $N \geq 1$ such that $f^N \in (b)$. Consequently $f^N = bc$ for some $c \in A$, and hence

$$g = \frac{a}{b} = \frac{ac}{f^N}.$$

□

Taking $f = 1$, we obtain that the rational functions geometrically regular on all of $\overline{\Bbbk}^n$ are precisely the polynomial functions.

We now pass from affine space to an affine algebraic set. Let $X = \mathbf{V}(\mathfrak{a}) \subset \overline{\Bbbk}^n$, where $\mathfrak{a} \subset A$ is prime. Then

$$\Bbbk[X] = A/\mathfrak{a}$$

is an integral domain.

Definition I.C.21. The fraction field

$$K(X) = \mathrm{Frac}(\Bbbk[X])$$

is called the *field of rational functions on* X. A rational function $g \in K(X)$ is geometrically regular on a subset $P \subset X$ if it admits a representation

$$g = \frac{u}{v}, \qquad u, v \in \Bbbk[X],$$

such that $v(p) \neq 0$ for every $p \in P$.

If $f \in \Bbbk[X]$, we write

$$D_X(f) = \{p \in X \mid f(p) \neq 0\}.$$

Choosing a polynomial representative $\widetilde{f} \in A$, this is simply

$$D_X(f) = X \cap D(\widetilde{f}),$$

and the right-hand side does not depend on the chosen representative.

Exactly as above, a rational function on X is geometrically regular on $D_X(f)$ if and only if it can be written

$$\frac{u}{f^m}, \qquad u \in \Bbbk[X], \quad m \geq 0.$$

Later, this construction will be formulated systematically using localisation. For the moment, the important point is geometric: allowing powers of f in the denominator gives precisely the rational functions which remain regular wherever f does not vanish.

I.D Noether normalisation

Further reading. See [Har77, Chapter I, §1] for the classical geometric use of Noether normalisation, and [Liu02] for a systematic algebraic treatment.

Let $A \to B$ be a ring homomorphism. Then B is equipped with an A-module structure. We say that B is *finite* over A if B is a finitely generated A-module. We will use the notion of *algebraically independent elements* $f_1, \ldots, f_n$ of a $\Bbbk$-algebra A, namely, there is no non-zero polynomial $g \in \Bbbk[x_1, \ldots, x_n]$ such that $g(f_1, \ldots, f_n) = 0$.

In this case, the subring $\Bbbk[f_1, \ldots, f_n] \subset A$ generated by these elements is isomorphic to $\Bbbk[x_1, \ldots, x_n]$, where $x_1, \ldots, x_n$ are indeterminates. Indeed, any element in the kernel of the map $\Bbbk[x_1, \ldots, x_n] \to A$ sending x_i to f_i for $1 \leq i \leq n$ is a polynomial $g \in \Bbbk[x_1, \ldots, x_n]$ such that $g(f_1, \ldots, f_n) = 0$.

I.D.1 The algebraic statement

Theorem I.D.1. *Let $\Bbbk$ be a field and let $B \neq 0$ be a finitely generated $\Bbbk$-algebra. Then there exist an integer m and algebraically independent elements $y_1, \ldots, y_m \in B$ such that B is finite over $\Bbbk[y_1, \ldots, y_m]$.*

Proof. Since B is a finitely generated $\Bbbk$-algebra, there is a surjection $\varphi : \Bbbk[x_1, \ldots, x_n] \to B$, where n is the number of generators of B as a $\Bbbk$-algebra and $x_1, \ldots, x_n$ are indeterminates. We write simply $\Bbbk[x] = \Bbbk[x_1, \ldots, x_n]$. Then we have $B \simeq \Bbbk[x]/\mathfrak{a}$, where $\mathfrak{a} = \ker(\varphi)$. We replace B with $\Bbbk[x]/\mathfrak{a}$ and we prove the statement by induction on n. If $n = 0$, then $B \simeq \Bbbk$ and there is nothing to prove. Also, if $\mathfrak{a} = 0$, then $B \simeq \Bbbk[x_1, \ldots, x_n]$ and the statement is clear.

So let $n \geq 1$, $\mathfrak{a} \neq 0$. We may assume that the statement holds for $\Bbbk$-algebras generated by $n-1$ elements. Consider a non-zero element $f \in \mathfrak{a}$ and write

$$f = \sum_{\mu \in \mathbb{N}^n} a_\mu x^\mu, \qquad \text{for some } a_\mu \in \Bbbk, \text{ for all } \mu \in \mathbb{N}^n, \text{ with } x^\mu = x_1^{\mu_1} \cdots x_n^{\mu_n}.$$

For every $(\nu_1, \dots, \nu_{n-1}) \in \mathbb{N}^{n-1}$, we consider $\nu = (\nu_1, \dots, \nu_{n-1}, 1) \in \mathbb{N}^n$ and we introduce:

$$s_i = x_i - x_n^{\nu_i}, \qquad \text{for } 1 \leq i \leq n-1.$$

We write $s = (s_1, \dots, s_{n-1})$. Since for $1 \leq i \leq n-1$, x_i can be expressed in terms of s_i and x_n, we have

$$\Bbbk[x] = \Bbbk[s][x_n].$$

The point is that, for a suitable choice of ν, writing f in the variables $s_1, \dots, s_{n-1}, x_n$ results in a monic polynomial in x_n, up to a non-zero scalar. Indeed, we consider the lexicographic order on $\mathbb{N}^n$, so that for any $\mu, \mu' \in \mathbb{N}^n$, we let $\mu < \mu'$ if there is $1 \leq j \leq n$ such that $\mu_i = \mu'_i$ for $1 \leq i \leq j-1$ and $\mu_j < \mu'_j$. Then we put $d = \deg(f)$, $N = d+1$ and we let

$$\mu^0 = \max\{\mu \in \mathbb{N}^n \mid a_\mu \neq 0\}, \qquad \nu_i = N^{n-i}, \qquad 1 \leq i \leq n.$$

We claim that, after the change of variables

$$x_i = s_i + x_n^{\nu_i}, \qquad 1 \leq i \leq n-1,$$

the polynomial $f(x_1, \dots, x_n)$, viewed as an element of $\Bbbk[s_1, \dots, s_{n-1}][x_n]$, is monic in x_n, up to the coefficient a_{μ^0}. More precisely,

$$f(x_1, \dots, x_n) = f(s_1 + x_n^{\nu_1}, \dots, s_{n-1} + x_n^{\nu_{n-1}}, x_n) = \sum_{\mu \in \mathbb{N}^n} a_\mu \prod_{1 \leq i \leq n-1} (s_i + x_n^{\nu_i})^{\mu_i} x_n^{\mu_n}.$$

In this expression, the largest power of x_n contributed by the monomial x^μ is $x_n^{\langle \nu | \mu \rangle}$, where

$$\langle \nu \mid \mu \rangle = \nu_1 \mu_1 + \cdots + \nu_{n-1} \mu_{n-1} + \mu_n.$$

Now, take $\mu \neq \mu^0$ with $a_\mu \neq 0$. We have $\mu < \mu^0$, so there is $1 \leq j \leq n$ such that $\mu_i = \mu_i^0$ for $1 \leq i \leq j-1$ and $\mu_j < \mu_j^0$. Then our choice of ν gives

$$\langle \nu \mid \mu^0 - \mu \rangle = (\mu_j^0 - \mu_j) N^{n-j} + \sum_{j+1 \leq i \leq n} (\mu_i^0 - \mu_i) N^{n-i}.$$

The first term is at least N^{n-j}. The second term has absolute value at most

$$d(N^{n-j-1} + \cdots + 1) = d \frac{N^{n-j} - 1}{N - 1} = N^{n-j} - 1,$$

because $d = N - 1$. Therefore

$$\langle \nu \mid \mu^0 - \mu \rangle > 0,$$

hence $\langle \nu \mid \mu \rangle < \langle \nu \mid \mu^0 \rangle$. In conclusion, only μ^0 contributes to the coefficient of the highest power of x_n, namely $x_n^{\langle \nu | \mu^0 \rangle}$, and the coefficient of this power in the expression of $f(x_1, \dots, x_n)$ in the variables $s_1, \dots, s_{n-1}, x_n$ is a_{μ^0}. Dividing by this non-zero scalar, we obtain a monic polynomial in x_n.

Having set up this expression, we consider $\mathfrak{b} = \mathfrak{a} \cap \Bbbk[s]$ and $C = \Bbbk[s]/\mathfrak{b}$. Since C is generated by $n-1$ elements, by induction, there are algebraically independent elements $y_1, \dots, y_m$ of C such that C is finite over $\Bbbk[y] = \Bbbk[y_1, \dots, y_m]$. By construction, we have an injection $C \hookrightarrow B$, and B is generated by the image b_n of x_n in B, as a C-algebra. Since the equation $f = 0$ gives a monic equation in x_n, the element b_n is integral over C, so B is finitely generated as a C-module by Lemma I.C.10. Also, C is finitely generated as a $\Bbbk[y]$-module, so B is finitely generated as a $\Bbbk[y]$-module too. $\square$

Example I.D.2. Let
$$B = \Bbbk[x, y]/(y^2 - x^3).$$
The image of x is algebraically independent over $\Bbbk$, and the image of y is integral over $\Bbbk[x]$, since it satisfies $T^2 - x^3 = 0$. Hence B is finite over $\Bbbk[x]$.

Example I.D.3. Let
$$B = \Bbbk[x, y]/(xy - 1).$$
This algebra is isomorphic to $\Bbbk[x, x^{-1}]$. It is not finite over $\Bbbk[x]$, since x^{-1} is not integral over $\Bbbk[x]$. However, Noether normalisation says that, after choosing a better coordinate, B is finite over a polynomial subring in one variable.

Exercise I.D.4. Show that $\Bbbk[x, y]/(y^2 - x)$ is finite over $\Bbbk[y]$ and also finite over $\Bbbk[x]$. Compare the two module structures.

Solution. Let $B = \Bbbk[x, y]/(y^2 - x)$. Since $x = y^2$ in B, we have $B = \Bbbk[y]$. Therefore B is finite over $\Bbbk[y]$, in fact free of rank 1. It is also finite over $\Bbbk[x]$: the element y satisfies the monic equation $T^2 - x = 0$, so B is generated by $1, y$ as a $\Bbbk[x]$-module. Thus the first module structure has rank 1, while the second one has rank 2.

I.D.2 Geometric version of Noether normalisation

Consider an ideal $\mathfrak{a} \subset \Bbbk[x_1, \ldots, x_n]$ and the algebraic set $X = \mathbf{V}(\mathfrak{a}) \subset \Bbbk^n$. Put
$$B = \Bbbk[x_1, \ldots, x_n]/\mathfrak{a}.$$
By Noether normalisation, there are algebraically independent elements $y_1, \ldots, y_m \in B$ such that, if
$$\Bbbk[y_1, \ldots, y_m] \subset B,$$
then B is finite as an A-module. In other words, setting $A = \Bbbk[t_1, \ldots, t_m]$, where $t_1, \ldots, t_m$ are indeterminates, we have an injective morphism of rings $A \to B$ defined by sending t_i to y_i for $1 \leq i \leq m$, and B is a finitely generated A-module.

Choose polynomial representatives $f_1, \ldots, f_m \in \Bbbk[x_1, \ldots, x_n]$ of $y_1, \ldots, y_m$. These functions induce a map of algebraic sets
$$f : X \to \Bbbk^m, \qquad b \mapsto (f_1(b), \ldots, f_m(b)).$$

Proposition I.D.5. *For all $a \in \Bbbk^m$, the fibre $f^{-1}(a)$ is finite. If the field $\Bbbk$ is algebraically closed, this fibre is also non-empty.*

Proof. Let $a = (a_1, \ldots, a_m) \in \Bbbk^m$. We know that B is finitely generated as an A-module. For $1 \leq j \leq n$, the image $\bar{x}_j$ of x_j in B is therefore integral over A. Hence, for each j, there is a monic polynomial
$$h_j = h_j(t_1, \ldots, t_m)(T) \in A[T]$$
such that
$$h_j(y_1, \ldots, y_m)(\bar{x}_j) = 0$$
in B. By mapping t_i to f_i, for $1 \leq i \leq m$, we get a polynomial
$$H_j = h_j(f_1, \ldots, f_m)(T) \in \Bbbk[x_1, \ldots, x_n, T]$$
which is monic in T and satisfies
$$H_j(x_1, \ldots, x_n)(x_j) \in \mathfrak{a}.$$

Now let $b = (b_1, \dots, b_n) \in f^{-1}(a)$. Then $b \in X$, so $g(b) = 0$ for all $g \in \mathfrak{a}$, and also

$$f_i(b) = a_i \qquad \text{for } 1 \le i \le m.$$

Since $H_j(x_1, \dots, x_n)(x_j) \in \mathfrak{a}$, evaluating at b gives

$$H_j(b_1, \dots, b_n)(b_j) = 0, \qquad \text{for } 1 \le j \le n.$$

In other words, if we set $\ell_j(T) = h_j(a_1, \dots, a_m)(T) \in \Bbbk[T]$, for $1 \le j \le n$, then

$$\ell_j(b_j) = H_j(b_1, \dots, b_n)(b_j) = h_j(f_1(b), \dots, f_m(b))(b_j) = 0, \qquad \text{for } 1 \le j \le n.$$

Since, for $1 \le j \le n$, each $\ell_j(T) = h_j(a_1, \dots, a_m)(T)$ is a monic polynomial in $\Bbbk[T]$, it has only finitely many roots. Hence each coordinate b_j can take only finitely many values. Therefore there are only finitely many possible points $b = (b_1, \dots, b_n)$ in the fibre $f^{-1}(a)$.

Let us now prove non-emptiness, assuming that $\Bbbk$ is algebraically closed. By the Nullstellensatz, it is enough to prove that

$$\mathfrak{a} + (f_1 - a_1, \dots, f_m - a_m) \neq \Bbbk[x_1, \dots, x_n].$$

Equivalently, it is enough to prove that

$$(\bar{f}_1 - a_1, \dots, \bar{f}_m - a_m)B \neq B. \tag{I.1}$$

Let $\mathfrak{m}_a = (y_1 - a_1, \dots, y_m - a_m) \subset A$. Under the inclusion $A \subset B$, the ideal $\mathfrak{m}_a B$ is exactly

$$(\bar{f}_1 - a_1, \dots, \bar{f}_m - a_m)B.$$

Since B is a finite A-module and $B \neq 0$, Nakayama's lemma (see the next exercise) gives $\mathfrak{m}_a B \neq B$. Therefore (I.1) holds, and the proof is complete. $\square$

Exercise I.D.6. Prove the following form of Nakayama's lemma. Let $A \to B$ be an injective homomorphism of rings, with $B \neq 0$, and assume that B is a finitely generated A-module. If $\mathfrak{a} \subsetneq A$ is a proper ideal, then $\mathfrak{a}B \neq B$.

Solution. Since B is a finitely generated A-module, there are $b_1, \dots, b_n \in B$ such that $B = b_1 A + \cdots + b_n A$. Suppose, for a contradiction, that $\mathfrak{a}B = B$. Then there are elements $a_{i,j} \in \mathfrak{a}$, for $1 \le i, j \le n$, such that

$$b_i = \sum_{1 \le j \le n} a_{i,j} b_j, \qquad \text{for } 1 \le i \le n.$$

Hence the matrix

$$M = (\delta_{i,j} - a_{i,j})_{1 \le i,j \le n}$$

annihilates the column vector $(b_1, \dots, b_n)^{\mathrm{t}}$. By the determinant trick, $d = \det(M)$ annihilates each b_i, and therefore annihilates B. Since $A \to B$ is injective, the A-module B is faithful: if $cB = 0$, then in particular $c \cdot 1_B = 0$, so $c = 0$ in A. Thus $d = 0$ in A.

On the other hand, expanding the determinant gives

$$d = 1 + \alpha$$

for some $\alpha \in \mathfrak{a}$. Since $d = 0$, we get $1 = -\alpha \in \mathfrak{a}$, contradicting the fact that $\mathfrak{a}$ is proper. Hence $\mathfrak{a}B \neq B$.

Exercise I.D.7. Again in the spirit of Example I.D.3.

$$B = \Bbbk[x, y]/(xy - 1) \simeq \Bbbk[x, x^{-1}]$$

and set $u = x + y = x + x^{-1}$.

i) Show that the homomorphism $\Bbbk[u] \hookrightarrow B$ is injective.

ii) Show that x is integral over $\Bbbk[u]$ and that B is generated by $1, x$ as a $\Bbbk[u]$-module.

iii) Describe the fibres of the corresponding map $\mathbf{V}(xy-1) \to \Bbbk$, $(x,y) \mapsto x+y$.

Solution. The element x is not integral over $\Bbbk[x]$. Indeed, if there was a monic polynomial $f \in \Bbbk[x]$ of degree m vanishing at x^{-1}, then looking at $x^{m-1}f$ we see that x^{-1} would then lie in $\Bbbk[x]$, which is not the case. Then, $u = x + x^{-1}$ is also transcendental: if $P(u) = 0$ for a non-zero polynomial P, multiplication by a sufficiently high power of x would give a non-zero polynomial relation for x. ***

Since $y = u - x$ and $xy = 1$, the element x satisfies

$$x^2 - ux + 1 = 0.$$

Thus x is integral over $\Bbbk[u]$ and every element of $B = \Bbbk[u][x]$ reduces to $a(u) + b(u)x$. Over $u = c$, the possible values of x are the roots of

$$x^2 - cx + 1 = 0,$$

and $y = x^{-1}$. Hence every fibre has at most two points; over an algebraically closed field it has two points counted with multiplicity.

Example I.D.8 (An elliptic curve as a branched double cover). Assume that $\Bbbk$ is algebraically closed of characteristic different from 2, and choose $\lambda \in \Bbbk \setminus \{0,1\}$. Consider

$$B = \Bbbk[x,y]/\big(y^2 - x(x-1)(x-\lambda)\big).$$

The image of x is algebraically independent over $\Bbbk$, while the image of y is integral over $\Bbbk[x]$, since it satisfies the monic equation

$$T^2 - x(x-1)(x-\lambda) = 0.$$

Hence B is finite over $\Bbbk[x]$. More precisely,

$$B = \Bbbk[x] \oplus \Bbbk[x]y$$

as a $\Bbbk[x]$-module. Geometrically, the corresponding map

$$\mathbf{V}\big(y^2 - x(x-1)(x-\lambda)\big) \longrightarrow \Bbbk, \qquad (x,y) \longmapsto x,$$

is generically two-to-one. Indeed, the fibre over $a \in \Bbbk$ is determined by

$$y^2 = a(a-1)(a-\lambda).$$

It consists of two distinct points when

$$a \notin \{0,1,\lambda\},$$

and of a single point when $a \in \{0,1,\lambda\}$. Thus the affine double cover is branched over 0, 1, and λ.

II Projective algebraic sets

Further reading. For projective algebraic sets, homogeneous coordinates, and morphisms, see [Har77, Chapter I, §§2–3]; [Hul03] and [Hol12] contain many concrete examples.

Throughout this section, $\Bbbk$ is a field. We consider the polynomial ring

$$A = \Bbbk[x_0, \ldots, x_n],$$

endowed with its standard grading by total degree.

II.A Projective space and projective subspaces

Definition II.A.1. Let $n \geq 0$. The n-dimensional *projective space* over $\Bbbk$ is the quotient

$$\mathbf{P}^n_{\Bbbk} = \big(\Bbbk^{n+1} \setminus \{0\}\big)/\Bbbk^*,$$

where $\lambda \in \Bbbk^*$ acts by

$$\lambda \cdot (a_0, \dots, a_n) = (\lambda a_0, \dots, \lambda a_n).$$

The class of a non-zero vector $(a_0, \dots, a_n)$ is denoted by

$$[a_0 : \cdots : a_n].$$

The entries $a_0, \dots, a_n$ are called homogeneous coordinates. By convention, $\mathbf{P}(\{0\}) = \emptyset$.

Thus, a point of $\mathbf{P}^n_{\Bbbk}$ is identified with a one-dimensional linear subspace of $\Bbbk^{n+1}$. For $0 \leq i \leq n$, set

$$U_i = \{[a_0 : \cdots : a_n] \in \mathbf{P}^n_{\Bbbk} \mid a_i \neq 0\}.$$

Remark II.A.2. Every point of U_i has a unique representative whose i-th coordinate is equal to 1. We therefore obtain a bijection

$$\varphi_i : U_i \longrightarrow \Bbbk^n, \qquad [a_0 : \cdots : a_n] \longmapsto \left(\frac{a_0}{a_i}, \dots, \frac{\widehat{a_i}}{a_i}, \dots, \frac{a_n}{a_i}\right).$$

The subsets $U_0, \dots, U_n$ cover $\mathbf{P}^n_{\Bbbk}$ and are called the standard affine charts of projective space.

Remark II.A.3. In particular,

$$\mathbf{P}^n_{\Bbbk} = U_0 \sqcup H_\infty, \qquad H_\infty = \{[a_0 : \cdots : a_n] \mid a_0 = 0\} \simeq \mathbf{P}^{n-1}_{\Bbbk}.$$

Under the identification $U_0 \simeq \Bbbk^n$, the subset H_∞ is called the hyperplane at infinity.

Definition II.A.4. Let $V = \Bbbk^{n+1}$ and let $W \subset V$ be a non-zero vector subspace. The associated projective subspace is

$$\mathbf{P}(W) = \{[w] \in \mathbf{P}(V) \mid 0 \neq w \in W\}.$$

If $\dim(W) = r + 1$, then $\mathbf{P}(W)$ is called a projective r-plane. Projective 1-planes and projective $(n-1)$-planes are called, respectively, projective lines and projective hyperplanes.

If W is the common kernel of linear forms $\ell_1, \dots, \ell_s \in V^\vee$, then

$$\mathbf{P}(W) = \{[a] \in \mathbf{P}(V) \mid \ell_1(a) = \cdots = \ell_s(a) = 0\}.$$

Thus projective subspaces are described by homogeneous linear equations.

Example II.A.5. We have

$$\mathbf{P}^1_{\Bbbk} = U_0 \sqcup \{[0 : 1]\} \simeq \Bbbk \sqcup \{\infty\}.$$

The point $[0 : 1]$ is the point at infinity of the affine line $U_0 = \{[1 : t] \mid t \in \Bbbk\}$.

Exercise II.A.6. Show that each map $\varphi_i : U_i \to \Bbbk^n$ is a bijection. Describe explicitly the change of coordinates from the chart U_i to the chart U_j on $U_i \cap U_j$.

Solution. The inverse of φ_i inserts the coordinate 1 in the i-th position. For instance, for $i = 0$ it is

$$(u_1, \dots, u_n) \longmapsto [1 : u_1 : \cdots : u_n].$$

On $U_i \cap U_j$, write $u_\ell = a_\ell/a_i$, with $u_i = 1$. Since $u_j \neq 0$, the coordinates $v_\ell = a_\ell/a_j$ in the chart U_j are

$$v_i = \frac{1}{u_j}, \qquad v_\ell = \frac{u_\ell}{u_j} \quad \text{for } \ell \neq i, j, \qquad v_j = 1.$$

Exercise II.A.7. Let

$$p = [a_0 : \cdots : a_n], \qquad q = [b_0 : \cdots : b_n]$$

be distinct points of $\mathbf{P}^n_{\Bbbk}$.

i) Show that there is a unique projective line L containing p and q, namely $L = \mathbf{P}(\langle a, b\rangle)$.

ii) Show that a point $[x_0 : \cdots : x_n]$ lies on L if and only if the matrix

$$\begin{pmatrix} a_0 & \cdots & a_n \\ b_0 & \cdots & b_n \\ x_0 & \cdots & x_n \end{pmatrix}$$

has rank at most 2. Deduce that L is defined by its 3×3 minors.

iii) When $n = 2$, write the resulting single linear equation explicitly as a determinant.

Solution. Since $p \neq q$, the vectors a and b are linearly independent. Hence $\mathbf{P}(\langle a, b\rangle)$ is a projective line containing p and q. Any projective line containing them comes from a two-dimensional vector subspace containing a and b, so it is the same line.

A vector x belongs to $\langle a, b\rangle$ precisely when the three rows a, b, x are linearly dependent, which is equivalent to the vanishing of all 3×3 minors. For $n = 2$ this gives the equation

$$\det \begin{pmatrix} a_0 & a_1 & a_2 \\ b_0 & b_1 & b_2 \\ x_0 & x_1 & x_2 \end{pmatrix} = 0.$$

II.B Projective equations and homogenisation

We see here how to define algebraic subsets of the projective space. These are associated with homogenous ideals, the graded ring structure of the polynomial ring acquire thus a particular prominence when dealing with subsets of the projective space.

II.B.1 Homogeneous ideals and projective subsets

We start by recalling the definition of homogeneous polynomials and homogeneous ideals. This refers to the usual grading of the polynomial ring, where scalars have degree zero and each variable has degree one.

Definition II.B.1. A polynomial $F \in A$ is *homogeneous of degree d* if every monomial appearing in F has total degree d. Equivalently,

$$F(\lambda x_0, \dots, \lambda x_n) = \lambda^d F(x_0, \dots, x_n)$$

is a polynomial identity. An ideal $\mathfrak{a} \subset A$ is *homogeneous* if it is generated by homogeneous polynomials. Equivalently, whenever $F \in \mathfrak{a}$ and

$$F = F_0 + F_1 + \cdots + F_d$$

is its decomposition into homogeneous components, then every F_i belongs to $\mathfrak{a}$.

If F is homogeneous, the condition $F(a_0, \dots, a_n) = 0$ does not change when the vector $(a_0, \dots, a_n)$ is multiplied by a non-zero scalar. It therefore makes sense on projective space.

Definition II.B.2. Let $\mathfrak{a} \subset A$ be a homogeneous ideal. Its *projective zero locus* is

$$\mathbf{V}_+(\mathfrak{a}) = \big\{[a_0 : \dots : a_n] \in \mathbf{P}^n_{\Bbbk} \mid F(a_0, \dots, a_n) = 0 \text{ for every homogeneous } F \in \mathfrak{a}\big\}.$$

A subset $X \subset \mathbf{P}^n_{\Bbbk}$ of the form $X = \mathbf{V}_+(\mathfrak{a})$ is called a projective algebraic set.

For a homogeneous polynomial F, we use the notation

$$D_+(F) = \mathbf{P}^n_{\Bbbk} \setminus \mathbf{V}_+(F) = \{p \in \mathbf{P}^n_{\Bbbk} \mid F(p) \neq 0\}.$$

At this point, this is only notation for a subset of projective space. Later, after introducing the Zariski topology, we shall see that the subsets $\mathbf{V}_+(\mathfrak{a})$ are closed and that the subsets $D_+(F)$ are open. Notice that

$$U_i = D_+(x_i).$$

If $X \subset \mathbf{P}^n_{\Bbbk}$ is fixed, we also write $D_+(F)$ for $X \cap D_+(F)$ when no confusion can arise. Given a subset $X \subset \mathbf{P}^n_{\Bbbk}$, its homogeneous vanishing ideal is

$$\mathbf{I}_+(X) = \bigoplus_{d \geq 0} \{F \in A_d \mid F(p) = 0 \text{ for every } p \in X\}.$$

This is a homogeneous ideal of A. The quotient

$$\Bbbk[X] = A/\mathbf{I}_+(X)$$

is called the homogeneous coordinate ring of X. It is usually denoted in the same way as the coordinate ring of an affine algebraic subset.

Example II.B.3. A projective subspace is a projective algebraic set defined by homogeneous linear equations. The most familiar example of higher degree equations is provided by the projective plane conic:

$$C = \mathbf{V}_+(x_0x_2 - x_1^2) \subset \mathbf{P}^2_{\Bbbk}$$

On the chart U_0, with affine coordinates $u = x_1/x_0$ and $v = x_2/x_0$, its equation becomes $v = u^2$.

II.B.2 Dehomogenisation, affine charts and projective closure

Let $F \in A$ be homogeneous. Its dehomogenisation with respect to x_i is the polynomial obtained by setting $x_i = 1$. Let us denote it by f. Under the bijection $U_i \simeq \Bbbk^n$, the subset

$$\mathbf{V}(f) = \mathbf{V}_+(F) \cap U_i, \qquad \text{for } 0 \leq i \leq n.$$

is therefore the affine algebraic set defined by this dehomogenised polynomial. The same statement holds for homogeneous ideals. Here, the equality displayed above in really means that, under the identification $U_i \simeq \Bbbk^n$, the subset $\mathbf{V}_+(F) \cap U_i$ corresponds to $\mathbf{V}(f)$.

Let us look at the other direction. Let

$$R = \Bbbk[y_1, \dots, y_n]$$

and let $f \in R$ have degree d. The homogenisation of f is

$$f^{\mathrm{h}}(x_0, \dots, x_n) = x_0^d f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right).$$

This is a homogeneous polynomial of degree d and satisfies

$$f^{\mathrm{h}}(1, y_1, \dots, y_n) = f(y_1, \dots, y_n).$$

Definition II.B.4. For an ideal $\mathfrak{a} \subset R$, define its homogenisation by

$$\mathfrak{a}^{\mathrm{h}} = (f^{\mathrm{h}} \mid f \in \mathfrak{a}) \subset A.$$

It is important here to homogenise every element of $\mathfrak{a}$, and not merely an arbitrarily chosen family of generators.

Let us look at the geometric side of the story. Consider the map

$$\iota : \Bbbk^n \longrightarrow \mathbf{P}^n_{\Bbbk}, \qquad (a_1, \ldots, a_n) \longmapsto [1 : a_1 : \cdots : a_n].$$

For an affine algebraic set $X \subset \Bbbk^n$, define its projective homogenisation by

$$X^{\mathrm{h}} = \mathbf{V}_+\big(\mathbf{I}(X)^{\mathrm{h}}\big).$$

Remark II.B.5. For any affine algebraic subset, we have

$$X^{\mathrm{h}} \cap U_0 = \iota(X).$$

Indeed, dehomogenising the equations of $\mathbf{I}(X)^{\mathrm{h}}$ with respect to x_0 gives back the equations of $\mathbf{I}(X)$. When $\Bbbk$ is algebraically closed, once the Zariski topology has been introduced, we shall see that X^{h} is precisely the closure of $\iota(X)$ in $\mathbf{P}^n_{\Bbbk}$.

Exercise II.B.6. Compute the projective homogenisations in $\mathbf{P}^2_{\Bbbk}$ of the affine curves

$$y = x^2, \qquad xy = 1.$$

Determine their points on the line at infinity $\mathbf{V}_+(x_0)$.

Solution. We use homogeneous coordinates $[x_0 : x_1 : x_2]$, with $x = x_1/x_0$ and $y = x_2/x_0$ on U_0. The equation $y = x^2$ becomes

$$x_0x_2 - x_1^2 = 0.$$

On the line $x_0 = 0$, this gives $x_1 = 0$, hence the unique point at infinity is $[0 : 0 : 1]$.

The equation $xy = 1$ becomes

$$x_1x_2 - x_0^2 = 0.$$

On the line $x_0 = 0$, it gives $x_1x_2 = 0$, hence the two points at infinity are

$$[0 : 1 : 0] \qquad \text{and} \qquad [0 : 0 : 1].$$

Exercise II.B.7. In $\Bbbk[x, y]$, the ideal (x, y) is also generated by x and $y + x^2$. Homogenise these two different systems of generators and compare the resulting projective zero loci in $\mathbf{P}^2_{\Bbbk}$.

Solution. Using coordinates $[x_0 : x_1 : x_2]$, the homogenisations of x and y are x_1 and x_2. Their common zero locus is the single point

$$\mathbf{V}_+(x_1, x_2) = \{[1 : 0 : 0]\}.$$

On the other hand, the homogenisations of x and $y + x^2$ are

$$x_1, \qquad x_0x_2 + x_1^2.$$

Their common zero locus is defined by $x_1 = 0$ and $x_0x_2 = 0$, hence it is

$$\{[1 : 0 : 0], [0 : 0 : 1]\}.$$

Thus homogenising only a chosen system of generators may introduce extra points at infinity. Homogenising every element of the ideal removes this ambiguity.

Exercise II.B.8. In $\mathbf{P}^2_{\Bbbk}$ with coordinates $[x : y : z]$, let

$$M = \begin{pmatrix} x & 0 \\ y & y \\ 0 & z \end{pmatrix}.$$

Show that the 2×2 minors of M generate the homogeneous ideal

$$(xy, xz, yz),$$

and that its zero locus consists of the three coordinate points.

More generally, let $\ell_1, \ldots, \ell_4$ be four lines in $\mathbf{P}^2_{\Bbbk}$, no three concurrent. Show that their six pairwise intersection points are defined by the maximal minors of

$$\begin{pmatrix} \ell_1 & 0 & 0 \\ -\ell_2 & \ell_2 & 0 \\ 0 & -\ell_3 & \ell_3 \\ 0 & 0 & -\ell_4 \end{pmatrix}.$$

Solution. The minors of M are xy, xz, yz. Their common zero locus consists of points at which at most one coordinate is non-zero, hence of $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[0 : 0 : 1]$.

The four maximal minors of the second matrix are, up to signs,

$$\ell_2\ell_3\ell_4, \quad \ell_1\ell_3\ell_4, \quad \ell_1\ell_2\ell_4, \quad \ell_1\ell_2\ell_3.$$

At a common zero at least two of the ℓ_i must vanish. Conversely, every pairwise intersection of the four lines annihilates all four products. Since no three lines are concurrent, this gives exactly six points.

II.C Morphisms of projective algebraic sets

II.C.1 Rational functions on projective algebraic sets

A polynomial function of the homogeneous coordinates does not in general define a function on projective space: replacing $(x_0, \ldots, x_n)$ by $(\lambda x_0, \ldots, \lambda x_n)$ changes a homogeneous polynomial of degree d by the factor λ^d. Quotients of homogeneous polynomials of the same degree, however, are unchanged.

Definition II.C.1. A *rational function on* $\mathbf{P}^n_{\Bbbk}$ is an equivalence class of fractions

$$\frac{F}{G},$$

where $F, G \in \Bbbk[x_0, \ldots, x_n]$ are homogeneous of the same degree and $G \neq 0$. Two such fractions are equivalent when

$$FG' = F'G.$$

The fraction F/G is defined at a point $[a_0 : \cdots : a_n]$ whenever $G(a_0, \ldots, a_n) \neq 0$.

The equal-degree condition guarantees that

$$\frac{F(\lambda a_0, \ldots, \lambda a_n)}{G(\lambda a_0, \ldots, \lambda a_n)} = \frac{F(a_0, \ldots, a_n)}{G(a_0, \ldots, a_n)},$$

so the value does not depend on the chosen homogeneous coordinates.

More generally, let $X \subset \mathbf{P}^n_{\Bbbk}$ be a projective algebraic set. Two equal-degree fractions F/G and F'/G' define the same rational function on X when

$$FG' - F'G$$

vanishes identically on X. If X is irreducible, these rational functions form a field, denoted $K(X)$.

Let H be homogeneous of degree $d > 0$. On

$$D_+(H) = \mathbf{P}^n_{\Bbbk} \setminus \mathbf{V}_+(H)$$

we may divide by powers of H. Thus every expression

$$\frac{F}{H^m}, \qquad \deg(F) = md,$$

defines a geometrically regular function on $D_+(H)$. More generally, on $X \cap D_+(H)$ we use the same expressions, considered modulo the homogeneous equations of X.

For the standard chart $U_i = D_+(x_i)$, these functions are polynomials in the ratios

$$\frac{x_0}{x_i}, \ldots, \frac{\widehat{x_i}}{x_i}, \ldots, \frac{x_n}{x_i}.$$

Indeed, a monomial of degree m divided by x_i^m becomes a monomial in these ratios. This recovers the usual identification

$$U_i \simeq \Bbbk^n.$$

Later, after localisation has been introduced, the ring of these functions will be denoted

$$\Bbbk[x_0, \ldots, x_n][H^{-1}]_0.$$

The subscript 0 records exactly the equal-degree condition above.

A tuple of homogeneous polynomials of the same degree transforms in the same way under a rescaling of the homogeneous coordinates. This gives the basic form of a map between projective algebraic sets.

Definition II.C.2. Let $X \subset \mathbf{P}^n_{\Bbbk}$ and $Y \subset \mathbf{P}^m_{\Bbbk}$ be projective algebraic sets, and let $\varphi : X \to Y$ be a map. We say that φ is regular at a point $p \in X$ if there exist a homogeneous polynomial G and homogeneous polynomials

$$F_0, \ldots, F_m \in \Bbbk[x_0, \ldots, x_n]$$

of the same degree such that:

(i) $G(p) \neq 0$,

(ii) the polynomials $F_0, \ldots, F_m$ do not vanish simultaneously at any point of $X \cap D_+(G)$;

(iii) for every $q \in X \cap D_+(G)$, we have

$$\varphi(q) = [F_0(q) : \cdots : F_m(q)].$$

The map φ is a morphism if it is regular at every point of X.

The polynomial G is auxiliary: it does not enter the formula for φ, but only specifies a distinguished subset on which one homogeneous formula $[F_0 : \cdots : F_m]$ is valid. There is no further relation required between G and the F_i, except that the latter must have no common zero on $X \cap D_+(G)$. The condition $G(p) \neq 0$ is independent of the chosen homogeneous representative of p, because G is homogeneous.

Example II.C.3. Let $F_0, \dots, F_m$ be homogeneous polynomials of the same degree which do not vanish simultaneously at any point of X. If

$$[F_0(p) : \cdots : F_m(p)] \in Y$$

for every $p \in X$, then

$$X \longrightarrow Y, \qquad p \longmapsto [F_0(p) : \cdots : F_m(p)]$$

is a morphism.

Exercise II.C.4. Show that the identity map of a projective algebraic set is a morphism and that the composition of two morphisms of projective algebraic sets is a morphism.

Solution. The identity is represented by the homogeneous coordinates $x_0, \dots, x_n$.

Let $\varphi : X \to Y$ and $\psi : Y \to Z$ be morphisms. Near a point $p \in X$, write

$$\varphi = [F_0 : \cdots : F_m]$$

with the F_i homogeneous of the same degree. Near $\varphi(p)$, write

$$\psi = [G_0 : \cdots : G_r],$$

where the G_j are homogeneous of the same degree. Then, after restricting to a suitable subset containing p, the composition is given by

$$\psi \circ \varphi = [G_0(F_0, \dots, F_m) : \cdots : G_r(F_0, \dots, F_m)].$$

These polynomials are homogeneous of the same degree and do not vanish simultaneously on a sufficiently small distinguished subset containing p.

Example II.C.5. Let $u : \Bbbk^{n+1} \to \Bbbk^{m+1}$ be a linear map and let $X \subset \mathbf{P}^n_{\Bbbk}$ be a projective algebraic set such that

$$X \cap \mathbf{P}(\ker u) = \emptyset.$$

Then

$$X \longrightarrow \mathbf{P}^m_{\Bbbk}, \qquad [v] \longmapsto [u(v)]$$

is a morphism. In particular, an injective linear map u induces a morphism $\mathbf{P}^n_{\Bbbk} \to \mathbf{P}^m_{\Bbbk}$.

Remark II.C.6. The formula

$$[x_0 : x_1 : x_2] \longmapsto [x_1 : x_2]$$

does not define a morphism on all of $\mathbf{P}^2_{\Bbbk}$, since both coordinates vanish at $[1 : 0 : 0]$. At every other point, however, it is given by a valid local formula in the sense of the preceding definition. Later, this will be interpreted as a rational map

$$\mathbf{P}^2_{\Bbbk} \dashrightarrow \mathbf{P}^1_{\Bbbk},$$

called the projection from the point $[1 : 0 : 0]$.

Example II.C.7 (The quadratic Veronese morphism). Consider the morphism

$$\nu_2 : \mathbf{P}^1_{\Bbbk} \longrightarrow \mathbf{P}^2_{\Bbbk}, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$

Its image is contained in the conic

$$C = \mathbf{V}_+(x_0x_2 - x_1^2) \subset \mathbf{P}^2_{\Bbbk},$$

since $s^2t^2 - (st)^2 = 0$. Conversely, let $[x_0 : x_1 : x_2] \in C$. We define a map

$$\psi : C \longrightarrow \mathbf{P}^1_{\Bbbk}$$

using two local formulas.

- On the subset $C \cap D_+(x_0)$, set
$$\psi([x_0 : x_1 : x_2]) = [x_0 : x_1].$$
- On the subset $C \cap D_+(x_2)$, set
$$\psi([x_0 : x_1 : x_2]) = [x_1 : x_2].$$

These two subsets cover C. Indeed, if $x_0 = x_2 = 0$, then the equation
$$x_0x_2 = x_1^2$$
implies $x_1 = 0$, which is impossible for a point of projective space.

Moreover, the two formulas agree on their intersection. Indeed, $[x_0 : x_1] = [x_1 : x_2]$ if and only if $x_0x_2 = x_1^2$, which is precisely the equation defining C. The map ψ is inverse to ν_2. For instance, on $C \cap D_+(x_0)$, the relation $x_0x_2 = x_1^2$ gives
$$\nu_2([x_0 : x_1]) = [x_0^2 : x_0x_1 : x_1^2] = [x_0 : x_1 : x_2].$$
The computation on $C \cap D_+(x_2)$ is analogous. Thus
$$\mathbf{P}^1_{\Bbbk} \simeq C.$$

This example also illustrates the role of the polynomial G in Definition II.C.2. On $C \cap D_+(x_0)$, we take
$$G = x_0, \qquad (F_0, F_1) = (x_0, x_1),$$
whereas on $C \cap D_+(x_2)$, we take
$$G = x_2, \qquad (F_0, F_1) = (x_1, x_2).$$
Neither pair defines the inverse morphism everywhere on C: the pair (x_0, x_1) vanishes simultaneously at $[0 : 0 : 1]$, while the pair (x_1, x_2) vanishes simultaneously at $[1 : 0 : 0]$. Hence the inverse admits homogeneous coordinate formulas locally, but neither of these formulas works globally.

In general, no single pair of homogeneous polynomials defines ψ on the whole conic. Indeed, assume that $\Bbbk$ is algebraically closed. Suppose that homogeneous polynomials F_0, F_1 of degree d define ψ everywhere on C. Composing with ν_2, we obtain
$$[F_0(s^2, st, t^2) : F_1(s^2, st, t^2)] = [s : t].$$
Hence
$$F_0(s^2, st, t^2) = Hs, \qquad F_1(s^2, st, t^2) = Ht$$
for some homogeneous polynomial $H \in \Bbbk[s, t]$ of degree $2d - 1$. Since $\Bbbk$ is algebraically closed and $\deg(H) > 0$, the polynomial H vanishes at some point of $\mathbf{P}^1_{\Bbbk}$. At the corresponding point of C, both F_0 and F_1 vanish, a contradiction.

Example II.C.8. The quadratic Veronese map
$$\nu_2 : \mathbf{P}^1_{\Bbbk} \longrightarrow \mathbf{P}^2_{\Bbbk}, \qquad [s : t] \longmapsto [s^2 : st : t^2]$$
is a morphism. Its image is contained in the conic
$$C = \mathbf{V}_+(x_0x_2 - x_1^2).$$

Example II.C.9. Looking back at Example I.D.8, we see that the its projective homogenisation of the elliptic curve we say as branched cover of the affine line is the cubic curve

$$\mathbf{V}(x_2^2 x_0 - x_1(x_1 - x_0)(x_1 - \lambda x_0)) \subset \mathbf{P}^2_{\Bbbk}.$$

The projection onto the x-coordinate extends to a degree-two map to $\mathbf{P}^1_{\Bbbk}$, branched also at the point at infinity. This projective curve is an elliptic curve.

Exercise II.C.10. Here is one more exercice about elementary equations of Segre and Veronese varieties.

i) Consider the Segre map

$$\sigma : \mathbf{P}^1_{\Bbbk} \times \mathbf{P}^1_{\Bbbk} \longrightarrow \mathbf{P}^3_{\Bbbk}, \qquad ([s_0 : s_1], [t_0 : t_1]) \longmapsto [s_0 t_0 : s_0 t_1 : s_1 t_0 : s_1 t_1].$$

Compute the kernel of

$$\Bbbk[z_{00}, z_{01}, z_{10}, z_{11}] \longrightarrow \Bbbk[s_0, s_1, t_0, t_1], \qquad z_{ij} \longmapsto s_i t_j,$$

and deduce that the image is the quadric $z_{00}z_{11} - z_{01}z_{10} = 0$.

ii) Consider the quadratic Veronese map

$$\nu_2 : \mathbf{P}^2_{\Bbbk} \longrightarrow \mathbf{P}^5_{\Bbbk}, \quad [x_0 : x_1 : x_2] \longmapsto [x_0^2 : x_0x_1 : x_0x_2 : x_1^2 : x_1x_2 : x_2^2].$$

Arrange the six target coordinates as the symmetric matrix

$$Z = \begin{pmatrix} z_{00} & z_{01} & z_{02} \\ z_{01} & z_{11} & z_{12} \\ z_{02} & z_{12} & z_{22} \end{pmatrix}.$$

Verify that every 2×2 minor of Z vanishes on the image. Show that their common projective zero locus consists precisely of symmetric matrices of rank one, hence is the image of ν_2. The homogeneous ideal of the Veronese surface is generated by these minors.

Solution. For the Segre map, the determinant $z_{00}z_{11} - z_{01}z_{10}$ lies in the kernel. Conversely, modulo this relation every monomial can be reduced so that it does not contain both z_{00} and z_{11}; their images are distinct monomials in the s_i, t_j, which proves that the kernel is the principal ideal generated by the determinant. A point of the quadric is a non-zero 2×2 matrix of rank one, hence an outer product $(s_0, s_1)^{\mathrm{t}}(t_0, t_1)$.

For the Veronese map one has $Z = (x_i x_j)_{i,j}$, a matrix of rank one, so all 2×2 minors vanish. Conversely, a non-zero symmetric matrix of rank one is projectively of the form vv^{t} and therefore comes from $[v] \in \mathbf{P}^2_{\Bbbk}$. The standard determinantal calculation shows that the kernel of $z_{ij} \mapsto x_i x_j$ is generated by these minors.

III Localisation

Further reading. For further details on localisation, local rings, and residue fields, see [Liu02]. These constructions are used throughout [Har77, Chapter II].

We define the localisation of a module or ring at a multiplicative subset S.

III.A Localisation and inversion of elements

Definition III.A.1. A subset S of A is a *multiplicative set* if $1 \in S$ and S is closed under multiplication, i.e., given $a, b \in S$, we have $ab \in S$.

Remark III.A.2. If $\mathfrak{p}$ is a prime ideal of A, then $A \backslash \mathfrak{p}$ is multiplicative. Conversely, if $A \backslash \mathfrak{p}$ is multiplicative and $\mathfrak{p}$ is an ideal, then $\mathfrak{p}$ is prime.

Let M be an A-module and S a multiplicative subset of A. We describe the elements of the localisation $S^{-1}M$. We consider the equivalence relation

$$\frac{x}{s} \sim \frac{y}{t} \iff \exists u \in S, u(tx - sy) = 0.$$

Then the localisation is

$$S^{-1}M = \left\{ \frac{x}{s} \mid x \in M, s \in S \right\} / \sim .$$

When $M = A$, the localisation $S^{-1}A$ is a ring, with the usual operations on fractions. If $S = \{1, f, f^2, \ldots\}$, we write A_f for $S^{-1}A$; if $\mathfrak{p}$ is a prime ideal and $S = A \setminus \mathfrak{p}$, we write $A_{\mathfrak{p}}$.

Proposition III.A.3. *Let $\varphi : A \to B$ be a ring homomorphism such that $\varphi(s)$ is invertible in B for every $s \in S$. Then there is a unique ring homomorphism*

$$\overline{\varphi} : S^{-1}A \longrightarrow B, \qquad \frac{a}{s} \longmapsto \varphi(a)\varphi(s)^{-1},$$

such that $\overline{\varphi}(a/1) = \varphi(a)$ for every $a \in A$.

Proof. The displayed formula is forced by the requirement that every $s \in S$ become invertible. It is well defined by the equivalence relation defining $S^{-1}A$, and it plainly gives the required homomorphism. $\square$

Example III.A.4. Let $A = \Bbbk[x]$ and let

$$S = \{1, x, x^2, \ldots\}.$$

Then

$$S^{-1}A = \Bbbk[x, x^{-1}],$$

the ring of Laurent polynomials. Geometrically, this corresponds to removing the point 0 from the affine line.

Example III.A.5. Let $A = \Bbbk[x]$ and let $\mathfrak{p} = (x)$. Then

$$A_{\mathfrak{p}} = \Bbbk[x]_{(x)}$$

consists of fractions f/g with $g(0) \neq 0$. Thus we invert precisely the polynomials which do not vanish at the origin. The unique maximal ideal is generated by x.

Example III.A.6. Let

$$A = \Bbbk[x, y]/(xy)$$

and localise at the multiplicative set $S = \{1, x, x^2, \ldots\}$. In $S^{-1}A$, the element x becomes invertible. Since $xy = 0$, this forces $y = 0$. Hence

$$S^{-1}A \simeq \Bbbk[x, x^{-1}].$$

Thus, after making x invertible, the relation $xy = 0$ forces $y = 0$.

Exercise III.A.7. Let $A = \Bbbk[\varepsilon]/(\varepsilon^2)$.

i) Compute A_ε.

ii) Show that every element $a + b\varepsilon$ with $a \neq 0$ is a unit and deduce that $A_{(\varepsilon)} \simeq A$.

iii) Compute $A_{1+\varepsilon}$ and explain the contrast between localising at a nilpotent element and at a unit.

Solution. Since $\varepsilon^2 = 0$, the multiplicative set generated by ε contains 0. Hence $A_\varepsilon = 0$. If $a \neq 0$, then

$$(a + b\varepsilon)^{-1} = a^{-1} - a^{-2}b\varepsilon,$$

so the complement of (ε) already consists of units and $A_{(\varepsilon)} = A$. Finally $1 + \varepsilon$ is a unit, so localising at it changes nothing: $A_{1+\varepsilon} \simeq A$.

III.B Localisation and fraction field

In the next proposition $\mathrm{Frac}(A)$ denotes the fraction field of an integral domain A.

Exercise III.B.1. Let A be an integral domain and let $S = A \setminus \{0\}$. Show that $S^{-1}A$ is a field.

Solution. Every element of $S^{-1}A$ has the form a/s with $a \in A$ and $s \neq 0$. If $a \neq 0$, then

$$\frac{a}{s} \cdot \frac{s}{a} = 1.$$

Thus every non-zero element is invertible, so $S^{-1}A$ is a field. This field is precisely $\mathrm{Frac}(A)$.

Proposition III.B.2. *If A is an integral domain and $0 \notin S$, then $S^{-1}A$ is an integral domain, namely the subring of* $\mathrm{Frac}(A)$ *consisting of fractions with denominator in S.*

Proof. First, $0 \neq 1$ in $S^{-1}A$, since $0 \notin S$. We see that $S^{-1}A$ is an integral domain: assume $(a/s)\cdot(b/t) = 0$, then $u(ab) = 0$ for some $u \in S$, so $a = 0$ or $b = 0$. Also, set $T = A \setminus \{0\}$. By construction $T^{-1}A$ is identified with $\mathrm{Frac}(A)$. If a/s is an element of $S^{-1}A$, we can see a/s in $\mathrm{Frac}(A)$ and this defines an injective assignment $S^{-1}A \to \mathrm{Frac}(A)$ which is a ring homomorphism. Indeed, if b/t represents the same element as a/s then $u(at - bs) = 0$ for some $u \in S$, hence $at = bs$ because A is an integral domain, so $a/s = b/t$ in $\mathrm{Frac}(A)$. □

III.C Local rings

Recall that a *local ring* is a ring having a unique maximal ideal. For instance, any field is a local ring, with (0) as maximal ideal. Another classical example of local ring is the ring of power series in one variable $\Bbbk[\![T]\!]$ over a field $\Bbbk$. The maximal ideal consists of power series with vanishing constant term.

Proposition III.C.1. *Let $\mathfrak{p}$ be a prime ideal of A. Then an element $x/s \in A_\mathfrak{p}$ is invertible if and only if $x \notin \mathfrak{p}$. In addition, $\mathfrak{p}A_\mathfrak{p}$ is a maximal ideal of $A_\mathfrak{p}$, actually the only maximal ideal of this ring.*

Proof. Let $x \in A$ and $s \in A \setminus \mathfrak{p}$. If $x \notin \mathfrak{p}$, then s/x is an inverse of x/s in $A_\mathfrak{p}$.

Conversely, assume that $x \in \mathfrak{p}$ and that x/s is invertible. Then there is $y/t \in A_\mathfrak{p}$ such that

$$\frac{x}{s}\frac{y}{t} = 1.$$

Hence there exists $u \in A \setminus \mathfrak{p}$ such that

$$u(xy - st) = 0.$$

It follows that $ust = uxy \in \mathfrak{p}$. Since $u, s, t \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime, this is impossible. Thus x/s is invertible if and only if $x \notin \mathfrak{p}$.

The non-invertible elements of $A_\mathfrak{p}$ are therefore exactly the elements of $\mathfrak{p}A_\mathfrak{p}$. This set is an ideal, so it is the unique maximal ideal of $A_\mathfrak{p}$. □

Lemma III.C.2. *For any prime ideal* $\mathfrak{p}$ *of a ring* A*, we have* $\mathrm{Frac}(A/\mathfrak{p}) \simeq A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$.

Proof. The localisation map $\varphi : A \to A_{\mathfrak{p}}$, $x \mapsto x/1$, induces a map

$$\bar{\varphi} : A/\mathfrak{p} \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

This map is injective. Indeed, if $x/1 \in \mathfrak{p}A_{\mathfrak{p}}$, then there are $p \in \mathfrak{p}$ and $s \in A \setminus \mathfrak{p}$ such that $x/1 = p/s$ in $A_{\mathfrak{p}}$. Hence there exists $u \in A \setminus \mathfrak{p}$ such that

$$u(sx - p) = 0.$$

In particular, $usx \in \mathfrak{p}$. Since $u, s \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime, we get $x \in \mathfrak{p}$.

Since $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is a field and every non-zero element of $A/\mathfrak{p}$ becomes invertible in it, the map $\bar{\varphi}$ extends to an injective field homomorphism

$$\mathrm{Frac}(A/\mathfrak{p}) \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

It is surjective because every element of $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ can be written as the class of x/s, with $x \in A$ and $s \in A \setminus \mathfrak{p}$, and this is the image of $\bar{x}\bar{s}^{-1} \in \mathrm{Frac}(A/\mathfrak{p})$. □

Exercise III.C.3. Let $A = \Bbbk[x]$ and $S = \{1, x, x^2, \ldots\}$. Show that

$$S^{-1}A \simeq \Bbbk[x, x^{-1}].$$

Solution. The localisation $S^{-1}A$ is obtained by making x invertible. Hence every element of $S^{-1}A$ can be written as $f(x)/x^n$, with $f(x) \in \Bbbk[x]$. This is exactly a Laurent polynomial, i.e. an element of $\Bbbk[x, x^{-1}]$.

Exercise III.C.4. Let $A = \Bbbk[x]$ and $\mathfrak{p} = (x - a)$, for some $a \in \Bbbk$. Describe the elements of $A_{\mathfrak{p}}$ and its maximal ideal.

Solution. We have

$$A_{\mathfrak{p}} = \left\{ \frac{f}{g} \mid f, g \in \Bbbk[x],\ g(a) \neq 0 \right\}.$$

Indeed, the elements outside $\mathfrak{p}$ are precisely the polynomials which do not vanish at a. The unique maximal ideal is

$$\mathfrak{p}A_{\mathfrak{p}} = (x - a)A_{\mathfrak{p}}.$$

Exercise III.C.5. Let

$$A = \Bbbk[x, y]/(xy)$$

and let $S = \{1, x, x^2, \ldots\}$. Compute $S^{-1}A$.

Solution. In $S^{-1}A$, the element x is invertible. Since $xy = 0$ in A, we get $y = x^{-1}xy = 0$ in $S^{-1}A$. Therefore

$$S^{-1}A \simeq \Bbbk[x, x^{-1}].$$

IV Sheaves

Further reading. For sheaves, stalks, and sheafification, see [Har77, Chapter II, §1] and [Liu02]. The geometric role of sheaves in scheme theory is illustrated throughout [EH00].

IV.A Sheaves of abelian groups

Let X be a topological space.

Definition IV.A.1. A *presheaf of sets* on X is the data of a set $\mathcal{F}(U)$ for every open subset $U \subset X$, together with restriction maps

$$\rho_{UV} : \mathcal{F}(U) \longrightarrow \mathcal{F}(V), \qquad s \longmapsto s|_V$$

for every inclusion $V \subset U$, satisfying $s|_U = s, \qquad (s|_V)|_W = s|_W$ whenever $W \subset V \subset U$.

If for every open subset U of X, $\mathcal{F}(U)$ is a set equipped with some algebraic structure (typically an abelian group, a vector space, a ring, or a module over a ring) and for each inclusion of open subsets $V \subset U$, the restriction map ρ_{UV} is a morphism of that algebraic structure, then $\mathcal{F}$ is a presheaf with that structure. We say that $\mathcal{F}$ is a sheaf of abelian groups, of rings, etc. Note that these structures can be layered. For instance, if $\mathcal{F}$ is a sheaf of rings, then a sheaf of $\mathcal{F}$-modules is a sheaf $\mathcal{E}$ such that each $\mathcal{E}(U)$ is an $\mathcal{F}(U)$-module and, for each inclusion $V \subset U$, the restriction map $\mathcal{E}(U) \to \mathcal{E}(V)$ is compatible with the restriction map $\mathcal{F}(U) \to \mathcal{F}(V)$.

Definition IV.A.2. A presheaf $\mathcal{F}$ is a *sheaf* if it satisfies the following two conditions. Let $U = \bigcup_{i\in I} U_i$ be an open covering of an open subset U of X.

i) If $s, t \in \mathcal{F}(U)$ and $s|_{U_i} = t|_{U_i}$ for all $i \in I$, then $s = t$.

ii) If, for all $i \in I$, $s_i \in \mathcal{F}(U_i)$ are sections such that

$$s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}, \qquad \text{for all } i, j \in I,$$

then there exists a section $s \in \mathcal{F}(U)$ such that

$$s|_{U_i} = s_i, \qquad \text{for all } i.$$

A sheaf is a presheaf whose sections are determined locally and glue from compatible local data.

IV.A.1 The stalk

Let $\mathcal{F}$ be a sheaf of sets on a topological space X. Then, for all $x \in X$, we define the stalk of $\mathcal{F}$ at x as:

$$\mathcal{F}_x = \varinjlim_{U \ni x} \mathcal{F}(U).$$

Equivalently, an element of $\mathcal{F}_x$ is represented by a pair (U, s), where U is an open neighbourhood of x and $s \in \mathcal{F}(U)$. Two pairs (U, s) and (V, t) define the same element of $\mathcal{F}_x$ if there exists an open neighbourhood $W \subset U \cap V$ of x such that

$$s|_W = t|_W.$$

The class of (U, s) in $\mathcal{F}_x$ is called the *germ* of s at x and is denoted by s_x.

Exercise IV.A.3. Prove that this is well-defined and that for all open subsets $U \subset X$ containing x we get an induced map $\mathcal{F}(U) \to \mathcal{F}_x$.

Solution. The relation is reflexive and symmetric by definition. For transitivity, assume that (U, s) is equivalent to (V, t) and that (V, t) is equivalent to (W, r). Then there are neighbourhoods $U_1 \subset U \cap V$ and $U_2 \subset V \cap W$ of x such that $s|_{U_1} = t|_{U_1}$ and $t|_{U_2} = r|_{U_2}$. On $U_1 \cap U_2$, all three restrictions agree, so (U, s) is equivalent to (W, r).

If U is an open neighbourhood of x, the map $\mathcal{F}(U) \to \mathcal{F}_x$ sends s to the class of (U, s). It is compatible with restriction: if $V \subset U$ is another neighbourhood of x, then (U, s) and $(V, s|_V)$ define the same germ.

IV.A.2 Direct image

Let $f\colon X \to Y$ be a continuous map and let $\mathcal{F}$ be a sheaf of sets on X. The *direct image* of $\mathcal{F}$ is the sheaf $f_*\mathcal{F}$ on Y defined by

$$(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V))$$

for every open subset $V \subset Y$. The restriction maps are induced by the restriction maps of $\mathcal{F}$.

IV.A.3 The sheafification

There is a way to describe the sheaf associated with a presheaf, which goes through looking at the stalks of the presheaf at each point.

Lemma IV.A.4. *Let $\mathcal{F}$ be a presheaf of abelian groups on a topological space X. Then there is a sheaf $\mathcal{F}^+$ and a morphism $\mathcal{F} \to \mathcal{F}^+$ which is universal with respect to morphisms from $\mathcal{F}$ to sheaves.*

Proof. The definition of $\mathcal{F}^+$ goes as follows. Let U be an open subset of X. Then we consider the set of functions

$$s : U \to \coprod_{x \in U} \mathcal{F}_x$$

such that $s(x) \in \mathcal{F}_x$, for all $x \in U$. Among these functions, we consider those functions s such that, for all $x \in U$, there is an open subset $V \subset U$, with $x \in V$, and there is $t \in \mathcal{F}(V)$ such that, for any point $y \in V$, the canonical germ map $\mathcal{F}(V) \to \mathcal{F}_y$ takes t to $s(y)$.

This defines the set $\mathcal{F}^+(U)$ and we get a morphism $\mathcal{F} \to \mathcal{F}^+$ by sending $s \in \mathcal{F}(U)$ to its local values $(s_x)_{x\in U}$, where s_x is the image of s in $\mathcal{F}_x$. From the very definition of $\mathcal{F}^+(U)$, the image of s is in $\mathcal{F}^+(U)$. The resulting map is universal with respect to morphisms $\psi : \mathcal{F} \to \mathcal{G}$ of presheaves of abelian groups, where $\mathcal{G}$ is a sheaf. Indeed, we first observe that $\mathcal{F}^+_x \simeq \mathcal{F}_x$, for all $x \in X$. Then, one checks that, given ψ and an open subset $U \subset X$, we get maps $\psi : \mathcal{F}_x \to \mathcal{G}_x$ and an induced map $\mathcal{F}^+_x \to \mathcal{G}_x$, which gives a map $\mathcal{F}^+(U) \to \mathcal{G}(U)$, which is uniquely defined and commutes with our morphism $\mathcal{F} \to \mathcal{F}^+$. □

Exercise IV.A.5. Fill in the details of the above proof.

Solution. The restrictions on $\mathcal{F}^+$ are obtained by restricting functions $U \to \coprod_{x\in U} \mathcal{F}_x$. The locality condition is local by definition, so these restrictions are well defined. The sheaf axioms are also immediate from the description by functions: if two sections have the same restrictions to an open cover, then they have the same value in each stalk; and compatible local sections glue by defining the value at each point using any member of the cover containing that point.

The morphism $\mathcal{F} \to \mathcal{F}^+$ sends a section to its germs. If $\mathcal{G}$ is a sheaf and $\psi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves, then ψ induces maps on stalks $\psi_x : \mathcal{F}_x \to \mathcal{G}_x$. Given $s \in \mathcal{F}^+(U)$, define a section of $\mathcal{G}$ locally as follows: near every $x \in U$, the section s is represented by some $t \in \mathcal{F}(V)$, and we take $\psi(t) \in \mathcal{G}(V)$. These local sections agree on overlaps because they have the same germs and $\mathcal{G}$ is a sheaf. Hence they glue to a unique section of $\mathcal{G}(U)$. This gives the unique morphism $\mathcal{F}^+ \to \mathcal{G}$ extending ψ.

IV.A.4 Sheaves on a basis of the topology

Let X be a topological space and let $\mathcal{B}$ be a basis of its topology. In practice it is often much easier to prescribe sections only on the open sets belonging to $\mathcal{B}$.

Proposition IV.A.6. *Suppose that, for every $B \in \mathcal{B}$, an object $\mathcal{F}(B)$ is given, together with restriction maps for inclusions of basis elements, and suppose that the uniqueness and gluing axioms hold for coverings of an element of $\mathcal{B}$ by elements of $\mathcal{B}$. Then there exists a sheaf $\widetilde{\mathcal{F}}$ on X, unique up to unique isomorphism, whose restriction to $\mathcal{B}$ is $\mathcal{F}$.*

Construction. For an arbitrary open subset $U \subset X$, define $\widetilde{\mathcal{F}}(U)$ as the set of compatible families

$$(s_B)_{B \in \mathcal{B},\, B \subset U}, \qquad s_B \in \mathcal{F}(B),$$

where compatibility means that $s_B|_C = s_{B'}|_C$ whenever $C \subset B \cap B'$ is a basis element. The restriction maps are obtained by forgetting part of the family. The sheaf axioms follow directly from the corresponding axioms on the basis. If $U = B$ belongs to $\mathcal{B}$, the family is uniquely determined by its component on B, so $\widetilde{\mathcal{F}}(B) \simeq \mathcal{F}(B)$. □

Remark IV.A.7. It is enough to check compatibility on basis elements contained in intersections: every point of $B \cap B'$ has such a neighbourhood. This proposition will later be applied to the principal opens $D(f)$ of Spec(A), with sections given by localisations A_f or F_f.

IV.B Examples and exercises

Exercise IV.B.1. Let X be a topological space. For every open subset $U \subset X$, let

$$\mathcal{C}^0_X(U) = \{\text{continuous functions } U \to \mathbb{R}\}.$$

Show that $\mathcal{C}^0_X$ is a sheaf of abelian groups. If X is a smooth manifold, define similarly $\mathcal{C}^\infty_X(U)$ as the set of smooth functions $U \to \mathbb{R}$. Show that $\mathcal{C}^\infty_X$ is a sheaf.

Solution. Addition is pointwise, and restrictions are homomorphisms of abelian groups. If two continuous functions on U agree on every member of an open cover, they agree at every point of U. Conversely, compatible continuous functions $s_i : U_i \to \mathbb{R}$ glue to a function $s : U \to \mathbb{R}$ by setting $s(x) = s_i(x)$ for any i such that $x \in U_i$. This is well defined by compatibility, and it is continuous because continuity is local on the source.

The same proof works for smooth functions on a smooth manifold, since smoothness is also local on the source.

Exercise IV.B.2. Let X be a topological space and let A be an abelian group. Define a presheaf by

$$\mathcal{F}(U) = A$$

for every non-empty open subset $U \subset X$, put $\mathcal{F}(\emptyset) = 0$, and let all restriction maps between non-empty open subsets be the identity. Is this presheaf always a sheaf? What happens if U is disconnected?

Solution. This presheaf is not always a sheaf. Suppose that $U = U_1 \sqcup U_2$ is a disjoint union of two non-empty open subsets. Choose two distinct elements $a, b \in A$. The sections $a \in \mathcal{F}(U_1)$ and $b \in \mathcal{F}(U_2)$ are automatically compatible on the empty intersection, but they cannot be glued to a single element of A on U, because its restriction to both U_1 and U_2 would have to be the same element.

The associated sheaf is the sheaf of locally constant functions with values in A.

Exercise IV.B.3. Let $f \colon X \to Y$ be a continuous map and let $\mathcal{F}$ be a sheaf on X. Show directly from the definition that $f_*\mathcal{F}$ is a sheaf on Y.

Solution. Let $V = \bigcup_i V_i$ be an open cover in Y. Then

$$f^{-1}(V) = \bigcup_i f^{-1}(V_i)$$

is an open cover in X. Sections of $f_*\mathcal{F}$ on V are sections of $\mathcal{F}$ on $f^{-1}(V)$. The uniqueness and gluing axioms for $f_*\mathcal{F}$ therefore follow immediately from the corresponding axioms for $\mathcal{F}$ applied to the cover $(f^{-1}(V_i))$.

Exercise IV.B.4. Let X be a complex manifold. For every open subset $U \subset X$, let

$$\mathcal{O}_X(U) = \{\text{holomorphic functions } U \to \mathbb{C}\}.$$

Show that $\mathcal{O}_X$ is a sheaf of abelian groups. What is the stalk $\mathcal{O}_{X,x}$ at a point $x \in X$?

Solution. The sheaf axioms are proved exactly as for continuous functions, using the fact that holomorphicity is local. Thus compatible holomorphic functions on an open cover glue to a holomorphic function.

The stalk $\mathcal{O}_{X,x}$ is the ring of germs of holomorphic functions at x. If X has complex dimension n and $z_1, \ldots, z_n$ are local holomorphic coordinates centred at x, then $\mathcal{O}_{X,x}$ identifies with the ring of convergent power series

$$\mathbb{C}\{z_1, \ldots, z_n\}.$$

Exercise IV.B.5. Let $p\colon E \to X$ be a holomorphic vector bundle of rank r on a complex manifold X. For every open subset $U \subset X$, let

$$\mathcal{E}(U) = \Gamma(U, E)$$

be the space of holomorphic sections of E over U. Show that $\mathcal{E}$ is a sheaf of $\mathcal{O}_X$-modules. Describe the stalk $\mathcal{E}_x$ at a point $x \in X$, and prove that if E is trivial on an open neighbourhood U of x, then

$$\mathcal{E}|_U \simeq \mathcal{O}_U^{\oplus r} \qquad \text{and hence} \qquad \mathcal{E}_x \simeq \mathcal{O}_{X,x}^{\oplus r}.$$

Solution. Restrictions of holomorphic sections are holomorphic, and addition and multiplication by holomorphic functions are defined pointwise. Thus each $\mathcal{E}(U)$ is an $\mathcal{O}_X(U)$-module, compatibly with restriction.

Let $U = \bigcup_i U_i$ be an open cover. Two holomorphic sections of E which agree on every U_i agree pointwise on U. Conversely, compatible sections $s_i \in \Gamma(U_i, E)$ glue to a section $s\colon U \to E$ by setting $s(x) = s_i(x)$ whenever $x \in U_i$. Compatibility makes this well defined, and holomorphicity is local on U. Hence $\mathcal{E}$ is a sheaf of $\mathcal{O}_X$-modules.

The stalk $\mathcal{E}_x$ is the $\mathcal{O}_{X,x}$-module of germs at x of local holomorphic sections of E. If $E|_U \simeq U \times \mathbb{C}^r$, a section over an open subset $V \subset U$ is the same as an r-tuple of holomorphic functions on V. Therefore

$$\mathcal{E}|_U \simeq \mathcal{O}_U^{\oplus r},$$

and taking stalks at x gives

$$\mathcal{E}_x \simeq \mathcal{O}_{X,x}^{\oplus r}.$$

Exercise IV.B.6. Let $U \subset \mathbb{C}$ be a connected open subset containing at least two points.

i) Show that the ring $\mathcal{O}(U)$ of holomorphic functions has no zero divisors.

ii) Show that the ring $\mathcal{C}^0(U)$ of continuous functions usually has zero divisors: choose two disjoint closed discs contained in U and construct non-zero continuous functions with disjoint supports.

iii) Explain why connectedness alone does not force a ring of functions to be an integral domain.

Solution. If $fg = 0$ with $f, g \in \mathcal{O}(U)$ and $f \neq 0$, then the set on which f does not vanish is a non-empty open subset. The function g vanishes there, so the identity theorem gives $g = 0$ on the connected set U.

Choose disjoint closed discs $D_1, D_2 \subset U$. There are non-zero continuous functions f_i supported in small neighbourhoods of D_i. Their supports are disjoint, hence $f_1 f_2 = 0$. Thus $\mathcal{C}^0(U)$ has zero divisors even though U is connected. The difference is analytic rigidity, not topology alone.

Exercise IV.B.7. Let $\mathbf{P}^1_{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. For $d \in \mathbb{Z}$, define $\mathcal{O}(d) = \mathcal{O}(d\,\infty)$ as the sheaf whose sections on an open subset U are meromorphic functions on U which are holomorphic away from ∞ and have at most a pole of order d at ∞ when $d \geq 0$; for $d < 0$ they must vanish at ∞ to order at least $-d$.

i) Check directly that this is a sheaf.

ii) Prove that, for $d \geq 0$,
$$\Gamma(\mathbf{P}^1_{\mathbb{C}}, \mathcal{O}(d)) = \{a_0 + a_1 z + \cdots + a_d z^d\},$$
and therefore has dimension $d + 1$.

iii) Show that $\Gamma(\mathbf{P}^1_{\mathbb{C}}, \mathcal{O}(d)) = 0$ for $d < 0$.

Solution. The condition on the order of zeros and poles is local, and compatible meromorphic functions glue uniquely, so the assignment is a sheaf. A meromorphic function on $\mathbf{P}^1$ with no finite poles is a polynomial in the affine coordinate z. Its pole order at ∞ is its degree, so the sections of $\mathcal{O}(d)$ for $d \geq 0$ are exactly the polynomials of degree at most d. If $d < 0$, a global section is a holomorphic function on $\mathbf{P}^1$ which vanishes at ∞; global holomorphic functions are constant, hence the section is zero.

Exercise IV.B.8 (Transition functions of a vector bundle). Let X be a smooth manifold, respectively a complex manifold, and let $p : E \to X$ be a smooth, respectively holomorphic, vector bundle of rank r. Let $(U_i \mid i \in I)$ be a trivialising open covering of X for E, and choose trivialisations
$$\tau_i : E|_{U_i} \xrightarrow{\simeq} U_i \times \mathbb{K}^r,$$
where $\mathbb{K} = \mathbb{R}$ in the smooth case and $\mathbb{K} = \mathbb{C}$ in the holomorphic case.

i) Show that, on U_{ij}, the map
$$\tau_i \circ \tau_j^{-1} : U_{ij} \times \mathbb{K}^r \longrightarrow U_{ij} \times \mathbb{K}^r$$
has the form
$$(x, v) \longmapsto (x, g_{ij}(x)v),$$
where
$$g_{ij} : U_{ij} \longrightarrow \mathrm{GL}_r(\mathbb{K})$$
is smooth, respectively holomorphic.

ii) Show that the transition functions satisfy
$$g_{ii} = \mathbb{1}_r, \qquad g_{ji} = g_{ij}^{-1}, \qquad g_{ij}g_{jk} = g_{ik}$$
on every triple intersection U_{ijk}.

iii) Let $\mathcal{E}_i$ be the sheaf of smooth, respectively holomorphic, maps $U \longrightarrow \mathbb{K}^r$ on open subsets $U \subset U_i$. Show that multiplication by g_{ij} defines an isomorphism of sheaves
$$\varphi_{ij} : \mathcal{E}_j|_{U_{ij}} \xrightarrow{\simeq} \mathcal{E}_i|_{U_{ij}}, \qquad s \longmapsto g_{ij}s,$$
and that the cocycle condition for the φ_{ij} is precisely the identity
$$g_{ij}g_{jk} = g_{ik}.$$

iv) Conversely, suppose that smooth, respectively holomorphic, maps $g_{ij} : U_{ij} \longrightarrow \mathrm{GL}_r(\mathbb{K})$ are given and satisfy the cocycle condition. Show that the trivial bundles $U_i \times \mathbb{K}^r$ can be glued to a smooth, respectively holomorphic, vector bundle of rank r on X.

Solution. On U_{ij}, the map $\tau_i \circ \tau_j^{-1}$ preserves the projection to U_{ij} and is linear on each fibre. Hence it is of the form

$$(x, v) \longmapsto (x, g_{ij}(x)v)$$

for a map $g_{ij} : U_{ij} \to \mathrm{GL}_r(\mathbb{K})$. Its regularity follows from that of the trivialisations. We have

$$g_{ii} = \mathbb{1}_r$$

because $\tau_i \circ \tau_i^{-1}$ is the identity, and

$$g_{ji} = g_{ij}^{-1}$$

because $\tau_j \circ \tau_i^{-1}$ is the inverse of $\tau_i \circ \tau_j^{-1}$. On a triple intersection,

$$(\tau_i \circ \tau_j^{-1}) \circ (\tau_j \circ \tau_k^{-1}) = \tau_i \circ \tau_k^{-1},$$

and therefore

$$g_{ij}g_{jk} = g_{ik}.$$

If $s : U \to \mathbb{K}^r$ is a local section on $U \subset U_{ij}$, define

$$\varphi_{ij}(s)(x) = g_{ij}(x)s(x).$$

This gives an isomorphism of sheaves, with inverse given by multiplication by g_{ji}. Moreover,

$$\varphi_{ij} \circ \varphi_{jk}(s) = g_{ij}g_{jk}s,$$

so the cocycle condition for the sheaves is equivalent to

$$g_{ij}g_{jk} = g_{ik}.$$

Conversely, define an equivalence relation on the disjoint union

$$\coprod_{i \in I} U_i \times \mathbb{K}^r$$

by declaring

$$(x, v)_j \sim (x, g_{ij}(x)v)_i$$

whenever $x \in U_{ij}$. The cocycle condition shows that this is an equivalence relation. The quotient carries natural local trivialisations over the U_i, and these endow it with the structure of a smooth, respectively holomorphic, vector bundle of rank r.

IV.C Morphisms, germs and functorial constructions

The definitions above describe the local sections of a sheaf. We now explain how sheaves are compared and transported. These constructions will later enter the definition of a morphism of ringed spaces and, in particular, of a morphism of schemes.

IV.C.1 Morphisms of presheaves and sheaves

Definition IV.C.1. Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on a topological space X. A *morphism of presheaves*

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

is a collection of maps

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U),$$

one for every open subset $U \subset X$, such that for every inclusion $V \subset U$ the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \end{array}$$

commutes. If $\mathcal{F}$ and $\mathcal{G}$ are sheaves, we call φ a morphism of sheaves.

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ induces, for every $x \in X$, a map on stalks

$$\varphi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x, \qquad s_x \longmapsto \varphi_U(s)_x,$$

where s is any representative of the germ s_x on a neighbourhood U of x. Compatibility with restriction makes this map well defined.

Example IV.C.2 (The de Rham differential). Let X be a smooth manifold. For each $p \geq 0$, let Ω^p_X denote the sheaf of smooth differential p-forms. Exterior differentiation defines a morphism of sheaves of real vector spaces

$$d : \Omega^p_X \longrightarrow \Omega^{p+1}_X, \qquad \omega \longmapsto d\omega,$$

and $d \circ d = 0$. For $p = 0$ this is

$$d : \mathcal{C}^\infty_X \longrightarrow \Omega^1_X, \qquad f \longmapsto df.$$

It is not a morphism of $\mathcal{C}^\infty_X$-modules, since

$$d(hf) = h\, df + f\, dh.$$

Thus, when speaking of a morphism of sheaves, it is important to specify which algebraic structure is being preserved.

Example IV.C.3 (The $\bar{\partial}$-operator). Let X be a complex manifold and let $\mathcal{A}^{0,q}_X$ be the sheaf of smooth $(0,q)$-forms. The operator

$$\bar{\partial} : \mathcal{A}^{0,q}_X \longrightarrow \mathcal{A}^{0,q+1}_X$$

is a morphism of sheaves of complex vector spaces and satisfies $\bar{\partial}^2 = 0$. In degree zero,

$$\mathcal{O}_X = \ker\big(\bar{\partial} : \mathcal{A}^{0,0}_X \to \mathcal{A}^{0,1}_X\big).$$

Thus holomorphic functions form the sheaf of solutions of a local differential equation.

IV.C.2 Concrete germs of functions

Example IV.C.4 (Different representatives of the same germ). At the origin of $\mathbb{C}$, consider

$$f(z) = \frac{1}{1-z}$$

on $\mathbb{C} \setminus \{1\}$ and

$$g(z) = \sum_{n \geq 0} z^n$$

on the unit disc. The two functions agree on a neighbourhood of 0, hence they define the same element of the stalk $\mathcal{O}_{\mathbb{C},0}$, although their domains and their descriptions are different.

Example IV.C.5 (A germ remembers more than a value). The holomorphic functions $f(z) = 0$ and $g(z) = z$ have the same value at the origin, but they do not define the same germ there. Indeed, they do not agree on any neighbourhood of 0. The stalk $\mathcal{O}_{\mathbb{C},0}$ therefore remembers the behaviour of a function near 0, not merely its value at 0.

Every germ of a holomorphic function at the origin has a convergent power series expansion, and two germs are equal if and only if their power series are equal. Thus

$$\mathcal{O}_{\mathbb{C},0} \simeq \mathbb{C}\{z\},$$

the ring of convergent power series. Its unique maximal ideal is

$$\mathfrak{m}_0 = \{f_0 \mid f(0) = 0\} = (z),$$

and a germ is invertible precisely when its constant term is non-zero.

Remark IV.C.6 (A contrast with smooth germs). The smooth function

$$h(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

has zero Taylor series at the origin, but its germ is not zero. Holomorphic germs are determined by their Taylor series; smooth germs are not.

Exercise IV.C.7. In $\mathcal{O}_{\mathbb{C},0}$, determine which of the following germs are invertible:

$$z, \qquad 1+z, \qquad e^z - 1, \qquad \cos z.$$

For each non-invertible germ, determine its order of vanishing.

Solution. A holomorphic germ is invertible exactly when its value at 0 is non-zero. Thus $1+z$ and $\cos z$ are invertible. The germs z and $e^z - 1$ are not; both have order of vanishing one, since

$$e^z - 1 = z + \frac{z^2}{2!} + \cdots .$$

IV.C.3 Examples of direct images

Let $f : X \to Y$ be continuous and let $\mathcal{F}$ be a sheaf on X. Recall that

$$(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V))$$

for every open subset $V \subset Y$.

Example IV.C.8 (The map to a point). Let $p : X \to \{*\}$ be the unique map. Then

$$(p_*\mathcal{F})(\{*\}) = \mathcal{F}(X).$$

Thus direct image to a point is the operation of taking global sections.

Example IV.C.9 (A skyscraper sheaf). Let $i : \{0\} \hookrightarrow \mathbb{C}$ be the inclusion and let W be a complex vector space, regarded as a sheaf on the one-point space. Then

$$(i_*W)(U) = \begin{cases} W, & 0 \in U, \\ 0, & 0 \notin U. \end{cases}$$

Its stalk is W at the origin and zero at every other point. This is the *skyscraper sheaf* supported at 0 with fibre W.

For $W = \mathbb{C}$, evaluation at the origin gives a morphism of sheaves

$$\mathcal{O}_{\mathbb{C}} \longrightarrow i_*\mathbb{C}, \qquad f \longmapsto f(0).$$

On the stalk at 0 it becomes

$$\mathcal{O}_{\mathbb{C},0} \longrightarrow \mathbb{C},$$

whose kernel is the maximal ideal (z).

Example IV.C.10 (An open inclusion). Let $j : \mathbb{C}^* \hookrightarrow \mathbb{C}$ be the inclusion. Then

$$(j_*\mathcal{O}_{\mathbb{C}^*})(U) = \mathcal{O}(U \cap \mathbb{C}^*).$$

The stalk $(j_*\mathcal{O}_{\mathbb{C}^*})_0$ consists of germs of holomorphic functions on punctured neighbourhoods of the origin. It contains, for instance, the germs of $1/z$ and $e^{1/z}$. Therefore direct image by an open inclusion is not the same as extension by zero.

Example IV.C.11 (A double cover). Consider locally the map

$$\pi : \mathbb{C} \longrightarrow \mathbb{C}, \qquad w \longmapsto z = w^2.$$

At the branch point, the map of local rings is

$$\mathbb{C}\{z\} \longrightarrow \mathbb{C}\{w\}, \qquad z \longmapsto w^2.$$

Every convergent power series in w has a unique decomposition into its even and odd parts,

$$h(w) = a(w^2) + w\,b(w^2).$$

Consequently

$$\mathbb{C}\{w\} = \mathbb{C}\{z\} \oplus w\mathbb{C}\{z\}$$

as a $\mathbb{C}\{z\}$-module. Sheaf-theoretically, $\pi_*\mathcal{O}_{\mathbb{C}}$ is locally a free $\mathcal{O}_{\mathbb{C}}$-module of rank two. This is the local analytic form of the algebraic decomposition

$$\Bbbk[x,y]/(y^2 - f(x)) = \Bbbk[x] \oplus \Bbbk[x]y$$

which appeared in the elliptic-curve example.

Exercise IV.C.12. Let $i : Z \hookrightarrow X$ be the inclusion of a closed subset and let $\mathcal{F}$ be a sheaf on Z. Show that

$$(i_*\mathcal{F})_x \simeq \begin{cases} \mathcal{F}_x, & x \in Z, \\ 0, & x \notin Z, \end{cases}$$

when $\mathcal{F}$ is a sheaf of abelian groups.

Solution. If $x \notin Z$, there is an open neighbourhood U of x disjoint from Z, so $(i_*\mathcal{F})(U) = \mathcal{F}(\emptyset) = 0$. Hence the stalk is zero. If $x \in Z$, the open neighbourhoods $U \cap Z$ form a neighbourhood basis of x in Z, and the definition of the two stalks gives $(i_*\mathcal{F})_x \simeq \mathcal{F}_x$.

IV.C.4 Inverse image and pullback

Let $f : X \to Y$ be continuous and let $\mathcal{G}$ be a sheaf on Y. The assignment

$$U \longmapsto \varinjlim_{U \subset f^{-1}(V)} \mathcal{G}(V),$$

where V runs through the open subsets of Y, is in general only a presheaf. Its sheafification is called the *inverse-image sheaf* and is denoted by $f^{-1}\mathcal{G}$. Its stalks have the particularly simple description

$$(f^{-1}\mathcal{G})_x \simeq \mathcal{G}_{f(x)}.$$

Thus inverse image transports germs near $f(x)$ to germs near x.

If X and Y are smooth manifolds and f is smooth, or if they are complex manifolds and f is holomorphic, composition of functions gives a natural morphism

$$f^{-1}\mathcal{O}_Y \longrightarrow \mathcal{O}_X, \qquad g \longmapsto g \circ f.$$

For general ringed spaces, a morphism of this form is additional data: it is not determined by the underlying continuous map alone.

Finally, if $\mathcal{E}$ is a sheaf of $\mathcal{O}_Y$-modules, its pullback is

$$f^*\mathcal{E} = f^{-1}\mathcal{E} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

For a vector bundle, this construction agrees with the sheaf of sections of the usual pulled-back vector bundle. We shall return to it after morphisms of ringed spaces have been introduced.

Remark IV.C.13. The inverse-image sheaf $f^{-1}\mathcal{G}$ and the pullback $f^*\mathcal{E}$ are different constructions. The former is defined for an arbitrary sheaf; the latter uses the module structures and the morphism $f^{-1}\mathcal{O}_Y \to \mathcal{O}_X$.

Chapter 2

Affine schemes

This chapter contains a review of the basic concepts and constructions concerning affine schemes. Throughout this chapter, rings are commutative and unital, and we allow the zero ring, whose unit is 0. For every ring A, we denote by

$$\gamma_A : \mathbb{Z} \longrightarrow A, \qquad n \longmapsto n1_A,$$

the unique unital ring homomorphism from $\mathbb{Z}$ to A.

I Prime spectrum of a ring

Further reading. For affine spectra and the Zariski topology, see [Har77, Chapter II, §2], [EH00], and [Liu02, Chapter 2].

We introduce the set of prime ideals of a ring and basic properties of the spectrum.

I.A Prime ideals

We define prime ideals and the prime spectrum. An ideal $\mathfrak{p}$ of A is *prime* if $\mathfrak{p} \neq A$ and, for all $x, y \in A$, whenever $xy \in \mathfrak{p}$, we must have $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. This is equivalent to the fact that $A/\mathfrak{p}$ is an integral domain.

Definition I.A.1. Let A be a ring. We denote by $\mathrm{Spec}(A)$ the set of prime ideals of A, called the *prime spectrum* of A. When we speak of a prime ideal, we exclude the ideal A, sometimes called the *unit ideal.* In particular, if A is the zero ring, then $\mathrm{Spec}(A) = \emptyset$.

Example I.A.2. Let us describe the prime spectrum of some well-known rings.

1. The prime spectrum of $\mathbb{Z}$ consists of the ideal (0), together with all ideals of the form (p), where p is a prime number.

2. More generally, the prime spectrum of a principal ideal domain A consists of the ideal (0), as well as ideals of the form (f), where f is irreducible in A.

3. The prime spectrum of a field $\Bbbk$ is given by the single ideal (0).

Example I.A.3. Let $A = \Bbbk[x]/(x^2)$ be the ring of dual numbers. The prime ideals of A correspond to the prime ideals of $\Bbbk[x]$ containing (x^2). Since every such prime ideal contains x, there is only one prime ideal:

$$(\bar{x}) \subset A.$$

Thus $\mathrm{Spec}(\Bbbk[x]/(x^2))$ has a single point.

I.B Functoriality of the prime spectrum

We explain how a ring homomorphism induces a map of spectra. Let $\varphi : A \to B$ be a ring homomorphism. Then we define:

$$\mathrm{Spec}(B) \to \mathrm{Spec}(A), \qquad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}).$$

This is well defined. Indeed, if $\mathfrak{q}$ is a prime ideal of B, then $\varphi^{-1}(\mathfrak{q})$ is also a prime ideal. To see this, note that for any ideal $\mathfrak{q}$ of B, $\varphi^{-1}(\mathfrak{q})$ is the kernel of the composition $A \xrightarrow{\varphi} B \xrightarrow{\pi} B/\mathfrak{q}$. Since $B/\mathfrak{q}$ is an integral domain, this kernel is prime.

Exercise I.B.1. Let $\varphi : A \to B$ be a morphism of rings. Prove that if φ and ψ are composable ring homomorphisms, then

$$f_{\psi\circ\varphi} = f_\varphi \circ f_\psi.$$

Next, show that if φ is surjective, then

$$f_\varphi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$$

is injective.

Solution. Let $\varphi : A \to B$ and $\psi : B \to C$, and let $\mathfrak{r} \in \mathrm{Spec}(C)$. Then

$$f_{\psi\circ\varphi}(\mathfrak{r}) = (\psi \circ \varphi)^{-1}(\mathfrak{r}) = \varphi^{-1}(\psi^{-1}(\mathfrak{r})) = f_\varphi(f_\psi(\mathfrak{r})).$$

Hence $f_{\psi\circ\varphi} = f_\varphi \circ f_\psi$.

If φ is surjective and $\mathfrak{q}_1, \mathfrak{q}_2 \in \mathrm{Spec}(B)$ have the same inverse image in A, then every element of $\mathfrak{q}_1$ is of the form $\varphi(a)$ with $a \in A$. It lies in $\mathfrak{q}_1$ if and only if $a \in \varphi^{-1}(\mathfrak{q}_1) = \varphi^{-1}(\mathfrak{q}_2)$, hence if and only if $\varphi(a) \in \mathfrak{q}_2$. Thus $\mathfrak{q}_1 = \mathfrak{q}_2$.

II The Zariski topology

The prime spectrum of a ring A is naturally equipped with the Zariski topology. The closed sets of this topology are the subsets of the form $\mathbb{V}(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A.

II.A Closed subsets of the affine spectrum

We define closed subsets of the spectrum and study basic set-theoretic properties.

II.A.1 Closed sets and ideals

Let C be a subset of A.

Definition II.A.1. Let C be any subset of A. Define

$$\mathbb{V}(C) := \{\mathfrak{p} \in \mathrm{Spec}(A) \mid C \subset \mathfrak{p}\}.$$

Let $\mathfrak{a}$ be the ideal of A generated by C. Then, given a prime ideal $\mathfrak{p}$ of A, we have $\mathfrak{a} \subset \mathfrak{p}$ if and only if $C \subset \mathfrak{p}$. For this reason we usually consider subsets $\mathbb{V}(\mathfrak{a})$ where $\mathfrak{a}$ is an ideal of A. We have

$$\mathbb{V}(\{0\}) = \mathrm{Spec}(A), \qquad \mathbb{V}(A) = \emptyset.$$

Remark II.A.2. Also, given $\mathfrak{a}$, an ideal of A, we have

$$\mathbb{V}(\mathfrak{a}) := \mathrm{Im}\big(\mathrm{Spec}(A/\mathfrak{a}) \xrightarrow{f_\pi} \mathrm{Spec}(A)\big),$$

where $\pi : A \to A/\mathfrak{a}$ is the projection. Indeed, on one hand, $\mathfrak{p} \in \mathbb{V}(\mathfrak{a})$ if and only if $\mathfrak{p} \supset \mathfrak{a}$. On the other hand, if $\mathfrak{p} \supset \mathfrak{a}$, then we see that $\pi^{-1}(\pi(\mathfrak{p})) = \mathfrak{p}$, because for any element $f \in \pi^{-1}(\pi(\mathfrak{p}))$, we have $\pi(f) \in \pi(\mathfrak{p})$ so $\pi(f) = \pi(g)$ for some $g \in \mathfrak{p}$, so $\pi(f - g) \in \mathfrak{a} \subset \mathfrak{p}$ and thus $f \in \mathfrak{p}$. Also, $\pi(\mathfrak{p})$ is a prime ideal of $A/\mathfrak{a}$ in this case. So given $\mathfrak{p} \in \mathrm{Spec}(A)$, the condition that $f \supset \mathfrak{a}$ is equivalent to $\mathfrak{p} = \pi^{-1}(\mathfrak{q})$, where $\mathfrak{q}$ is a prime ideal of $A/\mathfrak{a}$.

Remark II.A.3. The notion of Zariski closed subset of $\mathrm{Spec}(A)$ parallels the definition of the affine algebraic subsets of $\Bbbk^n$. Indeed, let $A = \Bbbk[x_1, \ldots, x_n]$. Then, among the points of $\mathrm{Spec}(A)$, it is natural to consider the maximal ideals of the form $\mathfrak{m}_a = (x_1 - a_1, \ldots, x_n - a_n)$, where $a = (a_1, \ldots, a_n)$ lies in $\Bbbk^n$. The scheme $\mathrm{Spec}(A)$ is called the n-th affine space over $\Bbbk$, denoted by $\mathbb{A}^n_{\Bbbk}$.

For a given ideal $\mathfrak{a}$ of A, a point $\mathfrak{m}_a$ lies in $\mathbb{V}(\mathfrak{a})$ if and only if $\mathfrak{a} \subset \mathfrak{m}_a$, which means that every $f \in \mathfrak{a}$ lies in $\mathfrak{m}_a$, in other words, that $f(a) = 0$. So we have

$$\mathfrak{m}_a \in \mathbb{V}(\mathfrak{a}) \Leftrightarrow a \in \mathbf{V}(\mathfrak{a}).$$

Example II.A.4. If $A = \Bbbk[x]$, then

$$\mathbb{V}(x) = \{(x)\}, \qquad \mathbb{V}(0) = \mathrm{Spec}(\Bbbk[x]).$$

If $\Bbbk$ is algebraically closed, the closed points of $\mathrm{Spec}(\Bbbk[x])$ are the ideals $(x - a)$, for $a \in \Bbbk$. On the other hand, (0) is a point whose closure is the whole space, since every prime ideal contains 0.

II.A.2 First properties of closed sets

Associating $\mathbb{V}(\mathfrak{a})$ to $\mathfrak{a}$ is an inclusion-reversing operation, namely, given ideals $\mathfrak{a}$ and $\mathfrak{b}$ of A, we have $\mathbb{V}(\mathfrak{b}) \subset \mathbb{V}(\mathfrak{a})$ if $\mathfrak{a} \subset \mathfrak{b}$. The next properties are crucial to define the subsets of the form $\mathbb{V}(\mathfrak{a})$, where $\mathfrak{a}$ runs in the set of all ideals of A, as the closed subsets of a topology.

Lemma II.A.5. *We have the following equalities of closed subsets.*

1. *For all ideals $\mathfrak{a}$, $\mathfrak{b}$ of A : $\mathbb{V}(\mathfrak{a}) \cup \mathbb{V}(\mathfrak{b}) = \mathbb{V}(\mathfrak{a}\mathfrak{b})$, where $\mathfrak{a}\mathfrak{b} = (xy \mid x \in \mathfrak{a}, y \in \mathfrak{b})$.*
2. *If $(\mathfrak{a}_i)_{i \in I}$ is a family of ideals and $\mathfrak{a} = \sum_{i \in I} \mathfrak{a}_i$, then $\mathbb{V}(\mathfrak{a}) = \bigcap_{i \in I} \mathbb{V}(\mathfrak{a}_i)$.*

Proof. For the first item, of course $\mathbb{V}(\mathfrak{a}\mathfrak{b}) \supset \mathbb{V}(\mathfrak{a}) \cup \mathbb{V}(\mathfrak{b})$. For the reverse inclusion, let $\mathfrak{p} \supset \mathfrak{a}\mathfrak{b}$ be a prime ideal. If $\mathfrak{p}$ contains neither $\mathfrak{a}$ nor $\mathfrak{b}$, then we may pick $x \in \mathfrak{a} \setminus \mathfrak{p}$, $y \in \mathfrak{b} \setminus \mathfrak{p}$. Then $xy \in \mathfrak{a}\mathfrak{b} \subset \mathfrak{p}$, a contradiction. About the second item, note that a prime ideal contains $\sum_{i \in I} \mathfrak{a}_i$ if and only if it contains each $\mathfrak{a}_i$, for all $i \in I$. $\square$

Proposition II.A.6. *Let $\mathfrak{p} \in \mathrm{Spec}(A)$. Then $\overline{\{\mathfrak{p}\}} = \mathbb{V}(\mathfrak{p})$. The ideal $\mathfrak{p}$ is maximal if and only if $\{\mathfrak{p}\}$ is closed in $\mathrm{Spec}(A)$, which is to say, if and only if $\mathbb{V}(\mathfrak{p}) = \{\mathfrak{p}\}$.*

Proof. A point $\mathfrak{q}$ lies in $\overline{\{\mathfrak{p}\}}$ if and only if, for all ideals $\mathfrak{a}$ of A with $\mathfrak{p} \in \mathbb{V}(\mathfrak{a})$, one has $\mathfrak{q} \in \mathbb{V}(\mathfrak{a})$. In other words, $\mathfrak{q}$ lies in $\overline{\{\mathfrak{p}\}}$ if and only if, for all ideals $\mathfrak{a}$ of A with $\mathfrak{a} \subset \mathfrak{p}$, one has $\mathfrak{a} \subset \mathfrak{q}$. Therefore, any point $\mathfrak{q} \in \mathbb{V}(\mathfrak{p})$ lies in $\overline{\{\mathfrak{p}\}}$, as $\mathfrak{q}$ satisfies $\mathfrak{p} \subset \mathfrak{q}$, so $\mathfrak{a} \subset \mathfrak{p}$ implies $\mathfrak{a} \subset \mathfrak{q}$. Conversely, given $\mathfrak{q} \in \overline{\{\mathfrak{p}\}}$, choosing $\mathfrak{a} = \mathfrak{p}$, we have $\mathfrak{a} \subset \mathfrak{a} = \mathfrak{p}$, so $\mathfrak{p} = \mathfrak{a} \subset \mathfrak{q}$. Therefore $\mathfrak{q} \in \mathbb{V}(\mathfrak{p})$.

For the second statement, if $\mathfrak{p}$ is maximal, then the only point of $\overline{\{\mathfrak{p}\}}$ is $\mathfrak{p}$. Indeed, if a prime ideal $\mathfrak{q}$ lies in $\overline{\{\mathfrak{p}\}}$, using the above characterisation and choosing $\mathfrak{a} = \mathfrak{p}$ we have $\mathfrak{a} = \mathfrak{p} \subset \mathfrak{q}$, so $\mathfrak{q} = \mathfrak{p}$ since $\mathfrak{p}$ is maximal. Conversely, if $\mathfrak{p}$ is not maximal, then there is a maximal ideal $\mathfrak{q}$ strictly containing $\mathfrak{p}$, so that $\mathfrak{q} \in \mathbb{V}(\mathfrak{p}) \setminus \{\mathfrak{p}\}$. □

Lemma II.A.7. *Let $\varphi : A \to B$ be a ring morphism. Then $f_\varphi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is continuous.*

Proof. Given an ideal $\mathfrak{a}$ of A, we have

$$f_\varphi^{-1}(\mathbb{V}(\mathfrak{a})) = \mathbb{V}(\varphi(\mathfrak{a})).$$

Indeed, on one hand, by definition a prime ideal $\mathfrak{q}$ of B contains $\varphi(\mathfrak{a})$ if and only if $\mathfrak{q} \in \mathbb{V}(\varphi(\mathfrak{a}))$. On the other hand $\mathfrak{q}$ lies in $f_\varphi^{-1}(\mathbb{V}(\mathfrak{a}))$ if and only if $f_\varphi(\mathfrak{q}) \in \mathbb{V}(\mathfrak{a})$, which happens if and only if $\varphi^{-1}(\mathfrak{q})$ contains $\mathfrak{a}$ and applying φ we see that this amounts precisely to $\mathfrak{q} \supset \varphi(\mathfrak{a})$. □

II.B Intersection of primes and the radical

It turns out that the radical of an ideal is the intersection of prime ideals containing it.

Lemma II.B.1. *Let $\mathfrak{a}$ be an ideal of a ring A. Then:*

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{p} \in \mathbb{V}(\mathfrak{a})} \mathfrak{p}.$$

In particular, the nilradical $\mathrm{Rad}(A) = \sqrt{0}$ is the intersection of all prime ideals of A.

Proof. Let us check that the left-hand side is contained in the right-hand side. Given a prime ideal $\mathfrak{p} \in \mathbb{V}(\mathfrak{a})$, by definition $\mathfrak{a} \subset \mathfrak{p}$, so $\sqrt{\mathfrak{a}} \subset \sqrt{\mathfrak{p}} = \mathfrak{p}$.

To check the opposite inclusion, we first consider the ring $B = A/\mathfrak{a}$ and the projection $\varphi : A \to B$. Then $\varphi^{-1}(\mathrm{Rad}(B)) = \sqrt{\mathfrak{a}}$ and sending $\mathfrak{q} \in \mathbb{V}(0)$ to $\varphi^{-1}(\mathfrak{q})$ gives a bijection of $\mathbb{V}(0)$ and $\mathbb{V}(\mathfrak{a})$. Therefore, up to replacing A by B and $\mathfrak{a}$ by 0, it suffices to prove the statement for $\mathfrak{a} = 0$.

So let us show that an element $a \in A$ lying in all prime ideals $\mathfrak{p}$ of A is nilpotent. Suppose the contrary, so that the localisation A_a is a non-zero ring. Then this ring has a prime ideal $\mathfrak{q}$. Pulling back $\mathfrak{q}$ to A, we get a prime ideal $\mathfrak{p}$, which does not contain a. A contradiction. □

Proposition II.B.2 (The image of a morphism of spectra). *Let $\varphi : A \to B$ be a ring homomorphism and let $f_\varphi : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the induced map. Then*

$$f_\varphi(\mathrm{Spec}(B)) \subset \mathbb{V}(\ker\varphi) \qquad \textit{and} \qquad \overline{f_\varphi(\mathrm{Spec}(B))} = \mathbb{V}(\ker\varphi).$$

In general, the first inclusion need not be an equality.

Proof. For every $\mathfrak{q} \in \mathrm{Spec}(B)$, the contracted prime $\varphi^{-1}(\mathfrak{q})$ contains $\ker\varphi$, so the image is contained in $\mathbb{V}(\ker\varphi)$. The smallest closed subset containing the image is defined by the ideal

$$\bigcap_{\mathfrak{q} \in \mathrm{Spec}(B)} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\left(\bigcap_{\mathfrak{q} \in \mathrm{Spec}(B)} \mathfrak{q}\right) = \varphi^{-1}(\mathrm{Rad}(B)).$$

An element $a \in A$ belongs to this last ideal if and only if $\varphi(a)$ is nilpotent, equivalently if and only if $a^n \in \ker\varphi$ for some $n \geq 1$. Hence

$$\varphi^{-1}(\mathrm{Rad}(B)) = \sqrt{\ker\varphi},$$

and the corresponding closed subset is $\mathbb{V}(\ker\varphi)$. $\square$

Exercise II.B.3. Compute the image of the map on spectra in each of the following cases, and compare it with $\mathbb{V}(\ker\varphi)$.

(i) The quotient map $A \to A/\mathfrak{a}$.

(ii) The localisation map $A \to S^{-1}A$.

(iii) The homomorphism $\gamma_{\mathbb{Q}} : \mathbb{Z} \hookrightarrow \mathbb{Q}$.

Solution. For a quotient, contraction gives a bijection between the primes of $A/\mathfrak{a}$ and the primes of A containing $\mathfrak{a}$. Thus the image is exactly $\mathbb{V}(\mathfrak{a}) = \mathbb{V}(\ker\varphi)$.

The image of $\mathrm{Spec}(S^{-1}A) \to \mathrm{Spec}(A)$ is

$$\{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\}.$$

It is usually not closed, although its closure is $\mathbb{V}(\ker(A \to S^{-1}A))$.

Finally, $\mathrm{Spec}(\mathbb{Q})$ has one point, whose image in $\mathrm{Spec}(\mathbb{Z})$ is (0). Since $\ker(\gamma_{\mathbb{Q}}) = 0$, Proposition II.B.2 gives

$$\overline{\{(0)\}} = \mathbb{V}(0) = \mathrm{Spec}(\mathbb{Z}).$$

Thus the image consists only of the generic point but is dense.

Exercise II.B.4. Show the equality of subsets $\mathbb{V}(\mathfrak{a}) = \mathbb{V}(\sqrt{\mathfrak{a}})$ for every ideal $\mathfrak{a} \subset A$.

Solution. A prime ideal $\mathfrak{p}$ contains $\mathfrak{a}$ if and only if it contains $\sqrt{\mathfrak{a}}$. One implication is immediate from $\mathfrak{a} \subset \sqrt{\mathfrak{a}}$. Conversely, if $\mathfrak{a} \subset \mathfrak{p}$, then $\sqrt{\mathfrak{a}} \subset \sqrt{\mathfrak{p}} = \mathfrak{p}$ because $\mathfrak{p}$ is radical.

Lemma II.B.5. *Given ideals $\mathfrak{a}$ and $\mathfrak{b}$ of A, we have $\mathbb{V}(\mathfrak{a}) \subset \mathbb{V}(\mathfrak{b})$ if and only if $\mathfrak{b} \subset \sqrt{\mathfrak{a}}$. Moreover:*

$$\mathbb{V}(\mathfrak{a}) = \emptyset \Leftrightarrow \mathfrak{a} = A, \qquad \mathbb{V}(\mathfrak{a}) = \mathrm{Spec}(A) \Leftrightarrow \mathfrak{a} \subset \mathrm{Rad}(A).$$

Proof. For the statement about inclusions, assume $\mathbb{V}(\mathfrak{a}) \subset \mathbb{V}(\mathfrak{b})$. Then $\mathfrak{b} \subset \sqrt{\mathfrak{b}} \subset \sqrt{\mathfrak{a}}$. Conversely, suppose $\mathfrak{b} \subset \sqrt{\mathfrak{a}}$. Then, given a prime ideal $\mathfrak{p} \in \mathbb{V}(\mathfrak{a})$, we have $\mathfrak{b} \subset \sqrt{\mathfrak{a}} \subset \sqrt{\mathfrak{p}} = \mathfrak{p}$, so $\mathfrak{p} \in \mathbb{V}(\mathfrak{b})$.

Let us show the statement appearing in the last display. If $\mathbb{V}(\mathfrak{a}) = \emptyset$, then there is no prime ideal containing $\mathfrak{a}$, so $\mathfrak{a} = A$, because any ideal different from A is contained in some maximal (hence prime) ideal. Conversely, if $\mathfrak{a} = A$, then there is no prime ideal containing $\mathfrak{a}$, so $\mathbb{V}(\mathfrak{a}) = \emptyset$. Also, if $\mathbb{V}(\mathfrak{a}) = \mathrm{Spec}(A)$ then any prime ideal of A contains $\mathfrak{a}$, so $\mathfrak{a}$ is contained in the intersection of all prime ideals of A, i.e., in $\mathrm{Rad}(A)$. Conversely, if $\mathfrak{a} \subset \mathrm{Rad}(A)$, then $\mathfrak{a}$ is contained in every prime ideal of A so $\mathbb{V}(\mathfrak{a}) = \mathrm{Spec}(A)$. $\square$

II.C Dominant morphisms

In the next proposition, a *dominant morphism* refers to a morphism of topological spaces such that the image is dense in the target. We use the notion of *radical* of a ring.

Example II.C.1. The inclusion $\Bbbk[x] \subset \Bbbk[x, y]$ induces the projection

$$\mathrm{Spec}(\Bbbk[x, y]) \to \mathrm{Spec}(\Bbbk[x]).$$

Its image is dense. Geometrically, this says that the projection from the affine plane to the affine line is dominant.

Proposition II.C.2. *Let $\varphi : A \to B$ be an injective morphism of rings. Then the induced map f_φ : $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is dominant.*

Proof. Since the nilradical of a ring is the set of nilpotent elements, we have $\mathrm{Rad}(B) \cap A \subset \mathrm{Rad}(A)$. Therefore, given an ideal $\mathfrak{a}$ of A, we have $\mathfrak{a} \subset \mathrm{Rad}(A)$, as soon as $\mathfrak{a} \subset \mathrm{Rad}(B) \cap A$.

Now, take a closed subset $\mathbb{V}(\mathfrak{a}) \subset \mathrm{Spec}(A)$, for some ideal $\mathfrak{a}$ of A, and assume that $\mathbb{V}(\mathfrak{a})$ contains the image of f_φ, so that $\mathfrak{a}$ is contained in $\mathfrak{q} \cap A$, for all prime ideals $\mathfrak{q}$ of B. Then, by Lemma II.B.1, we have

$$\mathfrak{a} \subset \bigcap_{\mathfrak{q} \in \mathrm{Spec}(B)} (\mathfrak{q} \cap A) = \mathrm{Rad}(B) \cap A \subset \mathrm{Rad}(A).$$

Then $\mathbb{V}(\mathfrak{a}) = \mathrm{Spec}(A)$, by Lemma II.B.5. So there is no closed strict subset of $\mathrm{Spec}(A)$ containing the image of f_φ, which says that f_φ is dominant. $\square$

Exercise II.C.3. Let $\varphi : A \to B$ be a morphism of rings and let $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the induced morphism. Generalise the previous proposition by showing that f is dominant if and only if

$$\ker(\varphi) \subset \mathrm{Rad}(A).$$

Deduce that, if A is reduced, then f is dominant if and only if φ is injective.

Solution. The image of f is contained in the closed subset $\mathbb{V}(\mathfrak{a})$ if and only if every prime ideal of B pulls back to a prime ideal of A containing $\mathfrak{a}$. This is equivalent to saying that

$$\mathfrak{a} \subset \varphi^{-1}(\mathfrak{q})$$

for every $\mathfrak{q} \in \mathrm{Spec}(B)$, or equivalently

$$\varphi(\mathfrak{a}) \subset \mathfrak{q}$$

for every prime ideal $\mathfrak{q}$ of B.

Thus $\mathbb{V}(\mathfrak{a})$ contains the image of f if and only if

$$\varphi(\mathfrak{a}) \subset \bigcap_{\mathfrak{q} \in \mathrm{Spec}(B)} \mathfrak{q} = \mathrm{Rad}(B).$$

In particular, the closure of the image of f is controlled by those elements of A whose image in B is nilpotent.

Now f is dominant if and only if the only closed subset of $\mathrm{Spec}(A)$ containing the image is the whole space. Equivalently, if an element $a \in A$ maps to a nilpotent element of B, then a must be nilpotent in A. In particular this gives

$$\ker(\varphi) \subset \mathrm{Rad}(A).$$

Conversely, if $\ker(\varphi) \subset \mathrm{Rad}(A)$, then

$$\mathrm{Spec}(B) \to \mathrm{Spec}(A)$$

has dense image. Indeed, after replacing A by $A/\ker(\varphi)$, the map $A/\ker(\varphi) \to B$ is injective. The induced map

$$\mathrm{Spec}(A/\ker(\varphi)) \to \mathrm{Spec}(A)$$

has image $\mathbb{V}(\ker(\varphi))$, which is dense in $\mathrm{Spec}(A)$ precisely because

$$\ker(\varphi) \subset \mathrm{Rad}(A).$$

Since $\mathrm{Spec}(B) \to \mathrm{Spec}(A/\ker(\varphi))$ is dominant by the injective case, the composite map is dominant.

If A is reduced, then $\mathrm{Rad}(A) = 0$, so the condition becomes

$$\ker(\varphi) = 0,$$

namely φ is injective.

Remark II.C.4. Dominance should not be checked only on maximal ideals in arbitrary rings. For example, the homomorphism

$$\gamma_{\mathbb{Q}} : \mathbb{Z} \hookrightarrow \mathbb{Q}$$

induces

$$\mathrm{Spec}(\mathbb{Q}) \to \mathrm{Spec}(\mathbb{Z}).$$

The image consists only of the generic point (0), but this point is dense in $\mathrm{Spec}(\mathbb{Z})$. Hence the morphism is dominant, although no closed point $(p) \in \mathrm{Spec}(\mathbb{Z})$ lies in the image.

Remark II.C.5. If $\Bbbk$ is algebraically closed and A, B are finitely generated reduced $\Bbbk$-algebras, then maximal ideals correspond to $\Bbbk$-rational points by the Nullstellensatz. In this setting, a morphism

$$f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$$

is dominant if and only if the image on closed points is Zariski dense. Equivalently, for affine varieties over $\Bbbk$, a morphism is dominant if its image contains a dense subset of classical points.

Exercise II.C.6. Let $\Bbbk$ be a field and consider the inclusion

$$\varphi : \Bbbk[x] \hookrightarrow \Bbbk[x, x^{-1}].$$

Describe the image of the induced morphism $\mathrm{Spec}(\Bbbk[x, x^{-1}]) \to \mathrm{Spec}(\Bbbk[x])$. What is its image on closed points when $\Bbbk$ is algebraically closed?

Solution. The ring $\Bbbk[x, x^{-1}]$ is the localisation of $\Bbbk[x]$ at the element x. Therefore the morphism

$$\mathrm{Spec}(\Bbbk[x, x^{-1}]) \to \mathrm{Spec}(\Bbbk[x])$$

identifies $\mathrm{Spec}(\Bbbk[x, x^{-1}])$ with the principal open subset

$$D(x) = \{\mathfrak{p} \in \mathrm{Spec}(\Bbbk[x]) \mid x \notin \mathfrak{p}\}.$$

Thus the image is precisely $D(x)$. The morphism is dominant because the ring morphism $\Bbbk[x] \hookrightarrow \Bbbk[x, x^{-1}]$ is injective.

Assume now that $\Bbbk$ is algebraically closed. The closed points of $\mathrm{Spec}(\Bbbk[x])$ are the maximal ideals

$$(x - a), \qquad a \in \Bbbk.$$

The point $(x - a)$ belongs to $D(x)$ if and only if

$$x \notin (x - a).$$

This happens if and only if $a \neq 0$. Hence the image on closed points is

$$\{(x - a) \mid a \in \Bbbk,\ a \neq 0\}.$$

So the closed point (x) is missing, but the image is still dense in $\mathrm{Spec}(\Bbbk[x])$.

Remark II.C.7. This example shows that a dominant morphism need not be surjective, even on closed points. Dominance only means that the image is dense.

II.D Open and principal open subsets

We define the Zariski topology on $\mathrm{Spec}(A)$ using the sets $\mathbb{V}(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A.

Definition II.D.1. We consider the topology on $\mathrm{Spec}(A)$ whose open subsets are $D(\mathfrak{a}) := \mathrm{Spec}(A) \setminus \mathbb{V}(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A. For $f \in A$, we have the set $D(f) = \mathrm{Spec}(A) \setminus \mathbb{V}(f)$. This is called a *principal open* subset. Let us spell out:

$$D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \notin \mathfrak{p}\}.$$

When f runs in A, the principal open subsets $D(f)$ form a base of the Zariski topology. Indeed, for any ideal $\mathfrak{a}$ of A we have

$$D(\mathfrak{a}) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \exists f \in \mathfrak{a} \setminus \mathfrak{p}\} = \bigcup_{f \in \mathfrak{a}} \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \notin \mathfrak{p}\} = \bigcup_{f \in \mathfrak{a}} D(f).$$

Example II.D.2 (The double point and localisation). Consider the ring of dual numbers

$$A = \Bbbk[x]/(x^2)$$

and write $\bar{x}$ for the class of x in A. The affine scheme

$$X = \mathrm{Spec}(A)$$

has a single point, namely the prime ideal

$$\mathfrak{p} = (\bar{x}).$$

Indeed, prime ideals of A correspond to prime ideals of $\Bbbk[x]$ containing (x^2), and any such prime ideal contains x.

Let us look at the principal open subsets of X. First,

$$D(0) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid 0 \notin \mathfrak{p}\} = \emptyset,$$

because every ideal contains 0. Correspondingly,

$$A_0 = 0,$$

since localising at 0 forces 0 to become invertible, hence forces $1 = 0$.

On the other hand,

$$D(\bar{x}) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \bar{x} \notin \mathfrak{p}\} = \emptyset,$$

because the unique prime ideal $\mathfrak{p} = (\bar{x})$ contains $\bar{x}$. This is also reflected algebraically by the localisation:

$$A_{\bar{x}} = 0.$$

Indeed, in $A_{\bar{x}}$, the element $\bar{x}$ becomes invertible. But

$$\bar{x}^2 = 0.$$

Multiplying by $\bar{x}^{-2}$, we get $1 = 0$. Thus the localisation is the zero ring.

Finally,

$$D(1) = X,$$

and

$$A_1 = A.$$

More generally, if $a \in A$, then $D(a) = X$ precisely when $a \notin (\bar{x})$, i.e. when a has non-zero constant term. In that case a is already invertible in A, so

$$A_a \simeq A.$$

If $a \in (\bar{x})$, then a is nilpotent, hence $A_a = 0$, and $D(a) = \emptyset$.

Lemma II.D.3. *Let $X = \mathrm{Spec}(A)$ and consider a collection $(f_j \mid j \in J)$ of elements of A. The open subsets $(D(f_j) \mid j \in J)$ cover X if and only if $(f_j \mid j \in J)$ generate the unit ideal. Therefore, X is quasi-compact.*

Proof. The subsets $(D(f_j) \mid j \in J)$ cover X if and only if, for any prime ideal $\mathfrak{p}$ of A, there is $j \in J$ such that f_j does not belong to $\mathfrak{p}$, i.e., no prime ideal $\mathfrak{p}$ of A contains all the f_j's. Let $\mathfrak{a}$ be the ideal generated by $(f_j \mid j \in J)$. If $\mathfrak{a}$ is the unit ideal, then it is contained in no prime ideal, so no prime ideal contains all the f_j's, and conversely, if $\mathfrak{a}$ is not the unit ideal then it is contained in some prime ideal $\mathfrak{p}$, which therefore contains all the f_j's. So the first statement is proved.

About quasi-compactness, let $(U_i \mid i \in I)$ be an open cover of X and write $U_i = X \setminus \mathbb{V}(\mathfrak{a}_i)$ for some ideal $\mathfrak{a}_i$ of A, for all $i \in I$. Then we write $D(\mathfrak{a}_i) = \cup_{f \in \mathfrak{a}_i} D(f)$, so

$$X = \bigcup_{i \in I} \bigcup_{f \in \mathfrak{a}_i} D(f).$$

Therefore the ideal generated by all elements belonging to the ideals $\mathfrak{a}_i$ is the unit ideal. By the definition of an ideal generated by a set, there are finitely many elements $f_1, \ldots, f_r$, with each f_j belonging to some $\mathfrak{a}_{i_j}$, and coefficients $u_1, \ldots, u_r \in A$ such that

$$1 = u_1 f_1 + \cdots + u_r f_r.$$

Hence $D(f_1), \ldots, D(f_r)$ cover X, and therefore $U_{i_1}, \ldots, U_{i_r}$ already cover X. Thus X is quasi-compact. □

Exercise II.D.4. Let A be a Noetherian ring. Then every open subset of $\mathrm{Spec}(A)$ is quasi-compact.

Solution. Since A is Noetherian, every ideal is finitely generated. Therefore every closed subset of $\mathrm{Spec}(A)$ is of the form $\mathbb{V}(\mathfrak{a})$ with $\mathfrak{a}$ finitely generated. This implies that every descending chain of closed subsets stabilizes, so $\mathrm{Spec}(A)$ is a Noetherian topological space.

In a Noetherian topological space, every open subset is quasi-compact. Indeed, if an open subset U had an open cover with no finite subcover, choose such a counterexample maximal among open subsets, or equivalently use the descending chain condition on closed complements. The standard Noetherian induction argument then gives a contradiction.

II.D.1 Localisation and principal open subsets

Let $f \in A$ and consider $D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \notin \mathfrak{p}\}$. Recall that the localised ring A_f is $A[f^{-1}] = S^{-1}A$ with $S = \{f^n \mid n \in \mathbb{N}\}$. Then sending a prime ideal $\mathfrak{p} \in D(f)$ to $\mathfrak{p}A_f$ gives a map:

$$D(f) \to \mathrm{Spec}(A_f).$$

We note that this is well-defined, since for all $\mathfrak{p} \in D(f)$, the ideal $\mathfrak{p}A_f$ is prime. Indeed, given elements $a, b \in A$ and integers $n, m \geq 0$, we have $a/f^n \cdot b/f^m \in \mathfrak{p}A_f$ if and only if there are $x_0, \ldots, x_N$ in $\mathfrak{p}$ and $a_0, \ldots, a_N$ in A such that

$$\frac{ab}{f^{n+m}} = \sum_{0 \leq i \leq N} \frac{x_i}{f^i},$$

where we may assume $N \geq n + m$. Then:

$$f^{N-n-m} ab = \sum_{0 \leq i \leq N} x_i f^{N-i} \in \mathfrak{p}.$$

So, since $\mathfrak{p}$ is prime and $f \notin \mathfrak{p}$, we get that $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$ so $a/f^n \in \mathfrak{p}A_f$ or $b/f^m \in \mathfrak{p}A_f$.

Exercise II.D.5. Show that $D(f) \cap D(g) = D(fg)$.

Solution. A prime ideal $\mathfrak{p}$ lies in $D(f) \cap D(g)$ if and only if $f \notin \mathfrak{p}$ and $g \notin \mathfrak{p}$. Since $\mathfrak{p}$ is prime, this is equivalent to $fg \notin \mathfrak{p}$, namely to $\mathfrak{p} \in D(fg)$.

Lemma II.D.6. *Let $f, g \in A$. Then $D(f) \subset D(g)$ if and only if g divides a power of f, i.e. if and only if $f^m \in (g)$ for some $m \geq 1$. In this case there is a morphism $A_g \to A_f$, compatible with the localisation maps from A.*

Proof. We have $D(f) \subset D(g)$ if and only if $\mathbb{V}(g) \subset \mathbb{V}(f)$. By Lemma II.B.1, this is equivalent to

$$\sqrt{(g)} \supset \sqrt{(f)}.$$

In turn, this means that $f^m \in (g)$ for some $m \geq 1$, i.e. that g divides a power of f.

If this is the case, write $f^m = gh$ for some $h \in A$. Then g becomes invertible in A_f, with inverse h/f^m. By the universal property of localisation, the localisation map $A \to A_f$ therefore factors uniquely through a ring morphism

$$A_g \longrightarrow A_f.$$

□

II.D.2 Residue field

Let A be a ring. We introduce a basic object attached to a point of $\mathrm{Spec}(A)$ given by a prime ideal $\mathfrak{p}$ of A, namely the fraction field of the integral domain $A/\mathfrak{p}$.

Definition II.D.7. Let $\mathfrak{p}$ be a prime ideal of A. Then the *residue field* $\kappa(\mathfrak{p})$ is defined as

$$\kappa(\mathfrak{p}) = \mathrm{Frac}(A/\mathfrak{p}).$$

Example II.D.8. For $A = \Bbbk[x]$, the residue field at the prime ideal (0) is

$$\kappa((0)) = \Bbbk(x),$$

while the residue field at $(x - a)$, for $a \in \Bbbk$, is

$$\kappa((x - a)) \simeq \Bbbk.$$

If $f \in \Bbbk[x]$ is irreducible, then the residue field at (f) is the field obtained by adjoining one root of f to $\Bbbk$. Indeed, since (f) is maximal,

$$\kappa((f)) = \Bbbk[x]/(f).$$

Lemma II.D.9. *Let S be a multiplicative subset of A and $\mathfrak{a}$ be an ideal of A. Set T for the image of S in $B = A/\mathfrak{a}$. Then there is an isomorphism of rings*

$$S^{-1}A/S^{-1}\mathfrak{a} \simeq T^{-1}B.$$

Proof. One first checks that T is a multiplicative subset of B. Then, we define a map $S^{-1}A \to T^{-1}B$ by sending a/s to $\bar{a}/\bar{s}$, where the canonical projection $A \to A/\mathfrak{a}$, a surjective morphism of rings, is denoted by $a \mapsto \bar{a}$. One verifies that this map is well-defined, for $a/s = b/t$ implies $u(at - bs) = 0$ for some $u \in S$ so that $\bar{u}(\bar{a}\bar{t} - \bar{b}\bar{s}) = \bar{0}$ with $\bar{u} \in T$ so that $\bar{a}/\bar{s} = \bar{b}/\bar{t}$.

The map $S^{-1}A \to T^{-1}B$ is surjective. An element a/s lies in its kernel if and only if $\bar{a}/\bar{s} = 0$ in $T^{-1}B$, namely if and only if there exists $u \in S$ such that $ua \in \mathfrak{a}$. This is exactly the condition that a/s belong to $S^{-1}\mathfrak{a}$. Hence the kernel is $S^{-1}\mathfrak{a}$.

□

The previous lemma recovers Lemma III.C.2. Indeed, when $S = A \setminus \mathfrak{p}$, with $\mathfrak{p}$ a prime ideal of A, by definition $S^{-1}A = A_\mathfrak{p}$ and $S^{-1}\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$ so we get that $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ is isomorphic to the localisation of $A/\mathfrak{p}$ at its zero prime ideal, which is the fraction field of $A/\mathfrak{p}$, i.e. the residue field $\kappa(\mathfrak{p})$. This says, in particular, that $\mathfrak{p}A_\mathfrak{p}$ is a maximal ideal of $A_\mathfrak{p}$. But we know that more is true about $A_\mathfrak{p}$. Namely, $A_\mathfrak{p}$ is a local ring, with maximal ideal $\mathfrak{p}A_\mathfrak{p}$, as we saw in Proposition III.C.1.

Exercise II.D.10. Compute the residue fields of the following points.

i) The points (0) and (p) of $\mathrm{Spec}(\mathbb{Z})$, where p is a prime number.

ii) The points (0) and (f) of $\mathrm{Spec}(\mathbb{F}_p[t])$, where $f \in \mathbb{F}_p[t]$ is irreducible.

iii) The points (p) and $(p, \bar{f})$ of $\mathrm{Spec}(\mathbb{Z}[t])$, where p is a prime number and $\bar{f} \in \mathbb{F}_p[t]$ is irreducible.

Solution. Recall that, for a prime ideal $\mathfrak{p} \subset A$, the residue field is

$$\kappa(\mathfrak{p}) = \mathrm{Frac}(A/\mathfrak{p}).$$

i) In $\mathrm{Spec}(\mathbb{Z})$, the ideal (0) is prime and

$$\kappa((0)) = \mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}.$$

If p is a prime number, then (p) is maximal and

$$\kappa((p)) = \mathrm{Frac}(\mathbb{Z}/(p)) = \mathbb{F}_p.$$

ii) In $\mathrm{Spec}(\mathbb{F}_p[t])$, the ideal (0) is prime and

$$\kappa((0)) = \mathrm{Frac}(\mathbb{F}_p[t]) = \mathbb{F}_p(t).$$

If $f \in \mathbb{F}_p[t]$ is irreducible, then (f) is maximal and

$$\kappa((f)) = \mathrm{Frac}(\mathbb{F}_p[t]/(f)) = \mathbb{F}_p[t]/(f).$$

This is a finite extension of $\mathbb{F}_p$ of degree $\deg(f)$.

iii) In $\mathrm{Spec}(\mathbb{Z}[t])$, the ideal (p) is prime because

$$\mathbb{Z}[t]/(p) \simeq \mathbb{F}_p[t],$$

which is an integral domain. Therefore

$$\kappa((p)) = \mathrm{Frac}(\mathbb{Z}[t]/(p)) = \mathrm{Frac}(\mathbb{F}_p[t]) = \mathbb{F}_p(t).$$

Now let $\bar{f} \in \mathbb{F}_p[t]$ be irreducible and consider the ideal

$$\mathfrak{p} = (p, \bar{f}) \subset \mathbb{Z}[t],$$

where, more precisely, we choose a lift $f \in \mathbb{Z}[t]$ of $\bar{f}$ and write

$$\mathfrak{p} = (p, f).$$

Then

$$\mathbb{Z}[t]/(p, f) \simeq \mathbb{F}_p[t]/(\bar{f}),$$

which is a field. Hence $\mathfrak{p}$ is maximal and

$$\kappa(\mathfrak{p}) = \mathbb{F}_p[t]/(\bar{f}).$$

II.D.3 Localisation and homeomorphism

We study how prime ideals and localisation interact, proving that the map on spectra is a homeomorphism onto its image. In the next result, $S \subset A$ is a multiplicative subset and a is an element of A.

Lemma II.D.11. *Let $\mathfrak{b}$ be an ideal of $S^{-1}A$ and $\mathfrak{a} = \varphi^{-1}(\mathfrak{b})$. Then*

$$\mathfrak{b} = S^{-1}\mathfrak{a}.$$

Also, if $\mathfrak{b} \neq S^{-1}A$, then $\mathfrak{a} \cap S = \emptyset$.

Proof. We have $\mathfrak{a} = \{a \in A \mid a/1 \in \mathfrak{b}\}$. So $S^{-1}\mathfrak{a} \subset \mathfrak{b}$. Conversely, let $a/s \in \mathfrak{b}$. Then $a/1 = (a/s)\cdot(s/1) \in \mathfrak{b}$, so $a \in \mathfrak{a}$. Hence $a/s \in S^{-1}\mathfrak{a}$, so $\mathfrak{b} \subset S^{-1}\mathfrak{a}$.

Assume now $\mathfrak{b} \neq S^{-1}A$ and, by contradiction, suppose $\mathfrak{a} \cap S \neq \emptyset$, so let $a \in \mathfrak{a} \cap S$. Then $\varphi(a) = a/1 \in \mathfrak{b}$ and, since $a \in S$, the ring $S^{-1}A$ contains the element $b = 1/a$. So $1 = a/1 \cdots b \in \mathfrak{b}$ and $\mathfrak{b} = S^{-1}A$, a contradiction. ☐

Proposition II.D.12. *Let $\varphi : A \to S^{-1}A$ be the map induced by localisation. Then $f_\varphi : \mathrm{Spec}(S^{-1}A) \to \mathrm{Spec}(A)$ is injective and gives a homeomorphism $\mathrm{Spec}(S^{-1}A) \simeq \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\}$. In particular, $D(f) \to \mathrm{Spec}(A_f)$ is a homeomorphism.*

Proof. Let $\mathfrak{b}$ be a prime ideal of $S^{-1}A$ and set $\mathfrak{a} = \varphi^{-1}(\mathfrak{b})$. By Lemma II.D.11, we have $\mathfrak{b} = S^{-1}\mathfrak{a}$, so $\mathfrak{b}$ is determined by its inverse image. This proves injectivity.

If $\mathfrak{b}$ is prime, then $\mathfrak{a} \cap S = \emptyset$, because every element of S becomes invertible in $S^{-1}A$. Conversely, if $\mathfrak{a} \in \mathrm{Spec}(A)$ and $\mathfrak{a} \cap S = \emptyset$, then $S^{-1}\mathfrak{a}$ is a prime ideal of $S^{-1}A$, since

$$S^{-1}A/S^{-1}\mathfrak{a} \simeq T^{-1}(A/\mathfrak{a}),$$

where T is the image of S in the domain $A/\mathfrak{a}$, and this localisation is again a domain. These two constructions are inverse to each other.

It remains to check the topology. For an ideal $\mathfrak{b} \subset S^{-1}A$, with inverse image $\mathfrak{a} \subset A$, the image of $\mathbb{V}(\mathfrak{b})$ is

$$\mathbb{V}(\mathfrak{a}) \cap \{\mathfrak{p} \in \mathrm{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\}.$$

Thus the bijection is closed onto its image, hence a homeomorphism. Taking $S = \{1, f, f^2, \ldots\}$ gives the homeomorphism $\mathrm{Spec}(A_f) \simeq D(f)$. ☐

Exercise II.D.13. Let A be a ring and let $f \in A$. Show that the localisation map $A \to A_f$ induces a homeomorphism

$$\mathrm{Spec}(A_f) \simeq D(f) \subset \mathrm{Spec}(A).$$

Then apply this to $A = \Bbbk[x]$ and describe the points of $\mathrm{Spec}(A_f)$ for

$$f = x, \qquad f = x(x-1), \qquad f = (x-a)^m,$$

where $a \in \Bbbk$ and $m \geq 1$.

Solution. By the localisation theorem for spectra, the prime ideals of A_f are in bijection with the prime ideals $\mathfrak{p} \subset A$ such that

$$\mathfrak{p} \cap \{1, f, f^2, \ldots\} = \emptyset.$$

This is equivalent to saying that $f \notin \mathfrak{p}$. Hence

$$\mathrm{Spec}(A_f) \simeq D(f) = \{\mathfrak{p} \in \mathrm{Spec}(A) \mid f \notin \mathfrak{p}\}.$$

Now take $A = \Bbbk[x]$. The prime ideals of $\Bbbk[x]$ are (0) and the ideals (p), where $p \in \Bbbk[x]$ is irreducible. If $f = x$, then $\mathrm{Spec}(A_x)$ corresponds to the points

$$(0), \qquad (p) \text{ with } p \neq x.$$

Thus we remove the closed point (x).

If $f = x(x-1)$, then $\mathrm{Spec}(A_f)$ corresponds to the points

$$(0), \qquad (p) \text{ with } p \neq x,\ x-1.$$

Thus we remove the two closed points (x) and $(x-1)$.

Finally, if $f = (x-a)^m$, then

$$D((x-a)^m) = D(x-a).$$

Therefore $\mathrm{Spec}(A_f)$ corresponds to the points

$$(0), \qquad (p) \text{ with } p \neq x-a.$$

So we remove only the closed point $(x-a)$.

Exercise II.D.14. Let $(A, \mathfrak{m})$ be a local ring.

i) Show that $\mathfrak{m}$ is the unique closed point of $\mathrm{Spec}(A)$.

ii) Show that, for every point $\mathfrak{p} \in \mathrm{Spec}(A)$, the point $\mathfrak{m}$ belongs to the closure of $\{\mathfrak{p}\}$.

Solution. i) A point $\mathfrak{p} \in \mathrm{Spec}(A)$ is closed if and only if $\mathfrak{p}$ is a maximal ideal. Since A is local, it has a unique maximal ideal, namely $\mathfrak{m}$. Therefore $\mathfrak{m}$ is the unique closed point of $\mathrm{Spec}(A)$.

ii) By Proposition II.A.6, we have

$$\overline{\{\mathfrak{p}\}} = \mathbb{V}(\mathfrak{p}) = \{\mathfrak{q} \in \mathrm{Spec}(A) \mid \mathfrak{p} \subset \mathfrak{q}\}.$$

Every prime ideal $\mathfrak{p}$ is contained in some maximal ideal. Since A is local, the only maximal ideal is $\mathfrak{m}$. Hence $\mathfrak{p} \subset \mathfrak{m}$. Therefore $\mathfrak{m} \in \mathbb{V}(\mathfrak{p}) = \overline{\{\mathfrak{p}\}}$.

Definition II.D.15. A *discrete valuation ring*, or DVR, is a principal ideal domain A which has a unique non-zero prime ideal. Equivalently, it is a local principal ideal domain which is not a field.

If A is a DVR, its unique maximal ideal is principal. A generator π of this maximal ideal is called a *uniformiser*.

Exercise II.D.16. Let A be a discrete valuation ring, with uniformiser π and fraction field K. Show that

$$\mathrm{Spec}(A) = \{(0), (\pi)\}.$$

Which point is closed? What is the closure of the point (0)?

Solution. Since A is a principal ideal domain, every non-zero prime ideal is generated by an irreducible element. Since A is a DVR, it has a unique non-zero prime ideal, namely its maximal ideal (π). Therefore the only prime ideals of A are

$$(0), \qquad (\pi).$$

Hence

$$\mathrm{Spec}(A) = \{(0), (\pi)\}.$$

The point (π) is closed because (π) is maximal. The point (0) is the generic point. Its closure is

$$\overline{\{(0)\}} = \mathbb{V}(0) = \mathrm{Spec}(A).$$

Thus $\mathrm{Spec}(A)$ has two points: a generic point and a closed point.

III The structure sheaf

Further reading. The structure sheaf of an affine scheme is treated in [Har77, Chapter II, §2] and [Liu02, Chapter 2]; [EH00] emphasises the geometric meaning through examples.

We construct the structure sheaf on $X = \mathrm{Spec}(A)$ and show that affine spectra are naturally locally ringed spaces.

We make one definition of the structure sheaf: first form a presheaf by localisation along multiplicative subsets, and then sheafify it. This uses the notions of stalk and sheafification introduced in the previous chapter. The later formula on principal open subsets is an equivalent computational description of this same sheaf, not a second definition.

III.A Definition by sheafification

Let $X = \mathrm{Spec}(A)$. Our definition of the sheaf of regular functions is the sheafification of the following localisation presheaf.

Definition III.A.1. Let A be a ring and put $X = \mathrm{Spec}(A)$. For an open subset $U \subset X$, define

$$S_U := \bigcap_{\mathfrak{p}\in U} (A \setminus \mathfrak{p}) = A \setminus \bigcup_{\mathfrak{p}\in U} \mathfrak{p}.$$

Then S_U is a multiplicative subset of A. If $U \subset U'$, then $S_{U'} \subset S_U$. This induces a ring homomorphism

$$S_{U'}^{-1}A \longrightarrow S_U^{-1}A.$$

Thus $U \mapsto S_U^{-1}A$ defines a presheaf of rings on the topological space X, denoted by ${}^{\mathrm{pre}}\mathcal{O}_X$. The associated sheaf of rings is denoted by $\mathcal{O}_X$.

Proposition III.A.2. *For every $f \in A$, there is a natural isomorphism*

$$S_{D(f)}^{-1}A \simeq A_f.$$

Consequently,

$$\mathcal{O}_X(D(f)) \simeq A_f,$$

and the definition by sheafification agrees with the construction on the basis of principal open subsets given below.

Proof. Since f belongs to $S_{D(f)}$, there is a natural map $A_f \to S_{D(f)}^{-1}A$. Conversely, if $g \in S_{D(f)}$, then $D(f) \subset D(g)$. By Lemma II.D.6, some power of f lies in (g), so g is invertible in A_f. The universal property of localisation gives a map $S_{D(f)}^{-1}A \to A_f$. The two maps are inverse. For compatibility with sheafification, use Lemma III.B.1 below: the assignment $D(f) \mapsto A_f$ satisfies the sheaf condition on the principal-open basis. The isomorphisms just constructed identify the restriction of ${}^{\mathrm{pre}}\mathcal{O}_X$ to that basis with this assignment. Hence its sheafification has $\mathcal{O}_X(D(f)) \simeq A_f$. □

Exercise III.A.3. For an open subset $U \subset X$, describe the elements of $\mathcal{O}_X(U)$ using Lemma IV.A.4.

Solution. A section of $\mathcal{O}_X(U)$ is a function

$$s : U \to \coprod_{\mathfrak{p}\in U} A_{\mathfrak{p}}$$

such that $s(\mathfrak{p}) \in A_{\mathfrak{p}}$ for every $\mathfrak{p} \in U$, and which is locally given by a fraction. More precisely, for every $\mathfrak{p} \in U$, there is an open neighbourhood $V \subset U$ of $\mathfrak{p}$ and elements $a, t \in A$, with $t \notin \mathfrak{q}$ for all $\mathfrak{q} \in V$, such that

$$s(\mathfrak{q}) = \frac{a}{t} \in A_{\mathfrak{q}} \qquad \text{for all } \mathfrak{q} \in V.$$

In other words, regular functions on U are functions on prime ideals which are locally represented by fractions with denominators not vanishing on the neighbourhood under consideration.

III.B The principal-open description and associated modules

The structure sheaf has already been defined by sheafification. Proposition III.A.2 identifies its values on the basis of principal opens:

$$\mathcal{O}_X(D(f)) \simeq A_f.$$

We now verify the sheaf condition directly on this basis, both to justify the computational description and to extend the construction from rings to modules. The principal opens $(D(f) \mid f \in A)$ form a basis, and an inclusion $D(f) \subset D(g)$ induces the compatible localisation map $A_g \to A_f$.

Lemma III.B.1. *Let F be an A-module. For all $f \in A$, put $\mathcal{F}(D(f)) = F_f$. Then $\mathcal{F}$ satisfies the sheaf condition on open coverings of $D(f)$ by principal open subsets. Therefore, it extends uniquely to a sheaf of $\mathcal{O}_X$-modules on X. For $F = A$, this extension is precisely the structure sheaf $\mathcal{O}_X$ defined above.*

Proof. Replacing A by A_f, it is enough to check the sheaf condition for a covering of $X = \operatorname{Spec}(A)$ by principal open subsets. By quasi-compactness, we may assume that the covering is finite, say

$$X = \bigcup_{1 \le i \le \ell} D(f_i).$$

Equivalently, the elements $f_1, \ldots, f_\ell$ generate the unit ideal.

First, let $v \in F$ be a section whose image in every F_{f_i} is zero. Then, for each i, there is an integer $n_i \ge 0$ such that $f_i^{n_i} v = 0$ in F. Since the opens $D(f_i^{n_i}) = D(f_i)$ still cover X, the elements $f_i^{n_i}$ generate the unit ideal. We can write

$$1 = \sum_{1 \le i \le \ell} a_i f_i^{n_i}$$

for suitable $a_i \in A$. Multiplying by v, we get $v = 0$. This proves uniqueness.

Now let $v_i \in F_{f_i}$ be sections which agree on the overlaps, i.e. whose images in $F_{f_i f_j}$ coincide for all i, j. Write

$$v_i = \frac{x_i}{f_i^r}$$

with $x_i \in F$, after increasing r if necessary. Since v_i and v_j agree in $F_{f_i f_j}$, there is an integer $n_{ij} \ge 0$ such that

$$(f_i f_j)^{n_{ij}} (f_j^r x_i - f_i^r x_j) = 0.$$

Choose an integer N such that $N \ge r + n_{ij}$ for all i, j. Since the opens $D(f_i^N)$ still cover X, we can write

$$1 = \sum_{1 \le i \le \ell} c_i f_i^N$$

for some $c_i \in A$. Define

$$v = \sum_{1 \le i \le \ell} c_i f_i^{N-r} x_i \in F.$$

For a fixed k, we compute
$$f_k^r v - x_k = \sum_{1\le i\le \ell} c_i f_i^{N-r}(f_k^r x_i - f_i^r x_k).$$
Multiplying by a sufficiently large power of f_k kills every summand by the compatibility relation above. Hence v maps to $x_k/f_k^r = v_k$ in F_{f_k}. This proves existence, and hence the sheaf condition on the basis. □

Definition III.B.2 (The sheaf associated with a module). Let A be a ring, let $X = \operatorname{Spec}(A)$, and let M be an A-module. The *sheaf associated with* M, denoted by $\widetilde{M}$, is the sheaf of $\mathcal{O}_X$-modules obtained from the prescription
$$\widetilde{M}(D(f)) = M_f$$
on the basis of principal open subsets. Equivalently, it is the sheafification of this presheaf on the principal-open basis.

By construction,
$$\widetilde{M}_{\mathfrak{p}} \simeq M_{\mathfrak{p}} \qquad \text{for every } \mathfrak{p} \in \operatorname{Spec}(A),$$
and $\widetilde{A} \simeq \mathcal{O}_X$. A homomorphism of A-modules $M \to N$ induces a morphism of $\mathcal{O}_X$-modules $\widetilde{M} \to \widetilde{N}$, obtained on $D(f)$ by localisation.

Example III.B.3. If $\mathfrak{a} \subset A$ is an ideal, then $\widetilde{\mathfrak{a}}$ is a sheaf of ideals of $\mathcal{O}_X$. The quotient map $A \to A/\mathfrak{a}$ sheafifies to an exact sequence
$$\widetilde{\mathfrak{a}} \longrightarrow \mathcal{O}_X \longrightarrow \widetilde{A/\mathfrak{a}} \longrightarrow 0.$$
In fact, the first arrow is injective and $\widetilde{A/\mathfrak{a}} \simeq \mathcal{O}_X/\widetilde{\mathfrak{a}}$.

Example III.B.4. Let $A = \Bbbk[x]$ and $X = \operatorname{Spec}(A)$. For the principal open subset
$$D(x) = X \setminus \mathbb{V}(x),$$
we have
$$\mathcal{O}_X(D(x)) \simeq A_x = \Bbbk[x, x^{-1}].$$
Thus the function x^{-1} is a regular function on $D(x)$, although it is not a regular function on all of X.

Exercise III.B.5. Let $A = \Bbbk[x]$ and $X = \operatorname{Spec}(A)$. Compute
$$\mathcal{O}_X(D(x)), \qquad \mathcal{O}_X(D(x-a)), \qquad \mathcal{O}_X(D(x(x-1))).$$

Solution. For every $f \in A$, we have
$$\mathcal{O}_X(D(f)) \simeq A_f.$$
Therefore
$$\mathcal{O}_X(D(x)) \simeq \Bbbk[x, x^{-1}],$$
and
$$\mathcal{O}_X(D(x-a)) \simeq \Bbbk[x, (x-a)^{-1}].$$
Finally,
$$\mathcal{O}_X(D(x(x-1))) \simeq \Bbbk[x, (x(x-1))^{-1}].$$
Equivalently, this is the subring of $\Bbbk(x)$ consisting of rational functions whose poles are allowed only at 0 and 1.

Exercise III.B.6. Let

$$A = \Bbbk[x]/(x^2)$$

and let $X = \mathrm{Spec}(A)$.

i) Describe the points of X.

ii) Describe the open subsets of X.

iii) Compute the stalk $\mathcal{O}_{X,\mathfrak{p}}$ at each point $\mathfrak{p} \in X$.

iv) Compute the residue field $\kappa(\mathfrak{p})$ at each point.

Solution. The prime ideals of $A = \Bbbk[x]/(x^2)$ correspond to the prime ideals of $\Bbbk[x]$ containing (x^2). Such a prime ideal must contain x, hence the only prime ideal is

$$\mathfrak{p} = (\bar{x}).$$

Therefore X has a single point. Since X has only one point, its open subsets are

$$\emptyset \qquad \text{and} \qquad X.$$

The stalk at the unique point $\mathfrak{p} = (\bar{x})$ is

$$\mathcal{O}_{X,\mathfrak{p}} \simeq A_{\mathfrak{p}}.$$

But A is already local: its unique maximal ideal is $(\bar{x})$, and the elements outside $(\bar{x})$ are precisely the classes of polynomials with non-zero constant term, which are invertible in A. Hence

$$A_{\mathfrak{p}} \simeq A = \Bbbk[x]/(x^2).$$

Thus

$$\mathcal{O}_{X,\mathfrak{p}} \simeq \Bbbk[x]/(x^2).$$

Finally, the residue field is

$$\kappa(\mathfrak{p}) = \mathrm{Frac}(A/\mathfrak{p}).$$

Since

$$A/\mathfrak{p} \simeq \Bbbk,$$

we get

$$\kappa(\mathfrak{p}) \simeq \Bbbk.$$

Remark III.B.7. The schemes

$$\mathrm{Spec}(\Bbbk) \qquad \text{and} \qquad \mathrm{Spec}(\Bbbk[x]/(x^2))$$

have the same underlying topological space: a single point. However, they are not isomorphic as schemes, because their local rings are different. In $\Bbbk[x]/(x^2)$, the class of x is a non-zero nilpotent element.

Exercise III.B.8. Let $A = \Bbbk[x, y]$ and $X = \mathrm{Spec}(A)$. Compute

$$\mathcal{O}_X(D(x)), \qquad \mathcal{O}_X(D(xy)).$$

Assume that $\Bbbk$ is algebraically closed. What do these open subsets look like on closed points?

Solution. We have

$$\mathcal{O}_X(D(x)) \simeq A_x = \Bbbk[x, y, x^{-1}].$$

Similarly,

$$\mathcal{O}_X(D(xy)) \simeq A_{xy} = \Bbbk[x, y, (xy)^{-1}].$$

Since inverting xy is the same as inverting both x and y, we also have

$$A_{xy} \simeq \Bbbk[x, y, x^{-1}, y^{-1}].$$

If $\Bbbk$ is algebraically closed, the closed points of X are the points $(a, b) \in \Bbbk^2$. The open subset $D(x)$ consists of the closed points with $a \neq 0$. The open subset $D(xy) = D(x) \cap D(y)$ consists of the closed points with

$$a \neq 0, \qquad b \neq 0.$$

Thus $D(x)$ is the affine plane with the line $x = 0$ removed, and $D(xy)$ is the affine plane with both coordinate axes removed.

Exercise III.B.9. Let $A = \Bbbk[x]$ and $X = \mathrm{Spec}(A)$. Compute the stalks

$$\mathcal{O}_{X,(0)}, \qquad \mathcal{O}_{X,(x-a)}$$

for $a \in \Bbbk$.

Solution. For every prime ideal $\mathfrak{p} \subset A$, we have

$$\mathcal{O}_{X,\mathfrak{p}} \simeq A_{\mathfrak{p}}.$$

Therefore

$$\mathcal{O}_{X,(0)} \simeq A_{(0)} = \Bbbk(x),$$

the rational function field of the affine line.

On the other hand,

$$\mathcal{O}_{X,(x-a)} \simeq A_{(x-a)} = \left\{ \frac{f}{g} \in \Bbbk(x) \;\middle|\; g(a) \neq 0 \right\}.$$

This is the local ring of rational functions which are regular at the point a.

Remark III.B.10. The double point is a first example showing why schemes are not determined by their underlying topological spaces. The nilpotent element $\bar{x}$ is invisible as a point, but visible in the ring of functions.

IV Affine schemes as locally ringed spaces

We define a category of spaces, called locally ringed spaces, where schemes live. A locally ringed space is a topological space equipped with a sheaf of rings such that the stalk at each point is a local ring. Morphisms in this category are defined in a subtle way, in that they should combine continuous maps of topological spaces and a ring homomorphism on local sections of the structure sheaves, which respects the locality condition about maximal ideals on stalks. Schemes are locally ringed spaces which are locally isomorphic to the spectrum of a ring.

IV.A The affine spectrum as a locally ringed space

Definition IV.A.1. A *ringed space* is a topological space equipped with a sheaf of rings. A *locally ringed space* is a ringed space such that all stalks of its sheaf of rings are local rings.

Given a ringed space $(X, \mathcal{O}_X)$, we often write just X, meaning that X is silently equipped with some sheaf of rings, usually denoted by $\mathcal{O}_X$. In case we want to refer to X as the topological space only, without the datum of the structure sheaf, sometimes we write $|X|$. Given a point x of the ringed space X, we write $\mathcal{O}_{X,x}$ for the stalk at x of the sheaf of rings on X, so by definition,

$$\mathcal{O}_{X,x} = \varinjlim_{U \ni x} \mathcal{O}_X(U).$$

Exercise IV.A.2. Show that the stalk of the structure sheaf of $\mathrm{Spec}(A)$ at a prime ideal $\mathfrak{p}$ is naturally isomorphic to $A_{\mathfrak{p}}$.

Solution. The principal open subsets $D(f)$ with $f \notin \mathfrak{p}$ form a basis of neighbourhoods of $\mathfrak{p}$. On such an open subset, the structure sheaf gives A_f. Therefore

$$\mathcal{O}_{\mathrm{Spec}(A),\mathfrak{p}} = \varinjlim_{\mathfrak{p} \in U} \mathcal{O}_{\mathrm{Spec}(A)}(U) \simeq \varinjlim_{f \notin \mathfrak{p}} A_f.$$

The last direct limit is $A_{\mathfrak{p}}$, because localising successively at all elements outside $\mathfrak{p}$ is exactly localisation at $A \setminus \mathfrak{p}$.

Lemma IV.A.3. *For any ring A, the pair $X = (\mathrm{Spec}(A), \mathcal{O}_X)$ is a locally ringed space.*

Proof. By the previous exercise, the stalk of $\mathcal{O}_X$ at a prime ideal $\mathfrak{p}$ is naturally isomorphic to $A_{\mathfrak{p}}$. This ring is local, with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, by Proposition III.C.1. Hence all stalks of $\mathcal{O}_X$ are local rings. $\square$

Exercise IV.A.4. Let X be either a smooth manifold, with sheaf $\mathcal{C}^\infty_X$, or a complex manifold, with sheaf $\mathcal{O}_X$. Show that the stalk at a point $x \in X$ is a local ring. Describe its maximal ideal and its residue field.

Solution. In the smooth case, the stalk $\mathcal{C}^\infty_{X,x}$ is the ring of germs of smooth real-valued functions at x. Evaluation at x gives a surjective morphism

$$\mathcal{C}^\infty_{X,x} \to \mathbb{R}.$$

Its kernel is the ideal of germs vanishing at x:

$$\mathfrak{m}_x = \{f \in \mathcal{C}^\infty_{X,x} \mid f(x) = 0\}.$$

A germ f is invertible if and only if $f(x) \neq 0$, because then $1/f$ is defined and smooth near x. Hence $\mathfrak{m}_x$ is the unique maximal ideal, and

$$\mathcal{C}^\infty_{X,x}/\mathfrak{m}_x \simeq \mathbb{R}.$$

In the complex case, the same argument applies with holomorphic functions. The maximal ideal of $\mathcal{O}_{X,x}$ is

$$\mathfrak{m}_x = \{f \in \mathcal{O}_{X,x} \mid f(x) = 0\},$$

and the residue field is

$$\mathcal{O}_{X,x}/\mathfrak{m}_x \simeq \mathbb{C}.$$

IV.B Morphisms of locally ringed spaces

Given a continuous map $f : X \to Y$, where $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are ringed spaces, recall that we have the direct image sheaf on Y, whose sections on an open subset U of Y are given by:

$$f_*(\mathcal{O}_X) : U \mapsto \mathcal{O}_X(f^{-1}(U)).$$

There is a map

$$f_*(\mathcal{O}_X)_{f(x)} \to \mathcal{O}_{X,x}. \tag{IV.1}$$

Indeed, we have, by definition:

$$\mathcal{O}_{X,x} = \varinjlim_{U \ni x} \mathcal{O}_X(U). \tag{IV.2}$$

Also, by definition, we have

$$f_*(\mathcal{O}_X)_{f(x)} = \varinjlim_{V \ni f(x)} f_*(\mathcal{O}_X(V)) = \varinjlim_{V \ni f(x)} \mathcal{O}_X(f^{-1}(V)). \tag{IV.3}$$

Indeed, for any open subset $V \subset Y$ with $f(x) \in V$, we have that $U = f^{-1}(V)$ is an open subset of X containing x. So any element of (IV.3) occurs in (IV.2), hence we get maps $\mathcal{O}_X(f^{-1}(V)) \to \mathcal{O}_{X,x}$, which are compatible with the maps induced by inclusions of the form $V \subset V'$. This gives (IV.1).

To relate the structure sheaves of X and Y, we use a morphism $f^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$. In other words, for each open subset V of Y, we need:

$$f^\#_V : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V)).$$

Definition IV.B.1. A *morphism of ringed spaces* $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$ where $f : X \to Y$ is a continuous map of topological spaces and $f^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$ is a morphism of sheaves of rings. A *morphism of locally ringed spaces* $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces $(f, f^\#)$ such that, for all $x \in X$, the induced local map $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ sends the maximal ideal of $\mathcal{O}_{Y,f(x)}$ into the maximal ideal of $\mathcal{O}_{X,x}$.

Example IV.B.2 (A morphism of ringed spaces which is not local). Let $X = Y = \{*\}$ be the one-point topological space. Put

$$\mathcal{O}_Y = \Bbbk[t]_{(t)}, \qquad \mathcal{O}_X = \Bbbk(t),$$

viewed as sheaves of rings on the one-point space. The inclusion

$$\Bbbk[t]_{(t)} \hookrightarrow \Bbbk(t)$$

defines a morphism of ringed spaces

$$(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y).$$

However, this morphism is not a morphism of locally ringed spaces.

Indeed, the ring $\Bbbk[t]_{(t)}$ is local, with maximal ideal

$$(t)\Bbbk[t]_{(t)}.$$

The ring $\Bbbk(t)$ is a field, hence its maximal ideal is (0). The induced map on stalks is

$$\Bbbk[t]_{(t)} \hookrightarrow \Bbbk(t).$$

But the element t belongs to the maximal ideal of $\Bbbk[t]_{(t)}$, while its image in $\Bbbk(t)$ is non-zero, hence invertible. Therefore

$$t \notin (0).$$

So the maximal ideal of $\Bbbk[t]_{(t)}$ is not sent into the maximal ideal of $\Bbbk(t)$, and the morphism is not local.

Exercise IV.B.3. Let $X = Y = \{*\}$ be the one-point space. Let R and S be local rings, and view them as sheaves of rings on Y and X respectively. Show that a ring homomorphism

$$\varphi : R \to S$$

defines a morphism of ringed spaces

$$(X, S) \to (Y, R).$$

Show that this morphism is a morphism of locally ringed spaces if and only if the ring homomorphism $R \to S$ is local, i.e. if and only if

$$\varphi(\mathfrak{m}_R) \subset \mathfrak{m}_S.$$

Apply this to the inclusion

$$\Bbbk[t]_{(t)} \hookrightarrow \Bbbk(t).$$

Solution. Since X and Y each have only one open subset different from the empty set, a sheaf of rings on such a space is just a ring. A morphism of ringed spaces

$$(X, S) \to (Y, R)$$

therefore amounts to a ring homomorphism

$$R \to S.$$

The condition for this morphism to be a morphism of locally ringed spaces is that the induced map on stalks sends the maximal ideal of R into the maximal ideal of S:

$$\varphi(\mathfrak{m}_R) \subset \mathfrak{m}_S.$$

This is precisely the usual definition of a local homomorphism of local rings.

Now take

$$R = \Bbbk[t]_{(t)}, \qquad S = \Bbbk(t).$$

Then R is local with maximal ideal (t), while S is a field, with maximal ideal (0). The inclusion sends t to a non-zero element of $\Bbbk(t)$, hence to an invertible element. Thus

$$\varphi((t)) \not\subset (0),$$

so the corresponding morphism of ringed spaces is not a morphism of locally ringed spaces.

Proposition IV.B.4. *Every ring homomorphism $\varphi : A \to B$ induces a morphism of locally ringed spaces*

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A).$$

This construction is contravariantly functorial in the ring.

Proof. Let A be a ring and let X be the ringed space $\mathrm{Spec}(A)$. So X is a topological space equipped with the sheaf of rings $\mathcal{O}_X$. We said that X is a locally ringed space as for all $\mathfrak{p} \in X$, the stalk $\mathcal{O}_{X,\mathfrak{p}} \simeq A_{\mathfrak{p}}$ is a local ring, with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Let $\varphi : A \to B$ be a ring homomorphism and set $X = \mathrm{Spec}(B)$, $Y = \mathrm{Spec}(A)$. Then we proved that the map $f_\varphi : X \to Y$ sending a prime ideal $\mathfrak{q} \subset B$ to $\varphi^{-1}(\mathfrak{q})$ is continuous for the Zariski topology.

We need to define a morphism of sheaves of rings $f_\varphi^\# : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$. To do so, we may define $f_\varphi^\#$ over principal open subsets of Y. So let $a \in A$ and consider the principal open subsets $D(a) \subset Y$ and $D(b) \subset X$, where we put $b = \varphi(a)$. We note that $f_\varphi^{-1}(D(a)) = D(b)$. Indeed:

$$\begin{aligned} f_\varphi^{-1}(D(a)) &= \{\mathfrak{p} \in X \mid f_\varphi(\mathfrak{p}) \in D(a)\} = \\ &= \{\mathfrak{p} \in X \mid a \notin \varphi^{-1}(\mathfrak{p})\} = \\ &= \{\mathfrak{p} \in X \mid \varphi(a) \notin \mathfrak{p}\} = D(b). \end{aligned}$$

Then we should define:

$$f^{\#}_{D(a)} : A_a \simeq \mathcal{O}_Y(D(a)) \to \mathcal{O}_X(f_\varphi^{-1}(D(a))) \simeq \mathcal{O}_X(D(b)) \simeq B_b$$

This map is nothing but the localisation of φ at a. Since these maps are compatible with restriction to further principal open subsets, we get the desired map of sheaves $f^{\#}_\varphi$. For any prime ideal $\mathfrak{p} \subset B$ and $\mathfrak{q} = \varphi^{-1}(\mathfrak{p}) \subset A$, by construction, the induced map

$$A_{\mathfrak{q}} \simeq \mathcal{O}_{Y,\mathfrak{q}} \to B_{\mathfrak{p}} \simeq \mathcal{O}_{X,\mathfrak{p}}$$

is given by localising φ at $\mathfrak{q}$. Since $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$, we also have $\mathfrak{q}A_{\mathfrak{q}} = \varphi^{-1}(\mathfrak{p}B_{\mathfrak{p}})$. In other words, the maximal ideals correspond to each other.

So $(f_\varphi, f^{\#}_\varphi)$ defines a morphism of locally ringed spaces. The compatibility with composition of ring homomorphisms follows directly from the construction on principal open subsets.

□

Exercise IV.B.5. Let $f : X \to Y$ be a continuous map.

i) Assume that X and Y are smooth manifolds and that f is a smooth map. Show that pullback of smooth functions defines a morphism of locally ringed spaces

$$(X, \mathcal{C}^\infty_X) \longrightarrow (Y, \mathcal{C}^\infty_Y).$$

ii) Assume that X and Y are complex manifolds and that f is a holomorphic map. Show that pullback of holomorphic functions defines a morphism of locally ringed spaces

$$(X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y).$$

Solution. We treat the smooth case first. For every open subset $V \subset Y$, define

$$f^{\#}_V : \mathcal{C}^\infty_Y(V) \longrightarrow \mathcal{C}^\infty_X(f^{-1}(V))$$

by

$$f^{\#}_V(g) = g \circ f.$$

Since f is smooth, the pullback $g \circ f$ is a smooth function on $f^{-1}(V)$. These maps are compatible with restrictions: if $V' \subset V$, then

$$(g|_{V'}) \circ f = (g \circ f)|_{f^{-1}(V')}.$$

Therefore the maps $f^{\#}_V$ define a morphism of sheaves of rings

$$f^{\#} : \mathcal{C}^\infty_Y \longrightarrow f_*\mathcal{C}^\infty_X.$$

Together with the continuous map $f : X \to Y$, this gives a morphism of ringed spaces.

We now check that this morphism is local. Let $x \in X$. The induced map on stalks is

$$f^{\#}_x : \mathcal{C}^\infty_{Y,f(x)} \longrightarrow \mathcal{C}^\infty_{X,x}, \qquad [g]_{f(x)} \longmapsto [g \circ f]_x.$$

The maximal ideal of $\mathcal{C}^\infty_{Y,f(x)}$ is

$$\mathfrak{m}_{f(x)} = \{[g]_{f(x)} \mid g(f(x)) = 0\}.$$

Similarly, the maximal ideal of $\mathcal{C}^\infty_{X,x}$ is

$$\mathfrak{m}_x = \{[h]_x \mid h(x) = 0\}.$$

If $[g]_{f(x)} \in \mathfrak{m}_{f(x)}$, then $g(f(x)) = 0$. Hence

$$(g \circ f)(x) = g(f(x)) = 0,$$

so

$$[g \circ f]_x \in \mathfrak{m}_x.$$

Thus

$$f_x^{\#}(\mathfrak{m}_{f(x)}) \subset \mathfrak{m}_x,$$

which is precisely the condition for a morphism of locally ringed spaces.

The holomorphic case is identical. If $f : X \to Y$ is holomorphic and $g \in \mathcal{O}_Y(V)$, then $g \circ f$ is holomorphic on $f^{-1}(V)$. Thus we obtain maps

$$f_V^{\#} : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_X(f^{-1}(V)), \qquad g \longmapsto g \circ f,$$

compatible with restrictions, hence a morphism of sheaves

$$f^{\#} : \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X.$$

On stalks,

$$f_x^{\#} : \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x}$$

sends the germ of a holomorphic function g at $f(x)$ to the germ of $g \circ f$ at x. Since

$$g(f(x)) = 0 \quad \Longrightarrow \quad (g \circ f)(x) = 0,$$

the maximal ideal of germs vanishing at $f(x)$ is sent into the maximal ideal of germs vanishing at x. Hence this is again a morphism of locally ringed spaces.

Theorem IV.B.6. *Let A and B be rings. Taking global sections gives a natural bijection*

$$\mathrm{Hom}_{(\mathrm{LRS})}\big(\mathrm{Spec}(B), \mathrm{Spec}(A)\big) \simeq \mathrm{Hom}_{(\mathrm{Ring})}(A, B).$$

Under this bijection, a ring homomorphism $\varphi : A \to B$ corresponds to the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *constructed in Proposition IV.B.4.*

Proof. A morphism $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ induces on global sections a ring homomorphism

$$\varphi = f^{\#}_{\mathrm{Spec}(A)} : A \longrightarrow B.$$

Let $\mathfrak{q} \in \mathrm{Spec}(B)$. The locality of the map on stalks

$$A_{f(\mathfrak{q})} \longrightarrow B_{\mathfrak{q}}$$

shows that an element $a \in A$ belongs to $f(\mathfrak{q})$ if and only if $\varphi(a) \in \mathfrak{q}$. Hence

$$f(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}).$$

Thus the underlying continuous map is the one induced by φ. On a principal open subset $D(a) \subset \mathrm{Spec}(A)$, the map on sections is then the localisation

$$A_a \longrightarrow B_{\varphi(a)},$$

by compatibility with the map on global sections. Since principal open subsets form a basis, the morphism of sheaves is uniquely determined. Therefore f is the morphism induced by φ. $\square$

Definition IV.B.7. An *affine scheme* is a locally ringed space isomorphic to $\mathrm{Spec}(A)$ for some ring A. A *scheme* is a locally ringed space which admits an open covering by affine schemes. Morphisms of schemes are morphisms of locally ringed spaces.

V Affine subschemes

Further reading. For open and closed subschemes and their scheme structures, see [Har77, Chapter II, §§2–3], [Liu02, Chapter 2], and the examples in [EH00].

Let us review how to define subschemes of an affine scheme.

V.A Open subsets

Let A be a ring and let $X = \mathrm{Spec}(A)$. For a principal open subset $D(a) \subset X$, the localisation map $A \to A_a$ induces an isomorphism of schemes

$$\mathrm{Spec}(A_a) \xrightarrow{\simeq} D(a).$$

Thus principal open subsets of affine schemes are affine schemes.

More generally, an arbitrary open subset $U \subset X$ inherits a scheme structure by restriction:

$$(U, \mathcal{O}_X|_U).$$

Such an open subscheme need not be affine. The inclusion $j : U \to X$ is a morphism of schemes. It is moreover a monomorphism: if $f_1, f_2 : Y \to U$ are morphisms of schemes such that $j \circ f_1 = j \circ f_2$, then $f_1 = f_2$ set-theoretically, and the maps on structure sheaves agree because they agree after composition with the inclusion.

Example V.A.1. In $X = \mathrm{Spec}(\Bbbk[x])$, the principal open subset $D(x)$ is the complement of the closed point (x). As a scheme,

$$D(x) \simeq \mathrm{Spec}(\Bbbk[x, x^{-1}]).$$

Example V.A.2. Let $A = \Bbbk[x, y]$ and $X = \mathrm{Spec}(A) = \mathbb{A}^2_{\Bbbk}$. The principal open subset

$$D(x) \subset X$$

is affine and isomorphic to

$$\mathrm{Spec}(\Bbbk[x, y, x^{-1}]).$$

On closed points, assuming that $\Bbbk$ is algebraically closed, this is the affine plane with the line $x = 0$ removed.

V.B Closed affine subschemes

Let A be a ring and set $X = \mathrm{Spec}(A)$. Then, given an ideal $\mathfrak{a} \subset A$, we have a ring $B = A/\mathfrak{a}$ and a surjective morphism of rings $\varphi : A \to B$. Put $Y = \mathrm{Spec}(B)$. The morphism φ induces a homeomorphism $i : Y \to \mathbb{V}(\mathfrak{a}) \subset X$. This also gives a morphism of sheaves:

$$i^\# : \mathcal{O}_X \to i_*(\mathcal{O}_Y).$$

On global sections, this morphism is the quotient map

$$\varphi : A = \mathcal{O}_X(X) \to i_*(\mathcal{O}_Y)(i^{-1}(X)) = \mathcal{O}_Y(Y) = B.$$

The morphism of sheaves $i^\#$ is obtained by localising this quotient map on principal open subsets $D(a) \subset X$ for $a \in A$. Indeed, we have

$$\begin{aligned} i^{-1}(D(a)) &= \{\mathfrak{q} \in \mathrm{Spec}(B) \mid a \notin i(\mathfrak{q})\} = \\ &= \{\mathfrak{q} \in \mathrm{Spec}(B) \mid a \notin \varphi^{-1}(\mathfrak{q})\} = \\ &= \{\mathfrak{q} \in \mathrm{Spec}(B) \mid \varphi(a) \notin \mathfrak{q}\} = D(\varphi(a)). \end{aligned}$$

Then for each $a \in A$, we just consider $i^{\#}$ to be defined at $D(a)$ by the localisation $A_a \to B_{\varphi(a)}$ of φ at a.

A closed affine subscheme Y of an affine scheme $X = \mathrm{Spec}(A)$ is, by definition, a closed subscheme arising in the above form, namely $Y = \mathrm{Spec}(B)$, for some surjective morphism of rings $\varphi : A \to B$. The ideal $\mathfrak{a} = \ker(\varphi)$ is the *ideal defining Y as a subscheme of X*.

Example V.B.1. Let $X = \mathrm{Spec}(\Bbbk[x, y]) = \mathbb{A}^2_{\Bbbk}$. The ideal (x, y) defines the origin:

$$\mathrm{Spec}(\Bbbk[x, y]/(x, y)) \simeq \mathrm{Spec}(\Bbbk).$$

The ideal (x^2, y) defines a non-reduced closed subscheme supported at the origin:

$$\mathrm{Spec}(\Bbbk[x, y]/(x^2, y)).$$

Its underlying closed subset is the same point, but the scheme remembers an infinitesimal direction along the x-axis.

Exercise V.B.2. Describe the closed subscheme of $\mathbb{A}^1_{\Bbbk}$ defined by the ideal (x^2). What is its underlying topological space?

Solution. The closed subscheme is

$$\mathrm{Spec}(\Bbbk[x]/(x^2)).$$

Its underlying topological space has one point, corresponding to the unique prime ideal $(x)/(x^2)$ of $\Bbbk[x]/(x^2)$. Its structure sheaf remembers the nilpotent element x, whose square is zero. Thus this is a non-reduced double point supported at the origin.

Exercise V.B.3. Let $X = \mathrm{Spec}(A)$, and let $Y, Z \subset X$ be closed subschemes defined by ideals $\mathfrak{a}, \mathfrak{b} \subset A$. Show that the closed subscheme defined by $\mathfrak{a} \cap \mathfrak{b}$ is the smallest closed subscheme of X containing both Y and Z. This closed subscheme is called the *scheme-theoretic union* of Y and Z.

Show also that $\mathfrak{a} \cap \mathfrak{b}$ and $\mathfrak{a}\mathfrak{b}$ define the same underlying closed subset of X. Give an example where the corresponding closed subschemes are not isomorphic.

Solution. Let $W \subset X$ be a closed subscheme defined by an ideal $\mathfrak{c} \subset A$. Then W contains Y if and only if the closed immersion

$$Y = \mathrm{Spec}(A/\mathfrak{a}) \longrightarrow X$$

factors through $W = \mathrm{Spec}(A/\mathfrak{c})$. Algebraically, this is equivalent to the existence of a surjective morphism

$$A/\mathfrak{c} \longrightarrow A/\mathfrak{a},$$

or equivalently to

$$\mathfrak{c} \subset \mathfrak{a}.$$

Similarly, W contains Z if and only if $\mathfrak{c} \subset \mathfrak{b}$. Therefore W contains both Y and Z if and only if

$$\mathfrak{c} \subset \mathfrak{a} \cap \mathfrak{b}.$$

Thus the largest ideal defining a closed subscheme containing both Y and Z is $\mathfrak{a} \cap \mathfrak{b}$. Since larger ideals give smaller closed subschemes, the closed subscheme defined by $\mathfrak{a} \cap \mathfrak{b}$ is the smallest closed subscheme of X containing both Y and Z.

Now compare $\mathfrak{a} \cap \mathfrak{b}$ and $\mathfrak{a}\mathfrak{b}$. Since

$$\mathfrak{a}\mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{b},$$

we have

$$\mathbb{V}(\mathfrak{a} \cap \mathfrak{b}) \subset \mathbb{V}(\mathfrak{a}\mathfrak{b}).$$

Conversely, let $\mathfrak{p} \in \mathbb{V}(\mathfrak{a}\mathfrak{b})$. Then $\mathfrak{a}\mathfrak{b} \subset \mathfrak{p}$. Since $\mathfrak{p}$ is prime, we must have

$$\mathfrak{a} \subset \mathfrak{p} \qquad \text{or} \qquad \mathfrak{b} \subset \mathfrak{p}.$$

In either case,

$$\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{p}.$$

Thus

$$\mathbb{V}(\mathfrak{a}\mathfrak{b}) \subset \mathbb{V}(\mathfrak{a} \cap \mathfrak{b}).$$

Hence

$$\mathbb{V}(\mathfrak{a}\mathfrak{b}) = \mathbb{V}(\mathfrak{a} \cap \mathfrak{b}).$$

So the two ideals define the same underlying closed subset of X.

However, they need not define the same closed subscheme. For example, take

$$A = \Bbbk[x], \qquad \mathfrak{a} = \mathfrak{b} = (x).$$

Then

$$\mathfrak{a} \cap \mathfrak{b} = (x), \qquad \mathfrak{a}\mathfrak{b} = (x^2).$$

The closed subscheme defined by $\mathfrak{a} \cap \mathfrak{b}$ is

$$\operatorname{Spec}(\Bbbk[x]/(x)) \simeq \operatorname{Spec}(\Bbbk),$$

whereas the closed subscheme defined by $\mathfrak{a}\mathfrak{b}$ is

$$\operatorname{Spec}(\Bbbk[x]/(x^2)).$$

These schemes are not isomorphic, because $\Bbbk[x]/(x) \simeq \Bbbk$ is reduced, whereas $\Bbbk[x]/(x^2)$ has a non-zero nilpotent element, namely the class of x.

VI Subschemes of the affine space

Let B be a commutative ring and $A = B[x_1, \ldots, x_n]$. Then $\mathbb{A}^n_B = \operatorname{Spec}(A)$ is the *affine n-space over B*. The inclusion $B \subset A$ induces a surjective morphism $\mathbb{A}^n_B \to \operatorname{Spec}(B)$. We will mostly consider the case when B is a field.

VI.A Subschemes of the affine line over a field

Definition VI.A.1. An *affine scheme of finite type over* $\Bbbk$ is the spectrum of a finitely generated $\Bbbk$-algebra, namely a scheme of the form

$$\operatorname{Spec}(\Bbbk[x_1, \ldots, x_n]/\mathfrak{a})$$

for some ideal $\mathfrak{a} \subset \Bbbk[x_1, \ldots, x_n]$. Equivalently, it is a closed subscheme of an affine space over $\Bbbk$.

We call such a scheme an *affine variety* when its coordinate ring is an integral domain. Some authors use the word "variety" for more general reduced schemes; the convention should always be stated explicitly.

Proposition VI.A.2. *A closed subscheme $X \subset \mathbb{A}^1_{\Bbbk}$ is either $\emptyset$, or $\mathbb{A}^1_{\Bbbk}$, or the disjoint union of schemes of the form $\operatorname{Spec}(\Bbbk[x]/(f^m))$, where $f \in \Bbbk[x]$ is an irreducible polynomial in $\Bbbk[x]$ and $m \geq 1$.*

Proof. By definition, a closed subscheme of $\mathbb{A}^1_{\Bbbk}$ has the form $\operatorname{Spec}(\Bbbk[x]/\mathfrak{a})$ for some ideal $\mathfrak{a}$ of $\Bbbk[x]$, so that $\mathfrak{a} = (g)$, for some $g \in \Bbbk[x]$, since $\Bbbk[x]$ is a principal ideal domain. If $g = 0$ then $X = \mathbb{A}^1_{\Bbbk}$ and, if g is a non-zero constant, then $X = \emptyset$. If g is non-constant, then we can write $g = f_1^{m_1} \cdots f_s^{m_s}$, in an essentially unique way, where $m_i \geq 1$ are integers and f_i are pairwise coprime irreducible polynomials in $\Bbbk[x]$, for $1 \leq i \leq s$. Then $\mathbb{V}(g)$ is the disjoint union of the subschemes $\mathbb{V}(f_i^{m_i})$, for $1 \leq i \leq s$. Indeed, by the Chinese remainder theorem, we have

$$\Bbbk[x]/(g) \simeq \prod_{1\leq i\leq s} \Bbbk[x]/(f_i^{m_i}).$$

So taking spectra (see the next exercise), we get

$$X = \operatorname{Spec}(\Bbbk[x]/g) \simeq \bigsqcup_{1\leq i\leq s} \operatorname{Spec}(\Bbbk[x]/(f_i^{m_i})).$$

□

Exercise VI.A.3. If A and B are rings, then

$$\operatorname{Spec}(A \times B) \simeq \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B).$$

Solution. Let $e_A = (1,0)$, and $e_B = (0,1)$ be the two standard idempotents of $A \times B$. They satisfy

$$e_A e_B = 0, \qquad e_A + e_B = 1.$$

If $\mathfrak{p} \subset A \times B$ is a prime ideal, then from $e_A e_B = 0 \in \mathfrak{p}$ we get $e_A \in \mathfrak{p}$ or $e_B \in \mathfrak{p}$. Since $e_A + e_B = 1$, they cannot both belong to $\mathfrak{p}$.

If $e_B \in \mathfrak{p}$, then $\mathfrak{p}$ is of the form

$$\mathfrak{q} \times B$$

for a unique prime ideal $\mathfrak{q} \subset A$. Similarly, if $e_A \in \mathfrak{p}$, then $\mathfrak{p}$ is of the form

$$A \times \mathfrak{r}$$

for a unique prime ideal $\mathfrak{r} \subset B$. Thus, as sets,

$$\operatorname{Spec}(A \times B) = \operatorname{Spec}(A) \sqcup \operatorname{Spec}(B).$$

Moreover, these two pieces are open and closed. Indeed,

$$D(e_A) \simeq \operatorname{Spec}(A), \qquad D(e_B) \simeq \operatorname{Spec}(B),$$

and

$$D(e_A) \cap D(e_B) = \varnothing, \qquad D(e_A) \cup D(e_B) = \operatorname{Spec}(A \times B).$$

Finally, the structure sheaf restricts correctly, because

$$(A \times B)_{e_A} \simeq A, \qquad (A \times B)_{e_B} \simeq B.$$

Therefore the above bijection is an isomorphism of schemes.

More generally, for finitely many rings $A_1, \ldots, A_s$, one has

$$\operatorname{Spec}\left(\prod_{i=1}^{s} A_i\right) \simeq \bigsqcup_{i=1}^{s} \operatorname{Spec}(A_i).$$

VI.B Prime ideals in the affine plane over a field

Example VI.B.1. In $\Bbbk[x, y]$, the ideal $(y - x^2)$ is prime because

$$\Bbbk[x, y]/(y - x^2) \simeq \Bbbk[x]$$

is an integral domain. It corresponds to the parabola. The ideal (x, y) is maximal and corresponds to the origin.

Let $\Bbbk$ be an algebraically closed field and consider $\mathbb{A}^2_{\Bbbk} = \mathrm{Spec}(A)$, with $A = \Bbbk[x, y]$. Then, a subscheme of $\mathbb{A}^2_{\Bbbk}$ is given by an ideal $\mathfrak{a}$ of A. By the Nullstellensatz, we have $\mathbf{V}(\mathfrak{a}) = \emptyset$ if and only if $\sqrt{\mathfrak{a}} = A$, i.e. if there is $n \in \mathbb{N}$ such that $1^n \in \mathfrak{a}$, which of course means that $1 \in \mathfrak{a}$, so that $\mathfrak{a} = A$. Therefore, for any proper ideal $\mathfrak{a} \subsetneq A$, we have $\mathbf{V}(\mathfrak{a}) \neq \emptyset$ and any given point $a \in \mathbf{V}(\mathfrak{a})$ gives a maximal ideal $\mathfrak{m}_a$ which contains $\mathfrak{a}$.

For the affine plane, we have a classification of prime ideals. It relies on the fact that $A = R[x_2]$ where $R = \Bbbk[x_1]$, the argument being strongly dependent on the fact that R is a principal ideal domain.

Proposition VI.B.2. *Let R be a principal ideal domain, let y be an indeterminate, and put $B = R[y]$. Let $\mathfrak{p}$ be a prime ideal of B.*

i) If $\mathfrak{p} \cap R = (0)$, then $\mathfrak{p}$ is principal.

ii) If $\mathfrak{p} \cap R = (r) \neq (0)$, then r is irreducible and either

$$\mathfrak{p} = (r),$$

or

$$\mathfrak{p} = (r, g),$$

where, setting $\kappa = R/(r)$, the image $\bar{g} \in \kappa[y]$ is irreducible. In the latter case, $\mathfrak{p}$ is maximal.

Proof. Set $K = \mathrm{Frac}(R)$.

Assume first that $\mathfrak{p} \cap R = (0)$. If $\mathfrak{p} = (0)$ there is nothing to prove. Otherwise, $\mathfrak{p} \cap (\setminus\{0\}) = \emptyset$ so localising at $R \setminus \{0\}$ gives a non-zero prime ideal $\mathfrak{p}K[y]$ of the principal ideal domain $K[y]$. Hence

$$\mathfrak{p}K[y] = (F)$$

for some irreducible polynomial $F \in K[y]$. After multiplying by a non-zero scalar, we may choose a primitive polynomial $f \in R[y]$ associated with F. Since $f \in \mathfrak{p}K[y]$, there exists $s \in R \setminus \{0\}$ such that $sf \in \mathfrak{p}$. As $s \notin \mathfrak{p}$ and $\mathfrak{p}$ is prime, we get $f \in \mathfrak{p}$. Thus $(f) \subset \mathfrak{p}$.

Conversely, if $g \in \mathfrak{p}$, then $g \in (f)$ in $K[y]$. Since f is primitive, Gauss's lemma implies that f divides g already in $R[y]$. Therefore $\mathfrak{p} \subset (f)$, and hence $\mathfrak{p} = (f)$. Recall how this works: write $g = fh$ for some $h \in K[y]$. Then, for some $c = a/b \in K$, with $a, b \in R$ coprime, we have $h = ch_0$ with $h_0 \in R[y]$ primitive. Then $g = cfh_0$ gives that b must divide all coefficients of fh_0, because $g \in R[y]$. But fh_0 is primitive, so b must be a unit. Hence c lies in R and $g = ch_0 f$ happens in $R[y]$, so f divides g in $R[y]$.

Now assume that $\mathfrak{p} \cap R = (r) \neq (0)$. Since this is a non-zero prime ideal of the PID R, the element r is irreducible. Also, κ is a field. Modulo (r), the ideal

$$\bar{\mathfrak{p}} = \mathfrak{p}/(r)$$

is a prime ideal of the PID

$$\kappa[y].$$

Thus either $\bar{\mathfrak{p}} = (0)$, in which case $\mathfrak{p} = (r)$, or $\bar{\mathfrak{p}} = (\bar{g})$ for an irreducible polynomial $\bar{g}$. Choosing a lift $g \in R[y]$, we obtain $\mathfrak{p} = (r, g)$. In this case

$$B/\mathfrak{p} \simeq \kappa[y]/(\bar{g})$$

is a field, so $\mathfrak{p}$ is maximal. $\square$

Corollary VI.B.3. *Let $\Bbbk$ be an algebraically closed field. Then any prime ideal $\mathfrak{p}$ of $A = \Bbbk[x_1, x_2]$ is (0), or (f) with $f \in A$ irreducible, or $\mathfrak{p} = (x_1 - a_1, x_2 - a_2)$ with $a = (a_1, a_2) \in \Bbbk^2$.*

- If $\mathfrak{p}$ is maximal, then we have $\mathfrak{p} = \mathfrak{m}_a$ for some $a \in \Bbbk^2$. We have $A_{\mathfrak{m}_a} = \{f/g \in \Bbbk(x_1, x_2) \mid g(a) \neq 0\}$. Also, we have $\kappa(\mathfrak{m}_a) \simeq \Bbbk$, for all $a \in \Bbbk^2$.
- Principal prime ideals are precisely the ideals of the form $\mathfrak{p} = (p)$ where $p \in A$ is an irreducible polynomial. We have $A_{(p)} = \{f/g \in \Bbbk(x_1, x_2) \mid p \nmid g\}$. The residue field $\kappa((p)) \simeq \mathrm{Frac}(A/(p))$ depends strongly on p itself.
- The ideal $(0) = \{0\}$ is also prime and $A_{(0)} = \Bbbk(x_1, x_2) = \kappa((0))$.

VI.C Arbitrary closed subschemes of affine space over a field

Let $\Bbbk$ be a field and let

$$X = \mathrm{Spec}(\Bbbk[x_1, \ldots, x_n]/\mathfrak{a})$$

be a closed subscheme of $\mathbb{A}^n_{\Bbbk}$. If the ideal $\mathfrak{a}$ is complicated, one way to understand X is to write $\mathfrak{a}$ as an intersection of simpler ideals.

Definition VI.C.1. Let A be a ring and let

$$Y_1, \ldots, Y_r \subset \mathrm{Spec}(A)$$

be closed subschemes defined by ideals

$$\mathfrak{q}_1, \ldots, \mathfrak{q}_r \subset A.$$

The *scheme-theoretic union* of $Y_1, \ldots, Y_r$ is the closed subscheme defined by the ideal

$$\mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r.$$

Proposition VI.C.2. *Let A be a ring and let $\mathfrak{q}_1, \ldots, \mathfrak{q}_r \subset A$ be ideals. Put*

$$\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r.$$

Then the closed subscheme

$$X = \mathrm{Spec}(A/\mathfrak{a})$$

is the scheme-theoretic union of the closed subschemes

$$X_i = \mathrm{Spec}(A/\mathfrak{q}_i).$$

Moreover, the underlying closed subset of X is the union of the underlying closed subsets of the X_i:

$$\mathbb{V}(\mathfrak{a}) = \mathbb{V}(\mathfrak{q}_1) \cup \cdots \cup \mathbb{V}(\mathfrak{q}_r).$$

Proof. The first statement is just the definition of the scheme-theoretic union.

Let us prove the statement about the underlying closed subsets. It is enough to treat the case of two ideals. Let $\mathfrak{q}_1, \mathfrak{q}_2 \subset A$. Since

$$\mathfrak{q}_1\mathfrak{q}_2 \subset \mathfrak{q}_1 \cap \mathfrak{q}_2,$$

we have

$$\mathbb{V}(\mathfrak{q}_1 \cap \mathfrak{q}_2) \subset \mathbb{V}(\mathfrak{q}_1\mathfrak{q}_2).$$

Conversely, let $\mathfrak{p} \in \mathbb{V}(\mathfrak{q}_1\mathfrak{q}_2)$. Then

$$\mathfrak{q}_1\mathfrak{q}_2 \subset \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, we have

$$\mathfrak{q}_1 \subset \mathfrak{p} \qquad \text{or} \qquad \mathfrak{q}_2 \subset \mathfrak{p}.$$

Hence $\mathfrak{p} \in \mathbb{V}(\mathfrak{q}_1) \cup \mathbb{V}(\mathfrak{q}_2)$. Since

$$\mathbb{V}(\mathfrak{q}_1) \cup \mathbb{V}(\mathfrak{q}_2) \subset \mathbb{V}(\mathfrak{q}_1 \cap \mathfrak{q}_2)$$

is immediate, we get

$$\mathbb{V}(\mathfrak{q}_1 \cap \mathfrak{q}_2) = \mathbb{V}(\mathfrak{q}_1) \cup \mathbb{V}(\mathfrak{q}_2).$$

The general case follows by induction. □

Remark VI.C.3. The scheme-theoretic union depends on the ideals themselves, not only on their radicals. Thus it remembers possible nilpotent structure. For instance, the ideals

$$(x) \qquad \text{and} \qquad (x^2)$$

in $\Bbbk[x]$ define the same underlying closed subset of $\mathbb{A}^1_{\Bbbk}$, namely the origin, but the corresponding closed subschemes are different.

Example VI.C.4. Let

$$A = \Bbbk[x, y], \qquad \mathfrak{a} = (xy).$$

Then

$$(xy) = (x) \cap (y).$$

Therefore

$$\operatorname{Spec}(A/(xy))$$

is the scheme-theoretic union of

$$\operatorname{Spec}(A/(x)) \qquad \text{and} \qquad \operatorname{Spec}(A/(y)).$$

Geometrically, this is the union of the two coordinate axes.

Example VI.C.5. Let

$$A = \Bbbk[x], \qquad \mathfrak{a} = (x^2(x-1)).$$

Since the ideals (x^2) and $(x-1)$ are coprime, we have

$$(x^2(x-1)) = (x^2) \cap (x-1).$$

Hence

$$\operatorname{Spec}(A/(x^2(x-1)))$$

is the scheme-theoretic union of the double point at 0,

$$\operatorname{Spec}(\Bbbk[x]/(x^2)),$$

and the reduced point at 1,

$$\operatorname{Spec}(\Bbbk[x]/(x-1)).$$

Remark VI.C.6. In a Noetherian ring, every proper ideal admits a primary decomposition

$$\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r.$$

This gives a way to view a closed subscheme

$$\operatorname{Spec}(A/\mathfrak{a})$$

as a scheme-theoretic union of closed subschemes defined by the ideals $\mathfrak{q}_i$. Later, after introducing irreducible components and generic points, we will see how primary decomposition also detects the components of the underlying topological space and possible embedded components.

Exercise VI.C.7. Let
$$A = \Bbbk[x, y].$$
Show that
$$(xy) = (x) \cap (y).$$
Deduce that $\operatorname{Spec}(A/(xy))$ is the scheme-theoretic union of the two coordinate axes.

Solution. The inclusion
$$(xy) \subset (x) \cap (y)$$
is clear. Conversely, if $f \in (x) \cap (y)$, then f is divisible by both x and y. Since x and y are coprime in $\Bbbk[x, y]$, the polynomial f is divisible by xy. Thus
$$(x) \cap (y) = (xy).$$
Therefore the closed subscheme defined by (xy) is the scheme-theoretic union of the closed subschemes defined by (x) and (y), namely the two coordinate axes.

Exercise VI.C.8. Let
$$A = \Bbbk[x]$$
and let
$$\mathfrak{a} = (x^2(x-1)^3).$$
Show that
$$\mathfrak{a} = (x^2) \cap ((x-1)^3).$$
Describe the corresponding closed subscheme of $\mathbb{A}^1_{\Bbbk}$.

Solution. The ideals (x^2) and $((x-1)^3)$ are coprime, because x and $x-1$ are coprime. Hence
$$(x^2) \cap ((x-1)^3) = (x^2)((x-1)^3) = (x^2(x-1)^3).$$
Thus
$$\operatorname{Spec}(\Bbbk[x]/(x^2(x-1)^3))$$
is the scheme-theoretic union of the double point at 0 and the triple point at 1.

Exercise VI.C.9. Let
$$A = \Bbbk[x]$$
and compare the closed subschemes of $\mathbb{A}^1_{\Bbbk}$ defined by
$$(x) \cap (x) \qquad \text{and} \qquad (x)(x).$$

Solution. We have
$$(x) \cap (x) = (x),$$
whereas
$$(x)(x) = (x^2).$$
Thus $(x) \cap (x)$ defines the reduced origin
$$\operatorname{Spec}(\Bbbk[x]/(x)) \simeq \operatorname{Spec}(\Bbbk),$$
while $(x)(x)$ defines the double point
$$\operatorname{Spec}(\Bbbk[x]/(x^2)).$$
The two closed subschemes have the same underlying closed subset, but they are not isomorphic as schemes: $\Bbbk[x]/(x)$ is reduced, whereas $\Bbbk[x]/(x^2)$ has a non-zero nilpotent element.

VI.D Arithmetic affine schemes

Further reading. For schemes over $\mathrm{Spec}(\mathbb{Z})$, reduction modulo primes, and the arithmetic viewpoint, see [Liu02]; [EH00] offers complementary geometric examples.

Every ring A is a $\mathbb{Z}$-algebra through γ_A. Thus every affine scheme has a canonical morphism

$$\pi_A = \mathrm{Spec}(\gamma_A) : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(\mathbb{Z}).$$

This single map records the characteristics in which the scheme has points.

Proposition VI.D.1. *Let A be an integral domain and let $\pi_A = \mathrm{Spec}(\gamma_A) : \mathrm{Spec}(A) \to \mathrm{Spec}(\mathbb{Z})$ be its structure morphism.*

(i) If $\mathrm{char}(A) = p > 0$, then the image of π_A is $\{(p)\}$.

(ii) If $\mathrm{char}(A) = 0$, then
$$\mathrm{Im}(\pi_A) = \{(0)\} \cup \{(p) \mid pA \neq A\}.$$
In particular, π_A is dominant.

Proof. The image of a prime $\mathfrak{q} \subset A$ is its contraction $\gamma_A^{-1}(\mathfrak{q})$. If A has characteristic p, every such contraction is (p). Suppose that A has characteristic zero. The prime $(0) \subset A$ maps to $(0) \subset \mathbb{Z}$. Moreover, a prime of A maps to (p) precisely when it contains $\gamma_A(p) = p1_A$, or equivalently when it contains pA. Such a prime exists precisely when pA is proper. Finally, (0) is dense in $\mathrm{Spec}(\mathbb{Z})$. □

For a general $\mathbb{Z}$-algebra, reduction modulo a prime packages all points of a fixed residue characteristic.

Proposition VI.D.2. *Let A be a ring and let p be a prime number. The closed immersion*

$$\mathrm{Spec}(A/pA) \longrightarrow \mathrm{Spec}(A)$$

has image
$$\mathbb{V}(pA) = \{\mathfrak{q} \in \mathrm{Spec}(A) \mid \mathrm{char}\,\kappa(\mathfrak{q}) = p\}.$$
Equivalently, these are exactly the points mapping to (p) in $\mathrm{Spec}(\mathbb{Z})$.

Proof. Prime ideals of A/pA correspond to prime ideals of A containing pA. For $\mathfrak{q} \in \mathrm{Spec}(A)$, the contraction $\gamma_A^{-1}(\mathfrak{q})$ is the kernel of the composite

$$\mathbb{Z} \xrightarrow{\gamma_A} A \longrightarrow \kappa(\mathfrak{q}),$$

which is precisely $\gamma_{\kappa(\mathfrak{q})}$. It is (p) exactly when $\kappa(\mathfrak{q})$ has characteristic p. □

Example VI.D.3. Let
$$X = \mathrm{Spec}\big(\mathbb{Z}[x]/(x^2-1)\big).$$
For an odd prime p,
$$X_p = \mathrm{Spec}\big(\mathbb{F}_p[x]/(x^2-1)\big) \simeq \mathrm{Spec}(\mathbb{F}_p) \coprod \mathrm{Spec}(\mathbb{F}_p),$$
because $x^2 - 1 = (x-1)(x+1)$ has two distinct factors. At $p = 2$ the two factors coincide:

$$X_2 = \mathrm{Spec}\big(\mathbb{F}_2[x]/((x-1)^2)\big).$$

Its underlying space has one point, but its ring contains a nonzero nilpotent. Thus reduction may change both the number of points and the scheme structure.

Example VI.D.4 (An arithmetic subscheme of the plane). Let

$$Y = \mathrm{Spec}\big(\mathbb{Z}[x,y]/(xy-2)\big) \subset \mathbb{A}^2_{\mathbb{Z}}.$$

For $p \neq 2$, reduction gives

$$Y_p = \mathrm{Spec}\big(\mathbb{F}_p[x,y]/(xy-2)\big) \simeq \mathrm{Spec}\big(\mathbb{F}_p[x,x^{-1}]\big),$$

a hyperbola. At $p=2$ the equation becomes $xy=0$, so

$$Y_2 = \mathrm{Spec}\big(\mathbb{F}_2[x,y]/(xy)\big)$$

is the union of the two coordinate axes. This example exhibits, in one equation, why a family over $\mathrm{Spec}(\mathbb{Z})$ must be studied prime by prime.

VI.E Rings of integers in number fields

VI.E.1 Number fields

Definition VI.E.1. A *number field* is a finite field extension $K/\mathbb{Q}$.

The basic examples are $\mathbb{Q}$, quadratic fields $\mathbb{Q}(\sqrt{d})$ for squarefree $d \in \mathbb{Z}$, and cyclotomic fields $\mathbb{Q}(\zeta_n)$. By contrast, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{Q}(t)$ are not number fields: none is finite over $\mathbb{Q}$.

VI.E.2 Rings of integers of number fields

Definition VI.E.2. Let K be a number field. An element $\alpha \in K$ is an *algebraic integer* if it satisfies a monic polynomial with coefficients in $\mathbb{Z}$. The algebraic integers in K form a ring, denoted by $\mathcal{O}_K$ and called the *ring of integers* of K.

For example,

$$\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}, \qquad \mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i].$$

More generally, if d is squarefree, then

$$\mathcal{O}_{\mathbb{Q}(\sqrt{d})} = \begin{cases} \mathbb{Z}\left[\dfrac{1+\sqrt{d}}{2}\right], & d \equiv 1 \pmod 4, \\ \mathbb{Z}[\sqrt{d}], & d \not\equiv 1 \pmod 4. \end{cases}$$

The homomorphism $\gamma_{\mathcal{O}_K} : \mathbb{Z} \to \mathcal{O}_K$ induces

$$\mathrm{Spec}(\gamma_{\mathcal{O}_K}) : \mathrm{Spec}(\mathcal{O}_K) \longrightarrow \mathrm{Spec}(\mathbb{Z}).$$

Its generic point has residue field K, while the closed points over (p) are the prime ideals of

$$\mathcal{O}_K/p\mathcal{O}_K.$$

Thus factorisation modulo p translates directly into the geometry over the prime (p).

Example VI.E.3 (The Gaussian integers). For $K = \mathbb{Q}(i)$, write

$$\mathcal{O}_K = \mathbb{Z}[i] \simeq \mathbb{Z}[x]/(x^2+1).$$

Three reductions display the basic possibilities:

p	x^2+1 in $\mathbb{F}_p[x]$	$\mathrm{Spec}(\mathcal{O}_K/p\mathcal{O}_K)$
2	$(x+1)^2$	one non-reduced point with residue field $\mathbb{F}_2$
3	irreducible	one point with residue field $\mathbb{F}_9$
5	$(x-2)(x+2)$	two points with residue field $\mathbb{F}_5$.

Repeated, irreducible and split factorisations therefore produce three different kinds of reductions.

Exercise VI.E.4. Let $K = \mathbb{Q}(\sqrt{2})$, so that $\mathcal{O}_K = \mathbb{Z}[\sqrt{2}]$. Describe

$$\mathrm{Spec}(\mathcal{O}_K/p\mathcal{O}_K)$$

for $p = 2, 3, 7$, including the residue fields and any non-reduced structure.

Solution. We have

$$\mathcal{O}_K/p\mathcal{O}_K \simeq \mathbb{F}_p[x]/(x^2-2).$$

For $p = 2$, this is $\mathbb{F}_2[x]/(x^2)$, so there is one non-reduced point with residue field $\mathbb{F}_2$. Over $\mathbb{F}_3$, the polynomial x^2-2 is irreducible, so there is one point with residue field $\mathbb{F}_9$. Finally,

$$x^2-2 = (x-3)(x+3) \qquad \text{in } \mathbb{F}_7[x],$$

so the reduction at 7 is the disjoint union of two reduced $\mathbb{F}_7$-points.

Chapter 3

Schemes

I Schemes

Further reading. For the basic language of schemes, local rings, closed subschemes and morphisms, see [Liu02, Chapter 2] and [Har77, Chapter II, §2]; [EH00, Chapters I–II] gives a particularly geometric introduction.

I.A Locally ringed spaces

A *locally ringed space* is a topological space X equipped with a sheaf of rings $\mathcal{O}_X$ such that, for every point $x \in X$, the stalk $\mathcal{O}_{X,x}$ is a local ring.

Definition I.A.1. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ admitting an open covering $X = \bigcup_{i \in I} U_i$ such that, for every $i \in I$, the locally ringed space $(U_i, \mathcal{O}_X|_{U_i})$ is isomorphic to $\mathrm{Spec}(A_i)$ for some ring A_i.

The maximal ideal of the local ring $\mathcal{O}_{X,x}$ is denoted by $\mathfrak{m}_{X,x}$, and the residue field at x is

$$\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_{X,x}.$$

Remark I.A.2. Let $U \simeq \mathrm{Spec}(A)$ be an affine open subset of a scheme X, and let $x \in U$ correspond to a prime ideal $\mathfrak{p} \subset A$. Then

$$\mathcal{O}_{X,x} \simeq A_{\mathfrak{p}}, \qquad \mathfrak{m}_{X,x} \simeq \mathfrak{p}A_{\mathfrak{p}}, \qquad \kappa(x) \simeq \mathrm{Frac}(A/\mathfrak{p}).$$

Thus the local ring and the residue field may be computed in any affine neighbourhood of the point.

Example I.A.3 (The scheme of dual numbers). Let $A = \Bbbk[\varepsilon]/(\varepsilon^2)$ and set $X = \mathrm{Spec}(A)$. The ring A has a unique prime ideal, namely (ε), so the underlying topological space of X consists of a single point x. Nevertheless,

$$\mathcal{O}_{X,x} = A$$

is not a field: it contains the nonzero nilpotent element ε. Hence the structure sheaf contains information which is invisible in the underlying topological space.

Example I.A.4 (Smooth manifolds as locally ringed spaces). Let M be a smooth manifold and let $\mathcal{C}^\infty_M$ be the sheaf of real-valued smooth functions. For $x \in M$, the stalk $\mathcal{C}^\infty_{M,x}$ is the ring of germs of smooth functions at x. It is a local ring: its unique maximal ideal is

$$\mathfrak{m}_x = \{[g]_x \mid g(x) = 0\}.$$

Indeed, a germ $[g]_x$ is invertible precisely when $g(x) \neq 0$, since then g is nonzero on a neighbourhood of x and $1/g$ is smooth there. Thus $(M, \mathcal{C}^\infty_M)$ is a locally ringed space.

A smooth map $f : M \to N$ induces a morphism of locally ringed spaces by pullback of smooth functions,

$$f^\# : \mathcal{C}^\infty_N \longrightarrow f_* \mathcal{C}^\infty_M, \qquad g \longmapsto g \circ f.$$

This is an important model for the definition, although a smooth manifold with this sheaf is generally not a scheme.

Exercise I.A.5. Let $U \subset X$ be an open subset of a scheme. Prove that $(U, \mathcal{O}_X|_U)$ is a scheme.

Solution. Let $x \in U$. Since X is a scheme, there is an affine open neighbourhood $V = \operatorname{Spec}(A)$ of x in X. The intersection $U \cap V$ is an open neighbourhood of x in V. Since the principal open subsets form a basis of the topology of $\operatorname{Spec}(A)$, there is an element $f \in A$ such that

$$x \in D(f) \subset U \cap V.$$

Moreover,

$$(D(f), \mathcal{O}_X|_{D(f)}) \simeq \operatorname{Spec}(A_f),$$

so $D(f)$ is an affine open neighbourhood of x contained in U. Therefore U is covered by affine open subschemes and is a scheme.

I.B Quasi-coherent sheaves

Let $(X, \mathcal{O}_X)$ be a scheme. A *sheaf of $\mathcal{O}_X$-modules* is a sheaf $\mathcal{F}$ such that $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module for every open subset $U \subset X$, compatibly with restriction maps.

Definition I.B.1. A sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ is *quasi-coherent* if every point of X has an affine open neighbourhood $U = \operatorname{Spec}(A)$ for which there is an A-module M and an isomorphism

$$\mathcal{F}|_U \simeq \widetilde{M}.$$

Here $\widetilde{M}$ is the associated sheaf of Definition III.B.2. Equivalently, on such an affine open,

$$\mathcal{F}(D(f)) \simeq M_f \qquad (f \in A),$$

compatibly with restrictions.

Thus quasi-coherence is a local condition: it says that the sheaf is obtained locally from modules by localisation. The structure sheaf $\mathcal{O}_X$ and every finite direct sum $\mathcal{O}_X^{\oplus r}$ are quasi-coherent.

I.B.1 Generating sections

Let $\mathcal{F}$ be an $\mathcal{O}_X$-module and let $s_1, \ldots, s_r \in \Gamma(\mathcal{F})$. Their *evaluation morphism* is

$$\operatorname{ev}_{s_1,\ldots,s_r} : \mathcal{O}_X^{\oplus r} \longrightarrow \mathcal{F}, \qquad (a_1, \ldots, a_r) \longmapsto \sum_{i=1}^{r} a_i s_i.$$

The sections $s_1, \ldots, s_r$ *generate* $\mathcal{F}$ if this morphism is surjective.

Lemma I.B.2 (Generation may be checked on stalks). *The sections $s_1, \ldots, s_r$ generate $\mathcal{F}$ if and only if, for every $x \in X$, their germs $(s_1)_x, \ldots, (s_r)_x$ generate the $\mathcal{O}_{X,x}$-module $\mathcal{F}_x$.*

Proof. The stalk at x of the evaluation morphism is

$$\mathcal{O}_{X,x}^{\oplus r} \longrightarrow \mathcal{F}_x, \qquad (a_1, \dots, a_r) \longmapsto \sum_i a_i (s_i)_x.$$

A morphism of sheaves of modules is surjective if and only if it is surjective on every stalk, which gives the assertion. □

Example I.B.3. Let $X = \operatorname{Spec}(A)$ and let M be generated by $m_1, \dots, m_r$. The elements $m_i \in M = \widetilde{M}(X)$ define global sections of $\widetilde{M}$, and their germs generate

$$\widetilde{M}_{\mathfrak{p}} \simeq M_{\mathfrak{p}}$$

for every $\mathfrak{p} \in X$. Hence they generate $\widetilde{M}$. In particular, if $A = \Bbbk[x, y]$, the sections x and y generate the quasi-coherent ideal sheaf $\widetilde{(x, y)}$.

Exercise I.B.4 (Generating the structure sheaf). Let $f_1, \dots, f_r \in \Gamma(\mathcal{O}_X)$. Show that they generate $\mathcal{O}_X$ if and only if the open subsets

$$X_{f_i} = \{x \in X \mid (f_i)_x \text{ is invertible in } \mathcal{O}_{X,x}\}$$

cover X. If $X = \operatorname{Spec}(A)$ and the f_i come from elements of A, show that this is equivalent to $(f_1, \dots, f_r) = A$.

Solution. The germs $(f_i)_x$ generate the local ring $\mathcal{O}_{X,x}$ precisely when at least one of them is a unit: a proper ideal of a local ring is contained in its maximal ideal. Lemma I.B.2 proves the first assertion. On $\operatorname{Spec}(A)$ one has $X_{f_i} = D(f_i)$, and these principal opens cover X if and only if

$$\mathbb{V}(f_1, \dots, f_r) = \varnothing,$$

equivalently $(f_1, \dots, f_r) = A$.

I.B.2 Finite type and finite presentation

An $\mathcal{O}_X$-module $\mathcal{F}$ is *of finite type* if every point has an open neighbourhood U on which there is a surjection

$$\mathcal{O}_U^{\oplus r} \twoheadrightarrow \mathcal{F}|_U$$

for some finite r. It is *of finite presentation* if locally there is an exact sequence

$$\mathcal{O}_U^{\oplus m} \longrightarrow \mathcal{O}_U^{\oplus r} \longrightarrow \mathcal{F}|_U \longrightarrow 0$$

with m and r finite. Thus finite presentation asks for finitely many local generators and finitely many relations among them.

Example I.B.5. If M is a finitely presented A-module with presentation

$$A^{\oplus m} \xrightarrow{P} A^{\oplus r} \longrightarrow M \longrightarrow 0,$$

then localisation gives the corresponding presentation of M_f over A_f for every $f \in A$. The associated sheaf therefore has a presentation

$$\mathcal{O}_{\operatorname{Spec}(A)}^{\oplus m} \xrightarrow{\widetilde{P}} \mathcal{O}_{\operatorname{Spec}(A)}^{\oplus r} \longrightarrow \widetilde{M} \longrightarrow 0.$$

For $M = A/(f_1, \dots, f_m)$, the first map is the row $(f_1, \dots, f_m)$.

I.B.3 Quasi-coherent sheaves on an affine scheme

Write (Mod_A) for the category of A-modules and $(\mathrm{QCoh}(X))$ for the category of quasi-coherent $\mathcal{O}_X$-modules.

Theorem I.B.6 (The affine equivalence for quasi-coherent sheaves). *Let $X = \mathrm{Spec}(A)$. The functors*

$$(\mathrm{Mod}_A) \xrightarrow{\sim} (\mathrm{QCoh}(X)), \qquad (\mathrm{QCoh}(X)) \xrightarrow{\Gamma} (\mathrm{Mod}_A)$$

are quasi-inverse equivalences. More precisely,

$$\Gamma(\widetilde{M}) \simeq M, \qquad \widetilde{\Gamma(\mathcal{F})} \simeq \mathcal{F},$$

naturally in M and $\mathcal{F}$. Both functors are exact. Under this equivalence, finitely generated modules correspond to quasi-coherent sheaves of finite type, and finitely presented modules correspond to quasi-coherent sheaves of finite presentation.

Proof. By construction,

$$\Gamma(\widetilde{M}) = \widetilde{M}(D(1)) = M.$$

The key affine localisation statement is that, for a quasi-coherent sheaf $\mathcal{F}$ and $f \in A$, the natural homomorphism

$$\Gamma(\mathcal{F})_f \longrightarrow \mathcal{F}(D(f))$$

is an isomorphism. To see this, choose a finite principal-open covering on which $\mathcal{F}$ is associated with modules. The sheaf axiom expresses global sections as the equaliser of the sections on this covering and on its pairwise intersections. Localising that finite equaliser at f gives the corresponding equaliser over $D(f)$, because localisation is exact. This is precisely the displayed homomorphism.

Taking $M = \Gamma(\mathcal{F})$, these isomorphisms on all principal opens assemble to a natural isomorphism $\widetilde{M} \simeq \mathcal{F}$. Thus the two functors are quasi-inverse. The functor $M \mapsto \widetilde{M}$ is exact because its maps on stalks are localisations, and localisation is exact. Its quasi-inverse Γ is therefore exact on quasi-coherent sheaves over X.

Finally, finite generators and finite relations sheafify to local finite presentations. Conversely, a finite principal-open cover and the usual clearing-of-denominators argument turn local finite generators and relations into a finite presentation of $\Gamma(\mathcal{F})$. This proves the last assertion. □

Exercise I.B.7 (Modules and sheaves on an affine scheme). Let $X = \mathrm{Spec}(A)$ and let M, N be A-modules.

(i) Prove that the natural map

$$\mathrm{Hom}_A(M, N) \longrightarrow \mathrm{Hom}_{\mathcal{O}_X}(\widetilde{M}, \widetilde{N})$$

is a bijection.

(ii) If $M = A/(f_1, \ldots, f_r)$, compute the stalk $\widetilde{M}_{\mathfrak{p}}$ and show that it is zero precisely when some f_i does not belong to $\mathfrak{p}$.

(iii) Explain why kernels, cokernels and short exact sequences of quasi-coherent sheaves on X may be computed on the corresponding A-modules.

Solution. For a sheaf morphism $\widetilde{M} \to \widetilde{N}$, take global sections and use Theorem I.B.6; this is inverse to sheafification and proves (i). For (ii),

$$\widetilde{M}_{\mathfrak{p}} \simeq M_{\mathfrak{p}} \simeq A_{\mathfrak{p}}/(f_1, \ldots, f_r)A_{\mathfrak{p}}.$$

This module is zero exactly when the ideal generated by the f_i becomes the unit ideal in $A_{\mathfrak{p}}$, equivalently when at least one f_i is not in $\mathfrak{p}$. Part (iii) follows from the exactness of the two quasi-inverse functors.

Exercise I.B.8. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module.

(i) Show that $\mathcal{F}|_U$ is quasi-coherent for every open subset $U \subset X$.

(ii) Let $X = \bigcup_i U_i$ be an open covering. Suppose that $\mathcal{F}|_{U_i}$ is quasi-coherent for every i. Show that $\mathcal{F}$ is quasi-coherent.

Solution. Both assertions follow from the fact that quasi-coherence may be checked on an affine open covering. For the first, if $x \in U$, choose an affine open neighbourhood V of x on which $\mathcal{F}$ is associated with a module. A principal open neighbourhood $D(f)$ of x contained in $U \cap V$ is affine, and

$$\mathcal{F}|_{D(f)} \simeq \widetilde{M_f}.$$

For the second, cover every U_i by affine opens on which $\mathcal{F}|_{U_i}$ is associated with a module. Together these affine opens cover X, proving that $\mathcal{F}$ is quasi-coherent.

I.C Closed subschemes

Definition I.C.1. A morphism $i\colon Z \to X$ is a *closed immersion* if X admits an affine open covering $U_\alpha = \mathrm{Spec}(A_\alpha)$ such that $i^{-1}(U_\alpha)$ is affine and the homomorphism

$$A_\alpha \longrightarrow \Gamma\big(i^{-1}(U_\alpha), \mathcal{O}_Z\big)$$

induced by i is surjective. Equivalently, on every such chart the restriction of i is isomorphic over U_α to

$$\mathrm{Spec}(A_\alpha/\mathfrak{a}_\alpha) \longrightarrow \mathrm{Spec}(A_\alpha)$$

for some ideal $\mathfrak{a}_\alpha \subset A_\alpha$.

A *closed subscheme* of X is a scheme Z equipped with a closed immersion $i\colon Z \hookrightarrow X$.

Let X be a scheme. A *sheaf of ideals* of $\mathcal{O}_X$ is a subsheaf $\mathcal{I} \subset \mathcal{O}_X$ such that $\mathcal{I}(U)$ is an ideal of $\mathcal{O}_X(U)$ for every open subset $U \subset X$. It is *quasi-coherent* if it is quasi-coherent as an $\mathcal{O}_X$-module. Thus, on an affine open $U = \mathrm{Spec}(A)$,

$$\mathcal{I}|_U \simeq \widetilde{\mathfrak{a}}$$

for an ideal $\mathfrak{a} \subset A$, and $\mathcal{I}(D(f)) = \mathfrak{a}A_f$.

Proposition I.C.2. *Closed subschemes of X, up to unique isomorphism over X, are in bijection with quasi-coherent sheaves of ideals of $\mathcal{O}_X$. The correspondence sends a closed immersion $i\colon Z \hookrightarrow X$ to*

$$\mathcal{I}_{Z/X} = \ker\big(\mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z\big).$$

Conversely, if $\mathcal{I}|_U \simeq \widetilde{\mathfrak{a}}$ on $U = \mathrm{Spec}(A)$, the corresponding closed subscheme restricts to $\mathrm{Spec}(A/\mathfrak{a}) \hookrightarrow \mathrm{Spec}(A)$.

Proof. Let $i\colon Z \hookrightarrow X$ be a closed immersion. On an affine chart $U = \mathrm{Spec}(A)$ as in the definition, write

$$i^{-1}(U) = \mathrm{Spec}(A/\mathfrak{a}).$$

The morphism $\mathcal{O}_X \to i_*\mathcal{O}_Z$ restricts to the sheafification of $A \twoheadrightarrow A/\mathfrak{a}$. It is therefore surjective, and its kernel on U is $\widetilde{\mathfrak{a}}$. Hence $\mathcal{I}_{Z/X}$ is a quasi-coherent ideal sheaf and

$$i_*\mathcal{O}_Z \simeq \mathcal{O}_X/\mathcal{I}_{Z/X}.$$

Conversely, let $\mathcal{I} \subset \mathcal{O}_X$ be quasi-coherent. Define a closed subset

$$|Z| = \{x \in X \mid \mathcal{I}_x \subset \mathfrak{m}_{X,x}\}.$$

On $U = \operatorname{Spec}(A)$ with $\mathcal{I}|_U \simeq \widetilde{\mathfrak{a}}$, this subset is $\mathbb{V}(\mathfrak{a})$. If $j : |Z| \hookrightarrow X$ is the inclusion, put

$$\mathcal{O}_Z = j^{-1}(\mathcal{O}_X/\mathcal{I}).$$

Locally, $(|Z|, \mathcal{O}_Z)$ is $\operatorname{Spec}(A/\mathfrak{a})$, so it is a scheme and $j : Z \hookrightarrow X$ is a closed immersion. The two constructions are inverse on every affine open chart, and therefore globally. $\square$

Example I.C.3 (Restricting a closed subscheme). Let $i : Z \hookrightarrow X$ be the closed subscheme defined by a quasi-coherent ideal sheaf $\mathcal{I}$, and let $U \subset X$ be open. Then $Z \cap U = i^{-1}(U)$ is a closed subscheme of U, defined by $\mathcal{I}|_U$. Hence closed subschemes restrict naturally to open subsets; on an affine chart $V = \operatorname{Spec}(A)$ this is just the familiar passage from an ideal $\mathfrak{a} \subset A$ to its localisations $\mathfrak{a}A_f$.

Exercise I.C.4 (Gluing closed subschemes). Let $X = \bigcup_i U_i$ be an open covering. For every i, let $Z_i \hookrightarrow U_i$ be a closed subscheme, and suppose that the restrictions of Z_i and Z_j to $U_i \cap U_j$ agree. Show that there is a unique closed subscheme $Z \hookrightarrow X$ whose restriction to U_i is Z_i.

Solution. Let $\mathcal{I}_i \subset \mathcal{O}_{U_i}$ be the quasi-coherent ideal sheaf defining Z_i. The hypothesis identifies $\mathcal{I}_i|_{U_i \cap U_j}$ with $\mathcal{I}_j|_{U_i \cap U_j}$. The sheaf gluing axiom therefore gives a unique sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ such that $\mathcal{I}|_{U_i} = \mathcal{I}_i$. It is quasi-coherent because this property is local. The ideal sheaf $\mathcal{I}$ defines the required closed subscheme. Uniqueness follows from the uniqueness of the glued ideal sheaf and the correspondence between quasi-coherent ideal sheaves and closed subschemes.

Definition I.C.5. A morphism $j\colon Z \to X$ is an *immersion* (or *locally closed immersion*) if it factors as

$$Z \xrightarrow{i} U \xrightarrow{u} X,$$

where i is a closed immersion and u is an open immersion. A *subscheme* of X is a scheme Z equipped with an immersion $Z \to X$. Thus open and closed subschemes are the two special cases.

Remark I.C.6. The factorisation is local on X: the image of an immersion is a locally closed subset, and Z is a closed subscheme of a suitable open subscheme $U \subset X$.

Exercise I.C.7 (A closed subscheme and its first thickening). Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent ideal sheaf. Denote by $Z = \mathbb{V}(\mathcal{I})$ and $Z^{(2)} = \mathbb{V}(\mathcal{I}^2)$ the corresponding closed subschemes.

(i) Show that $\mathcal{I}^2$ is quasi-coherent.

(ii) Prove that Z and $Z^{(2)}$ have the same underlying closed subset.

(iii) Explain why they are different as closed subschemes whenever $\mathcal{I}^2 \neq \mathcal{I}$.

(iv) Apply the construction to $X = \mathbb{A}^1_{\Bbbk}$ and $\mathcal{I} = (t)$.

Solution. On an affine open $U = \operatorname{Spec}(A)$, write $\mathcal{I}|_U \simeq \widetilde{\mathfrak{a}}$. Then

$$\mathcal{I}^2|_U \simeq \widetilde{\mathfrak{a}^2},$$

so $\mathcal{I}^2$ is quasi-coherent. Since $\sqrt{\mathfrak{a}^2} = \sqrt{\mathfrak{a}}$, the closed subsets $\mathbb{V}(\mathfrak{a})$ and $\mathbb{V}(\mathfrak{a}^2)$ agree on every affine chart. Thus $|Z| = |Z^{(2)}|$.

Closed subschemes correspond to their ideal sheaves, so $\mathcal{I}^2 \neq \mathcal{I}$ implies that Z and $Z^{(2)}$ are different as closed subschemes of X. For $X = \operatorname{Spec}(\Bbbk[t])$ and $\mathcal{I} = (t)$, they are

$$\operatorname{Spec}(\Bbbk[t]/(t)) \quad \text{and} \quad \operatorname{Spec}(\Bbbk[t]/(t^2)).$$

The first is a reduced point and the second is a non-reduced double point with the same support.

Exercise I.C.8 (The local form of a closed immersion). Let $i\colon Z \hookrightarrow X$ be the closed subscheme defined by a quasi-coherent ideal sheaf $\mathcal{I}$.

(i) Show that $|i| : |Z| \to |X|$ is a homeomorphism onto the closed subset

$$\{x \in X \mid \mathcal{I}_x \subset \mathfrak{m}_{X,x}\}.$$

(ii) If $U \subset X$ is open, show that $i^{-1}(U)$ is the closed subscheme of U defined by $\mathcal{I}|_U$.

(iii) For $z \in Z$, prove that

$$\mathcal{O}_{Z,z} \simeq \mathcal{O}_{X,i(z)}/\mathcal{I}_{i(z)} \qquad \text{and} \qquad \kappa(z) \simeq \kappa(i(z)).$$

Solution. All three assertions are local on X. Take an affine neighbourhood $U = \operatorname{Spec}(A)$ of $i(z)$ on which $\mathcal{I}$ corresponds to an ideal $\mathfrak{a} \subset A$. Then

$$i^{-1}(U) = \operatorname{Spec}(A/\mathfrak{a}) \longrightarrow \operatorname{Spec}(A).$$

Prime ideals of $A/\mathfrak{a}$ correspond to the primes of A containing $\mathfrak{a}$, and this identifies $\operatorname{Spec}(A/\mathfrak{a})$ homeomorphically with $\mathbb{V}(\mathfrak{a})$. Restricting to an open subset localises both A and $\mathfrak{a}$, which proves the second assertion. Finally, if z corresponds to $\bar{\mathfrak{p}} = \mathfrak{p}/\mathfrak{a} \subset A/\mathfrak{a}$, then

$$(A/\mathfrak{a})_{\bar{\mathfrak{p}}} \simeq A_{\mathfrak{p}}/\mathfrak{a}A_{\mathfrak{p}},$$

where $\bar{\mathfrak{p}}$ denotes the image of $\mathfrak{p}$ in $A/\mathfrak{a}$. Taking the quotient by the maximal ideal gives the same residue field $\operatorname{Frac}(A/\mathfrak{p})$ on both sides.

Exercise I.C.9 (Unions and intersections of closed subschemes). Let $Z_1, Z_2 \hookrightarrow X$ be closed subschemes with quasi-coherent ideal sheaves $\mathcal{I}_1, \mathcal{I}_2$.

(i) Show that $\mathcal{I}_1 + \mathcal{I}_2$ and $\mathcal{I}_1 \cap \mathcal{I}_2$ are quasi-coherent.

(ii) Define the scheme-theoretic intersection and union by

$$Z_1 \cap Z_2 = \mathbb{V}(\mathcal{I}_1 + \mathcal{I}_2), \qquad Z_1 \cup Z_2 = \mathbb{V}(\mathcal{I}_1 \cap \mathcal{I}_2),$$

and show that their underlying subsets are respectively $|Z_1| \cap |Z_2|$ and $|Z_1| \cup |Z_2|$.

(iii) For the two coordinate axes in $\mathbb{A}^2_{\Bbbk}$, compute both closed subschemes and compare their local rings at the origin.

Solution. On an affine open $U = \operatorname{Spec}(A)$, let $\mathcal{I}_1$ and $\mathcal{I}_2$ correspond to ideals $\mathfrak{a}_1, \mathfrak{a}_2 \subset A$. Localisation commutes with finite sums and intersections, so the two sheaves restrict to

$$\widetilde{\mathfrak{a}_1 + \mathfrak{a}_2} \qquad \text{and} \qquad \widetilde{\mathfrak{a}_1 \cap \mathfrak{a}_2}.$$

They are therefore quasi-coherent. The identities

$$\mathbb{V}(\mathfrak{a}_1 + \mathfrak{a}_2) = \mathbb{V}(\mathfrak{a}_1) \cap \mathbb{V}(\mathfrak{a}_2), \qquad \mathbb{V}(\mathfrak{a}_1 \cap \mathfrak{a}_2) = \mathbb{V}(\mathfrak{a}_1) \cup \mathbb{V}(\mathfrak{a}_2)$$

prove the assertion about underlying subsets.

For the axes, $\mathfrak{a}_1 = (x)$ and $\mathfrak{a}_2 = (y)$. Hence their intersection as closed subschemes is defined by

$$(x) + (y) = (x, y),$$

whereas their union is defined by

$$(x) \cap (y) = (xy).$$

At the origin the union has local ring

$$\Bbbk[x, y]_{(x,y)}/(xy),$$

while the two axes have local rings $\Bbbk[y]_{(y)}$ and $\Bbbk[x]_{(x)}$. The first ring has two branches and zero divisors; the other two are domains.

Example I.C.10 (A sheaf of ideals which is not quasi-coherent). Let

$$A = \Bbbk[x]_{(x)}, \qquad X = \operatorname{Spec}(A).$$

The scheme X has two points: the generic point $\eta = (0)$ and the closed point $s = (x)$. Its nonempty open subsets are

$$U_0 = D(x) = \{\eta\}, \qquad X,$$

with

$$\mathcal{O}_X(U_0) = \Bbbk(x), \qquad \mathcal{O}_X(X) = A.$$

Define a sheaf of ideals $\mathcal{I}$ by

$$\mathcal{I}(X) = 0, \qquad \mathcal{I}(U_0) = \Bbbk(x),$$

with the only possible restriction map. This is a sheaf of ideals, but it is not quasi-coherent: if $\mathcal{I} \simeq \widetilde{\mathfrak{a}}$ for an ideal $\mathfrak{a} \subset A$, then $\mathfrak{a} = \mathcal{I}(X) = 0$, hence $\mathcal{I}(U_0) = \mathfrak{a}A_x = 0$, a contradiction.

I.D Morphisms of schemes

The category (Sch) of schemes is the full subcategory of the category of locally ringed spaces whose objects are schemes.

Definition I.D.1. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be schemes. A *morphism of schemes*

$$f \colon X \longrightarrow Y$$

is a morphism of locally ringed spaces. Thus it consists of a continuous map $f \colon X \to Y$ and a morphism of sheaves of rings

$$f^{\#} \colon \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X$$

such that, for every $x \in X$, the induced homomorphism

$$f_x^{\#} \colon \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x}$$

is local, that is,

$$(f_x^{\#})^{-1}(\mathfrak{m}_{X,x}) = \mathfrak{m}_{Y,f(x)}.$$

Exercise I.D.2 (Morphisms are local on the source). Let $X = \bigcup_i U_i$ be an open covering and let Y be a scheme. Suppose that morphisms $f_i : U_i \to Y$ are given and that

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$$

for all i, j. Prove that there is a unique morphism $f : X \to Y$ whose restriction to U_i is f_i.

Solution. The continuous maps underlying the f_i glue uniquely to a continuous map $f : X \to Y$. For every open subset $V \subset Y$, the maps on sections

$$\mathcal{O}_Y(V) \longrightarrow \mathcal{O}_X(f^{-1}(V) \cap U_i)$$

agree on pairwise overlaps. The sheaf axiom for $\mathcal{O}_X$ glues them uniquely to a homomorphism

$$\mathcal{O}_Y(V) \longrightarrow \mathcal{O}_X(f^{-1}(V)).$$

These homomorphisms commute with restriction and define $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$. Locality on stalks can be checked on the open cover, so $(f, f^\#)$ is a morphism of schemes. Every part of the construction is forced by the restrictions to the U_i, which proves uniqueness.

I.E Gluing schemes

Further reading. For gluing and the basic non-affine examples, see [EH00, Chapter I] and [Liu02, Chapter 2]. The construction of projective space by affine charts is also discussed in [Har77, Chapter II, §2].

Many schemes are constructed by gluing simpler schemes along open subschemes.

I.E.1 The gluing lemma

Lemma I.E.1 (Gluing lemma). *Let $(X_i)_{i\in I}$ be a family of schemes. For every $i, j \in I$, let $X_{ij} \subset X_i$ be an open subscheme and let*

$$\varphi_{ij} : X_{ij} \xrightarrow{\sim} X_{ji}$$

be an isomorphism. Assume that

(i) $X_{ii} = X_i$ *and* $\varphi_{ii} = \mathrm{id}_{X_i}$*;*

(ii) $\varphi_{ji} = \varphi_{ij}^{-1}$*;*

(iii) for all i, j, k, one has

$$\varphi_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}$$

and, on $X_{ij} \cap X_{ik}$:

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$$

Then there exists a scheme X with an open covering $X = \bigcup_{i\in I} U_i$ and isomorphisms $X_i \simeq U_i$ which identify the transition maps on $U_i \cap U_j$ with the maps φ_{ij}. The scheme X is unique up to unique isomorphism compatible with these identifications.

Proof. As a set, define

$$X = \coprod_{i\in I} X_i / \sim,$$

where $x \in X_{ij}$ is identified with $\varphi_{ij}(x) \in X_{ji}$. The cocycle condition implies that this is an equivalence relation. Give X the quotient topology. The image U_i of X_i is open, and the natural map $X_i \to U_i$ is a homeomorphism.

The structure sheaves $\mathcal{O}_{X_i}$ are identified on overlaps by the isomorphisms φ_{ij}. For an open subset $V \subset X$, define $\mathcal{O}_X(V)$ to be the set of families

$$(s_i)_{i\in I}, \qquad s_i \in \mathcal{O}_{X_i}(V \cap U_i),$$

whose restrictions correspond under all transition isomorphisms. The sheaf axioms follow from the sheaf axioms on the X_i. Moreover,

$$\mathcal{O}_X|_{U_i} \simeq \mathcal{O}_{X_i},$$

so every point of X has an affine neighbourhood. Hence X is a scheme. Uniqueness follows from the corresponding gluing statement for morphisms. ☐

Remark I.E.2 (Gluing morphisms). Let X and Y be schemes and let $X = \bigcup_{i\in I} U_i$ be an open covering. Giving a morphism $f\colon X \to Y$ is equivalent to giving morphisms $f_i\colon U_i \to Y$ such that

$$f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$$

for every $i, j \in I$. This is the morphism-gluing statement proved in Exercise I.D.2.

Exercise I.E.3. In the construction of Lemma I.E.1, verify that the relation $\sim$ is transitive and that each $X_i \to U_i$ is a homeomorphism.

Solution. Suppose that $x \in X_i$ is identified with $y \in X_j$, and that y is identified with $z \in X_k$. Thus

$$x \in X_{ij}, \qquad y = \varphi_{ij}(x), \qquad y \in X_{jk}, \qquad z = \varphi_{jk}(y).$$

The equality

$$\varphi_{ij}(X_{ij} \cap X_{ik}) = X_{ji} \cap X_{jk}$$

shows that $x \in X_{ik}$. The cocycle condition then gives

$$z = \varphi_{jk}(\varphi_{ij}(x)) = \varphi_{ik}(x).$$

Hence x is identified with z, proving transitivity. Reflexivity follows from $\varphi_{ii} = \mathrm{id}$, and symmetry from $\varphi_{ji} = \varphi_{ij}^{-1}$.

Let $q : \coprod_i X_i \to X$ be the quotient map and let $U_i = q(X_i)$. The restriction $q_i : X_i \to U_i$ is surjective. It is injective because two points of the same X_i which are equivalent are identified by $\varphi_{ii} = \mathrm{id}_{X_i}$, and hence are equal.

It remains to check the topology. If $V \subset X_i$ is open, its saturation in the disjoint union has intersection with X_j equal to

$$\varphi_{ij}(V \cap X_{ij}),$$

which is open in X_j. Thus the saturation of V is open, so $q_i(V)$ is open in U_i for the quotient topology. Hence q_i is an open continuous bijection, and therefore a homeomorphism.

I.E.2 The line with two origins

Let $\Bbbk$ be a field and let

$$X_1 = \mathrm{Spec}(\Bbbk[x_1]), \qquad X_2 = \mathrm{Spec}(\Bbbk[x_2]).$$

The principal opens $D(x_1)$ and $D(x_2)$ have coordinate rings

$$\Bbbk[x_1, x_1^{-1}], \qquad \Bbbk[x_2, x_2^{-1}].$$

We identify $D(x_1)$ with $D(x_2)$ by $x_1 \mapsto x_2$. The resulting scheme X is called the *affine line with two origins*. Let $o_i \in X_i$ be the origin. The points o_1 and o_2 remain distinct in X, while the complements $X_i \setminus \{o_i\}$ are identified.

A global function on X is a pair

$$(f_1, f_2) \in \Bbbk[x_1] \times \Bbbk[x_2]$$

whose restrictions agree on the common punctured line. Hence

$$\Gamma(\mathcal{O}_X) = \{(f_1, f_2) \mid f_1 = f_2 \text{ in } \Bbbk[x, x^{-1}]\} \simeq \Bbbk[x].$$

The two origins cannot be distinguished by global functions. We shall interpret this observation through the affinisation morphism below and use it to prove that X is not affine.

Exercise I.E.4. Let X be the affine line with two origins.

(i) Show that deleting either origin gives a scheme isomorphic to $\mathbb{A}^1_{\Bbbk}$.

(ii) Show that any open neighbourhood of o_1 meets every open neighbourhood of o_2.

(iii) Compute the local rings $\mathcal{O}_{X,o_1}$ and $\mathcal{O}_{X,o_2}$.

Solution. (i) Let X_1 and X_2 be the two affine charts, with origins o_1 and o_2. After deleting o_1, the remaining part of X_1 is the common punctured line, which is already contained in the image of X_2. Hence

$$X \setminus \{o_1\} = X_2 \simeq \mathbb{A}^1_{\Bbbk}.$$

Similarly,

$$X \setminus \{o_2\} = X_1 \simeq \mathbb{A}^1_{\Bbbk}.$$

(ii) Let V_1 and V_2 be open neighbourhoods of o_1 and o_2 in X. Then $V_i \cap X_i$ is a nonempty open subset of the irreducible scheme $X_i \simeq \operatorname{Spec}(\Bbbk[x_i])$. Every nonempty open subset of X_i contains its generic point. The generic points of X_1 and X_2 are identified in the common punctured line. Therefore this common generic point belongs to $V_1 \cap V_2$, and

$$V_1 \cap V_2 \neq \emptyset.$$

(iii) Since X_i is an open neighbourhood of o_i, the local ring in X is the same as the local ring in that chart:

$$\mathcal{O}_{X,o_1} \simeq \Bbbk[x_1]_{(x_1)}, \qquad \mathcal{O}_{X,o_2} \simeq \Bbbk[x_2]_{(x_2)}.$$

I.E.3 Projective line

We start with the same ingredients as for the line with two origins, so we let $\Bbbk$ be a field and let

$$X_1 = \operatorname{Spec}(\Bbbk[x_1]), \qquad X_2 = \operatorname{Spec}(\Bbbk[x_2]).$$

The principal opens $D(x_1)$ and $D(x_2)$ have coordinate rings

$$\Bbbk[x_1, x_1^{-1}], \qquad \Bbbk[x_2, x_2^{-1}].$$

Now identify $D(x_1)$ with $D(x_2)$ by $x_1 \mapsto x_2^{-1}$. The resulting scheme is the projective line $\mathbb{P}^1_{\Bbbk}$.

Its global functions will be computed together with those of $\mathbb{P}^n_B$ in Proposition I.E.7.

Exercise I.E.5. Let U_0 and U_1 be the two standard affine charts of $\mathbb{P}^1_{\Bbbk}$.

(i) Show that $U_0 \cap U_1 \simeq \mathbb{G}_{m,\Bbbk}$.

(ii) Let ∞ be the origin in the second chart. Show that $\mathbb{P}^1_{\Bbbk} \setminus \{\infty\} \simeq \mathbb{A}^1_{\Bbbk}$ and compute $\kappa(\infty)$.

(iii) Describe all closed points of $\mathbb{P}^1_{\Bbbk}$ and their residue fields. Which of them are $\Bbbk$-rational?

Solution. Write

$$U_0 = \mathrm{Spec}(\Bbbk[x]), \qquad U_1 = \mathrm{Spec}(\Bbbk[y]),$$

with $x = y^{-1}$ on the overlap.

(i) The overlap is

$$D(x) = \mathrm{Spec}(\Bbbk[x, x^{-1}]) = \mathbb{G}_{m,\Bbbk}.$$

(ii) Every point of U_1 other than its origin belongs to $D(y)$ and hence to U_0. Therefore

$$\mathbb{P}^1_{\Bbbk} \setminus \{\infty\} = U_0 \simeq \mathbb{A}^1_{\Bbbk}.$$

Moreover,

$$\mathcal{O}_{\mathbb{P}^1_{\Bbbk},\infty} \simeq \Bbbk[y]_{(y)}, \qquad \kappa(\infty) \simeq \Bbbk.$$

(iii) Besides ∞, the closed points are the points of U_0 defined by monic irreducible polynomials $f \in \Bbbk[x]$. Their residue fields are

$$\kappa(x_f) = \Bbbk[x]/(f).$$

Consequently, a closed point is $\Bbbk$-rational precisely when it is ∞ or is defined by a linear polynomial $x - a$.

I.E.4 Higher-dimensional projective space

More generally, consider a ring B and the polynomial algebra

$$A = B[x_0, \ldots, x_n].$$

For $0 \le i \le n$, set

$$R_i = \left(A[x_i^{-1}]\right)_0 = B\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right] \simeq B[t_1, \ldots, t_n]$$

and

$$U_i = \mathrm{Spec}(R_i) \simeq \mathbb{A}^n_B.$$

For $i \ne j$, the principal open subset

$$D\left(\frac{x_j}{x_i}\right) \subset U_i$$

has coordinate ring

$$\left(A[(x_i x_j)^{-1}]\right)_0,$$

and the same ring is obtained from the j-th chart. These identifications satisfy the cocycle condition, so the U_i glue to a scheme denoted by $\mathbb{P}^n_B$. For the moment, the subscript records the coefficient ring used in the construction. In Subsection II.B we shall glue the compatible maps from the charts to $\mathrm{Spec}(B)$ and thereby make precise in what sense $\mathbb{P}^n_B$ is a scheme over B.

Example I.E.6 (Transition functions on $\mathbb{P}^2_B$). On the chart U_0, use coordinates

$$u = \frac{x_1}{x_0}, \qquad v = \frac{x_2}{x_0}.$$

On U_1, use

$$s = \frac{x_0}{x_1}, \qquad t = \frac{x_2}{x_1}.$$

On the overlap $u \ne 0$, the transition functions are

$$s = u^{-1}, \qquad t = vu^{-1}.$$

The subscript 0 means the degree-zero part of the graded ring obtained after inverting the indicated homogeneous element. Explicitly,

$$\big(A[x_i^{-1}]\big)_0 = \left\{ \frac{a}{x_i^m} \,\middle|\, a \in A_m \right\}.$$

This notation will be used systematically below; in particular, it is different from the local ring $A_{(f)}$ at the prime ideal (f).

Proposition I.E.7 (Global functions on projective space). *Assume $n \geq 1$. Then*

$$\Gamma(\mathcal{O}_{\mathbb{P}^n_B}) = B.$$

In particular, $\Gamma(\mathcal{O}_{\mathbb{P}^1_{\Bbbk}}) = \Bbbk$.

Proof. Regard all the rings R_i as subrings of

$$B[x_0^{\pm 1}, \dots, x_n^{\pm 1}]_0.$$

Let

$$L = B[x_0^{\pm 1}, \dots, x_n^{\pm 1}]_0.$$

The Laurent monomials

$$x_0^{a_0} \cdots x_n^{a_n}, \qquad a_0 + \cdots + a_n = 0,$$

form a B-basis of L. The subring R_i is generated by the monomials x_j/x_i, with $j \neq i$. Thus a Laurent monomial belongs to R_i only if

$$a_j \geq 0 \qquad \text{for every } j \neq i.$$

If a Laurent monomial belongs to every R_i, then, since $n \geq 1$, each exponent a_j is nonnegative: choose some $i \neq j$. Since their sum is zero, all the exponents are zero. Consequently, the only Laurent monomial which can occur in an element of every R_i is 1, and therefore

$$\bigcap_{i=0}^{n} R_i = B.$$

A global section on $\mathbb{P}^n_B$ is a compatible family

$$(s_i), \qquad s_i \in R_i.$$

After restricting to the common open subset on which all the x_i are invertible, all the s_i define the same element of L. Conversely, an element of $\bigcap_i R_i$ gives such a compatible family. Hence

$$\Gamma(\mathcal{O}_{\mathbb{P}^n_B}) \simeq \bigcap_{i=0}^{n} R_i = B.$$

□

I.F Affine targets and affinisation

The following adjunction explains why global functions control every morphism from an arbitrary scheme to an affine scheme.

I.F.1 The affine-target adjunction

Theorem I.F.1 (Morphisms to an affine scheme). *Let X be a scheme and let A be a ring. There is a natural bijection*

$$\mathrm{Hom}_{(\mathrm{Sch})}(X, \mathrm{Spec}(A)) \simeq \mathrm{Hom}_{(\mathrm{Ring})}\big(A, \Gamma(\mathcal{O}_X)\big).$$

Proof. A morphism $f\colon X \to \mathrm{Spec}(A)$ gives, by taking global sections, a ring homomorphism

$$\varphi_f\colon A = \Gamma(\mathcal{O}_{\mathrm{Spec}(A)}) \longrightarrow \Gamma(\mathcal{O}_X).$$

Conversely, let

$$\varphi\colon A \longrightarrow B = \Gamma(\mathcal{O}_X)$$

be a ring homomorphism. For $x \in X$, let

$$\rho_x\colon B \longrightarrow \mathcal{O}_{X,x}$$

be the canonical map. We define

$$f_\varphi(x) = (\rho_x \circ \varphi)^{-1}(\mathfrak{m}_{X,x}) \in \mathrm{Spec}(A).$$

Let $U = \mathrm{Spec}(R)$ be an affine open subset of X, and let $\rho_U\colon B \to R$ be the restriction map. For $a \in A$, set $b = \rho_U(\varphi(a))$. Then

$$f_\varphi^{-1}(D(a)) \cap U = D(b).$$

Indeed, a prime $\mathfrak{p} \subset R$ belongs to the left-hand side if and only if a does not belong to

$$(A \xrightarrow{\varphi} B \xrightarrow{\rho_U} R \longrightarrow R_{\mathfrak{p}})^{-1}(\mathfrak{p}R_{\mathfrak{p}}),$$

which is equivalent to $b \notin \mathfrak{p}$. Hence f_φ is continuous.

On the open subset $D(b) = f_\varphi^{-1}(D(a)) \cap U$, the homomorphism $\rho_U \circ \varphi$ localizes to

$$A_a \longrightarrow R_b = \Gamma(D(b), \mathcal{O}_X).$$

These maps are compatible on intersections of affine open subsets and therefore glue to a morphism of sheaves

$$f_\varphi^\#\colon \mathcal{O}_{\mathrm{Spec}(A)} \longrightarrow (f_\varphi)_*\mathcal{O}_X.$$

The induced homomorphisms on stalks are local by the definition of $f_\varphi(x)$. Thus $(f_\varphi, f_\varphi^\#)$ is a morphism of schemes. The two constructions are inverse to one another. □

Example I.F.2 (A polynomial map). The homomorphism

$$\Bbbk[u, v] \longrightarrow \Bbbk[t], \qquad u \longmapsto t, \qquad v \longmapsto t^2,$$

defines a morphism

$$\mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^2_{\Bbbk}.$$

On closed points, it is the familiar map

$$a \longmapsto (a, a^2).$$

Its image is contained in the closed subscheme defined by $v - u^2$.

Exercise I.F.3. Let $\varphi\colon A \to B$ be a ring homomorphism and let $f\colon \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the associated morphism.

(i) Prove that $f(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q})$ for every prime ideal $\mathfrak{q} \subset B$.

(ii) Show that $f^{-1}(D(a)) = D(\varphi(a))$.

(iii) Verify directly that the induced map $A_a \to B_{\varphi(a)}$ is the map on sections over these principal open subsets.

Solution. (i) By construction of the morphism associated with a ring homomorphism, its underlying map is contraction of prime ideals:

$$f(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}).$$

The inverse image is prime because $A/\varphi^{-1}(\mathfrak{q})$ embeds into the integral domain $B/\mathfrak{q}$.

(ii) Let $a \in A$. Then

$$\begin{aligned} \mathfrak{q} \in f^{-1}(D(a)) &\iff a \notin f(\mathfrak{q}) \\ &\iff a \notin \varphi^{-1}(\mathfrak{q}) \\ &\iff \varphi(a) \notin \mathfrak{q} \\ &\iff \mathfrak{q} \in D(\varphi(a)). \end{aligned}$$

Therefore

$$f^{-1}(D(a)) = D(\varphi(a)).$$

(iii) We have

$$\mathcal{O}_{\mathrm{Spec}(A)}(D(a)) = A_a, \qquad \mathcal{O}_{\mathrm{Spec}(B)}(D(\varphi(a))) = B_{\varphi(a)}.$$

The map on sections sends

$$\frac{c}{a^n} \longmapsto \frac{\varphi(c)}{\varphi(a)^n}.$$

It is exactly the localisation of $\varphi : A \to B$ at the multiplicative set generated by a.

Exercise I.F.4. Fill in the remaining details in the proof of Theorem I.F.1: verify compatibility on overlaps and check explicitly that the two constructions are inverse.

Solution. Let

$$\varphi : A \longrightarrow \Gamma(\mathcal{O}_X)$$

be a ring homomorphism. For an affine open subset $U = \mathrm{Spec}(R) \subset X$, let $\rho_U : \Gamma(\mathcal{O}_X) \to R$ be restriction. If $a \in A$ and $b_U = \rho_U(\varphi(a))$, the construction gives a homomorphism

$$A_a \longrightarrow R_{b_U} = \Gamma(D(b_U), \mathcal{O}_X).$$

We first verify compatibility. Let $U = \mathrm{Spec}(R)$ and $V = \mathrm{Spec}(S)$ be affine open subsets, and let W be an affine open subset of

$$D(b_U) \cap D(b_V) \subset U \cap V.$$

Both maps, after restriction to W, send

$$\frac{c}{a^n} \longmapsto \frac{\varphi(c)|_W}{\varphi(a)|_W^n}.$$

They therefore agree on W. Since such affine open subsets cover the overlap, the local morphisms of sheaves agree on overlaps and glue to a morphism

$$f_\varphi^\# : \mathcal{O}_{\mathrm{Spec}(A)} \longrightarrow (f_\varphi)_* \mathcal{O}_X.$$

We now check that the two constructions are inverse. Start with a morphism

$$f : X \longrightarrow \operatorname{Spec}(A)$$

and let

$$\varphi = f^{\#}_{\operatorname{Spec}(A)} : A \longrightarrow \Gamma(\mathcal{O}_X)$$

be the induced map on global sections. For $x \in X$, put $\mathfrak{p} = f(x)$. The map

$$A \longrightarrow \mathcal{O}_{X,x}$$

factors through the local homomorphism

$$A_{\mathfrak{p}} = \mathcal{O}_{\operatorname{Spec}(A),\mathfrak{p}} \longrightarrow \mathcal{O}_{X,x}.$$

Since this homomorphism is local, the inverse image of $\mathfrak{m}_{X,x}$ in $A_{\mathfrak{p}}$ is $\mathfrak{p}A_{\mathfrak{p}}$. Its inverse image in A is $\mathfrak{p}$. Hence

$$f_{\varphi}(x) = \varphi_x^{-1}(\mathfrak{m}_{X,x}) = \mathfrak{p} = f(x).$$

Thus the two maps agree on points. On each principal open subset $D(a)$, both maps on sections are obtained by localising the same global homomorphism φ. Therefore the morphisms of sheaves also agree, and $f_{\varphi} = f$.

Conversely, start with φ. In the construction of f_{φ}, take the principal open subset $D(1) = \operatorname{Spec}(A)$. The induced map on sections is precisely

$$A = A_1 \longrightarrow \Gamma(\mathcal{O}_X),$$

namely φ. Thus taking global sections recovers the original ring homomorphism.

Exercise I.F.5. Prove that, for every scheme X,

$$\operatorname{Hom}_{(\mathrm{Sch})}(X, \mathbb{A}^1_{\mathbb{Z}}) \simeq \Gamma(\mathcal{O}_X).$$

Under this bijection, which morphism corresponds to a global section $s \in \Gamma(\mathcal{O}_X)$?

Solution. Since

$$\mathbb{A}^1_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[t]),$$

Theorem I.F.1 gives

$$\operatorname{Hom}_{(\mathrm{Sch})}(X, \mathbb{A}^1_{\mathbb{Z}}) \simeq \operatorname{Hom}_{(\mathrm{Ring})}\big(\mathbb{Z}[t], \Gamma(\mathcal{O}_X)\big).$$

A ring homomorphism from $\mathbb{Z}[t]$ to $\Gamma(\mathcal{O}_X)$ restricts to $\gamma_{\Gamma(\mathcal{O}_X)}$ on $\mathbb{Z}$ and is uniquely determined by the image of t, which may be any element of $\Gamma(\mathcal{O}_X)$. Hence

$$\operatorname{Hom}_{(\mathrm{Sch})}(X, \mathbb{A}^1_{\mathbb{Z}}) \simeq \Gamma(\mathcal{O}_X).$$

The global section s corresponds to the unique morphism

$$f_s : X \longrightarrow \mathbb{A}^1_{\mathbb{Z}}$$

whose pullback of the coordinate function t is s:

$$f_s^{\#}(t) = s.$$

Corollary I.F.6. *The scheme* $\operatorname{Spec}(\mathbb{Z})$ *is a final object of the category* (Sch).

Proof. For every scheme X, there is the unique unital ring homomorphism

$$\gamma_{\Gamma(\mathcal{O}_X)} : \mathbb{Z} \longrightarrow \Gamma(\mathcal{O}_X).$$

The result follows from Theorem I.F.1. □

I.F.2 Global sections as left adjoint of affine spectrum

Here is a categorical version of the result concerning morphisms towards an affine scheme.

Proposition I.F.7 (Categorical form of the affine-target adjunction). *Taking global sections and taking spectra define functors*

$$\Gamma : (\mathrm{Sch}) \longrightarrow (\mathrm{Ring}^{\mathrm{op}}), \qquad \mathrm{Spec} : (\mathrm{Ring}^{\mathrm{op}}) \longrightarrow (\mathrm{Sch}).$$

The functor Γ *is left adjoint to* Spec*, and* Spec *is fully faithful.*

Proof. The adjunction is Theorem I.F.1, written as

$$\mathrm{Hom}_{(\mathrm{Sch})}(X, \mathrm{Spec}(A)) \simeq \mathrm{Hom}_{(\mathrm{Ring}^{\mathrm{op}})}\big(\Gamma(\mathcal{O}_X), A\big).$$

Taking $X = \mathrm{Spec}(B)$ and using $\Gamma(\mathcal{O}_{\mathrm{Spec}(B)}) \simeq B$ gives

$$\mathrm{Hom}_{(\mathrm{Sch})}(\mathrm{Spec}(B), \mathrm{Spec}(A)) \simeq \mathrm{Hom}_{(\mathrm{Ring}^{\mathrm{op}})}(B, A),$$

which is the full faithfulness of Spec. $\square$

I.F.3 The affinisation morphism

For every scheme X, the identity of $\Gamma(\mathcal{O}_X)$ corresponds under the affine-target adjunction to a canonical morphism

$$\eta_X : X \longrightarrow \mathrm{Spec}\big(\Gamma(\mathcal{O}_X)\big).$$

Remark I.F.8. The morphism η_X is called the *affinisation morphism* of X; it is the canonical morphism to the affine scheme of global functions. By the construction in Theorem I.F.1, it sends a point $x \in X$ to

$$\eta_X(x) = \ker\big(\Gamma(\mathcal{O}_X) \longrightarrow \kappa(x)\big).$$

Thus x is sent to the ideal of global functions vanishing at x.

Example I.F.9. If $X = \mathrm{Spec}(B)$ is affine, then $\Gamma(\mathcal{O}_X) = B$, and η_X is the identity of $\mathrm{Spec}(B)$ under this identification.

Proposition I.F.10 (Affinisation). *The morphisms* η_X *are natural in* X*. Moreover, the following conditions are equivalent:*

i) X *is affine;*

ii) η_X *is an isomorphism;*

iii) X *belongs to the essential image of* Spec.

Proof. If $f : X \to Y$ is a morphism, both composites in

$$\begin{tikzcd} X \arrow[r, "\eta_X"] \arrow[d, "f"'] & \mathrm{Spec}(\Gamma(\mathcal{O}_X)) \arrow[d, "\mathrm{Spec}(\Gamma(f))"] \\ Y \arrow[r, "\eta_Y"'] & \mathrm{Spec}(\Gamma(\mathcal{O}_Y)) \end{tikzcd}$$

correspond to the same homomorphism $\Gamma(\mathcal{O}_Y) \to \Gamma(\mathcal{O}_X)$, proving naturality. If η_X is an isomorphism, then X is affine. Conversely, for $X = \mathrm{Spec}(B)$ both η_X and the identity correspond to id_B. The last condition is simply the definition of the essential image. $\square$

Exercise I.F.11. Let X be a scheme. Prove directly, using Theorem I.F.1, that every morphism from X to an affine scheme factors uniquely through η_X. More precisely, for every ring A and every morphism

$$f : X \longrightarrow \operatorname{Spec}(A),$$

show that there is a unique morphism

$$\widetilde{f} : \operatorname{Spec}\big(\Gamma(\mathcal{O}_X)\big) \longrightarrow \operatorname{Spec}(A)$$

such that

$$f = \widetilde{f} \circ \eta_X.$$

Solution. Let

$$B = \Gamma(\mathcal{O}_X).$$

By Theorem I.F.1, the morphism

$$f : X \longrightarrow \operatorname{Spec}(A)$$

corresponds to a ring homomorphism

$$\alpha : A \longrightarrow B.$$

Let

$$\widetilde{f} = \operatorname{Spec}(\alpha) : \operatorname{Spec}(B) \longrightarrow \operatorname{Spec}(A).$$

The morphism $\eta_X : X \to \operatorname{Spec}(B)$ corresponds to id_B. Consequently, the composite

$$\widetilde{f} \circ \eta_X : X \longrightarrow \operatorname{Spec}(A)$$

corresponds, on global sections, to

$$A \xrightarrow{\alpha} B \xrightarrow{\mathrm{id}_B} B,$$

which is α. Hence

$$f = \widetilde{f} \circ \eta_X.$$

For uniqueness, a morphism

$$g : \operatorname{Spec}(B) \longrightarrow \operatorname{Spec}(A)$$

is uniquely determined by the corresponding ring homomorphism $A \to B$. If $g \circ \eta_X = f$, then the induced homomorphism on global sections is necessarily α. Therefore $g = \operatorname{Spec}(\alpha) = \widetilde{f}$.

I.F.4 What affinisation forgets

The examples constructed by gluing now give immediate tests for non-affineness.

Example I.F.12 (The doubled origin and projective space). Let L be the line with two origins. We computed $\Gamma(\mathcal{O}_L) \simeq \Bbbk[x]$. The morphism

$$\eta_L : L \longrightarrow \mathbb{A}^1_{\Bbbk}$$

restricts to the usual coordinate morphism on both affine charts and hence identifies the two origins. It is not an isomorphism, so L is not affine.

For $n \geq 1$, we also computed

$$\Gamma(\mathcal{O}_{\mathbb{P}^n_{\Bbbk}}) = \Bbbk.$$

Thus $\eta_{\mathbb{P}^n_{\Bbbk}}$ is the unique morphism to $\operatorname{Spec}(\Bbbk)$. Since projective space has more than one point, this morphism is not an isomorphism, and $\mathbb{P}^n_{\Bbbk}$ is not affine.

II Schemes over a base

II.A The category of schemes over a base

Further reading. For schemes over a base, rational points and the functorial viewpoint, see [Liu02, Chapter 2] and [Har77, Chapter II, §3]; the examples in [EH00, Chapter I] are a useful complement.

Let S be a scheme. The category $(\mathrm{Sch})_S$ of *schemes over* S has as objects morphisms $X \to S$. A morphism

$$(X \to S) \longrightarrow (Y \to S)$$

is a morphism $X \to Y$ commuting with the structure morphisms to S. Since $\mathrm{Spec}(\mathbb{Z})$ is final in (Sch), the category $(\mathrm{Sch})_{\mathrm{Spec}(\mathbb{Z})}$ is naturally equivalent to (Sch).

II.B Schemes over a ring

II.B.1 Standard schemes and their structure morphisms

Let R be a ring. An R-*scheme* is a scheme over $\mathrm{Spec}(R)$. By Theorem I.F.1, giving an R-scheme structure on X is equivalent to giving a ring homomorphism

$$R \longrightarrow \Gamma(\mathcal{O}_X).$$

If $U = \mathrm{Spec}(B)$ is an affine open subset of X, restriction gives a ring homomorphism $R \to B$, so B is naturally an R-algebra.

Example II.B.1. The affine line over R is

$$\mathbb{A}^1_R = \mathrm{Spec}(R[t]),$$

with structure morphism induced by the inclusion $R \to R[t]$. More generally,

$$\mathbb{A}^n_R = \mathrm{Spec}(R[t_1, \ldots, t_n]).$$

The projective space constructed by gluing also has a canonical structure morphism. Indeed, for

$$R_i = R\left[\frac{x_0}{x_i}, \ldots, \widehat{\frac{x_i}{x_i}}, \ldots, \frac{x_n}{x_i}\right],$$

the inclusions $R \to R_i$ define morphisms $U_i = \mathrm{Spec}(R_i) \to \mathrm{Spec}(R)$. They agree on the overlaps used in the construction of $\mathbb{P}^n_R$, and hence glue to a morphism

$$\mathbb{P}^n_R \longrightarrow \mathrm{Spec}(R).$$

This makes $\mathbb{P}^n_R$ an R-scheme and justifies the notation introduced earlier.

Remark II.B.2 (The affine line over a base). For the structure morphism

$$\pi : \mathbb{A}^1_R = \mathrm{Spec}(R[t]) \longrightarrow \mathrm{Spec}(R),$$

a point $\mathfrak{q}$ maps to $\mathfrak{q} \cap R$. For $\mathfrak{p} \in \mathrm{Spec}(R)$, the points mapping to $\mathfrak{p}$ are governed by

$$\mathrm{Spec}\big(\kappa(\mathfrak{p})[t]\big).$$

Thus they consist of its generic point and the closed points defined by irreducible polynomials over $\kappa(\mathfrak{p})$. This is precisely the polynomial-ring calculation of Chapter 2, now interpreted relative to the base; we shall use it without repeating the affine classification.

II.B.2 The relative adjunction

Let $S = \mathrm{Spec}(R)$. The adjunction restricts to an adjunction

$$\mathrm{Spec}\colon (R\text{-Alg})^{\mathrm{op}} \longrightarrow (\mathrm{Sch})_S, \qquad \Gamma\colon (\mathrm{Sch})_S \longrightarrow (R\text{-Alg})^{\mathrm{op}}.$$

Thus, if X is an R-scheme and A is an R-algebra, then

$$\mathrm{Hom}_{(\mathrm{Sch})_S}(X, \mathrm{Spec}(A)) \simeq \mathrm{Hom}_{(R\text{-Alg})}(A, \Gamma(\mathcal{O}_X)).$$

Exercise II.B.3. Let A and B be R-algebras. Prove that

$$\mathrm{Hom}_{(\mathrm{Sch})_R}(\mathrm{Spec}(B), \mathrm{Spec}(A)) \simeq \mathrm{Hom}_{(R\text{-Alg})}(A, B).$$

Explain precisely where compatibility with the maps from R enters.

Solution. Let

$$\alpha : R \longrightarrow A, \qquad \beta : R \longrightarrow B$$

be the given R-algebra structures. By the affine correspondence, a morphism of schemes

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

is equivalent to a ring homomorphism

$$\varphi : A \longrightarrow B.$$

It is a morphism over $\mathrm{Spec}(R)$ precisely when the diagram

$$\begin{tikzcd} \mathrm{Spec}(B) \arrow[r] \arrow[dr] & \mathrm{Spec}(A) \arrow[d] \\ & \mathrm{Spec}(R) \end{tikzcd}$$

commutes. Passing to rings reverses the arrows, and this condition becomes

$$\varphi \circ \alpha = \beta.$$

This is exactly the condition that φ be a homomorphism of R-algebras. Hence

$$\mathrm{Hom}_{(\mathrm{Sch})_R}(\mathrm{Spec}(B), \mathrm{Spec}(A)) \simeq \mathrm{Hom}_{(R\text{-Alg})}(A, B).$$

II.B.3 Schemes over a field

When $R = \Bbbk$ is a field, a $\Bbbk$-scheme is simply a scheme equipped with a morphism to $\mathrm{Spec}(\Bbbk)$. The principal examples in this course are

$$\mathbb{A}^n_{\Bbbk} = \mathrm{Spec}(\Bbbk[t_1, \ldots, t_n]) \qquad \text{and} \qquad \mathbb{P}^n_{\Bbbk} \longrightarrow \mathrm{Spec}(\Bbbk).$$

Their affine charts and all their transition morphisms are morphisms of $\Bbbk$-schemes.

The expression $\Bbbk$-*scheme* by itself imposes no finiteness condition. In particular, it is not yet synonymous with *algebraic variety*. The finite-type condition, and the precise convention for affine and projective varieties, will be introduced later.

II.C The functor of points

Let $p : X \to S$ be an S-scheme.

Definition II.C.1. An *S-valued point* of X is a morphism of S-schemes

$$s : S \longrightarrow X.$$

Equivalently, it is a section of the structure morphism p, that is,

$$p \circ s = \mathrm{id}_S .$$

The set of S-valued points is denoted by

$$X_S(S) = \mathrm{Hom}_{(\mathrm{Sch})_S}(S, X).$$

More generally, if $T \to S$ is an S-scheme, a *T-valued point* of X is a morphism

$$T \longrightarrow X$$

over S. We write

$$X_S(T) = \mathrm{Hom}_{(\mathrm{Sch})_S}(T, X).$$

When the base scheme S is understood, we abbreviate $X_S(T)$ to $X(T)$.

Let X be an S-scheme. The valued-point sets introduced above are functorial in the test scheme. They form the *functor of points*

$$h_X : (\mathrm{Sch})_S^{\mathrm{op}} \longrightarrow (\mathrm{Set}), \qquad T \longmapsto X(T).$$

If $g : T' \to T$ is a morphism of S-schemes, then

$$h_X(g) : X(T) \longrightarrow X(T'), \qquad u \longmapsto u \circ g.$$

A morphism $f : X \to Y$ induces a natural transformation

$$h_f : h_X \longrightarrow h_Y, \qquad u \longmapsto f \circ u.$$

Yoneda's lemma says that every natural transformation $h_X \to h_Y$ arises in this way from a unique morphism $X \to Y$. Thus a scheme is completely encoded by all its sets of T-valued points, together with their functoriality in T.

If X is a $\Bbbk$-scheme, restricting to affine test schemes gives a covariant functor

$$(\Bbbk\text{-Alg}) \longrightarrow (\mathrm{Set}), \qquad R \longmapsto X(R).$$

It is important to allow arbitrary $\Bbbk$-algebras, not only fields: points with values in non-reduced rings detect infinitesimal information which is invisible on ordinary rational points.

Example II.C.2 (Affine valued points). For an affine $\Bbbk$-scheme $X = \mathrm{Spec}(A)$ and a $\Bbbk$-algebra R,

$$X(R) \simeq \mathrm{Hom}_{(\Bbbk\text{-Alg})}(A, R).$$

Thus equations defining A may be evaluated in R; in particular, $\mathbb{A}^n_{\Bbbk}(R) = R^n$. This is the affine dictionary developed in Chapter 2, and we now turn to examples which are not affine.

Definition II.C.3 (The multiplicative group). For a ring R, set

$$\mathbb{G}_{m,R} = \operatorname{Spec}(R[t,t^{-1}]).$$

For every R-algebra A,

$$\mathbb{G}_{m,R}(A) \simeq A^\times.$$

This is a useful affine example of the functorial viewpoint: multiplication and inversion of units come from morphisms of schemes.

Exercise II.C.4. Let

$$D = \operatorname{Spec}\big(\Bbbk[\varepsilon]/(\varepsilon^2)\big).$$

Show that $D(\Bbbk)$ consists of a single point. If

$$R = \Bbbk[\delta]/(\delta^2),$$

show that

$$D(R) = \{b\delta \mid b \in \Bbbk\}.$$

Conclude that the functor of points distinguishes D from $\operatorname{Spec}(\Bbbk)$, even though the two schemes have the same underlying topological space and the same set of $\Bbbk$-rational points.

Solution. An element of $D(R)$ is a $\Bbbk$-algebra homomorphism

$$\Bbbk[\varepsilon]/(\varepsilon^2) \longrightarrow R,$$

and is therefore determined by an element $r \in R$ satisfying $r^2 = 0$. For $R = \Bbbk$, the only such element is 0. For $R = \Bbbk[\delta]/(\delta^2)$, an element has the form $a + b\delta$, and its square is zero only if $a = 0$. Hence the square-zero elements are precisely $b\delta$, with $b \in \Bbbk$.

II.D Field-valued and rational points

Definition II.D.1. Let $\Bbbk$ be a field and let X be a $\Bbbk$-scheme. A $\Bbbk$-*rational point* of X is a $\Bbbk$-valued point, namely an element of

$$X(\Bbbk) = \operatorname{Hom}_{(\mathrm{Sch})_\Bbbk}(\operatorname{Spec}(\Bbbk), X).$$

More generally, for a $\Bbbk$-algebra R, we put

$$X(R) = \operatorname{Hom}_{(\mathrm{Sch})_\Bbbk}(\operatorname{Spec}(R), X).$$

A $\Bbbk$-rational point determines a point $x \in X$ together with a $\Bbbk$-algebra homomorphism

$$\kappa(x) \longrightarrow \Bbbk.$$

Such a homomorphism is necessarily an isomorphism. Thus the $\Bbbk$-rational points of X correspond to the points $x \in X$ whose residue field is $\Bbbk$, compatibly with the given $\Bbbk$-scheme structure. In particular, a point of the underlying topological space need not be $\Bbbk$-rational.

Proposition II.D.2 (Closed points of a variety). *Let X be a scheme of finite type over $\Bbbk$. For every closed point $x \in X$, the residue field $\kappa(x)$ is a finite extension of $\Bbbk$. The canonical morphism*

$$\operatorname{Spec}\big(\kappa(x)\big) \longrightarrow X$$

is therefore a $\kappa(x)$-valued point of X, or equivalently a $\kappa(x)$-rational point of the base change $X_{\kappa(x)}$.

Conversely, if $K/\Bbbk$ is finite, the image in X of every K-valued point is a closed point.

Proof. Choose an affine open neighbourhood $U = \mathrm{Spec}(A)$ of a closed point x, and let $\mathfrak{m} \subset A$ be the corresponding maximal ideal. The field

$$\kappa(x) = A/\mathfrak{m}$$

is a finitely generated $\Bbbk$-algebra, so Zariski's lemma implies that it is finite over $\Bbbk$. The quotient homomorphism $A \to \kappa(x)$ gives the displayed canonical morphism.

Conversely, let $u : \mathrm{Spec}(K) \to X$ be a K-valued point and let x be its image. On an affine neighbourhood $\mathrm{Spec}(A)$ of x, the morphism is induced by a homomorphism $A \to K$. If $K/\Bbbk$ is finite, its image is a finite-dimensional domain over $\Bbbk$, hence a field. The kernel is therefore maximal, so x is closed. □

Remark II.D.3. The finiteness hypothesis on the value field matters. The inclusion

$$\Bbbk[t] \longrightarrow \Bbbk(t)$$

defines a $\Bbbk(t)$-valued point of $\mathbb{A}^1_{\Bbbk}$ whose image is the generic point (0), which is not closed.

Example II.D.4 (Points of projective space over a field). The projective scheme $\mathbb{P}^n_{\Bbbk}$, constructed below from its standard affine charts, satisfies

$$\mathbb{P}^n_{\Bbbk}(K) = (K^{n+1} \setminus \{0\})/K^{\times}$$

for every field extension $K/\Bbbk$. We write its elements as $[a_0 : \cdots : a_n]$.

Projective schemes also have closed points which are not rational. In the standard affine chart $\mathbb{A}^1_{\mathbb{Q}} \subset \mathbb{P}^1_{\mathbb{Q}}$, the irreducible polynomial $t^2 - 2$ defines a closed point with residue field $\mathbb{Q}(\sqrt{2})$, not a $\mathbb{Q}$-rational point. Similarly, the non-real closed points of $\mathbb{P}^1_{\mathbb{R}}$ defined by irreducible quadratic polynomials have residue field $\mathbb{C}$.

Exercise II.D.5 (A projective conic). Let

$$C = \{[X : Y : Z] \in \mathbb{P}^2_{\mathbb{Q}} \mid X^2 + Y^2 = Z^2\}.$$

For every field extension $K/\mathbb{Q}$, interpret $C(K)$ as projective solutions of this equation. Show that

$$\mathbb{P}^1_{\mathbb{Q}} \longrightarrow C, \qquad [s : t] \longmapsto [s^2 - t^2 : 2st : s^2 + t^2]$$

parametrises the rational points of C.

Solution. A K-valued point of C is a nonzero triple $(a, b, c) \in K^3$ satisfying $a^2 + b^2 = c^2$, modulo simultaneous multiplication by an element of $K^{\times}$. The displayed formula gives such a triple.

Conversely, let $[a : b : c] \in C(K)$. If $c + a \neq 0$, take

$$[s : t] = [c + a : b].$$

Using $b^2 = c^2 - a^2$, the image of this point is

$$[2a(c + a) : 2b(c + a) : 2c(c + a)] = [a : b : c].$$

If $c + a = 0$, then $b = 0$ and the point is $[-1 : 0 : 1]$, which is the image of $[0 : 1]$. Hence every rational point occurs.

III Projective schemes

Further reading. For the Proj construction and projective schemes, see [Har77, Chapter II, §2], [Liu02, Chapter 2], and the concrete treatment in [EH00, Chapter II].

The projective spectrum construction gives an intrinsic description of the projective spaces introduced above and, more generally, of closed projective subschemes.

III.A Graded rings and the homogeneous spectrum

Let B be a ring and let

$$A = \bigoplus_{d \geq 0} A_d$$

be a positively graded B-algebra with $A_0 = B$ and $A_p A_q \subset A_{p+q}$.

An element of A_d is called *homogeneous of degree d*. An ideal $\mathfrak{a} \subset A$ is *homogeneous* if it is generated by homogeneous elements. Equivalently, whenever

$$a = a_0 + a_1 + \cdots + a_m \in \mathfrak{a}, \qquad a_d \in A_d,$$

then every homogeneous component a_d belongs to $\mathfrak{a}$.

The ideal

$$A_+ = \bigoplus_{d > 0} A_d$$

is called the *irrelevant ideal*. A homogeneous prime ideal is *relevant* if it does not contain A_+.

Definition III.A.1. The *projective spectrum* of A is

$$\mathrm{Proj}(A) = \{\mathfrak{p} \subset A \mid \mathfrak{p} \text{ is homogeneous and prime, and } A_+ \not\subset \mathfrak{p}\}.$$

Example III.A.2. Let $A = \Bbbk[x_0, x_1]$ with the standard grading. The ideals

$$(0), \qquad (x_0), \qquad (x_1)$$

are homogeneous and relevant, while (x_0, x_1) is irrelevant and $(x_0 - 1)$ is not homogeneous. If $\Bbbk$ is algebraically closed, the closed points of $\mathrm{Proj}(A)$ are the ideals

$$(bx_0 - ax_1), \qquad [a : b] \in \mathbf{P}^1_{\Bbbk}.$$

The ideal (0) is the generic point.

Exercise III.A.3. Prove that an ideal $\mathfrak{a} \subset A$ is homogeneous if and only if it contains every homogeneous component of each of its elements.

Solution. Assume first that $\mathfrak{a}$ is generated by homogeneous elements. Let $a \in \mathfrak{a}$. We may write

$$a = \sum_{j=1}^{r} b_j h_j,$$

where the h_j are homogeneous elements of $\mathfrak{a}$. Decompose every b_j into its homogeneous components and collect terms of the same degree. Each homogeneous component of a is then a sum of multiples of the h_j, and therefore belongs to $\mathfrak{a}$.

Conversely, suppose that $\mathfrak{a}$ contains every homogeneous component of each of its elements. Let H be the set of homogeneous elements of $\mathfrak{a}$. If $a \in \mathfrak{a}$ and

$$a = a_0 + \cdots + a_m$$

is its homogeneous decomposition, then every a_d belongs to H and

$$a = a_0 + \cdots + a_m.$$

Thus every element of $\mathfrak{a}$ belongs to the ideal generated by H. Hence $\mathfrak{a}$ is generated by homogeneous elements and is homogeneous.

Exercise III.A.4. Determine the projective spectra of the following graded rings.

(i) $\Bbbk[x]$ with $\deg(x) = 1$;

(ii) $\Bbbk[x_0, x_1]/(x_0x_1)$ with the standard grading;

(iii) $\Bbbk[x_0, x_1]/(x_0^2)$ with the standard grading.

Compare the underlying topological spaces in (ii) and (iii) with their scheme-theoretic structures.

Solution. (i) Let $A = \Bbbk[x]$ with $\deg(x) = 1$. Its homogeneous prime ideals are (0) and (x), but (x) contains the irrelevant ideal $A_+ = (x)$ and is therefore not relevant. Thus $\mathrm{Proj}(A)$ has one point. Moreover,

$$D_+(x) = \mathrm{Proj}(A), \qquad \big(A[x^{-1}]\big)_0 = \Bbbk.$$

Hence

$$\mathrm{Proj}(\Bbbk[x]) \simeq \mathrm{Spec}(\Bbbk).$$

(ii) Put

$$A = \Bbbk[x_0, x_1]/(x_0x_1).$$

The standard opens satisfy

$$\big(A[x_0^{-1}]\big)_0 \simeq \Bbbk, \qquad \big(A[x_1^{-1}]\big)_0 \simeq \Bbbk.$$

Indeed, after inverting x_0, the relation forces $x_1 = 0$, and similarly with the variables exchanged. Their intersection is empty because $x_0x_1 = 0$. Therefore

$$\mathrm{Proj}(A) \simeq \mathrm{Spec}(\Bbbk) \coprod \mathrm{Spec}(\Bbbk).$$

Its two points correspond to the homogeneous prime ideals (x_0) and (x_1), and both are reduced.

(iii) Put

$$A = \Bbbk[x_0, x_1]/(x_0^2).$$

Every prime ideal contains the nilpotent element x_0. A relevant prime must not contain x_1, so the only relevant homogeneous prime is (x_0). Thus the underlying topological space consists of one point. The open $D_+(x_1)$ is the whole projective spectrum, and

$$\big(A[x_1^{-1}]\big)_0 \simeq \Bbbk\left[\frac{x_0}{x_1}\right] \Big/ \left(\left(\frac{x_0}{x_1}\right)^2\right).$$

Writing $u = x_0/x_1$, we obtain

$$\mathrm{Proj}(A) \simeq \mathrm{Spec}(\Bbbk[u]/(u^2)).$$

Hence (ii) is the disjoint union of two reduced points, whereas (iii) is a single non-reduced double point. In particular, the nilpotent structure in (iii) is invisible in its one-point underlying topological space.

III.B The Zariski topology and the structure sheaf

For a homogeneous ideal $\mathfrak{a} \subset A$, define

$$\mathbb{V}_+(\mathfrak{a}) = \{\mathfrak{p} \in \mathrm{Proj}(A) \mid \mathfrak{a} \subset \mathfrak{p}\}.$$

These subsets are the closed subsets of the Zariski topology on $\mathrm{Proj}(A)$. If $f \in A$ is homogeneous of positive degree, set

$$D_+(f) = \mathrm{Proj}(A) \setminus \mathbb{V}_+(f)$$

and write

$$\big(A[f^{-1}]\big)_0$$

for the degree-zero part of the graded localisation $A[f^{-1}]$. If f has degree d, its elements are represented by fractions a/f^m with $a \in A_{md}$.

Lemma III.B.1 (Affine charts of Proj**).** *For every homogeneous element $f \in A$ of positive degree, there is a natural homeomorphism*

$$D_+(f) \simeq \mathrm{Spec}\big((A[f^{-1}])_0\big).$$

Under this homeomorphism, the principal open subsets on both sides correspond.

Proof. A homogeneous prime ideal $\mathfrak{p}$ not containing f extends to a homogeneous prime ideal $\mathfrak{p}A_f$ of A_f. Taking its degree-zero part gives a prime ideal

$$\big(\mathfrak{p}A[f^{-1}]\big)_0 \subset \big(A[f^{-1}]\big)_0.$$

The homogeneous localisation correspondence gives a bijection between homogeneous prime ideals of A not containing f and prime ideals of $\big(A[f^{-1}]\big)_0$. This bijection identifies the basic open subsets, hence is a homeomorphism. □

Lemma III.B.2 (The standard-open sheaf on Proj**).** *On the basis of standard open subsets of* $\mathrm{Proj}(A)$*, the prescription*

$$D_+(f) \longmapsto \big(A[f^{-1}]\big)_0$$

satisfies the sheaf condition and therefore extends uniquely to a sheaf of rings on $\mathrm{Proj}(A)$*. On every* $D_+(f)$*, the resulting ringed space is the affine scheme*

$$\mathrm{Spec}\big((A[f^{-1}])_0\big).$$

Proof. Put $R_f = (A[f^{-1}])_0$. If f and g have respective positive degrees d and e, then

$$h_{f,g} = \frac{g^d}{f^e} \in R_f.$$

Under the homeomorphism of Lemma III.B.1, the intersection $D_+(f) \cap D_+(g)$ corresponds to the principal open $D(h_{f,g})$ in $\mathrm{Spec}(R_f)$, and homogeneous localisation gives

$$(R_f)_{h_{f,g}} \simeq \big(A[(fg)^{-1}]\big)_0.$$

Thus, on every chart $D_+(f)$, the displayed prescription is exactly the usual affine structure sheaf on its principal-open basis. The lemma for defining a sheaf on a basis applies, and the identifications agree on triple overlaps by functoriality of localisation. Hence the affine sheaves glue uniquely. □

The sheaf furnished by the lemma is denoted $\mathcal{O}_X$, where $X = \mathrm{Proj}(A)$. It equips the projective spectrum with a scheme structure.

III.C Projective schemes and closed subschemes

Definition III.C.1. A *projective B-scheme* is a B-scheme isomorphic to $\mathrm{Proj}(A)$, where A is a positively graded B-algebra generated over B by finitely many homogeneous elements of positive degree. Equivalently, it is a closed B-subscheme of some $\mathbb{P}^n_B$.

The finite-type condition is part of the definition. After replacing A by a suitable Veronese subalgebra, it may be taken to be generated in degree one, hence to be a homogeneous quotient of $B[x_0, \dots, x_n]$. Notice that $B \to A_0$ is allowed to be surjective rather than an isomorphism; this includes projective schemes supported over a closed subscheme of $\mathrm{Spec}(B)$. This differs from the affine case: every B-algebra R, without a finiteness hypothesis, defines the affine B-scheme $\mathrm{Spec}(R)$. The construction $\mathrm{Proj}(A)$ still makes sense for more general positively graded algebras, but it need not be a projective B-scheme in the sense above.

If A is generated in degree one by $x_0, \dots, x_n$, then

$$A \simeq B[x_0, \dots, x_n]/\mathfrak{a}$$

for a homogeneous ideal $\mathfrak{a}$, and $\mathrm{Proj}(A)$ is a closed subscheme of $\mathbb{P}^n_B$.

Example III.C.2 (The standard charts of projective space). Let $A = B[x_0, \dots, x_n]$ with the standard grading. Then

$$\left(A[x_i^{-1}]\right)_0 = B\left[\frac{x_0}{x_i}, \dots, \widehat{\frac{x_i}{x_i}}, \dots, \frac{x_n}{x_i}\right].$$

Thus

$$D_+(x_i) \simeq \mathbb{A}^n_B,$$

and the gluing maps are exactly those used earlier to construct $\mathbb{P}^n_B$. Hence

$$\mathrm{Proj}(B[x_0, \dots, x_n]) \simeq \mathbb{P}^n_B.$$

Example III.C.3 (A projective conic). Let

$$A = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2)$$

with the standard grading. Then $C = \mathrm{Proj}(A)$ is a closed subscheme of $\mathbb{P}^2_{\mathbb{k}}$. On $D_+(x_0)$, putting

$$u = \frac{x_1}{x_0}, \qquad v = \frac{x_2}{x_0},$$

the relation becomes $v = u^2$, so this chart is isomorphic to $\mathbb{A}^1_{\mathbb{k}}$. There is a morphism

$$\pi : C \longrightarrow \mathbb{P}^1_{\mathbb{k}}$$

obtained geometrically by projection from the point $p = [1 : 0 : 0] \in C$. The naive formula $[x_0 : x_1 : x_2] \mapsto [x_1 : x_2]$ is undefined at p, but it extends across p: on the two standard charts of C put

$$\pi|_{D_+(x_0)} = [x_0 : x_1], \qquad \pi|_{D_+(x_2)} = [x_1 : x_2].$$

On the overlap these expressions agree because $x_0x_2 = x_1^2$. The gluing statement for morphisms therefore gives a well-defined morphism to the non-affine scheme $\mathbb{P}^1_{\mathbb{k}}$.

Its inverse is the quadratic Veronese morphism

$$\nu : \mathbb{P}^1_{\mathbb{k}} \longrightarrow C, \qquad [s : t] \longmapsto [s^2 : st : t^2].$$

The formulas on the standard charts show that $\pi \circ \nu = \mathrm{id}_{\mathbb{P}^1_{\mathbb{k}}}$ and $\nu \circ \pi = \mathrm{id}_C$. Hence $C \simeq \mathbb{P}^1_{\mathbb{k}}$.

Exercise III.C.4. Prove the identities

$$\mathbb{V}_+(\mathfrak{a}) \cup \mathbb{V}_+(\mathfrak{b}) = \mathbb{V}_+(\mathfrak{a}\mathfrak{b}), \qquad \bigcap_\lambda \mathbb{V}_+(\mathfrak{a}_\lambda) = \mathbb{V}_+\left(\sum_\lambda \mathfrak{a}_\lambda\right).$$

Deduce that the sets $D_+(f)$ form a basis of the Zariski topology.

Solution. Let $\mathfrak{p} \in \operatorname{Proj}(A)$. Since $\mathfrak{p}$ is prime,

$$\mathfrak{a}\mathfrak{b} \subset \mathfrak{p} \iff \mathfrak{a} \subset \mathfrak{p} \text{ or } \mathfrak{b} \subset \mathfrak{p}.$$

Therefore

$$\mathbb{V}_+(\mathfrak{a}\mathfrak{b}) = \mathbb{V}_+(\mathfrak{a}) \cup \mathbb{V}_+(\mathfrak{b}).$$

Similarly,

$$\sum_\lambda \mathfrak{a}_\lambda \subset \mathfrak{p} \iff \mathfrak{a}_\lambda \subset \mathfrak{p} \text{ for every } \lambda,$$

so

$$\mathbb{V}_+\left(\sum_\lambda \mathfrak{a}_\lambda\right) = \bigcap_\lambda \mathbb{V}_+(\mathfrak{a}_\lambda).$$

If $\mathfrak{a}$ is homogeneous, it is generated by homogeneous elements. Hence

$$\operatorname{Proj}(A) \setminus \mathbb{V}_+(\mathfrak{a}) = \{\mathfrak{p} \mid \mathfrak{a} \not\subset \mathfrak{p}\} = \bigcup_{\substack{f \in \mathfrak{a} \\ f \text{ homogeneous}}} D_+(f).$$

Thus every open subset is a union of subsets of the form $D_+(f)$. Moreover,

$$D_+(f) \cap D_+(g) = D_+(fg),$$

so these subsets form a basis of the Zariski topology.

Exercise III.C.5. Let $\mathfrak{a} \subset A$ be a homogeneous ideal. Show that

$$\operatorname{Proj}(A/\mathfrak{a}) \longrightarrow \operatorname{Proj}(A)$$

is a closed immersion with image $\mathbb{V}_+(\mathfrak{a})$.

Solution. The quotient map $A \to A/\mathfrak{a}$ sends a homogeneous prime ideal of $A/\mathfrak{a}$ to its inverse image in A. This gives a bijection between the relevant homogeneous prime ideals of $A/\mathfrak{a}$ and the relevant homogeneous prime ideals of A containing $\mathfrak{a}$. Hence the image on underlying topological spaces is

$$\mathbb{V}_+(\mathfrak{a}).$$

To check the scheme structure, let $f \in A$ be homogeneous of positive degree and write $\bar{f}$ for its image in $A/\mathfrak{a}$. On the standard affine charts, the morphism is

$$\operatorname{Spec}\bigl(((A/\mathfrak{a})[\bar{f}^{-1}])_0\bigr) \longrightarrow \operatorname{Spec}\bigl((A[f^{-1}])_0\bigr).$$

There is a natural isomorphism

$$\bigl((A/\mathfrak{a})[\bar{f}^{-1}]\bigr)_0 \simeq \bigl(A[f^{-1}]\bigr)_0 / \bigl(\mathfrak{a}A[f^{-1}]\bigr)_0.$$

Thus the map on every standard affine chart is induced by a surjective ring homomorphism, and is a closed immersion. These local closed immersions agree on overlaps and glue. Therefore

$$\operatorname{Proj}(A/\mathfrak{a}) \longrightarrow \operatorname{Proj}(A)$$

is a closed immersion with image $\mathbb{V}_+(\mathfrak{a})$.

Exercise III.C.6. Continue the computation for the conic

$$C = \operatorname{Proj}\big(\Bbbk[x_0, x_1, x_2]/(x_0x_2 - x_1^2)\big).$$

Compute the chart $D_+(x_2)$ and its transition map with $D_+(x_0)$. Verify directly that the two formulas defining the projection $\pi : C \to \mathbb{P}^1_{\Bbbk}$ agree on the overlap, and that

$$\nu : \mathbb{P}^1_{\Bbbk} \longrightarrow C, \qquad [s : t] \longmapsto [s^2 : st : t^2]$$

is an isomorphism. Deduce a description of $C(K)$ for every field extension $K/\Bbbk$, and describe the residue fields of its closed points.

Solution. Let

$$A = \Bbbk[x_0, x_1, x_2]/(x_0x_2 - x_1^2).$$

On $D_+(x_2)$, put

$$v = \frac{x_1}{x_2}, \qquad w = \frac{x_0}{x_2}.$$

The relation becomes $w = v^2$, so

$$\big(A[x_2^{-1}]\big)_0 \simeq \Bbbk[v].$$

On $D_+(x_0)$ use $u = x_1/x_0$, so that $x_2/x_0 = u^2$. On the overlap,

$$v = \frac{x_1/x_0}{x_2/x_0} = u^{-1}.$$

The two charts are therefore affine lines glued by inversion, exactly as in the construction of $\mathbb{P}^1_{\Bbbk}$. On their overlap,

$$[x_0 : x_1] = [x_1 : x_2] \quad \Longleftrightarrow \quad x_0x_2 = x_1^2,$$

so the two formulas for π agree. The inverse morphism is induced by

$$[s : t] \longmapsto [s^2 : st : t^2].$$

For every field extension $K/\Bbbk$,

$$C(K) = \{[a : b : c] \in \mathbb{P}^2_{\Bbbk}(K) \mid ac = b^2\},$$

and the isomorphism identifies this set with $\mathbb{P}^1_{\Bbbk}(K)$. It also preserves local rings and residue fields. Hence the closed points of C correspond to ∞ and to monic irreducible polynomials $f \in \Bbbk[t]$; their residue fields are respectively $\Bbbk$ and $\Bbbk[t]/(f)$. In particular, the $\Bbbk$-rational points are exactly those corresponding to ∞ and to linear polynomials.

III.D Graded maps and rational maps

Definition III.D.1 (Rational map). Let X and Y be schemes. A *rational map* from X to Y is an equivalence class of pairs (U, f), where $U \subset X$ is a dense open subscheme and $f : U \to Y$ is a morphism. Two pairs (U, f) and (V, g) are equivalent if f and g agree on a dense open subscheme of $U \cap V$. We write

$$X \dashrightarrow Y.$$

Thus a formula defined only on an open subset represents a rational map only when that open subset is dense.

Definition III.D.2 (Birational map). Let X and Y be integral schemes. A rational map $\varphi : X \dashrightarrow Y$ is *birational* if it admits a rational inverse. Equivalently, there are non-empty open subschemes $U \subset X$ and $V \subset Y$ such that φ is represented by an isomorphism

$$U \xrightarrow{\sim} V.$$

The schemes X and Y are then said to be birational.

Not every rational map between integral schemes is birational. A constant map $\mathbb{A}^1_{\Bbbk} \dashrightarrow \mathbb{A}^1_{\Bbbk}$ is already a counterexample; even the dominant morphism $t \mapsto t^2$ is not birational. By contrast, the standard Cremona transformation below is birational because it is its own inverse on the dense torus.

For the graph construction, suppose in addition that X is integral, as in the examples below. Let X and Y be schemes over a common base S, and let the S-morphism $f : U \to Y$ represent a rational map over S. Its *graph morphism* is

$$\Gamma_f = (\mathrm{id}_U, f) : U \longrightarrow X \times_S Y.$$

It factors through $U \times_S Y$. When Y is projective over S, the resulting morphism $U \to U \times_S Y$ is a closed immersion (a special case of the closed-graph property for separated morphisms). The scheme-theoretic closure of its image in $X \times_S Y$,

$$\overline{\Gamma}_f \subset X \times_S Y$$

depends only on the rational map. The first projection is an isomorphism over U, while the second projection extends f. In this sense the graph closure eliminates indeterminacy. It may be singular, so it should not automatically be called a resolution of indeterminacies.

The contravariance of Spec has a projective analogue, with an important difference: a homomorphism of graded rings need not define a morphism on the whole projective spectrum.

Let

$$A = \bigoplus_{m \geq 0} A_m, \qquad B = \bigoplus_{m \geq 0} B_m$$

be positively graded algebras, and let $\varphi : A \to B$ be a homogeneous homomorphism of positive degree d, meaning that

$$\varphi(A_m) \subset B_{dm} \qquad \text{for every } m \geq 0.$$

Its *base ideal* is the homogeneous ideal

$$\mathfrak{b}_\varphi = \varphi(A_+)B \subset B,$$

and its *base locus* is $\mathbb{V}_+(\mathfrak{b}_\varphi) \subset \mathrm{Proj}(B)$.

Proposition III.D.3 (The map defined by a graded homomorphism). *The homomorphism φ defines a morphism*

$$\mathrm{Proj}(B) \setminus \mathbb{V}_+(\mathfrak{b}_\varphi) \longrightarrow \mathrm{Proj}(A).$$

If its domain is dense in $\mathrm{Proj}(B)$*, this morphism determines a rational map*

$$\mathrm{Proj}(B) \dashrightarrow \mathrm{Proj}(A).$$

Proof. The complement of the base locus is covered by the open subsets $D_+(\varphi(f))$, where $f \in A_+$ is homogeneous. Homogeneous localisation gives a homomorphism

$$\big(A[f^{-1}]\big)_0 \longrightarrow \big(B[\varphi(f)^{-1}]\big)_0,$$

and hence a morphism

$$D_+(\varphi(f)) \longrightarrow D_+(f).$$

These morphisms agree on pairwise overlaps and therefore glue. If their common domain is dense, Definition III.D.1 applies. $\square$

Thus the word "rational" requires a density hypothesis. Without it, the same construction still gives a perfectly good morphism on the displayed open subset, which could even be empty.

Example III.D.4 (Linear projection). Let

$$A = \Bbbk[u_1, \ldots, u_n], \qquad B = \Bbbk[x_0, x_1, \ldots, x_n], \qquad u_i \longmapsto x_i.$$

The resulting rational map is

$$\mathbb{P}^n_{\Bbbk} \dashrightarrow \mathbb{P}^{n-1}_{\Bbbk}, \qquad [x_0 : x_1 : \cdots : x_n] \longmapsto [x_1 : \cdots : x_n].$$

Its base locus is the point $[1 : 0 : \cdots : 0]$. For $n = 2$, restricting this projection to the conic $x_0x_2 = x_1^2$ gives the map studied in Example III.C.3; on the conic its apparent indeterminacy is removable.

Example III.D.5 (The standard quadratic transformation). The degree-two homomorphism

$$\Bbbk[u_0, u_1, u_2] \longrightarrow \Bbbk[x_0, x_1, x_2], \qquad (u_0, u_1, u_2) \longmapsto (x_1x_2, x_0x_2, x_0x_1)$$

defines the standard Cremona transformation

$$\mathrm{Cr} : \mathbb{P}^2_{\Bbbk} \dashrightarrow \mathbb{P}^2_{\Bbbk}, \qquad [x_0 : x_1 : x_2] \longmapsto [x_1x_2 : x_0x_2 : x_0x_1].$$

Its base locus consists of the three coordinate points. On the dense torus $x_0x_1x_2 \neq 0$ it is the inversion map $[x_0 : x_1 : x_2] \mapsto [x_0^{-1} : x_1^{-1} : x_2^{-1}]$, and applying it twice gives the identity.

Suppose now that A and B are graded algebras over a common ring R, that $\mathrm{Proj}(B)$ is integral, and that the domain U of the rational map is dense. The preceding construction gives the graph model

$$\overline{\Gamma}_{\varphi} \subset \mathrm{Proj}(B) \times_{\mathrm{Spec}(R)} \mathrm{Proj}(A).$$

For the standard Cremona transformation, one can show that this graph model is the blow-up of the three base points and is nonsingular; in that example it really is a resolution of indeterminacies.

Exercise III.D.6 (Projection and its graph). For the projection $\mathbb{P}^2_{\Bbbk} \dashrightarrow \mathbb{P}^1_{\Bbbk}$ from $p = [1 : 0 : 0]$:

(i) compute the base ideal and base locus;

(ii) show that the graph closure in $\mathbb{P}^2_{\Bbbk} \times \mathbb{P}^1_{\Bbbk}$ is defined by

$$x_1v - x_2u = 0,$$

where $[u : v]$ are the coordinates on $\mathbb{P}^1$;

(iii) show that the first projection has one-point fibres away from p and fibre $\mathbb{P}^1_{\Bbbk}$ over p.

Solution. The graded homomorphism sends u to x_1 and v to x_2, so its base ideal is (x_1, x_2) and its base locus is p. On the domain of the projection, the equality $[u : v] = [x_1 : x_2]$ is equivalent to $x_1v - x_2u = 0$. The same bihomogeneous equation defines its closure. If $[x_0 : x_1 : x_2] \neq p$, it determines the unique point $[u : v] = [x_1 : x_2]$. Over p the equation becomes $0 = 0$, so every $[u : v] \in \mathbb{P}^1_{\Bbbk}$ occurs.

Exercise III.D.7 (The standard Cremona transformation). For the map in Example III.D.5:

(i) determine the base scheme and its underlying points;

(ii) verify directly that Cr^2 is the identity wherever both sides are defined;

(iii) determine the image of each coordinate line away from the base points.

Solution. The base ideal is

$$(x_1x_2, x_0x_2, x_0x_1),$$

whose zero set consists of the three coordinate points. Applying the formula twice gives

$$[x_0x_1x_2x_0 : x_0x_1x_2x_1 : x_0x_1x_2x_2] = [x_0 : x_1 : x_2]$$

on the dense torus, hence wherever the two rational maps are defined. The line $x_i = 0$, with its two base points removed, is sent to the i-th coordinate point: for example, $x_0 = 0$ gives $[x_1x_2 : 0 : 0] = [1 : 0 : 0]$.

IV Invertible sheaves and projective space

We now develop the sheaf-theoretic language needed for the intrinsic description of morphisms to projective space.

IV.A Invertible sheaves

Definition IV.A.1. An $\mathcal{O}_X$-module $\mathcal{L}$ is *invertible* if every point of X has an open neighbourhood U such that

$$\mathcal{L}|_U \simeq \mathcal{O}_U.$$

Equivalently, $\mathcal{L}$ is locally free of rank one. An invertible sheaf is also called a *line bundle.*

Choose a covering $X = \bigcup_i U_i$ and generators e_i of $\mathcal{L}|_{U_i}$. On an overlap there is a unique unit $g_{ij} \in \mathcal{O}_X(U_i \cap U_j)^\times$ such that

$$e_j = g_{ij}e_i.$$

The transition functions satisfy

$$g_{ii} = 1, \qquad g_{ji} = g_{ij}^{-1}, \qquad g_{ik} = g_{jk}g_{ij}.$$

Conversely, units satisfying these identities glue the modules $\mathcal{O}_{U_i}$ to an invertible sheaf.

The tensor product and dual are again invertible, with

$$\mathcal{L}^{-1} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X), \qquad \mathcal{L} \otimes \mathcal{L}^{-1} \simeq \mathcal{O}_X.$$

If $f : X \to Y$ is a morphism and $\mathcal{F}$ is an $\mathcal{O}_Y$-module, its *pullback* is

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

Pullback commutes with restriction and tensor products; in particular, the pullback of an invertible sheaf is invertible. A section of $\mathcal{F}$ pulls back to a section of $f^*\mathcal{F}$.

Definition IV.A.2. Sections $s_0, \ldots, s_n \in \Gamma(\mathcal{L})$ *generate* an invertible sheaf $\mathcal{L}$ if the evaluation morphism

$$\mathcal{O}_X^{\oplus(n+1)} \longrightarrow \mathcal{L}, \qquad (a_0, \ldots, a_n) \longmapsto \sum_i a_is_i$$

is surjective. Equivalently, at every point $x \in X$, at least one germ $(s_i)_x$ generates the free $\mathcal{O}_{X,x}$-module $\mathcal{L}_x$.

IV.B Twisting sheaves on projective space

Let $U_i = D_+(x_i)$ be the standard affine covering of $\mathbb{P}^n_B$. For $d \in \mathbb{Z}$, take a free generator $e_i^{(d)}$ on U_i and glue by

$$e_j^{(d)} = \left(\frac{x_j}{x_i}\right)^d e_i^{(d)} \qquad \text{on } U_i \cap U_j.$$

The resulting invertible sheaf is denoted by $\mathcal{O}_{\mathbb{P}^n_B}(d)$. The transition functions immediately give

$$\mathcal{O}(d) \otimes \mathcal{O}(e) \simeq \mathcal{O}(d+e), \qquad \mathcal{O}(d)^{-1} \simeq \mathcal{O}(-d).$$

If $F \in B[x_0, \dots, x_n]_d$ is homogeneous of degree $d \geq 0$, the local expressions

$$\frac{F}{x_i^d} e_i^{(d)}$$

agree on overlaps and define a global section of $\mathcal{O}(d)$. In fact, for $n \geq 1$,

$$\Gamma(\mathcal{O}_{\mathbb{P}^n_B}(d)) \simeq \begin{cases} B[x_0, \dots, x_n]_d, & d \geq 0, \\ 0, & d < 0. \end{cases}$$

For $\mathbb{P}^1_{\mathbb{C}}$ this recovers the computation of $\Gamma(\mathcal{O}_{\mathbb{P}^1_{\mathbb{C}}}(d))$ made in Chapter 1. In particular, $x_0, \dots, x_n$ are generating sections of $\mathcal{O}(1)$.

Exercise IV.B.1. Using the standard affine covering of $\mathbb{P}^1_B$, prove the displayed formula for $\Gamma(\mathcal{O}_{\mathbb{P}^1_B}(d))$. Show explicitly that the global sections of $\mathcal{O}(d)$ for $d \geq 0$ have basis

$$x_0^d, x_0^{d-1}x_1, \dots, x_1^d.$$

Solution. Write $t = x_1/x_0$ on U_0 and $s = x_0/x_1 = t^{-1}$ on U_1. A global section is a pair $f(t)e_0^{(d)}$, $g(s)e_1^{(d)}$ satisfying

$$f(t) = t^d g(t^{-1}).$$

If $d \geq 0$, the common Laurent polynomial has exponents between 0 and d, giving the indicated $d+1$ homogeneous monomials. If $d < 0$, the two allowed ranges of exponents are disjoint and the section is zero.

IV.C Morphisms to projective space

The affine-target adjunction describes a morphism to affine space by global functions. For projective space the homogeneous coordinates are local and may be multiplied simultaneously by a unit. Invertible sheaves record precisely these changes of scale.

Proposition IV.C.1 (Morphisms to projective space). *Let X be a B-scheme. Morphisms of B-schemes*

$$X \longrightarrow \mathbb{P}^n_B$$

are in natural bijection with equivalence classes of data

$$(\mathcal{L}; s_0, \dots, s_n),$$

where $\mathcal{L}$ is an invertible sheaf and $s_0, \dots, s_n$ generate it. Two such data are equivalent if an isomorphism of invertible sheaves carries each s_i to the corresponding section.

Proof. Given $(\mathcal{L}; s_0, \dots, s_n)$, let U_i be the open subset where s_i generates $\mathcal{L}$. On U_i the ratios s_j/s_i are regular functions and define a morphism

$$U_i \longrightarrow D_+(x_i) \subset \mathbb{P}^n_B.$$

On $U_i \cap U_j$, the two homogeneous tuples differ by multiplication by the unit s_j/s_i, so the local morphisms agree and glue.

Conversely, if $f : X \to \mathbb{P}^n_B$, take

$$\mathcal{L} = f^*\mathcal{O}_{\mathbb{P}^n_B}(1), \qquad s_i = f^*x_i.$$

Since the x_i generate $\mathcal{O}(1)$, the sections s_i generate $\mathcal{L}$. The two constructions are inverse up to the stated equivalence. □

Thus a tuple of global functions with no common zero describes only the special case $\mathcal{L} \simeq \mathcal{O}_X$. The line-bundle formulation is what makes the statement valid for an arbitrary, possibly non-affine source.

Example IV.C.2 (The Veronese morphism). The monomials

$$x_0^d, x_0^{d-1}x_1, \dots, x_1^d$$

generate $\mathcal{O}_{\mathbb{P}^1_{\Bbbk}}(d)$ and therefore define the degree-d Veronese morphism $\mathbb{P}^1_{\Bbbk} \to \mathbb{P}^d_{\Bbbk}$. For $d = 2$ its image is the conic $z_0z_2 = z_1^2$.

Chapter 4

First properties of schemes

I Connected and reduced schemes

Further reading. For connectedness, idempotents, reduction and the first topological properties of schemes, see [Liu02, Chapter 2], [Har77, Chapter II, §3], and the examples in [EH00, Chapter I].

Connectedness of a scheme is the connectedness of its underlying topological space. Reducedness, on the other hand, is an algebraic property: it requires the absence of nilpotent elements in all local rings.

I.A Connected schemes

Definition I.A.1. A scheme X is *connected* if its underlying topological space is connected.

Global idempotents detect connectedness even when the scheme is not affine.

Proposition I.A.2. *A scheme X is connected if and only if the ring $\Gamma(\mathcal{O}_X)$ has no non-trivial idempotents.*

Proof. If $X = U \amalg V$ is a decomposition into two non-empty open-and-closed subsets, the sheaf condition gives

$$\Gamma(\mathcal{O}_X) \simeq \Gamma(\mathcal{O}_U) \times \Gamma(\mathcal{O}_V).$$

The section corresponding to $(1, 0)$ is a non-trivial idempotent.

Conversely, let $e \in \Gamma(\mathcal{O}_X)$ satisfy $e^2 = e$. Every stalk $\mathcal{O}_{X,x}$ is local, so e_x is either 0 or 1. The loci

$$U_0 = \{x \in X \mid e_x = 0\}, \qquad U_1 = \{x \in X \mid e_x = 1\}$$

are open, because equality of germs holds on a neighbourhood. They are disjoint, open-and-closed, and cover X. If e is non-trivial, both are non-empty, so X is disconnected. □

Remark I.A.3 (The affine special case). For $X = \operatorname{Spec}(A)$ the proposition recovers the familiar equivalences

$$X \text{ disconnected} \iff A \text{ has a non-trivial idempotent} \iff A \simeq B \times C$$

with $B, C \neq 0$. The corresponding formula

$$\operatorname{Spec}(B \times C) \simeq \operatorname{Spec}(B) \amalg \operatorname{Spec}(C)$$

was established in Chapter 2 and will not be reproved here.

I.B Reduced schemes

Reducedness is studied through nilpotent elements. The topological space underlying a scheme can always be equipped with a canonical reduced scheme structure obtained by killing nilpotents.

Definition I.B.1. A scheme X is *reduced* if, for every $x \in X$, the local ring $\mathcal{O}_{X,x}$ is reduced.

I.B.1 Reduced structure on a scheme

Recall that the *nilradical* of a ring A, denoted by $\operatorname{Rad}(A)$, is the ideal of nilpotent elements of A. It is the intersection of all prime ideals of A.

Lemma I.B.2. *Let X be a scheme.*

i) The scheme X is reduced if and only if $\mathcal{O}_X(U)$ is reduced for every open subset $U \subset X$.

ii) There is a unique reduced closed subscheme $X_{\mathrm{red}} \subset X$ having the same underlying topological space as X.

Proof. For an open subset $U \subset X$, let

$$\mathcal{R}(U) = \{f \in \mathcal{O}_X(U) \mid f_x \in \operatorname{Rad}(\mathcal{O}_{X,x}) \text{ for every } x \in U\}.$$

This is a sheaf of ideals of $\mathcal{O}_X$. If $U = \operatorname{Spec}(A)$ is affine, then

$$\mathcal{R}(U) = \operatorname{Rad}(A).$$

Indeed, a nilpotent element remains nilpotent in every localisation. Conversely, assume that $f_{\mathfrak{p}}$ is nilpotent in $A_{\mathfrak{p}}$ for every $\mathfrak{p} \in \operatorname{Spec}(A)$. For each $\mathfrak{p}$, there are $n_{\mathfrak{p}} \geq 1$ and $s_{\mathfrak{p}} \notin \mathfrak{p}$ such that $s_{\mathfrak{p}} f^{n_{\mathfrak{p}}} = 0$. The elements $s_{\mathfrak{p}}$ generate the unit ideal. Choosing finitely many of them and taking the maximum of the corresponding exponents shows that some power of f is zero.

Moreover,

$$\operatorname{Rad}(A)_g = \operatorname{Rad}(A_g)$$

for every $g \in A$. Hence $\mathcal{R}$ is quasi-coherent. The first assertion now follows immediately.

The quotient sheaf $\mathcal{O}_X/\mathcal{R}$ defines a closed subscheme $X_{\mathrm{red}} \subset X$. On an affine open subset $U = \operatorname{Spec}(A)$, it is given by

$$X_{\mathrm{red}} \cap U = \operatorname{Spec}(A/\operatorname{Rad}(A)).$$

Every prime ideal contains $\operatorname{Rad}(A)$, so X_{red} and X have the same points and the same topology. The quotient $A/\operatorname{Rad}(A)$ is reduced.

For uniqueness, let $Y \subset X$ be a reduced closed subscheme with the same underlying topological space. On $U = \operatorname{Spec}(A)$, write $Y \cap U = \operatorname{Spec}(A/\mathfrak{a})$. Equality of the underlying closed subsets gives $\sqrt{\mathfrak{a}} = \operatorname{Rad}(A)$, while reducedness of $A/\mathfrak{a}$ gives $\mathfrak{a} = \sqrt{\mathfrak{a}}$. Thus $\mathfrak{a} = \operatorname{Rad}(A)$ on every affine open subset. □

Corollary I.B.3 (Universal property of the reduction). *Let Y be a reduced scheme and let $f : Y \to X$ be a morphism. Then there is a unique morphism $f_{\mathrm{red}} : Y \to X_{\mathrm{red}}$ such that*

$$f = i \circ f_{\mathrm{red}},$$

where $i : X_{\mathrm{red}} \hookrightarrow X$ is the canonical closed immersion.

Proof. The assertion is local on X. Write $X = \operatorname{Spec}(A)$ and $X_{\mathrm{red}} = \operatorname{Spec}(A/\operatorname{Rad}(A))$. The morphism f is locally induced by a ring homomorphism $A \to B$, where B is reduced. Every nilpotent element of A maps to a nilpotent element of B, hence to zero. Thus the homomorphism factors uniquely through $A/\operatorname{Rad}(A)$. These local factorisations agree by uniqueness. □

I.B.2 The Zariski tangent space

The maps from the space of dual numbers to a given $\Bbbk$-scheme are thought of as tangent directions at a given point of X.

Example I.B.4 (The dual numbers and their reduction). Let

$$D_{\Bbbk} = \operatorname{Spec}\big(\Bbbk[\epsilon]/(\epsilon^2)\big).$$

The ideal (ϵ) is the unique prime ideal, so $D_{\Bbbk}$ has one point, but it is not reduced. Its reduction is

$$(D_{\Bbbk})_{\mathrm{red}} = \operatorname{Spec}(\Bbbk).$$

Thus reduction preserves the point and forgets the infinitesimal element ϵ.

Definition I.B.5 (Zariski tangent space). Let X be a $\Bbbk$-scheme and let $x \in X(\Bbbk)$. Write $i : \operatorname{Spec}(\Bbbk) \hookrightarrow D_{\Bbbk}$ for the closed point. The *Zariski tangent space* of X at x is

$$T_xX = \{v : D_{\Bbbk} \longrightarrow X \mid v \text{ is a } \Bbbk\text{-morphism and } v \circ i = x\}.$$

We relate the Zariski tangent space to derivations of A with values in $\Bbbk$. Namely, a $\Bbbk$-rational point x of an affine $\Bbbk$-scheme $X = \operatorname{Spec}(A)$ gives a $\Bbbk$-algebra homomorphism $x^{\#} : A \to \Bbbk$. Via $x^{\#}$, the field $\Bbbk$ is an A-module. Using this structure, define derivations of A with values in $\Bbbk$ by

$$\operatorname{Der}_{\Bbbk}(A, \Bbbk) = \{\theta \in \operatorname{Hom}_{\Bbbk}(A, \Bbbk) \mid \theta(ab) = x^{\#}(a)\theta(b) + x^{\#}(b)\theta(a),\ \forall a, b \in A\}.$$

Proposition I.B.6. *Let $X = \operatorname{Spec}(A)$ be an affine $\Bbbk$-scheme and let $x^{\#} : A \to \Bbbk$ be a $\Bbbk$-rational point, with $\mathfrak{m} = \ker(x^{\#})$. There are natural bijections*

$$T_xX \simeq \operatorname{Der}_{\Bbbk}(A, \Bbbk) \simeq \operatorname{Hom}_{\Bbbk}(\mathfrak{m}/\mathfrak{m}^2, \Bbbk).$$

In particular, these identifications give T_xX a natural $\Bbbk$-vector-space structure.

Proof. A tangent vector is a $\Bbbk$-algebra homomorphism

$$\varphi : A \longrightarrow \Bbbk[\epsilon]/(\epsilon^2)$$

whose composite with $\epsilon \mapsto 0$ is $x^{\#}$. It is uniquely of the form

$$\varphi(a) = x^{\#}(a) + \theta(a)\epsilon,$$

for some $\Bbbk$-linear map $\theta : A \to \Bbbk$. Since $x^{\#}$ is a ring homomorphism, the condition that φ preserve products is exactly

$$\theta(ab) = x^{\#}(a)\theta(b) + x^{\#}(b)\theta(a),$$

so θ is a $\Bbbk$-derivation with values in $\Bbbk$, viewed as an A-module through $x^{\#}$.

Such a derivation vanishes on $\mathfrak{m}^2$, because $\mathfrak{m}^2$ is generated by elements ab with $a, b \in \mathfrak{m} = \ker(x^{\#})$, and then $\theta(ab) = 0$. Hence θ restricts to a linear form on $\mathfrak{m}/\mathfrak{m}^2$. Conversely, a linear form $\ell : \mathfrak{m}/\mathfrak{m}^2 \to \Bbbk$ defines a derivation by

$$\theta(a) = \ell\big(a - x^{\#}(a) \bmod \mathfrak{m}^2\big).$$

The constructions are inverse. $\square$

Example I.B.7 (Equations and tangent vectors). Let

$$X = \mathbb{V}(f_1, \ldots, f_r) \subset \mathbb{A}^n_{\Bbbk}$$

and let $a = (a_1, \ldots, a_n) \in X(\Bbbk)$. A tangent vector is represented by

$$x_i \longmapsto a_i + v_i\epsilon,$$

with $v = (v_1, \ldots, v_n) \in \Bbbk^n$. For a monomial $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, the relation $\epsilon^2 = 0$ gives

$$(a + v\epsilon)^\alpha = a^\alpha + \sum_{\alpha_i > 0} \alpha_i a_1^{\alpha_1} \cdots a_i^{\alpha_i - 1} \cdots a_n^{\alpha_n} v_i \epsilon.$$

By linearity, for every polynomial f one obtains the first-order Taylor formula

$$f(a + v\epsilon) = f(a) + \left(\sum_i \frac{\partial f}{\partial x_i}(a) v_i \right) \epsilon.$$

The tangent vector factors through X precisely when every f_j maps to zero. Since $a \in X$, its constant term already vanishes, and the remaining conditions are

$$\sum_i \frac{\partial f_j}{\partial x_i}(a)\, v_i = 0, \qquad 1 \le j \le r.$$

Thus T_aX is the kernel of the Jacobian matrix at a. For the parabola $y = x^2$ it is the line $v_y = 2av_x$ at (a, a^2). For the node $xy = 0$, the tangent space at the origin is all of $\Bbbk^2$, reflecting the two crossing branches.

Exercise I.B.8 (Tangent spaces at a smooth and a singular point). Assume $\mathrm{char}(\Bbbk) \neq 2, 3$ and let

$$C = \mathbb{V}(y^2 - x^3) \subset \mathbb{A}^2_{\Bbbk}.$$

(i) Compute $T_{(0,0)}C$.

(ii) Compute $T_{(1,1)}C$ and give an equation for the resulting line.

(iii) For a plane curve $\mathbb{V}(f)$, call a point *smooth* when (f_x, f_y) does not vanish there and *singular* otherwise. Explain the difference between the dimensions of the two tangent spaces above.

Solution. For $f = y^2 - x^3$, the Jacobian row is

$$(-3x^2, 2y).$$

At the origin it vanishes, so $T_{(0,0)}C = \Bbbk^2$ and the origin is singular. At $(1, 1)$ the tangent space is

$$T_{(1,1)}C = \{(u, v) \in \Bbbk^2 \mid -3u + 2v = 0\},$$

a one-dimensional line; the point is smooth. Thus at a smooth point the Zariski tangent space has the expected dimension of the curve, whereas at the cusp it jumps to dimension two.

Exercise I.B.9. Let $A = \Bbbk[x_1, x_2]$ and consider the ideals

$$\mathfrak{a} = (x_1, x_2), \qquad \mathfrak{b} = (x_1, x_2)^2, \qquad \mathfrak{c} = (x_1^2, x_1x_2).$$

i) Compute $\sqrt{\mathfrak{a}}$, $\sqrt{\mathfrak{b}}$ and $\sqrt{\mathfrak{c}}$.

ii) Determine the underlying closed subsets of $\operatorname{Spec}(A/\mathfrak{a})$, $\operatorname{Spec}(A/\mathfrak{b})$ and $\operatorname{Spec}(A/\mathfrak{c})$.

iii) Compute the reductions of these three affine schemes.

iv) Exhibit a non-zero nilpotent element in $A/\mathfrak{b}$ and in $A/\mathfrak{c}$.

Solution. We have
$$\sqrt{\mathfrak{a}} = \mathfrak{a}, \qquad \sqrt{\mathfrak{b}} = \mathfrak{a}, \qquad \sqrt{\mathfrak{c}} = (x_1).$$
Indeed, $\mathfrak{b}$ contains x_1^2 and x_2^2, while every element of $\mathfrak{c}$ is divisible by x_1 and $x_1^2 \in \mathfrak{c}$.

Consequently, the first two schemes are supported at the origin, whereas the third is supported on the line $\mathbb{V}(x_1)$. Their reductions are respectively
$$\operatorname{Spec}(A/\mathfrak{a}), \qquad \operatorname{Spec}(A/\mathfrak{a}), \qquad \operatorname{Spec}(A/(x_1)).$$
The classes of x_1 and x_2 are non-zero nilpotents in $A/\mathfrak{b}$. The class of x_1 is a non-zero nilpotent in $A/\mathfrak{c}$.

II Irreducible schemes

Irreducibility is a property of the underlying topological space. It is stronger than connectedness.

II.A Irreducible components

Definition II.A.1. A non-empty topological space X is *irreducible* if, whenever $X = Z_1 \cup Z_2$ with Z_1, Z_2 closed subsets of X, then $X = Z_1$ or $X = Z_2$.

Remark II.A.2. For a non-empty topological space X, the following conditions are equivalent.

i) The space X is irreducible.

ii) Any two non-empty open subsets of X meet.

iii) Every non-empty open subset of X is dense in X.

Proof. Taking complements transforms a decomposition of X into two proper closed subsets into a pair of disjoint non-empty open subsets. This proves the equivalence of i) and ii). A subset is dense if and only if it meets every non-empty open subset, which gives the equivalence of ii) and iii). ☐

Definition II.A.3. An *irreducible component* of a topological space X is a maximal irreducible closed subset of X.

Every point of X is contained in an irreducible component. Indeed, the closure of a point is irreducible. More generally, the union of a chain of irreducible subsets is irreducible, and so is its closure. Zorn's lemma therefore gives a maximal irreducible closed subset containing any prescribed irreducible subset.

Lemma II.A.4. *Let U be a non-empty open subset of a topological space X.*

i) If X is irreducible, then U is irreducible and dense in X.

ii) If $(X_i)_{i\in I}$ are the irreducible components of X, then the non-empty sets $U \cap X_i$ are precisely the irreducible components of U.

iii) If $X = X_1 \cup \cdots \cup X_r$, where the X_i are irreducible closed subsets and no X_i is contained in another, then $X_1, \ldots, X_r$ are the irreducible components of X.

Proof. The first assertion follows from Remark II.A.2.

If $U \cap X_i \neq \emptyset$, then it is irreducible and dense in X_i. If it were properly contained in an irreducible closed subset F of U, the closure of F in X would be an irreducible closed subset properly containing X_i, a contradiction. Conversely, if F is an irreducible component of U, then its closure in X is contained in some irreducible component X_i, and maximality gives $F = U \cap X_i$.

Finally, an irreducible subset contained in a finite union of closed subsets is contained in one of them. Hence every irreducible component of X is one of the X_i. The assumption that no X_i is contained in another shows that every X_i is maximal. □

Example II.A.5. Let
$$X = \operatorname{Spec}\bigl(\Bbbk[x_1, x_2]/(x_1 x_2)\bigr).$$
Then
$$X = \mathbb{V}(\bar{x}_1) \cup \mathbb{V}(\bar{x}_2).$$
Both closed subsets are isomorphic to $\mathbb{A}^1_{\Bbbk}$, hence irreducible, and neither is contained in the other. They are therefore the two irreducible components of X. Their intersection is the closed point defined by $(\bar{x}_1, \bar{x}_2)$.

Exercise II.A.6. Let
$$X = \mathbb{V}\bigl(x_1 x_2 (x_1 - x_2)\bigr) \subset \mathbb{A}^2_{\Bbbk}.$$

i) Show that X is the union of three affine lines.

ii) Prove that these lines are the irreducible components of X.

iii) Determine the irreducible components of the open subset $U = X \setminus \{(0,0)\}$.

iv) Is U connected?

Solution. We have
$$X = \mathbb{V}(x_1) \cup \mathbb{V}(x_2) \cup \mathbb{V}(x_1 - x_2).$$
Each subset is an affine line and is therefore irreducible. No one of them is contained in another, so they are the irreducible components.

Their pairwise intersections are all equal to the origin. Hence removing the origin produces the disjoint union
$$U = \bigl(\mathbb{V}(x_1) \setminus \{0\}\bigr) \amalg \bigl(\mathbb{V}(x_2) \setminus \{0\}\bigr) \amalg \bigl(\mathbb{V}(x_1 - x_2) \setminus \{0\}\bigr).$$
These three subsets are the irreducible components of U, and U is disconnected.

II.B Irreducible affine schemes and minimal primes

Let A be a ring. A prime ideal $\mathfrak{p}$ is *minimal* if it contains no strictly smaller prime ideal. More generally, a prime ideal minimal among those containing an ideal $\mathfrak{a}$ is called a minimal prime over $\mathfrak{a}$.

Lemma II.B.1. *Let A be a ring and let $\mathfrak{a} \subset A$ be an ideal.*

i) The closed subset $\mathbb{V}(\mathfrak{a})$ is irreducible if and only if $\sqrt{\mathfrak{a}}$ is prime.

ii) If $(\mathfrak{p}_i)_{i \in I}$ are the minimal primes over $\mathfrak{a}$, then the subsets $\mathbb{V}(\mathfrak{p}_i)$ are the irreducible components of $\mathbb{V}(\mathfrak{a})$.

iii) The space $\operatorname{Spec}(A)$ is irreducible if and only if A has a unique minimal prime, equivalently if and only if $\operatorname{Rad}(A)$ is prime.

Proof. Suppose first that $\sqrt{\mathfrak{a}}$ is prime and

$$\mathbb{V}(\mathfrak{a}) = \mathbb{V}(\mathfrak{b}_1) \cup \mathbb{V}(\mathfrak{b}_2) = \mathbb{V}(\mathfrak{b}_1\mathfrak{b}_2).$$

Then $\mathfrak{b}_1\mathfrak{b}_2 \subset \sqrt{\mathfrak{a}}$, so primality gives $\mathfrak{b}_1 \subset \sqrt{\mathfrak{a}}$ or $\mathfrak{b}_2 \subset \sqrt{\mathfrak{a}}$. In the first case,

$$\mathbb{V}(\mathfrak{a}) \subset \mathbb{V}(\mathfrak{b}_1),$$

and hence $\mathbb{V}(\mathfrak{a}) = \mathbb{V}(\mathfrak{b}_1)$; similarly in the second case.

Conversely, if $\sqrt{\mathfrak{a}}$ is not prime, choose $f_1, f_2 \notin \sqrt{\mathfrak{a}}$ with $f_1 f_2 \in \sqrt{\mathfrak{a}}$. Then

$$\mathbb{V}(\mathfrak{a}) = \mathbb{V}(\mathfrak{a} + (f_1)) \cup \mathbb{V}(\mathfrak{a} + (f_2)),$$

and both closed subsets are proper. Thus $\mathbb{V}(\mathfrak{a})$ is reducible.

The second assertion follows because inclusion of prime ideals is reversed by $\mathbb{V}$: minimal primes over $\mathfrak{a}$ correspond exactly to maximal irreducible closed subsets of $\mathbb{V}(\mathfrak{a})$. The last assertion is the special case $\mathfrak{a} = (0)$. □

Example II.B.2. The ring

$$A = \Bbbk[x_1, x_2]/(x_1^2)$$

is not reduced, but $\operatorname{Spec}(A)$ is irreducible. Indeed,

$$\operatorname{Rad}(A) = (\bar{x}_1),$$

and this ideal is prime because $A/(\bar{x}_1) \simeq \Bbbk[x_2]$ is an integral domain. Thus irreducibility does not imply reducedness.

Exercise II.B.3. For each of the following ideals of $\Bbbk[x_1, x_2]$, compute its radical, its minimal primes, and the irreducible components of its zero set:

$$\mathfrak{a}_1 = (x_1x_2), \qquad \mathfrak{a}_2 = (x_1^2x_2), \qquad \mathfrak{a}_3 = (x_1x_2(x_1 - x_2)).$$

Which of the quotient rings $\Bbbk[x_1, x_2]/\mathfrak{a}_i$ are reduced? Which spectra are irreducible?

Solution. We have

$$\sqrt{\mathfrak{a}_1} = \sqrt{\mathfrak{a}_2} = (x_1x_2) = (x_1) \cap (x_2),$$

so the minimal primes are (x_1) and (x_2). Both zero sets have the two coordinate axes as irreducible components. The first quotient is reduced, while the second is not: the class of x_1x_2 is non-zero and has square zero modulo $(x_1^2x_2)$.

Moreover,

$$\sqrt{\mathfrak{a}_3} = \mathfrak{a}_3 = (x_1) \cap (x_2) \cap (x_1 - x_2),$$

so its zero set has three irreducible components. This quotient is reduced. None of the three spectra is irreducible.

II.C Generic points and specialisations

Definition II.C.1. Let X be a topological space and let $x, y \in X$. We say that y is a *specialisation* of x, or that x is a *generalisation* of y, if

$$y \in \overline{\{x\}}.$$

If $Z \subset X$ is an irreducible closed subset, a point $\eta \in Z$ is a *generic point of Z* if

$$\overline{\{\eta\}} = Z.$$

The generic points of the irreducible components of X are also called the generic points of X.

Lemma II.C.2. *Let $X = \operatorname{Spec}(A)$ and let $\mathfrak{p}, \mathfrak{q} \in X$. Then*

$$\overline{\{\mathfrak{p}\}} = \mathbb{V}(\mathfrak{p}).$$

Consequently, $\mathfrak{q}$ is a specialisation of $\mathfrak{p}$ if and only if $\mathfrak{p} \subset \mathfrak{q}$. The generic points of X correspond to the minimal prime ideals of A.

Proposition II.C.3. *Let X be a scheme.*

i) Every irreducible closed subset of X has a unique generic point. In particular, generic points are in bijection with irreducible components.

ii) Let $x \in X$. The irreducible components of X containing x are in bijection with the irreducible components of $\operatorname{Spec}(\mathcal{O}_{X,x})$.

Proof. Let $Z \subset X$ be irreducible, equip it with its reduced induced scheme structure, and choose a non-empty affine open subset $U = \operatorname{Spec}(A) \subset Z$. Since U is irreducible, A has a unique minimal prime $\mathfrak{p}$. The corresponding point η is dense in U, hence dense in Z. Uniqueness follows because two generic points of Z lie in every non-empty open subset of Z; placing both in an affine open subset and using Lemma II.C.2 shows that they coincide.

For ii), choose an affine neighbourhood $U = \operatorname{Spec}(A)$ of x, and let $\mathfrak{p}$ correspond to x. Prime ideals of $A_{\mathfrak{p}}$ correspond to prime ideals of A contained in $\mathfrak{p}$. Under this correspondence, minimal primes of $A_{\mathfrak{p}}$ correspond to minimal primes of A contained in $\mathfrak{p}$, namely to the generic points of the irreducible components of X passing through x. $\square$

Example II.C.4. In $\mathbb{A}^1_{\Bbbk} = \operatorname{Spec}(\Bbbk[x_1])$, the point (0) is generic: its closure is all of $\mathbb{A}^1_{\Bbbk}$. Every closed point $(x_1 - a)$ is a specialisation of (0) because $(0) \subset (x_1 - a)$. A closed point has no proper specialisations.

Exercise II.C.5. Let

$$X = \operatorname{Spec}\big(\Bbbk[x_1, x_2]/(x_1x_2)\big)$$

and let o be the point defined by $(\bar{x}_1, \bar{x}_2)$.

i) Determine the generic points of X and their closures.

ii) Which points specialise to o?

iii) Compute the minimal primes of the local ring $\mathcal{O}_{X,o}$.

iv) Verify directly the bijection in Proposition II.C.3, ii).

Solution. The minimal primes of the coordinate ring are $(\bar{x}_1)$ and $(\bar{x}_2)$. They are the generic points η_1 and η_2 of the two coordinate axes, with

$$\overline{\{\eta_1\}} = \mathbb{V}(\bar{x}_1), \qquad \overline{\{\eta_2\}} = \mathbb{V}(\bar{x}_2).$$

Both ideals are contained in $(\bar{x}_1, \bar{x}_2)$, so both generic points specialise to o. Of course o also specialises to itself.

After localising at $(\bar{x}_1, \bar{x}_2)$, the minimal primes are $(\bar{x}_1)\mathcal{O}_{X,o}$ and $(\bar{x}_2)\mathcal{O}_{X,o}$. They correspond to the two irreducible components of X passing through o.

III Integral schemes

Integral schemes are the scheme-theoretic counterparts of integral domains.

Definition III..1. A scheme X is *integral* if it is reduced and irreducible. It is *integral at a point $x \in X$* if the local ring $\mathcal{O}_{X,x}$ is an integral domain.

III.A Integral schemes and integral domains

Proposition III.A.1. *For a scheme X, the following conditions are equivalent.*

i) The scheme X is integral.

ii) Every non-empty affine open subset $U = \operatorname{Spec}(A)$ of X has A an integral domain.

iii) For every non-empty open subset $U \subset X$, the ring $\mathcal{O}_X(U)$ is an integral domain.

Proof. If X is integral, every non-empty open subset is reduced and irreducible. Let $U = \operatorname{Spec}(A)$ be affine. Since U is reduced, $\operatorname{Rad}(A) = 0$; since it is irreducible, $\operatorname{Rad}(A)$ is prime. Hence (0) is prime and A is an integral domain. This proves i)⇒ii).

Assume ii), let $U \subset X$ be non-empty, and suppose that $fg = 0$ in $\mathcal{O}_X(U)$. For every affine open subset $V \subset U$, the ring $\mathcal{O}_X(V)$ is a domain, so either $f|_V = 0$ or $g|_V = 0$. Consider the open subsets

$$U_f = \{x \in U \mid f_x \neq 0\}, \qquad U_g = \{x \in U \mid g_x \neq 0\}.$$

They are disjoint. The hypothesis ii) also implies that X is irreducible. Indeed, if X were the union of two proper closed subsets, their complements would be disjoint non-empty open subsets. Choosing a non-empty affine open subset inside each complement, their disjoint union would be an affine open subset whose ring of sections is a product of two non-zero rings, contradicting ii). Hence U is irreducible. Thus one of U_f, U_g is empty. The sheaf property then gives $f = 0$ or $g = 0$. Hence iii) holds.

Finally, iii) implies that all affine coordinate rings are reduced, so X is reduced. If X were reducible, there would be non-empty disjoint open subsets U_1, U_2 of X. Then

$$\mathcal{O}_X(U_1 \amalg U_2) \simeq \mathcal{O}_X(U_1) \times \mathcal{O}_X(U_2)$$

would not be an integral domain. Thus X is irreducible. □

Example III.A.2. The cusp

$$X = \operatorname{Spec}\big(\Bbbk[x_1, x_2]/(x_2^2 - x_1^3)\big)$$

is integral provided $x_2^2 - x_1^3$ is irreducible in $\Bbbk[x_1, x_2]$; for instance, this holds over every field. Indeed, regarding the polynomial as a polynomial in x_2, a factorisation would force x_1^3 to be a square in $\Bbbk[x_1]$.

By contrast,

$$\operatorname{Spec}\big(\Bbbk[x_1, x_2]/(x_1 x_2)\big)$$

is reduced but reducible, while $\operatorname{Spec}(\Bbbk[x_1]/(x_1^2))$ is irreducible but non-reduced. Neither is integral.

Exercise III.A.3. Determine which of the following schemes are integral, reduced, or irreducible:

$$\begin{aligned}
X_1 &= \operatorname{Spec}\big(\Bbbk[x_1, x_2]/(x_1 x_2)\big),\\
X_2 &= \operatorname{Spec}\big(\Bbbk[x_1, x_2]/(x_1^2)\big),\\
X_3 &= \operatorname{Spec}\big(\Bbbk[x_1, x_2]/(x_1 x_2 - 1)\big),\\
X_4 &= \operatorname{Spec}\big(\Bbbk[x_1, x_2]/((x_2 - x_1^2)^2)\big).
\end{aligned}$$

Compute the reduction of each non-reduced scheme.

Solution. The scheme X_1 is reduced because $(x_1 x_2)$ is radical, but it is reducible, with components $\mathbb{V}(\bar{x}_1)$ and $\mathbb{V}(\bar{x}_2)$. Thus it is not integral.

The scheme X_2 is irreducible because its nilradical $(\bar{x}_1)$ is prime, but it is not reduced. Its reduction is $\operatorname{Spec}(\Bbbk[x_2])$.

For X_3, eliminating x_2 after inverting x_1 gives

$$\Bbbk[x_1, x_2]/(x_1x_2 - 1) \simeq \Bbbk[x_1, x_1^{-1}],$$

which is an integral domain. Hence X_3 is integral.

Finally, X_4 is irreducible but non-reduced. Its reduction is the parabola

$$\mathrm{Spec}\big(\Bbbk[x_1, x_2]/(x_2 - x_1^2)\big) \simeq \mathbb{A}^1_{\Bbbk}.$$

III.B The function field

Lemma III.B.1. *Let X be an integral scheme and let η be its generic point. Then $\mathcal{O}_{X,\eta}$ is a field. If $U = \mathrm{Spec}(A)$ is any non-empty affine open subset of X, then*

$$\mathcal{O}_{X,\eta} = \mathrm{Frac}(A).$$

Proof. The generic point belongs to every non-empty open subset of X. On $U = \mathrm{Spec}(A)$, it corresponds to the prime ideal (0), because A is an integral domain. Hence

$$\mathcal{O}_{X,\eta} = A_{(0)} = \mathrm{Frac}(A).$$

□

Definition III.B.2. The field

$$K(X) = \mathcal{O}_{X,\eta} = \kappa(\eta)$$

is called the *function field* of the integral scheme X.

Example III.B.3. Let A be an integral domain and put $K = \mathrm{Frac}(A)$. Then

$$K(\mathbb{A}^n_A) = K(x_1, \ldots, x_n).$$

Likewise,

$$K(\mathbb{P}^n_A) = K(y_1, \ldots, y_n), \qquad y_i = x_i/x_0,$$

using the affine chart $D_+(x_0)$.

Recall Definition III.D.2: a rational map between integral schemes is *birational* if it admits a rational inverse, equivalently if it restricts to an isomorphism between non-empty open subschemes. Such an isomorphism induces an isomorphism

$$K(Y) \xrightarrow{\sim} K(X).$$

More generally, a dominant rational map $X \dashrightarrow Y$ induces an inclusion $K(Y) \hookrightarrow K(X)$. Thus the map $t \mapsto t^2$ on the affine line is not birational: on function fields it gives the proper inclusion $\Bbbk(t^2) \subsetneq \Bbbk(t)$.

Example III.B.4. The map

$$\mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{V}(x_2 - x_1^2) \subset \mathbb{A}^2_{\Bbbk}, \qquad t \longmapsto (t, t^2),$$

is an isomorphism. Hence the function field of the parabola is $\Bbbk(t)$.

The standard Cremona transformation

$$\mathbb{P}^2_{\Bbbk} \dashrightarrow \mathbb{P}^2_{\Bbbk}, \qquad [x_0 : x_1 : x_2] \longmapsto [x_1x_2 : x_0x_2 : x_0x_1],$$

restricts to an automorphism of the open torus $x_0x_1x_2 \neq 0$. In the affine coordinates $y_i = x_i/x_0$, it induces

$$y_1 \longmapsto y_1^{-1}, \qquad y_2 \longmapsto y_2^{-1}$$

on $\Bbbk(y_1, y_2)$.

Lemma III.B.5. *Let X be an integral scheme and let $U \subset X$ be a non-empty open subset. Inside $K(X)$, we have*

$$\mathcal{O}_X(U) = \bigcap_{x \in U} \mathcal{O}_{X,x}.$$

In particular, for every $x \in U$, there are injective maps

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,x} \hookrightarrow K(X).$$

Proof. The injectivity of $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$ follows from integrality: a section whose germ vanishes at one point vanishes on a non-empty neighbourhood, hence on all of U. The second injection is checked on an affine neighbourhood.

It remains to prove the equality. The inclusion from left to right is clear. It is enough to treat $U = \mathrm{Spec}(A)$. Let $f \in \mathrm{Frac}(A)$ lie in $A_\mathfrak{p}$ for every prime $\mathfrak{p}$. Put

$$\mathfrak{b} = \{s \in A \mid sf \in A\}.$$

For every prime $\mathfrak{p}$, the condition $f \in A_\mathfrak{p}$ gives an element $s_\mathfrak{p} \in \mathfrak{b} \setminus \mathfrak{p}$. Thus $\mathfrak{b}$ is contained in no prime ideal, hence $\mathfrak{b} = A$. Therefore $1 \in \mathfrak{b}$ and $f \in A$. ☐

Exercise III.B.6. Let

$$X = \mathbb{V}(x_1x_2 - 1) \subset \mathbb{A}^2_{\Bbbk} \quad \text{and} \quad Y = \mathbb{V}(x_2 - x_1^2) \subset \mathbb{A}^2_{\Bbbk}.$$

i) Show that X and Y are integral.

ii) Compute $K(X)$ and $K(Y)$.

iii) Show that X is isomorphic to the principal open subset $D(x_1) \subset \mathbb{A}^1_{\Bbbk}$.

iv) Are X and Y birational?

Solution. We have

$$\Bbbk[x_1, x_2]/(x_1x_2 - 1) \simeq \Bbbk[x_1, x_1^{-1}], \qquad \Bbbk[x_1, x_2]/(x_2 - x_1^2) \simeq \Bbbk[x_1].$$

Both rings are integral domains. Their fraction fields are both isomorphic to $\Bbbk(x_1)$. The first isomorphism identifies X with $\mathrm{Spec}(\Bbbk[x_1, x_1^{-1}]) = D(x_1) \subset \mathbb{A}^1_{\Bbbk}$. Since $D(x_1)$ is a non-empty open subset of $\mathbb{A}^1_{\Bbbk} \simeq Y$, the two curves are birational.

IV Finiteness conditions and algebraic varieties

Most schemes studied in algebraic geometry satisfy finiteness conditions. Two basic ones are Noetherianity and finite generation.

IV.A Noetherian and locally Noetherian schemes

Recall that a ring is Noetherian if every ideal is finitely generated, equivalently if every ascending chain of ideals is stationary.

Definition IV.A.1. A scheme X is *locally Noetherian* if it has an affine open covering $U_i = \mathrm{Spec}(A_i)$ with all A_i Noetherian. It is *Noetherian* if it admits such a covering with finitely many members.

Example IV.A.2. Every finitely generated $\Bbbk$-algebra is Noetherian by Hilbert's basis theorem. Hence every affine algebraic $\Bbbk$-variety is Noetherian, and every algebraic $\Bbbk$-variety is a Noetherian scheme.

By contrast, the ring

$$\Bbbk[x_1, x_2, x_3, \ldots]$$

is not Noetherian, because

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$$

is a strictly increasing chain of ideals.

Exercise IV.A.3. Let X be a scheme.

i) Show that X is locally Noetherian if and only if $\mathcal{O}_X(U)$ is Noetherian for every affine open subset $U \subset X$.

ii) Show that every open subscheme of a locally Noetherian scheme is locally Noetherian.

iii) Show that a quasi-compact open subscheme of a Noetherian scheme is Noetherian.

Solution. Only the forward implication in i) needs proof. Let $U = \mathrm{Spec}(A)$ be an affine open subset and cover it by affine opens $U \cap U_i$. Refining, we may cover U by principal open subsets $D(f_j)$ such that each A_{f_j} is Noetherian. Since U is quasi-compact, finitely many suffice. The elements f_j generate the unit ideal. If $\mathfrak{a} \subset A$ is an ideal, each localisation $\mathfrak{a}_{f_j}$ is finitely generated. Choosing finitely many generators for all these localisations and clearing denominators gives a finitely generated ideal $\mathfrak{b} \subset \mathfrak{a}$ such that $\mathfrak{b}_{f_j} = \mathfrak{a}_{f_j}$ for every j. Since the $D(f_j)$ cover $\mathrm{Spec}(A)$, we obtain $\mathfrak{a} = \mathfrak{b}$. Thus A is Noetherian.

Assertion ii) follows by restricting a Noetherian affine covering. For iii), a quasi-compact open subset has a finite affine covering by Noetherian rings, hence is Noetherian.

Proposition IV.A.4. *A Noetherian scheme has only finitely many irreducible components.*

Proof. It is enough to treat $X = \mathrm{Spec}(A)$ with A Noetherian. Replacing A by $A/\mathrm{Rad}(A)$ does not change the irreducible components, so assume that A is reduced.

Let $\mathcal{F}$ be the set of radical ideals which are not intersections of finitely many prime ideals. If $\mathcal{F}$ were non-empty, the ascending chain condition would give a maximal element $\mathfrak{a} \in \mathcal{F}$. It is not prime, so there are $u, v \notin \mathfrak{a}$ with $uv \in \mathfrak{a}$. The radical ideals

$$\sqrt{\mathfrak{a} + (u)} \quad \text{and} \quad \sqrt{\mathfrak{a} + (v)}$$

strictly contain $\mathfrak{a}$, hence each is an intersection of finitely many prime ideals. Moreover,

$$\sqrt{\mathfrak{a} + (u)} \cap \sqrt{\mathfrak{a} + (v)} = \sqrt{\mathfrak{a} + (uv)} = \mathfrak{a}.$$

Thus $\mathfrak{a}$ is also a finite intersection of prime ideals, a contradiction. Therefore (0) is a finite intersection of prime ideals. It follows that A has only finitely many minimal primes, hence X has only finitely many irreducible components. ☐

IV.B Morphisms of finite type

Definition IV.B.1. A morphism $f : X \to Y$ is *quasi-compact* if $f^{-1}(V)$ is quasi-compact for every affine open subset $V \subset Y$.

Definition IV.B.2. A morphism $f : X \to Y$ is *locally of finite type* if, for every affine open subset $V = \mathrm{Spec}(A) \subset Y$, the inverse image $f^{-1}(V)$ can be covered by affine open subsets $U_i = \mathrm{Spec}(B_i)$ such that every B_i is a finitely generated A-algebra. It is *of finite type* if it is locally of finite type and quasi-compact.

Equivalently, f is of finite type if, for every affine open subset $V = \operatorname{Spec}(A) \subset Y$, the inverse image $f^{-1}(V)$ has a finite affine covering by schemes $\operatorname{Spec}(B_i)$ with B_i finitely generated over A.

Example IV.B.3. A morphism of affine schemes

$$\operatorname{Spec}(B) \longrightarrow \operatorname{Spec}(A)$$

is of finite type if and only if B is a finitely generated A-algebra. Thus the projection $\mathbb{A}^n_A \to \operatorname{Spec}(A)$ is of finite type.

Every closed immersion is of finite type, since a quotient $A \to A/\mathfrak{a}$ is generated as an A-algebra by 1. Every principal open immersion $D(f) = \operatorname{Spec}(A_f) \to \operatorname{Spec}(A)$ is also of finite type because

$$A_f \simeq A[t]/(ft-1).$$

Proposition IV.B.4. *Let $f : X \to Y$ be a morphism of finite type. If Y is Noetherian, then X is Noetherian.*

Proof. Choose a finite affine covering $Y = \bigcup_{j=1}^r V_j$, with $V_j = \operatorname{Spec}(A_j)$ and A_j Noetherian. Since f is of finite type, every $f^{-1}(V_j)$ has a finite affine covering by $U_{ij} = \operatorname{Spec}(B_{ij})$, where B_{ij} is a finitely generated A_j-algebra. Hilbert's basis theorem implies that every B_{ij} is Noetherian. These finitely many affine opens cover X. ☐

Exercise IV.B.5. i) Show that the composition of two morphisms of finite type is of finite type.

ii) Let A be a ring and $f \in A$. Show directly that the open immersion $D(f) \to \operatorname{Spec}(A)$ is of finite type.

iii) Show that
$$\operatorname{Spec}\big(A[x_1,x_2]/(x_1x_2 - f)\big) \longrightarrow \operatorname{Spec}(A)$$
is of finite type.

iv) Give an example of an A-algebra which is not of finite type over A.

Solution. Finite generation of algebras is transitive: if B is generated over A by $b_1, \ldots, b_r$ and C is generated over B by $c_1, \ldots, c_s$, then C is generated over A by the b_i and the c_j. Applying this on affine charts, and using quasi-compactness, proves i).

For ii), use
$$A_f \simeq A[t]/(ft-1).$$

The algebra in iii) is generated over A by the classes of x_1, x_2. Finally, $A[x_1, x_2, x_3, \ldots]$ is not finitely generated as an A-algebra.

V Dimension

The dimension of a scheme is defined topologically, through chains of irreducible closed subsets, and agrees in the affine case with Krull dimension.

V.A Dimension of schemes and rings

Definition V.A.1. Let X be a topological space. A chain of irreducible closed subsets of length n is a strictly increasing chain

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n.$$

The *dimension* of X, denoted by $\dim(X)$, is the supremum of the lengths of such chains. We put $\dim(\emptyset) = -\infty$.

If $Z \subset X$ is an irreducible closed subset, its *codimension* in X is

$$\operatorname{codim}_X(Z) = \sup\{n \mid Z = Z_0 \subsetneq \cdots \subsetneq Z_n \text{ is a chain of irreducible closed subsets}\}.$$

The dimension of a scheme is the dimension of its underlying topological space, and its dimension at x is $\dim(\mathcal{O}_{X,x})$.

Definition V.A.2. Let A be a ring. For a prime ideal $\mathfrak{p}$, its *height* is

$$\operatorname{ht}(\mathfrak{p}) = \sup\{n \mid \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p} \text{ are prime ideals}\}.$$

The *Krull dimension* of A is

$$\dim(A) = \sup_{\mathfrak{p} \in \operatorname{Spec}(A)} \operatorname{ht}(\mathfrak{p}).$$

We set $\dim(0) = -\infty$.

Example V.A.3. If A is a principal ideal domain which is not a field, then $\dim(A) = 1$. Its prime ideals are (0) and the ideals generated by irreducible elements. Hence the maximal chains have the form

$$(0) \subsetneq (p),$$

or, geometrically,

$$\mathbb{V}(p) \subsetneq \operatorname{Spec}(A).$$

In particular, $\dim(\operatorname{Spec}(\mathbb{Z})) = 1$.

Example V.A.4. In $\mathbb{A}^2_{\Bbbk}$, the chain

$$\{(0,0)\} = \mathbb{V}(x_1, x_2) \subsetneq \mathbb{V}(x_1) \subsetneq \mathbb{A}^2_{\Bbbk}$$

has length 2. We shall see below that no longer chain exists, so $\dim(\mathbb{A}^2_{\Bbbk}) = 2$.

Lemma V.A.5. *Let X be a topological space.*

i) If $Y \subset X$ is closed, then $\dim(Y) \leq \dim(X)$.

ii) If X is irreducible of finite dimension and $Y \subsetneq X$ is closed, then $\dim(Y) < \dim(X)$.

iii) If $(X_i)_{i\in I}$ are the irreducible components of X, then

$$\dim(X) = \sup_{i \in I} \dim(X_i).$$

Proof. A chain of irreducible closed subsets of a closed subset Y is also such a chain in X, proving i). If X is irreducible and $Y \subsetneq X$, any chain in Y can be extended by appending X, proving ii). Finally, every irreducible closed subset of X is contained in an irreducible component, which proves iii). ☐

Proposition V.A.6. *Let A be a ring and $X = \operatorname{Spec}(A)$.*

i) We have

$$\dim(X) = \dim(A) = \dim(A/\mathrm{Rad}(A)).$$

ii) For every $\mathfrak{p} \in \mathrm{Spec}(A)$,

$$\dim(A_{\mathfrak{p}}) = \mathrm{ht}(\mathfrak{p}) = \mathrm{codim}_X(\mathbb{V}(\mathfrak{p})).$$

Proof. The correspondence $\mathfrak{p} \mapsto \mathbb{V}(\mathfrak{p})$ reverses strict inclusions. Thus chains of prime ideals in A correspond bijectively, with the same length, to chains of irreducible closed subsets of X. This proves $\dim(A) = \dim(X)$. Passing to $A/\mathrm{Rad}(A)$ does not change the prime spectrum.

Prime ideals of $A_{\mathfrak{p}}$ correspond to prime ideals of A contained in $\mathfrak{p}$. Hence chains in $\mathrm{Spec}(A_{\mathfrak{p}})$ correspond to chains ending at $\mathfrak{p}$, proving $\dim(A_{\mathfrak{p}}) = \mathrm{ht}(\mathfrak{p})$. After applying $\mathbb{V}$, these are exactly the chains of irreducible closed subsets beginning with $\mathbb{V}(\mathfrak{p})$. □

Exercise V.A.7. Compute the dimensions of the following rings and schemes:

$$\begin{aligned} A_1 &= \Bbbk[x_1]/(x_1^m), \quad m \geq 1, \\ A_2 &= \Bbbk[x_1, x_2]/(x_1 x_2), \\ A_3 &= \Bbbk[x_1, x_2]/(x_1 x_2 - 1). \end{aligned}$$

For $X = \mathrm{Spec}(A_2)$, compute the dimension of the local ring at the origin and at the prime ideal $(\bar{x}_1)$.

Solution. The spectrum of A_1 has the same topology as $\mathrm{Spec}(\Bbbk)$, so $\dim(A_1) = 0$.

The irreducible components of $\mathrm{Spec}(A_2)$ are two affine lines, hence $\dim(A_2) = 1$. At the origin, there are chains

$$(\bar{x}_1) \subsetneq (\bar{x}_1, \bar{x}_2) \quad \text{and} \quad (\bar{x}_2) \subsetneq (\bar{x}_1, \bar{x}_2),$$

so the local ring has dimension 1. At the generic point $(\bar{x}_1)$ of one component, the local ring has dimension 0.

Finally,

$$A_3 \simeq \Bbbk[x_1, x_1^{-1}],$$

which is a localisation of the one-dimensional ring $\Bbbk[x_1]$. It has dimension 1.

V.B Dimension of algebraic varieties

Let $\Bbbk$ be a field. Recall that an algebraic $\Bbbk$-variety is a $\Bbbk$-scheme of finite type. In this subsection we assume varieties to be reduced when speaking of their irreducible components and function fields. We use Noether normalisation in the form proved in Theorem I.D.1, together with the standard facts that finite integral extensions preserve Krull dimension and that, for integral domains $B \subset A$ with A finite over B, the field extension $\mathrm{Frac}(A)/\mathrm{Frac}(B)$ is algebraic.

Proposition V.B.1. *Let X be a reduced algebraic $\Bbbk$-variety. Then X has finitely many irreducible components $X_1, \ldots, X_r$, and*

$$\dim(X_i) = \mathrm{trdeg}_{\Bbbk} K(X_i)$$

for every i.

Proof. The variety X is Noetherian, so it has finitely many irreducible components. Replacing X by a non-empty affine open subset of one component, we may assume that $X = \mathrm{Spec}(A)$ is integral. By Theorem I.D.1, A is finite over a polynomial ring

$$B = \Bbbk[y_1, \ldots, y_n].$$

Therefore
$$\dim(A) = \dim(B) = n.$$
Moreover, $K(X) = \mathrm{Frac}(A)$ is algebraic over $\mathrm{Frac}(B) = \Bbbk(y_1, \ldots, y_n)$, and hence
$$\mathrm{trdeg}_{\Bbbk} K(X) = n.$$
$\square$

Remark V.B.2 (The base field is part of the data). There is no ambiguity in the formula: an algebraic $\Bbbk$-variety includes its structure morphism to $\mathrm{Spec}(\Bbbk)$. When X is integral, this morphism determines the embedding $\Bbbk \hookrightarrow K(X)$. If $K/\Bbbk$ is a field extension, the scheme X does not automatically become a K-variety. The natural K-variety is the base change
$$X_K = X \times_{\mathrm{Spec}(\Bbbk)} \mathrm{Spec}(K).$$
Its Krull dimension is unchanged:
$$\dim(X_K) = \dim(X).$$
Indeed, on a non-empty affine open of each irreducible component, Noether normalisation gives a finite inclusion $\Bbbk[t_1, \ldots, t_n] \subset A$; base change turns it into the finite inclusion $K[t_1, \ldots, t_n] \subset A \otimes_{\Bbbk} K$.

If the same integral scheme carries compatible variety structures over both $\Bbbk$ and K, the two transcendence-degree formulas are related by the tower formula
$$\mathrm{trdeg}_{\Bbbk} K(X) = \mathrm{trdeg}_{\Bbbk} K + \mathrm{trdeg}_K K(X).$$
For example, when $K/\Bbbk$ is finite, the first term on the right is zero and the two formulas give the same dimension. By contrast, $\mathrm{Spec}(\Bbbk(t))$ is a zero-dimensional variety over $\Bbbk(t)$ but is not a variety over $\Bbbk$, since $\Bbbk(t)$ is not finitely generated as a $\Bbbk$-algebra.

Example V.B.3. The affine hypersurface
$$X = \mathbb{V}(x_1 x_2 - 1) \subset \mathbb{A}^2_{\Bbbk}$$
has function field $\Bbbk(x_1)$ and therefore dimension 1. More generally, a proper irreducible hypersurface in $\mathbb{A}^n_{\Bbbk}$ has dimension $n - 1$.

Corollary V.B.4. *For every $n \geq 0$,*
$$\dim(\mathbb{A}^n_{\Bbbk}) = \dim(\mathbb{P}^n_{\Bbbk}) = n.$$
Every proper closed subvariety of $\mathbb{A}^n_{\Bbbk}$ or $\mathbb{P}^n_{\Bbbk}$ has dimension at most $n - 1$.

Exercise V.B.5. Let
$$A = \Bbbk[x_1, x_2]/(x_2^2 - x_1^3).$$

i) Show that A is an integral domain.

ii) Show that A is a finitely generated module over the polynomial subring $\Bbbk[\bar{x}_1]$, generated by 1 and $\bar{x}_2$.

iii) Compute $\dim(A)$ and $\mathrm{trdeg}_{\Bbbk}\mathrm{Frac}(A)$.

iv) Put $t = \bar{x}_2/\bar{x}_1 \in \mathrm{Frac}(A)$. Show that $\bar{x}_1 = t^2$ and $\bar{x}_2 = t^3$, and deduce that $\mathrm{Frac}(A) \simeq \Bbbk(t)$.

Solution. The polynomial $x_2^2 - x_1^3$ is irreducible: viewed as a polynomial in x_2 over $\Bbbk(x_1)$, it would be reducible only if x_1^3 were a square in $\Bbbk(x_1)$. Hence the quotient A is an integral domain.

The relation $\bar{x}_2^2 = \bar{x}_1^3$ reduces every power of $\bar{x}_2$ to a $\Bbbk[\bar{x}_1]$-linear combination of 1 and $\bar{x}_2$. Thus A is finite over $\Bbbk[\bar{x}_1]$. Finite integral extensions preserve dimension, so

$$\dim(A) = 1.$$

Moreover $\mathrm{Frac}(A)$ is algebraic over $\Bbbk(\bar{x}_1)$ and therefore has transcendence degree 1.

In the fraction field,

$$t^2 = \frac{\bar{x}_2^2}{\bar{x}_1^2} = \bar{x}_1, \qquad t^3 = t\bar{x}_1 = \bar{x}_2.$$

Thus $\mathrm{Frac}(A) = \Bbbk(t)$.

VI Regular and smooth schemes

Further reading. For regular local rings, nonsingular varieties and the Jacobian criterion, see [Har77, Chapter I, §5] and [Har77, Chapter II, §8]. For the scheme-theoretic distinction between regularity and smoothness over a non-perfect field, see [Liu02].

The purpose of this section is to give a precise meaning to the comparison between the dimension of a scheme and the dimension of its Zariski tangent space that already appeared in Example I.B.7.

VI.A Regular local rings and the cotangent space

Let $(A, \mathfrak{m})$ be a Noetherian local ring and let $\kappa = A/\mathfrak{m}$ be its residue field. The vector space

$$\mathfrak{m}/\mathfrak{m}^2$$

is the *cotangent space* of A. By Nakayama's lemma, its dimension is the minimal number of generators of the maximal ideal.

Definition VI.A.1. The *embedding dimension* of A is

$$\mathrm{edim}(A) = \dim_\kappa(\mathfrak{m}/\mathfrak{m}^2).$$

The local ring A is *regular* if

$$\dim(A) = \mathrm{edim}(A).$$

A locally Noetherian scheme X is *regular at $x \in X$* if $\mathcal{O}_{X,x}$ is a regular local ring. It is a *regular scheme* if it is regular at every point.

The terminology is justified by a standard inequality from local algebra.

Proposition VI.A.2. *For every Noetherian local ring* $(A, \mathfrak{m})$*,*

$$\dim(A) \leq \dim_\kappa(\mathfrak{m}/\mathfrak{m}^2).$$

Thus regularity says that the cotangent space has the smallest possible dimension.

Idea of proof. Choose elements $a_1, \dots, a_e \in \mathfrak{m}$ whose classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$. Nakayama's lemma gives $\mathfrak{m} = (a_1, \dots, a_e)$. Krull's height theorem then gives $\dim(A) = \mathrm{ht}(\mathfrak{m}) \leq e$. □

For a scheme X and a point x, put

$$\mathfrak{m}_x = \mathfrak{m}_{X,x}, \qquad \kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x.$$

When X is a $\Bbbk$-scheme and $x \in X(\Bbbk)$, Proposition I.B.6 gives

$$T_x X \simeq \mathrm{Hom}_{\Bbbk}(\mathfrak{m}_x/\mathfrak{m}_x^2, \Bbbk).$$

Consequently

$$\dim_x X = \dim(\mathcal{O}_{X,x}) \leq \dim_{\Bbbk} T_x X,$$

and equality holds precisely when X is regular at x.

Example VI.A.3 (Three familiar singularities). At the origin of the cusp $y^2 = x^3$ and of the node $xy = 0$, the local dimension is 1, whereas the Jacobian vanishes and the tangent space has dimension 2. Both points are therefore non-regular. For the dual numbers

$$A = \Bbbk[\epsilon]/(\epsilon^2),$$

the unique local ring has dimension 0, while $(\epsilon)/(\epsilon^2)$ has dimension 1. Thus a non-reduced point is not regular either.

VI.B Smoothness over a field

Regularity is intrinsic: it only uses the local rings of X. Smoothness is relative: it is a property of a morphism. We restrict here to the structure morphism of a scheme of finite type over a field,

$$X \longrightarrow \mathrm{Spec}(\Bbbk).$$

Definition VI.B.1. Let X be a scheme of finite type over $\Bbbk$. We say that X is *smooth over* $\Bbbk$ if, for an algebraic closure $\overline{\Bbbk}$ of $\Bbbk$, the scheme

$$X_{\overline{\Bbbk}} = X \times_{\mathrm{Spec}(\Bbbk)} \mathrm{Spec}(\overline{\Bbbk})$$

is regular. We say that X is smooth at x if this holds on a neighbourhood of every point of $X_{\overline{\Bbbk}}$ lying over x; equivalently, $X_{\overline{\Bbbk}}$ is regular at all those points.

This is the notion usually called *geometric regularity.* For schemes of finite type over a field it is equivalent to the usual definition of a smooth structure morphism. The passage to an algebraic closure is essential: a regular scheme need not remain regular after extending the ground field.

Theorem VI.B.2 (Regular versus smooth). *Let X be a scheme of finite type over a field $\Bbbk$.*

i) If X is smooth over $\Bbbk$, then X is regular.

ii) If $\Bbbk$ is perfect, then X is smooth over $\Bbbk$ if and only if X is regular.

iii) The regular locus and the smooth locus are open subsets of X.

In particular, regularity and smoothness agree for varieties over a field of characteristic zero or over an algebraically closed field.

We state this theorem without proof. Its second assertion says that, over a perfect field, regularity is automatically preserved by every field extension. The example below shows exactly what can fail otherwise.

Example VI.B.3 (Regular but not smooth). Assume that $\mathrm{char}(\Bbbk) = p > 0$ and that $\Bbbk$ is not perfect. Choose $a \in \Bbbk$ which is not a p-th power and put

$$L = \Bbbk[t]/(t^p - a), \qquad X = \mathrm{Spec}(L).$$

The ring L is a field, so X is a regular zero-dimensional scheme. If $K = \Bbbk(a^{1/p})$, however, then

$$L \otimes_{\Bbbk} K \simeq K[u]/(u^p), \qquad u = t - a^{1/p}.$$

The base change is non-reduced and is not regular. Hence X is not smooth over $\Bbbk$. Equivalently, the derivative of $t^p - a$ is identically zero, so the Jacobian cannot have the rank expected of a zero-dimensional smooth scheme.

VI.C The Jacobian criterion

The preceding definitions recover the computation with dual numbers. Let

$$X = \mathbb{V}(f_1, \ldots, f_r) \subset \mathbb{A}^n_{\Bbbk}$$

and let x be a closed point. Write

$$J_x = \left(\frac{\partial f_i}{\partial x_j}(x) \right)$$

for the Jacobian matrix with entries in $\kappa(x)$. As in Example I.B.7, its kernel is the relative Zariski tangent space at x. Since $\Bbbk$ is perfect, the finite residue-field extension $\kappa(x)/\Bbbk$ is separable, and this tangent space is naturally dual to $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Theorem VI.C.1 (Jacobian criterion). *Assume that $\Bbbk$ is perfect and put $d = \dim(\mathcal{O}_{X,x})$. Then the following conditions are equivalent:*

i) X is smooth over $\Bbbk$ at x;

ii) $\mathcal{O}_{X,x}$ is regular;

iii) the Zariski tangent space at x has dimension d;

iv) the Jacobian matrix has rank $n - d$ at x.

Indeed, the tangent-space computation gives

$$\dim_{\kappa(x)} T_x X = n - \mathrm{rank}(J_x).$$

The equivalence of ii) and iii) is the definition of a regular local ring, while Theorem VI.B.2 gives the equivalence with i).

Example VI.C.2 (Affine and projective space). After extending the ground field to $\overline{\Bbbk}$, every closed point of $\mathbb{A}^n_{\Bbbk}$ has local parameters $x_1 - a_1, \ldots, x_n - a_n$; its local ring has dimension and embedding dimension n. The polynomial ring and all its localisations are regular, so affine space is smooth. The standard affine charts of $\mathbb{P}^n_{\Bbbk}$ are copies of $\mathbb{A}^n_{\Bbbk}$, so projective space is smooth as well.

Exercise VI.C.3. i) Use the Jacobian criterion to show that the projective conic

$$C = \mathbb{V}_+(xy - z^2) \subset \mathbb{P}^2_{\Bbbk}$$

is smooth over every field $\Bbbk$.

ii) Assume that char($\Bbbk$) $\neq 2, 3$. Determine the regular locus of the cuspidal curve $y^2 = x^3$.

iii) For the field $L/\Bbbk$ in Example VI.B.3, verify directly that Spec(L) has zero-dimensional local ring and zero-dimensional cotangent space, but that its base change to K has a one-dimensional cotangent space.

Solution. The partial derivatives of $xy - z^2$ are

$$y, \qquad x, \qquad -2z.$$

If they all vanished at a point of the conic, then $x = y = 0$; the equation would then force $z = 0$, which is not a projective point. This argument also works in characteristic 2, where the first two derivatives already suffice. The calculation is unchanged after every field extension, so applying it over an algebraic closure proves smoothness over an arbitrary ground field.

For $f = y^2 - x^3$, the gradient is $(-3x^2, 2y)$. Under the stated characteristic hypothesis it vanishes on the curve only at $(0, 0)$. Thus the regular locus is the complement of the cusp.

The local ring of Spec(L) is the field L, whose maximal ideal and cotangent space are zero. After base change, the local ring is $K[u]/(u^p)$ with maximal ideal (u); the quotient $(u)/(u^2)$ is one-dimensional over K. The base-changed point is therefore not regular.

Chapter 5

Global properties of morphisms

I Fibred products

One important construction is that of products, or fibred products, in the category of schemes.

I.A The universal property of the fibred product

Definition I.A.1. Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes. A scheme W is the *fibred product* of f and g, denoted by $X \times_S Y$, if W is equipped with morphisms $F : W \to X$ and $G : W \to Y$ such that $f \circ F = g \circ G$ and, given any scheme T and morphisms $u : T \to X$ and $v : T \to Y$ with $f \circ u = g \circ v$, there is a unique morphism $w : T \to W$ such that $F \circ w = u$ and $G \circ w = v$. This is expressed in the next diagram.

$$\begin{tikzcd}
T \arrow[dr, dotted] \arrow[drr, bend left, "v"] \arrow[ddr, bend right, "u"'] & & \\
& W \arrow[r, "F"'] \arrow[d, "G"] & Y \arrow[d, "g"] \\
& X \arrow[r, "f"'] & S
\end{tikzcd} \tag{I.1}$$

Remark I.A.2. The fibred product has the following immediate properties.

i) It is unique up to a unique isomorphism compatible with the two projections. Indeed, if W_1 and W_2 both satisfy the universal property, it gives unique morphisms

$$W_1 \longrightarrow W_2, \qquad W_2 \longrightarrow W_1$$

compatible with the projections. The two composites must be the identity by uniqueness.

ii) Let

$$p_X : X \times_S Y \longrightarrow X$$

be the first projection. For every open subscheme $U \subset X$, the inverse image $p_X^{-1}(U)$ is canonically isomorphic to $U \times_S Y$.

iii) Since every scheme has a unique morphism to $\mathrm{Spec}(\mathbb{Z})$, the product of two schemes is their fibred product over $\mathrm{Spec}(\mathbb{Z})$.

Exercise I.A.3. Prove item ii) of Remark I.A.2 directly from the universal property.

Solution. Let
$$W = X \times_S Y$$
and let $p_X : W \to X$ be the first projection. Put
$$W_U = p_X^{-1}(U).$$

The restrictions of the two projections give morphisms
$$W_U \longrightarrow U, \qquad W_U \longrightarrow Y$$
over S.

Let T be a scheme equipped with morphisms $u : T \to U$ and $v : T \to Y$ over S. Composing u with $U \hookrightarrow X$, the universal property of W gives a unique morphism $w : T \to W$. Since $p_X \circ w$ has image in U, the morphism w factors uniquely through the open subscheme W_U. This is exactly the universal property of $U \times_S Y$.

I.B Constructing the fibred product

The next result is crucial in the understanding of schemes and families of schemes.

Theorem I.B.1. *For every scheme S, the category* $(\mathrm{Sch})_S$ *admits finite products. Equivalently, fibred products of schemes exist.*

Proof. It is enough to construct the product of two S-schemes X and Y.

1. Assume first that
$$S = \mathrm{Spec}(R), \qquad X = \mathrm{Spec}(A), \qquad Y = \mathrm{Spec}(B).$$
Set
$$W = \mathrm{Spec}(A \otimes_R B).$$
The homomorphisms
$$A \longrightarrow A \otimes_R B, \quad a \longmapsto a \otimes 1, \qquad B \longrightarrow A \otimes_R B, \quad b \longmapsto 1 \otimes b$$
give morphisms $W \to X$ and $W \to Y$ over S.
Let T be a scheme. By Theorem I.F.1, giving morphisms $T \to X$ and $T \to Y$ over S is equivalent to giving homomorphisms
$$A \longrightarrow \Gamma(\mathcal{O}_T), \qquad B \longrightarrow \Gamma(\mathcal{O}_T)$$
which agree on R. By the universal property of the tensor product, this is equivalent to giving a unique homomorphism
$$A \otimes_R B \longrightarrow \Gamma(\mathcal{O}_T),$$
hence a unique morphism $T \to W$. Thus
$$X \times_S Y \simeq \mathrm{Spec}(A \otimes_R B).$$

2. Assume that S and Y are affine, while X is arbitrary. Choose an affine open covering
$$X = \bigcup_{i \in I} U_i.$$
By the affine case, the fibred products
$$W_i = U_i \times_S Y$$
exist. If $U_{ij} = U_i \cap U_j$, then
$$W_i|_{U_{ij}} = U_{ij} \times_S Y = W_j|_{U_{ij}}$$
canonically. These identifications satisfy the cocycle condition, so the W_i glue to a scheme W with morphisms
$$W \longrightarrow X, \qquad W \longrightarrow Y.$$
To verify the universal property, let $u : T \to X$ and $v : T \to Y$ be morphisms over S. The open subsets $T_i = u^{-1}(U_i)$ cover T, and the affine case gives compatible morphisms $T_i \to W_i$. They glue uniquely to a morphism $T \to W$. Hence $W \simeq X \times_S Y$.

3. Assume that S is affine and X, Y are arbitrary. Choose an affine open covering
$$Y = \bigcup_{j \in J} V_j.$$
By the preceding step, each $X \times_S V_j$ exists. Their restrictions to the overlaps $V_j \cap V_\ell$ are canonically isomorphic, so they glue to $X \times_S Y$.

4. Finally, let S be arbitrary. Choose an affine open covering
$$S = \bigcup_{k \in K} S_k$$
and put
$$U_k = X \times_S S_k = f^{-1}(S_k), \qquad V_k = Y \times_S S_k = g^{-1}(S_k).$$
By the preceding step,
$$W_k = U_k \times_{S_k} V_k$$
exists. It also represents $U_k \times_S Y$: if $T \to U_k$ and $T \to Y$ have the same composite with S, then the second morphism factors uniquely through V_k.

On $S_k \cap S_\ell$, the corresponding restrictions of W_k and W_ℓ are canonically isomorphic. These isomorphisms satisfy the cocycle condition, so the W_k glue to a scheme W. The universal property is local on T, and therefore follows from the universal properties of the W_k. Thus $W \simeq X \times_S Y$.

□

Example I.B.2 (Scheme-theoretic intersections). Let A be a ring and let $\mathfrak{a}, \mathfrak{b} \subset A$ be ideals. The two closed subschemes
$$Z_1 = \mathrm{Spec}(A/\mathfrak{a}), \qquad Z_2 = \mathrm{Spec}(A/\mathfrak{b})$$
of $\mathrm{Spec}(A)$ have fibred product
$$Z_1 \times_{\mathrm{Spec}(A)} Z_2 \simeq \mathrm{Spec}\big((A/\mathfrak{a}) \otimes_A (A/\mathfrak{b})\big) \simeq \mathrm{Spec}\big(A/(\mathfrak{a} + \mathfrak{b})\big).$$
Thus the fibred product recovers the scheme-theoretic intersection of two closed subschemes.

Exercise I.B.3. In $\mathbb{A}^2_{\Bbbk} = \operatorname{Spec}(\Bbbk[x,y])$, let

$$L = \mathbb{V}(y), \qquad P = \mathbb{V}(y - x^2).$$

Compute $L \times_{\mathbb{A}^2_{\Bbbk}} P$. Compare it with the set-theoretic intersection of the line and the parabola.

Solution. Using the preceding example,

$$L \times_{\mathbb{A}^2_{\Bbbk}} P \simeq \operatorname{Spec}\left(\frac{\Bbbk[x,y]}{(y) + (y - x^2)}\right) \simeq \operatorname{Spec}\bigl(\Bbbk[x]/(x^2)\bigr).$$

The underlying topological space consists only of the origin, which is the set-theoretic intersection. Scheme-theoretically, however, the intersection is a double point: the class of x is nonzero and has square zero.

I.C Properties of the fibred product

First observe that the fibred product need not agree with the set-theoretic product of X and Y along S.

Remark I.C.1. The underlying set of a fibred product need not be the set-theoretic fibred product. Let

$$S = \operatorname{Spec}(\mathbb{R}), \qquad X = \operatorname{Spec}(\mathbb{C}).$$

The set-theoretic fibred product of the one-point spaces $|X|$ with itself over $|S|$ has one point. On the other hand,

$$X \times_S X = \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}).$$

Using one copy of $\mathbb{C}$ as the coefficient field, we have

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathbb{C}[t]/(t^2+1) \simeq \mathbb{C} \times \mathbb{C},$$

where the last isomorphism follows from the factorisation

$$t^2 + 1 = (t - i)(t + i).$$

Thus $X \times_S X$ has two points.

Exercise I.C.2. Let X and Y be S-schemes and let Z be a Y-scheme. Prove the following properties.

1. $X \times_S S \simeq X$;
2. $X \times_S Y \simeq Y \times_S X$;
3. $(X \times_S Y) \times_Y Z \simeq X \times_S Z$.

Solution. All three isomorphisms follow from the universal property.

1. The morphisms
$$X \longrightarrow X, \qquad X \longrightarrow S$$
give a morphism $X \to X \times_S S$. Conversely, the first projection gives $X \times_S S \to X$. The two composites are identities by uniqueness.
2. Exchanging the two projections gives a morphism
$$X \times_S Y \longrightarrow Y \times_S X.$$
The same construction in the other direction gives its inverse.

3. A morphism $T \to (X \times_S Y) \times_Y Z$ is equivalent to morphisms

$$T \to X, \qquad T \to Y, \qquad T \to Z$$

such that the map to Y is both the composite $T \to X \to S$ together with the structure data and the composite $T \to Z \to Y$. Equivalently, it is the datum of morphisms $T \to X$ and $T \to Z$ having the same composite with S. This is the universal property of $X \times_S Z$.

I.D Base change

Let S be a scheme and consider a scheme X over S and a morphism of schemes $g : S' \to S$. Then, using the fibred product, we obtain an S'-scheme $X' = X \times_S S' \to S'$. We say that X' is obtained by base change from X via g.

Remark I.D.1. Let B be a ring and let A be a positively graded B-algebra. Let C be another B-algebra. Then one can show that

$$\mathrm{Proj}(A \otimes_B C) \simeq \mathrm{Proj}(A) \times_{\mathrm{Spec}(B)} \mathrm{Spec}(C).$$

For instance, we defined the projective space $\mathbb{P}^n_{\mathbb{Z}} = \mathrm{Proj}(\mathbb{Z}[x_0, \ldots, x_n])$. Then, given a scheme Y, by base change we obtain *the projective space over* Y

$$\mathbb{P}^n_Y = \mathbb{P}^n_{\mathbb{Z}} \times_{\mathrm{Spec}(\mathbb{Z})} Y.$$

Example I.D.2 (Extension of the ground field). Let $X = \mathrm{Spec}(A)$ be an affine $\Bbbk$-scheme and let $K/\Bbbk$ be a field extension. The base change of X to K is

$$X_K = X \times_{\mathrm{Spec}(\Bbbk)} \mathrm{Spec}(K) \simeq \mathrm{Spec}(A \otimes_{\Bbbk} K).$$

Thus base change replaces the coefficients in the equations defining X by coefficients in K.

Exercise I.D.3. Put $A = \mathbb{Z}[u]/(u^2 - 2)$ and let

$$X = \mathrm{Spec}(A)$$

with structure morphism $\mathrm{Spec}(\gamma_A) : X \to \mathrm{Spec}(\mathbb{Z})$. Compute the fibres over the generic point (0) and over the closed points (2), (3), and (7).

Solution. The generic fibre is

$$X_{(0)} \simeq \mathrm{Spec}\big(\mathbb{Q}[u]/(u^2 - 2)\big) = \mathrm{Spec}\big(\mathbb{Q}(\sqrt{2})\big).$$

Over 2, we obtain

$$X_{(2)} \simeq \mathrm{Spec}\big(\mathbb{F}_2[u]/(u^2)\big),$$

a non-reduced point.

Over 3, the polynomial $u^2 - 2 = u^2 + 1$ has no root in $\mathbb{F}_3$, so

$$X_{(3)} \simeq \mathrm{Spec}(\mathbb{F}_9)$$

is a single point with residue field $\mathbb{F}_9$.

Finally, 2 is a square modulo 7, since $3^2 = 2$. Hence

$$u^2 - 2 = (u - 3)(u + 3)$$

in $\mathbb{F}_7[u]$, and the Chinese remainder theorem gives

$$X_{(7)} \simeq \mathrm{Spec}(\mathbb{F}_7 \times \mathbb{F}_7),$$

the disjoint union of two $\mathbb{F}_7$-rational points.

II Fibres

II.A Fibres and residue fields

Let $f : X \to Y$ be a morphism of schemes and let $y \in Y$. The residue field at y is

$$\kappa(y) = \mathcal{O}_{Y,y}/\mathfrak{m}_{Y,y}.$$

There is a canonical morphism

$$i_y : \operatorname{Spec}(\kappa(y)) \longrightarrow Y$$

whose image is the point y. If $V = \operatorname{Spec}(A)$ is an affine open neighbourhood of y and y corresponds to the prime ideal $\mathfrak{p} \subset A$, then this morphism is induced by

$$A \longrightarrow A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} = \kappa(\mathfrak{p}).$$

Notice that i_y is a closed immersion if and only if y is a closed point.

Definition II.A.1. The *fibre of f over y* is the $\kappa(y)$-scheme

$$X_y = X \times_Y \operatorname{Spec}(\kappa(y)).$$

Proposition II.A.2. *Let $y \in Y$. The projection $X_y \to X$ induces a homeomorphism*

$$|X_y| \xrightarrow{\sim} f^{-1}(y),$$

where the right-hand side is equipped with the subspace topology.

Proof. The assertion is local on both X and Y. We may therefore assume that

$$Y = \operatorname{Spec}(A), \qquad X = \operatorname{Spec}(B),$$

that f is induced by a ring homomorphism $\varphi : A \to B$, and that y corresponds to a prime ideal $\mathfrak{p} \subset A$. Then

$$X_y = \operatorname{Spec}\big(B \otimes_A \kappa(\mathfrak{p})\big).$$

Since

$$B \otimes_A \kappa(\mathfrak{p}) \simeq B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}},$$

its prime ideals correspond first to the prime ideals of $B_{\mathfrak{p}}$ containing $\mathfrak{p}B_{\mathfrak{p}}$, and hence to the prime ideals $\mathfrak{q} \subset B$ satisfying

$$\varphi^{-1}(\mathfrak{q}) \subset \mathfrak{p} \qquad \text{and} \qquad \varphi(\mathfrak{p}) \subset \mathfrak{q}.$$

These two conditions are equivalent to

$$\varphi^{-1}(\mathfrak{q}) = \mathfrak{p}.$$

Thus the points of X_y correspond exactly to the points of X mapping to y. The localisation and quotient correspondences identify closed subsets on both sides, so this bijection is a homeomorphism. $\square$

Example II.A.3 (A family of hyperbolas degenerating to two lines). Let

$$X = \operatorname{Spec}\big(\Bbbk[t, x, y]/(xy - t)\big)$$

and let $f : X \to \mathbb{A}^1_{\Bbbk} = \operatorname{Spec}(\Bbbk[t])$ be induced by the inclusion $\Bbbk[t] \hookrightarrow \Bbbk[t, x, y]/(xy - t)$. For $a \in \Bbbk$, the fibre is

$$X_a \simeq \operatorname{Spec}\big(\Bbbk[x, y]/(xy - a)\big).$$

If $a \neq 0$, then x is invertible and

$$X_a \simeq \mathrm{Spec}(\Bbbk[x, x^{-1}]) = \mathbb{G}_{m,\Bbbk}.$$

The special fibre is

$$X_0 = \mathbb{V}(xy),$$

the union of the two coordinate axes. Thus the geometry of the fibres may change in a family.

Exercise II.A.4. Let

$$X = \mathrm{Spec}\big(\Bbbk[t, x]/(x^2 - t)\big)$$

and let $f : X \to \mathbb{A}^1_{\Bbbk}$ be the morphism induced by $t \mapsto x^2$. Compute the fibre over $a \in \Bbbk$. Show that the fibre over 0 is non-reduced.

Solution. The fibre over a is

$$X_a \simeq \mathrm{Spec}\big(\Bbbk[x]/(x^2 - a)\big).$$

For $a = 0$, this is

$$\mathrm{Spec}\big(\Bbbk[x]/(x^2)\big),$$

the double point, and is therefore non-reduced. For $a \neq 0$, the structure of the fibre depends on the factorisation of $x^2 - a$ over $\Bbbk$: it is the disjoint union of two rational points when a is a nonzero square and $\mathrm{char}(\Bbbk) \neq 2$, and otherwise it may be a single point with a quadratic residue field.

Example II.A.5 (The generic fibre of the affine-plane projection). Consider the projection

$$\pi\colon \mathbb{A}^2_{\mathbb{C}} = \mathrm{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}^1_{\mathbb{C}} = \mathrm{Spec}(\mathbb{C}[t]).$$

Let $\eta = (0)$ be the generic point of the base. Since $\kappa(\eta) = \mathbb{C}(t)$, the generic fibre is

$$(\mathbb{A}^2_{\mathbb{C}})_\eta = \mathrm{Spec}\big(\mathbb{C}[t, x] \otimes_{\mathbb{C}[t]} \mathbb{C}(t)\big) = \mathrm{Spec}(\mathbb{C}(t)[x]).$$

Thus the ordinary affine line over the function field $\mathbb{C}(t)$ appears naturally as the generic member of the family of affine lines parametrised by $\mathbb{A}^1_{\mathbb{C}}$. Its points correspond to prime ideals of $\mathbb{C}[t, x]$ lying over the generic point of the base.

II.B Dimension of fibres

We now combine the construction of fibres with the dimension theory developed in the previous chapter. Throughout this subsection, an *integral algebraic* $\Bbbk$*-variety* means an integral scheme of finite type over $\Bbbk$.

Let $f : X \to Y$ be a dominant morphism of integral schemes, and let η be the generic point of Y. Since

$$\kappa(\eta) = K(Y),$$

the fibre

$$X_\eta = X \times_Y \mathrm{Spec}(K(Y))$$

is called the *generic fibre* of f. Dominance implies that the generic point of X maps to η, and therefore induces an inclusion of function fields

$$K(Y) \hookrightarrow K(X).$$

Proposition II.B.1 (Dimension of the generic fibre). *Let $f : X \to Y$ be a dominant morphism of integral algebraic $\Bbbk$-varieties, and let η be the generic point of Y. Then X_η is an integral $K(Y)$-variety, its function field is naturally $K(X)$, and*

$$\dim(X_\eta) = \mathrm{trdeg}_{K(Y)} K(X) = \dim(X) - \dim(Y).$$

Proof. Choose non-empty affine open subsets

$$V = \mathrm{Spec}(A) \subset Y, \qquad U = \mathrm{Spec}(B) \subset f^{-1}(V).$$

Since X and Y are integral and f is dominant, the induced morphism $A \to B$ is injective. Put $S = A\backslash\{0\}$. The part of the generic fibre lying over U is

$$U_\eta = \mathrm{Spec}\big(B \otimes_A K(Y)\big) = \mathrm{Spec}(S^{-1}B).$$

This is integral, and its fraction field is $\mathrm{Frac}(B) = K(X)$. Hence the generic fibre is integral and has function field $K(X)$, viewed as an extension of $K(Y)$. By the dimension–transcendence-degree formula,

$$\dim(X_\eta) = \mathrm{trdeg}_{K(Y)} K(X).$$

Finally, the tower formula for transcendence degrees gives

$$\mathrm{trdeg}_{\Bbbk} K(X) = \mathrm{trdeg}_{\Bbbk} K(Y) + \mathrm{trdeg}_{K(Y)} K(X),$$

and therefore

$$\dim(X_\eta) = \dim(X) - \dim(Y).$$

□

The dimension of a special fibre may be larger than the dimension of the generic fibre. Nevertheless, the generic value occurs on a non-empty open subset of the base.

Theorem II.B.2 (Generic constancy of fibre dimension). *Let $f : X \to Y$ be a dominant morphism of integral algebraic $\Bbbk$-varieties. Then there exists a non-empty open subset $V \subset Y$ such that, for every $y \in V$, the fibre X_y is non-empty and*

$$\dim(X_y) = \dim(X) - \dim(Y).$$

Proof. Choose a non-empty affine open subset $V_0 = \mathrm{Spec}(A) \subset Y$. Since f is of finite type, $f^{-1}(V_0)$ is quasi-compact; choose a finite covering by non-empty affine opens

$$U_i = \mathrm{Spec}(B_i), \qquad 1 \le i \le m.$$

Every U_i contains the generic point of the integral scheme X. Hence the maps $A \to B_i$ are injective and

$$\mathrm{Frac}(B_i) = K(X).$$

Put

$$r = \mathrm{trdeg}_{K(Y)} K(X) = \dim(X) - \dim(Y).$$

For completeness, the relative form of Noether normalisation used here is obtained by applying Chapter 1 over $K(Y) = \mathrm{Frac}(A)$ to $B_i \otimes_A K(Y)$, then clearing the finitely many denominators in the chosen generators and in their monic integral equations. It gives a non-zero element $a_i \in A$ and elements $z_{i1}, \ldots, z_{ir} \in (B_i)_{a_i}$ such that

$$C_i = A_{a_i}[z_{i1}, \ldots, z_{ir}] \hookrightarrow (B_i)_{a_i}$$

is finite. Replacing all the a_i by their product, we may use one common principal open subset $D(a) \subset V_0$.

It remains to ensure that the displayed inclusions survive in every fibre. The quotient $(B_i)_a/C_i$ is a finite C_i-module. We use generic freeness in the following form: if A is a domain, C is a finite-type A-algebra and M is a finite C-module, then M_b is free, hence flat, over A_b for some non-zero $b \in A$.

Applying this to the finitely many quotients and multiplying a by the resulting elements, every quotient is flat over A_a. Tensoring

$$0 \longrightarrow C_i \longrightarrow (B_i)_a \longrightarrow (B_i)_a/C_i \longrightarrow 0$$

with $\kappa(y)$ is therefore still exact for every $y \in D(a)$. We obtain an injective finite homomorphism

$$\kappa(y)[z_{i1}, \dots, z_{ir}] \hookrightarrow B_i \otimes_A \kappa(y).$$

Consequently $(U_i)_y$ is finite and dominant, hence finite and surjective, over $\mathbb{A}^r_{\kappa(y)}$. It is non-empty and has dimension r.

The schemes $(U_i)_y$ form a finite open covering of X_y. Thus X_y is non-empty and

$$\dim(X_y) = \max_i \dim((U_i)_y) = r$$

for every $y \in D(a)$, which proves the theorem. □

Example II.B.3 (A projection). The projection

$$\pi : \mathbb{A}^{m+n}_{\Bbbk} \longrightarrow \mathbb{A}^m_{\Bbbk}$$

is dominant. Every fibre is isomorphic to $\mathbb{A}^n_{\kappa(y)}$, and hence

$$\dim(\pi^{-1}(y)) = n = (m+n) - m.$$

Here the generic dimension occurs over the whole base.

Example II.B.4 (A reducible special fibre). For the morphism

$$f : \mathrm{Spec}\big(\Bbbk[t,x,y]/(xy-t)\big) \longrightarrow \mathbb{A}^1_{\Bbbk},$$

the total space has dimension 2 and the base has dimension 1. Every fibre has dimension 1. For $a \neq 0$, the fibre $xy = a$ is isomorphic to $\mathbb{G}_{m,\Bbbk}$, whereas

$$X_0 = \mathbb{V}(xy)$$

is the union of the two coordinate axes. Fibre dimension is constant here, even though irreducibility is not.

Example II.B.5 (A jumping fibre). Consider

$$f : \mathbb{A}^2_{\Bbbk} \longrightarrow \mathbb{A}^2_{\Bbbk}, \qquad (x,y) \longmapsto (x, xy).$$

Equivalently, f is induced by

$$\Bbbk[u,v] \longrightarrow \Bbbk[x,y], \qquad u \longmapsto x, \quad v \longmapsto xy.$$

This ring map is injective, so f is dominant. Over the principal open $D(u)$, the fibre over (a,b) consists of the single point

$$x = a, \qquad y = b/a.$$

Thus the generic fibre has dimension $0 = 2 - 2$. However,

$$f^{-1}(0,0) = \mathbb{V}(x) \simeq \mathbb{A}^1_{\Bbbk},$$

so the fibre dimension jumps at the origin. If $a = 0$ and $b \neq 0$, the fibre is empty. This illustrates why the theorem only asserts constancy after restricting to a suitable non-empty open subset.

Remark II.B.6. Dimension does not detect nilpotent structure. For the power map

$$\mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}, \qquad t \longmapsto t^n,$$

the fibre over 0 is the zero-dimensional but non-reduced scheme

$$\mathrm{Spec}\big(\Bbbk[t]/(t^n)\big).$$

Thus the scheme structure of a fibre contains information that its dimension alone forgets.

Exercise II.B.7 (The affine generic fibre). Let $A \subset B$ be finitely generated integral $\Bbbk$-algebras and let

$$f : \mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

be the corresponding dominant morphism. Put $K = \mathrm{Frac}(A)$ and $S = A \setminus \{0\}$.

i) Show that the generic fibre is $\mathrm{Spec}(S^{-1}B)$.

ii) Show that it is integral and that its function field is $\mathrm{Frac}(B)$.

iii) Deduce that

$$\dim(X_\eta) = \dim(B) - \dim(A).$$

Solution. The generic point of $\mathrm{Spec}(A)$ has residue field $K = S^{-1}A$. Hence

$$X_\eta = \mathrm{Spec}(B \otimes_A K) = \mathrm{Spec}(S^{-1}B).$$

Since B is a domain and S contains no zero element of B, the localisation $S^{-1}B$ is a domain. Its fraction field is the same as that of B. The last assertion follows from

$$\dim(S^{-1}B) = \mathrm{trdeg}_K \mathrm{Frac}(B) = \mathrm{trdeg}_{\Bbbk} \mathrm{Frac}(B) - \mathrm{trdeg}_{\Bbbk} K = \dim(B) - \dim(A).$$

Exercise II.B.8 (The jumping-fibre example). For the morphism

$$f : \mathbb{A}^2_{\Bbbk} \to \mathbb{A}^2_{\Bbbk}, \qquad (x, y) \mapsto (x, xy),$$

compute scheme-theoretically the fibre over $(a, b) \in \mathbb{A}^2_{\Bbbk}(\Bbbk)$. Determine the locus on which the fibres are zero-dimensional and identify the fibre over $(0, 0)$.

Solution. The fibre over (a, b) is

$$\mathrm{Spec}\big(\Bbbk[x, y]/(x - a, xy - b)\big).$$

If $a \neq 0$, the equations give $x = a$ and $y = b/a$, so the fibre is one reduced point. If $a = 0$ and $b \neq 0$, the quotient is the zero ring and the fibre is empty. If $(a, b) = (0, 0)$, the quotient is

$$\Bbbk[x, y]/(x, xy) \simeq \Bbbk[y],$$

so the fibre is $\mathbb{A}^1_{\Bbbk}$. Thus the fibres are zero-dimensional on $D(u)$, while the dimension jumps at the origin.

Exercise II.B.9 (Dimension versus scheme structure). Assume that $n \geq 2$ and consider the morphism

$$f : \mathbb{A}^1_{\Bbbk} \to \mathbb{A}^1_{\Bbbk}, \qquad t \mapsto t^n.$$

Compute the fibre over $a \in \Bbbk$. Show that every non-empty fibre has dimension 0, but that the fibre over 0 is non-reduced.

Solution. The fibre over a is

$$\mathrm{Spec}\big(\Bbbk[t]/(t^n - a)\big).$$

Since this is finite-dimensional as a $\Bbbk$-vector space, every non-empty fibre has dimension 0. At $a = 0$ one obtains

$$\mathrm{Spec}\big(\Bbbk[t]/(t^n)\big),$$

which is non-reduced because the class of t is non-zero and nilpotent.

Further reading. For the classical dimension-of-fibres theorem and its relation with transcendence degree, see [Har77, Chapter I, §7]. For a systematic scheme-theoretic treatment of fibres and dimension, see [Liu02].

III Separated morphisms

The underlying topological space of a scheme is rarely Hausdorff. For instance, if an irreducible component of X contains two distinct points, then any neighbourhood of either point meets that component in a nonempty open subset and therefore contains its generic point. The two points cannot be separated by disjoint open neighbourhoods.

III.A Separated morphisms and the diagonal

Exercise III.A.1. Let X be a scheme.

i) Show that if the underlying topological space of X is Hausdorff, then $\dim(X) = 0$.

ii) Assume in addition that X is locally Noetherian. Show that the converse holds: if $\dim(X) = 0$, then the underlying topological space of X is discrete, hence Hausdorff.

Solution. i) In a Hausdorff space every point is closed. Let $U = \mathrm{Spec}(A)$ be an affine open subset of X. Every prime ideal of A is therefore a closed point of $\mathrm{Spec}(A)$, hence is maximal. Thus there is no strict inclusion of prime ideals, and $\dim(A) = 0$. Since this holds on every affine open subset, $\dim(X) = 0$.

ii) Let $x \in X$ and choose a Noetherian affine neighbourhood $U = \mathrm{Spec}(A)$. Since $\dim(A) = 0$, the Noetherian ring A is Artinian. Hence $\mathrm{Spec}(A)$ is a finite discrete space. In particular, $\{x\}$ is open in U, and therefore open in X. Thus every point of X is open, so X is discrete and hence Hausdorff.

Even if schemes are rarely Hausdorff spaces, there is a notion adapted to schemes that plays the role of the Hausdorff condition. To understand it, let us start with the following observation. The idea is that, for a morphism $X \to Y$, the diagonal $\Delta_{X/Y} \subset X \times_Y X$ plays the role of the set of pairs of points of X that map to the same point in Y. Asking it to be closed is the scheme-theoretic analogue of the Hausdorff condition.

Remark III.A.2. Let X be a topological space. We equip $X \times X$ with the product topology and consider $\Delta = \{(x_1, x_2) \in X \times X \mid x_1 = x_2\}$. Then X is Hausdorff if and only if Δ is closed in $X \times X$.

Indeed, let us assume that Δ is closed in $X \times X$. Let x_1, x_2 be distinct elements of X. Then (x_1, x_2) does not lie in Δ, so there is an open subset U of $X \times X$ containing (x_1, x_2) and disjoint from Δ. Since $X \times X$ is equipped with the product topology, there are open subsets U_1 and U_2 of X, with $x_1 \in U_1$ and $x_2 \in U_2$, such that $U_1 \times U_2 \subset U$. Then $U_1 \times U_2 \cap \Delta = \emptyset$, which implies $U_1 \cap U_2 = \emptyset$. So X is Hausdorff.

Conversely, assume that X is Hausdorff and let us check that $X \times X \setminus \Delta$ is open in $X \times X$. Let $(x_1, x_2) \in X \times X \setminus \Delta$, so that x_1 and x_2 are distinct elements of X. Then there are disjoint neighbourhoods U_1 and U_2 of x_1 and x_2 and $U_1 \times U_2$ is an open neighbourhood of (x_1, x_2) in $X \times X \setminus \Delta$.

Let $f : X \to Y$ be a morphism of schemes. Then we consider the fibred product $X \times_Y X$. Now, the identity of X gives, by the universal property of $X \times_Y X$, a unique morphism $X \to X \times_Y X$, which we call the diagonal morphism and denote by $\Delta_{X/Y}$. If $Y = \mathrm{Spec}(\mathbb{Z})$, we write simply $\Delta_X : X \to X \times X$.

Definition III.A.3. A morphism $f : X \to Y$ is separated if $\Delta_{X/Y}$ is a closed immersion. We also say that X *is separated over* Y. A scheme X is *separated* if it is separated over $\mathrm{Spec}(\mathbb{Z})$.

Remark III.A.4. Any morphism $f : X \to Y$ of affine schemes is separated. Indeed, let $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(A)$, so that f corresponds to a morphism of rings $A \to B$. Then the diagonal morphism $\Delta_{X/Y} : X \to X \times_Y X$ corresponds to the morphism of rings $B \otimes_A B \to B$. Since this ring map is surjective, the associated morphism of schemes is a closed immersion.

Proposition III.A.5 (Graphs). *Let $f : X \to Y$ be a morphism of S-schemes. Its graph is the morphism*

$$\Gamma_f = (\mathrm{id}_X, f) : X \longrightarrow X \times_S Y.$$

It is a monomorphism. If Y is separated over S, then Γ_f is a closed immersion.

Proof. The graph is obtained by base change from the diagonal of Y:

$$\begin{tikzcd} X \arrow[r, "\Gamma_f"] \arrow[d, "f"'] & X \times_S Y \arrow[d, "f\times \mathrm{id}_Y"] \\ Y \arrow[r, "\Delta_{Y/S}"'] & Y \times_S Y. \end{tikzcd}$$

The square is cartesian. Since a diagonal is always a monomorphism, so is its base change Γ_f. If $Y \to S$ is separated, then $\Delta_{Y/S}$ is a closed immersion, and closed immersions are preserved by base change. □

Example III.A.6. If $p(t) \in \Bbbk[t]$, the graph of the morphism

$$\mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}, \qquad t \longmapsto p(t),$$

is the closed subscheme of $\mathbb{A}^2_{\Bbbk} = \mathrm{Spec}(\Bbbk[x, y])$ defined by

$$y - p(x) = 0.$$

III.B Affine coverings of separated schemes

In a similar vein, we have the following lemma.

Lemma III.B.1. *Let X be a scheme. Then X is separated if and only if, for any pair of open affine subsets U, V of X, $U \cap V$ is affine and the multiplication map $\mu_{U,V} : \mathcal{O}_X(U) \otimes \mathcal{O}_X(V) \to \mathcal{O}_X(U \cap V)$ is surjective. This happens if and only if there is an affine open covering $\{U_i \mid i \in I\}$ of X, such that all $i, j \in I$, $U_i \cap U_j$ is affine and the map μ_{U_i,U_j} is surjective.*

Proof. Let us take $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$ affine open subsets of X. Then $U \times V$ is an open subset of $X \times X$ and $\Delta_X^{-1}(U \times V) = U \cap V$ as a topological space, as $\Delta_X(x) = (x, x)$. Since U and V are affine, we have $U \times V = \mathrm{Spec}(A \otimes_{\mathbb{Z}} B)$ and the morphism of schemes Δ_X, restricted to $U \cap V$, corresponds to the ring homomorphism $\mu_{U,V} : A \otimes_{\mathbb{Z}} B \to \mathcal{O}_X(U \cap V)$.

If X is separated, Δ_X is a closed immersion so this map is surjective and $U \cap V$ is a closed subscheme of the affine $U \times V$, so $U \cap V$ is affine.

Let $\{U_i \mid i \in I\}$ be an open affine covering of X. If X is separated, we can apply the above argument to any $i, j \in I$ and see that μ_{U_i,U_j} is surjective.

Conversely, assume that for all $i, j \in I$ the map μ_{U_i,U_j} is surjective and recall that $U_i \cap U_j = \Delta_X^{-1}(U_i \times U_j)$. Since the morphism $U_i \cap U_j \to U_i \times U_j$ corresponds to the map $\mu_{U_i,U_j} : \mathcal{O}_X(U_i) \otimes \mathcal{O}_X(U_j) \to \mathcal{O}_X(U_i \cap U_j)$ the map Δ_X, restricted to $U_i \cap U_j$, is a closed immersion. Since the subsets $\{U_i \times U_j \mid (i, j) \in I^2\}$ cover $X \times X$, the map Δ_X is a closed immersion. □

Exercise III.B.2. Let $\Bbbk$ be a field.

1. Show that the affine line over $\Bbbk$ with two origins is not a separated scheme.
2. Construct a scheme X by gluing two affine planes $Z_1 = \mathbb{A}^2_{\Bbbk}$ and $Z_2 = \mathbb{A}^2_{\Bbbk}$ along $Z_{1,2} = Z_1 \setminus \{(0,0)\}$ and $Z_{2,1} = Z_2 \setminus \{(0,0)\}$ using the identity map.
3. Check that the scheme X constructed in the previous item has a covering by two affine subschemes U_1 and U_2 isomorphic to affine planes, but that $U_1 \cap U_2$ is not affine.

Solution.

1. Let $U_1, U_2 \simeq \operatorname{Spec}(\Bbbk[x])$ be the two standard affine open subsets. Their intersection is $D(x) = \operatorname{Spec}(\Bbbk[x, x^{-1}])$. The multiplication map
$$\Bbbk[x] \otimes_{\Bbbk} \Bbbk[x] \longrightarrow \Bbbk[x, x^{-1}]$$
has image contained in $\Bbbk[x]$, and in particular does not contain x^{-1}. It is not surjective, so the affine criterion for separatedness shows that the line with two origins is not separated.
2. Glue Z_1 and Z_2 by identifying their punctured planes through the identity map. Since there are only two charts, the cocycle condition is automatic. The gluing lemma produces a scheme X containing the images U_1, U_2 of the two affine planes as open subschemes.
3. We have
$$U_1 \cap U_2 \simeq \mathbb{A}^2_{\Bbbk} \setminus \{(0,0)\} = D(x) \cup D(y).$$
Its ring of global functions is
$$\Bbbk[x, y]_x \cap \Bbbk[x, y]_y = \Bbbk[x, y]$$
inside $\Bbbk(x, y)$. If the punctured plane were affine, the canonical morphism to the spectrum of its global functions would identify it with $\operatorname{Spec}(\Bbbk[x, y]) = \mathbb{A}^2_{\Bbbk}$, which is impossible because the origin is missing. Hence $U_1 \cap U_2$ is not affine, and X is not separated.

Example III.B.3. Let Y be a scheme. Then the structure morphism
$$\mathbb{P}^n_Y \longrightarrow Y$$
is separated. Indeed, we first treat $\mathbb{P}^n_{\mathbb{Z}}$. Put
$$A = \mathbb{Z}[x_0, \ldots, x_n]$$
and let $U_i = D_+(x_i)$. Then
$$U_i = \operatorname{Spec}\big(A[1/x_i]_0\big)$$
is affine, and
$$U_i \cap U_j = \operatorname{Spec}\big(A[1/x_i, 1/x_j]_0\big).$$
The overlap ring is obtained from $A[1/x_i]_0$ by inverting x_j/x_i:
$$A[1/x_i, 1/x_j]_0 = A[1/x_i]_0 \left[\left(\frac{x_j}{x_i} \right)^{-1} \right].$$

The element $(x_j/x_i)^{-1} = x_i/x_j$ belongs to $A[1/x_j]_0$. Hence the multiplication map

$$A[1/x_i]_0 \otimes_{\mathbb{Z}} A[1/x_j]_0 \longrightarrow A[1/x_i, 1/x_j]_0$$

is surjective. The affine criterion therefore shows that $\mathbb{P}^n_{\mathbb{Z}}$ is separated. Since separatedness is preserved by base change and

$$\mathbb{P}^n_Y \simeq \mathbb{P}^n_{\mathbb{Z}} \times_{\mathrm{Spec}(\mathbb{Z})} Y,$$

the morphism $\mathbb{P}^n_Y \to Y$ is separated.

Exercise III.B.4. Prove the following facts.

1. Closed immersions are separated.
2. Let $f : X \to Y$ and $g : Y \to Z$ be separated morphisms. Show that $g \circ f$ is separated.
3. Let $X \to Y$ be a separated morphism of S-schemes and consider a morphism $S' \to S$. Set $X' = X \times_S S'$ and $Y' = Y \times_S S'$ and let $f' : X' \to Y'$ be the morphism induced by base change from S to S'. Then f' is separated.
4. Let X and Y be separated S-schemes. Then $X \times_S Y$ is a separated S-scheme.

Solution.

1. A closed immersion is a monomorphism, so its relative diagonal is an isomorphism. In particular, it is a closed immersion.
2. The morphism
$$X \times_Y X \longrightarrow X \times_Z X$$
is the inverse image of $\Delta_{Y/Z}$ under $f \times f$, hence is a closed immersion. The diagonal $\Delta_{X/Z}$ is the composite
$$X \xrightarrow{\Delta_{X/Y}} X \times_Y X \longrightarrow X \times_Z X.$$
It is therefore a closed immersion.
3. The diagonal of the base-changed morphism is the base change of $\Delta_{X/Y}$. Since closed immersions are preserved by base change, f' is separated.
4. The diagonal of $X \times_S Y$ identifies with the product of the two diagonals:
$$\Delta_{(X \times_S Y)/S} = \Delta_{X/S} \times_S \Delta_{Y/S}.$$
Equivalently, it is a composite of base changes of these diagonals. Hence it is a closed immersion.

III.C Projective morphisms

Definition III.C.1. A morphism of schemes $X \to Y$ is *projective* if it factors as $X \to \mathbb{P}^n_Y \to Y$, where $X \to \mathbb{P}^n_Y$ is a closed immersion and $\mathbb{P}^n_Y \to Y$ is the structure morphism.

From Example III.B.3 and Exercise III.B.4, we get the following result.

Proposition III.C.2. *Let $f : X \to Y$ be a projective morphism, namely, the composition of a closed immersion $X \to \mathbb{P}^n_Y$ with the structure map $\mathbb{P}^n_Y \to Y$. Then f is separated.*

Proposition III.C.3. *Let $f, g : X \to Y$ be morphisms of S-schemes. Assume that Y is separated over S, that X is reduced, and that there exists a dense open subset $U \subset X$ such that*

$$f|_U = g|_U.$$

Then $f = g$.

Proof. Let
$$H = (f, g) : X \longrightarrow Y \times_S Y.$$
Since Y is separated over S, the diagonal
$$\Delta_{Y/S} : Y \longrightarrow Y \times_S Y$$
is a closed immersion. The inverse image
$$E = H^{-1}\big(\Delta_{Y/S}(Y)\big)$$
is therefore closed in X. It contains U, so its underlying topological space is all of X. Hence f and g agree on points.

It remains to compare their maps on structure sheaves. This is local on X and Y. Let $V = \mathrm{Spec}(A) \subset Y$ be affine and let $W = \mathrm{Spec}(B) \subset f^{-1}(V) = g^{-1}(V)$ be affine. The restrictions of f and g correspond to ring homomorphisms
$$\varphi, \psi : A \longrightarrow B.$$
The open subset $U \cap W$ is dense in W. For $a \in A$, put
$$b = \varphi(a) - \psi(a).$$
For every $\mathfrak{p} \in U \cap W$, equality of the two morphisms on U gives $b/1 = 0$ in $B_{\mathfrak{p}}$. In particular $b \in \mathfrak{p}$, so the closed subset $\mathbb{V}(b)$ contains the dense subset $U \cap W$. Therefore
$$\mathbb{V}(b) = \mathrm{Spec}(B),$$
and b belongs to the nilradical of B. Since X, hence W, is reduced, we obtain $b = 0$. Thus $\varphi = \psi$, and the local maps glue to give $f = g$. □

Example III.C.4 (Why reducedness is needed). Let
$$X = \mathrm{Spec}\big(\Bbbk[x, \varepsilon]/(\varepsilon^2, x\varepsilon)\big)$$
and let $U = D(x)$. Since x becomes invertible on U, the relation $x\varepsilon = 0$ implies $\varepsilon|_U = 0$. The two morphisms
$$f, g : X \longrightarrow \mathbb{A}^1_{\Bbbk}$$
corresponding respectively to
$$t \longmapsto 0, \qquad t \longmapsto \varepsilon$$
agree on the dense open subset U, but they are not equal on X.

IV Proper morphisms

Proper schemes and morphisms are the algebraic analogue of the notion of compactness.

IV.A Closed and universally closed morphisms

Definition IV.A.1. Let $f : X \to Y$ be a morphism of schemes.

i) The morphism f is *closed* if $f(F)$ is closed in Y for every closed subset $F \subset X$.

ii) The morphism f is *universally closed* if, for every morphism $Y' \to Y$, the base-changed morphism

$$f' : X \times_Y Y' \longrightarrow Y'$$

is closed.

Note that not all closed maps are universally closed. For instance, if $\Bbbk$ is a field and X is a $\Bbbk$-scheme, then the structure map $X \to \mathrm{Spec}(\Bbbk)$ is closed. But very often this map is not universally closed.

Example IV.A.2. For $n \geq 1$, the affine space $\mathbb{A}^n_{\Bbbk} = \mathrm{Spec}(\Bbbk[x_1, \dots, x_n]) \to \mathrm{Spec}(\Bbbk)$ is not universally closed, for we can base change along $\mathbb{A}^1_{\Bbbk} = \mathrm{Spec}(\Bbbk[t]) \to \mathrm{Spec}(\Bbbk)$ to get a morphism $\mathbb{A}^{n+1}_{\Bbbk} \to \mathbb{A}^1_{\Bbbk}$, which is not closed. Indeed, this morphism is given by the obvious morphism of rings $\Bbbk[t] \to \Bbbk[t, x_1, \dots, x_n]$, and the closed subset $\mathbb{V}(x_1 t - 1)$ is mapped to:

$$f'(\mathbb{V}(x_1 t - 1)) = \{\mathfrak{q} \cap \Bbbk[t] \mid \mathfrak{q} \in \mathrm{Spec}(\Bbbk[t, x_1, \dots, x_n]), (x_1 t - 1) \in \mathfrak{q}\}.$$

We show that:

$$f'(\mathbb{V}(x_1 t - 1)) = D(t) \subset \mathbb{A}^1_{\Bbbk}.$$

Let $\mathfrak{p} = \mathfrak{q} \cap \Bbbk[t]$ be a prime ideal $f'(\mathbb{V}(x_1 t - 1))$. If $t \in \mathfrak{p}$, then $t \in \mathfrak{q}$, hence $1 = t x_1 - (x_1 t - 1) \in \mathfrak{q}$, which is not the case. So any $\mathfrak{p} \in f'(\mathbb{V}(x_1 t - 1))$ satisfies $t \notin \mathfrak{p}$, hence $f'(\mathbb{V}(x_1 t - 1)) \subset D(t)$.

Conversely, let $\mathfrak{p} \in D(t)$. Then, $\mathfrak{p}$ is obtained intersecting a prime ideal of $\Bbbk[t]_t$ with $\Bbbk[t]$. Under the isomorphism

$$\Bbbk[x_1, t]/(x_1 t - 1) \simeq \Bbbk[t, t^{-1}] \simeq \Bbbk[t]_t,$$

a prime ideal of $\Bbbk[t, t^{-1}]$ is sent to a prime ideal of $\Bbbk[t, x_1]/(x_1 t - 1)$, which pulls back in $\Bbbk[t, x_1]$ to a prime ideal containing $(x_1 t - 1)$. This further pulls-back to a prime ideal $\mathfrak{q}$ of $\Bbbk[t, x_1, \dots, x_n]$ which contains $(x_1 t - 1)$. So, $\mathfrak{p} = \mathfrak{q} \cap \Bbbk[t]$ and $\mathfrak{p} \in f'(\mathbb{V}(x_1 t - 1))$.

Example IV.A.3. A closed immersion $i : X \hookrightarrow Y$ is universally closed. Indeed, after any base change $Y' \to Y$, the induced morphism

$$X \times_Y Y' \longrightarrow Y'$$

is again a closed immersion, hence a closed map.

Remark IV.A.4. The terminology is motivated by a familiar topological fact. If K is a compact topological space, then the projection

$$\mathrm{pr}_1 : X \times K \longrightarrow X$$

is closed for every topological space X. Indeed, if $F \subset X \times K$ is closed and $x \notin \mathrm{pr}_1(F)$, then $\{x\} \times K$ is contained in the open subset $(X \times K) \setminus F$. By the tube lemma, there is an open neighbourhood U of x such that

$$U \times K \subset (X \times K) \setminus F.$$

Thus U is disjoint from $\mathrm{pr}_1(F)$, proving that the latter is closed.

The condition "universally closed" asks for the analogous property after every base change.

Definition IV.A.5. Let $f : X \to Y$ be a morphism of schemes. The morphism f is *proper* if it is

i) of finite type;

ii) separated;

iii) universally closed.

Recall that f is of finite type if, for every affine open subset $V = \mathrm{Spec}(A) \subset Y$, the inverse image $f^{-1}(V)$ admits a finite covering by affine open subsets

$$U_i = \mathrm{Spec}(B_i)$$

such that each B_i is a finitely generated A-algebra.

An A-scheme X is *proper over A* if its structure morphism

$$X \longrightarrow \mathrm{Spec}(A)$$

is proper. A scheme is *proper* if it is proper over $\mathrm{Spec}(\mathbb{Z})$.

Proposition IV.A.6. *Closed immersions are proper.*

Proof. Let $i : X \hookrightarrow Y$ be a closed immersion. It is universally closed because every base change of i is again a closed immersion. It is separated because its diagonal is a closed immersion.

Finally, if $V = \mathrm{Spec}(A) \subset Y$ is affine, then

$$i^{-1}(V) = \mathrm{Spec}(A/\mathfrak{a})$$

for some ideal $\mathfrak{a} \subset A$. The quotient $A/\mathfrak{a}$ is a finitely generated A-algebra, so i is of finite type. □

Exercise IV.A.7. Prove the following properties.

1. The composition of proper morphisms is proper.
2. The base change of a proper morphism is proper.
3. Suppose that $f : X \to S$ is proper and factors as
$$X \xrightarrow{f'} S' \longrightarrow S,$$
where $S' \to S$ is separated. Show that f' is proper.
4. Let X and Y be S-schemes. Assume that $X \to S$ is proper, that $Y \to S$ is separated and of finite type, and that there is a surjective S-morphism $X \to Y$. Show that $Y \to S$ is proper.

Solution.

1. Morphisms of finite type are stable under composition, as are separated morphisms. If $X \to Y$ and $Y \to Z$ are universally closed, then after any base change to $Z' \to Z$, the resulting composite is a composite of closed maps, hence is closed.
2. Finite type and separatedness are preserved by base change. Universal closedness is preserved by base change directly from its definition: a further base change of $X \times_Y Y' \to Y'$ is also a base change of $X \to Y$.
3. The graph
$$\Gamma_{f'} : X \longrightarrow X \times_S S'$$
is a closed immersion because $S' \to S$ is separated. The projection
$$X \times_S S' \longrightarrow S'$$
is proper, being a base change of $X \to S$. Since
$$f' = \mathrm{pr}_{S'} \circ \Gamma_{f'},$$
the first item shows that f' is proper.

4. It remains only to prove universal closedness. Let $S' \to S$ be a base change and write
$$h' : X_{S'} \longrightarrow Y_{S'}.$$
The morphism h' is still surjective. If $F \subset Y_{S'}$ is closed, then $(h')^{-1}(F)$ is closed in $X_{S'}$. Since $X_{S'} \to S'$ is closed,
$$\mathrm{im}\big((h')^{-1}(F) \to S'\big)$$
is closed. By surjectivity of h', this image is exactly the image of F in S'. Thus $Y_{S'} \to S'$ is closed. The assumptions already give finite type and separatedness.

Lemma IV.A.8. *Properness is local on the target: if $Y = \bigcup_i V_i$ is an open covering, then $f : X \to Y$ is proper if and only if every restriction*
$$f^{-1}(V_i) \longrightarrow V_i$$
is proper.

Proof. Finite type and separatedness are local on the target. For universal closedness, perform an arbitrary base change $Y' \to Y$. The inverse images of the V_i form an open covering of Y', and a subset of Y' is closed if and only if its intersection with each member of this covering is closed. Thus the base-changed morphism is closed if and only if all its restrictions over these open subsets are closed. □

IV.B The valuative criterion and projective morphisms

Definition IV.B.1. A *valuation ring* is an integral domain R with fraction field K such that, for every $a \in K^\times$, either $a \in R$ or $a^{-1} \in R$. Equivalently, the ideals of R are totally ordered by inclusion.

Proposition IV.B.2. *Every valuation ring R is local. Its maximal ideal is*
$$\mathfrak{m}_R = \{a \in R \mid a = 0 \text{ or } a^{-1} \notin R\},$$
the set of non-units of R.

Proof. The principal ideals of R are totally ordered. If a, b are non-units, assume after exchanging them that $aR \subset bR$, and write $a = bc$. Then $a + b = b(c + 1)$ is again a non-unit. A multiple of a non-unit is a non-unit as well. Hence the non-units form an ideal, which is necessarily the unique maximal ideal. □

Example IV.B.3 (Basic valuation rings). Every field is a valuation ring of itself; this is the trivial example. Every discrete valuation ring (Definition II.D.15) is a valuation ring. Indeed, if R has uniformiser π, every element of $K^\times$ has the form
$$u\pi^n, \qquad u \in R^\times, \quad n \in \mathbb{Z},$$
so either it or its inverse belongs to R. In particular,
$$\mathbb{Z}_{(p)} \subset \mathbb{Q}, \qquad \Bbbk[t]_{(t)} \subset \Bbbk(t), \qquad \Bbbk[\![t]\!] \subset \Bbbk(\!(t)\!)$$
are valuation rings. The first two are localisations of principal ideal domains, while the third records the order of vanishing of a Laurent series.

Remark IV.B.4 (The geometric picture). The morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(R)$ identifies $\mathrm{Spec}(K)$ with the generic point of $\mathrm{Spec}(R)$. The ring R is local, so it also has a unique closed point. For a discrete valuation ring these are the only two points; a general valuation ring may have a longer totally ordered chain of prime ideals. The valuative criterion asks whether a map defined at the generic point extends over this whole chain of specialisations.

Theorem IV.B.5 (Valuative criterion for properness). *Let $f : X \to Y$ be a morphism of finite type. Then f is proper if and only if, for every valuation ring R with fraction field K, every commutative diagram*

$$\begin{tikzcd} \mathrm{Spec}(K) \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \mathrm{Spec}(R) \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

admits a unique dotted arrow.

We use this standard criterion without proof. The existence of the lift expresses universal closedness, while its uniqueness expresses separatedness.

Exercise IV.B.6. Let $R = \Bbbk[\![\pi]\!]$ and $K = \Bbbk((\pi))$.

i) Show that R is a valuation ring.

ii) Describe $\mathrm{Spec}(R)$ and identify the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(R)$.

iii) Use the K-point $\pi^{-1} \in \mathbb{A}^1_{\Bbbk}(K)$ to see directly from the valuative criterion that $\mathbb{A}^1_{\Bbbk} \to \mathrm{Spec}(\Bbbk)$ is not proper.

Solution. i) Every nonzero Laurent series can be written uniquely as

$$\pi^m u, \qquad m \in \mathbb{Z},$$

with $u \in R^\times$. If $m \geq 0$, the element belongs to R; if $m < 0$, its inverse belongs to R.

ii) The ring R is a discrete valuation ring with maximal ideal (π). Hence

$$\mathrm{Spec}(R) = \{(0), (\pi)\}.$$

The morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(R)$ identifies its unique point with the generic point (0); the point (π) is closed.

iii) The element $\pi^{-1} \in K$ defines a morphism

$$\mathrm{Spec}(K) \longrightarrow \mathbb{A}^1_{\Bbbk}.$$

An extension to $\mathrm{Spec}(R)$ would correspond to an element of R whose image in K is π^{-1}, which is impossible. Hence the required lift does not exist.

Theorem IV.B.7. *For every scheme Y, the projection*

$$\mathbb{P}^n_Y \longrightarrow Y$$

is proper.

Proof. The morphism is of finite type and was proved above to be separated. We check the valuative criterion. Let R be a valuation ring with fraction field K, and consider a diagram

$$\begin{tikzcd} \mathrm{Spec}(K) \arrow[r] \arrow[d] & \mathbb{P}^n_Y \arrow[d] \\ \mathrm{Spec}(R) \arrow[r] & Y. \end{tikzcd}$$

After base change by $\mathrm{Spec}(R) \to Y$, the upper morphism is a K-point of $\mathbb{P}^n_R$, hence is represented by homogeneous coordinates

$$[a_0 : \cdots : a_n], \qquad a_i \in K,$$

not all zero.

The principal fractional ideals a_iR are totally ordered. Choose a_j such that a_jR contains all the a_iR. Then

$$a_i/a_j \in R \qquad \text{for every } i,$$

and the j-th ratio is 1. Therefore

$$[a_0/a_j : \cdots : a_n/a_j]$$

defines an R-point of $\mathbb{P}^n_R$ extending the original K-point. Uniqueness follows from separatedness. The valuative criterion now implies that $\mathbb{P}^n_Y \to Y$ is proper. □

Corollary IV.B.8 (Projective morphisms are proper). *Let $f : X \to Y$ factor as*

$$X \hookrightarrow \mathbb{P}^n_Y \longrightarrow Y$$

with the first morphism a closed immersion. Then f is proper.

Proof. Closed immersions are proper, the projection $\mathbb{P}^n_Y \to Y$ is proper, and proper morphisms are stable under composition. □

IV.C Finite morphisms

Definition IV.C.1. A morphism $f : X \to Y$ is *finite* if, for every affine open subset $V = \mathrm{Spec}(A) \subset Y$, the inverse image $f^{-1}(V)$ is affine, say

$$f^{-1}(V) = \mathrm{Spec}(B),$$

and B is a finitely generated A-module.

Proposition IV.C.2 (Finite morphisms and projective embeddings). *Let $A \to B$ be a finite ring homomorphism. Then*

$$\mathrm{Spec}(B) \longrightarrow \mathrm{Spec}(A)$$

is projective.

More generally, every finite morphism is projective locally on its target.

Proof. Choose A-module generators

$$b_0 = 1, b_1, \ldots, b_n$$

of B. Give $B[T]$ the grading for which B has degree 0 and T has degree 1. There is a surjective homomorphism of graded A-algebras

$$A[x_0, \ldots, x_n] \longrightarrow B[T], \qquad x_i \longmapsto b_iT.$$

Indeed, x_0 maps to T, and every element bT^d is an A-linear combination of the elements $b_iT^d = (b_iT)T^{d-1}$.

Taking Proj gives a closed immersion

$$\mathrm{Proj}(B[T]) \hookrightarrow \mathbb{P}^n_A.$$

Since the irrelevant ideal of $B[T]$ is generated by T, the single chart $D_+(T)$ is all of $\mathrm{Proj}(B[T])$, and

$$\mathrm{Proj}(B[T]) \simeq \mathrm{Spec}\big((B[T][T^{-1}])_0\big) = \mathrm{Spec}(B).$$

This proves the assertion over an affine base. Applying the construction on affine open subsets of an arbitrary target proves the local statement. □

Remark IV.C.3. With the more general definition of a projective morphism using relative projective spectra, every finite morphism is projective globally:

$$X \simeq \mathrm{Proj}_Y\big(f_*\mathcal{O}_X[T]\big).$$

With the present convention requiring one global closed immersion into a fixed $\mathbb{P}^n_Y$, the global conclusion follows whenever $f_*\mathcal{O}_X$ is generated by finitely many global sections; it is automatic when Y is affine.

Corollary IV.C.4. *Every finite morphism is proper.*

Proof. Properness is local on the target. On every affine open subset of the target, the preceding proposition realizes the finite morphism as a projective morphism, hence as a proper morphism. ☐

Proposition IV.C.5 (Affine proper morphisms). *An affine morphism is proper if and only if it is finite.*

Proof. The implication "finite implies proper" was just proved. The converse is the standard algebraic theorem that an affine morphism of finite type which is universally closed is finite. We use this result without proof. ☐

Example IV.C.6 (The power map). Let $n \geq 1$. The homomorphism

$$\Bbbk[t] \longrightarrow \Bbbk[u], \qquad t \longmapsto u^n$$

makes $\Bbbk[u]$ a free $\Bbbk[t]$-module with basis

$$1, u, \ldots, u^{n-1}.$$

Hence the induced morphism

$$\mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}$$

is finite, projective and proper.

Exercise IV.C.7. Assume that $\Bbbk$ is algebraically closed and that $\mathrm{char}(\Bbbk) \nmid n$. For the power map

$$f : \mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}, \qquad u \longmapsto u^n,$$

compute the fibres over 0 and over $a \neq 0$.

Solution. The fibre over $a \in \Bbbk$ is

$$\mathrm{Spec}\big(\Bbbk[u]/(u^n - a)\big).$$

For $a = 0$, this is the non-reduced point

$$\mathrm{Spec}\big(\Bbbk[u]/(u^n)\big).$$

If $a \neq 0$, the polynomial $u^n - a$ has n distinct roots because $\Bbbk$ is algebraically closed and its derivative nu^{n-1} does not vanish at a nonzero root. Hence the fibre is the disjoint union of n reduced $\Bbbk$-rational points.

Example IV.C.8 (Normalisation of the cusp). The inclusion

$$\Bbbk[t^2, t^3] \subset \Bbbk[t]$$

is finite, since $\Bbbk[t]$ is generated by $1, t$ as a $\Bbbk[t^2, t^3]$-module. It defines a finite, hence proper, morphism

$$\mathbb{A}^1_{\Bbbk} \longrightarrow \mathrm{Spec}(\Bbbk[t^2, t^3]),$$

which is the normalisation morphism of the cuspidal cubic.

Theorem IV.C.9 (Global functions on a proper scheme). *Let A be a ring and let X be a proper A-scheme. Then*

$$\Gamma(\mathcal{O}_X)$$

is integral over A. If X is a reduced proper k-scheme of finite type, where k is a field, then $\Gamma(\mathcal{O}_X)$ is a finite-dimensional k-vector space.

Proof. Put

$$B = \Gamma(\mathcal{O}_X)$$

and let $b \in B$. The section b determines an A-morphism

$$g : X \longrightarrow \mathbb{A}^1_A = \operatorname{Spec}(A[t])$$

corresponding to the homomorphism

$$A[t] \longrightarrow B, \qquad t \longmapsto b.$$

Let

$$C = \operatorname{im}(A[t] \to B)$$

and put $Z = \operatorname{Spec}(C)$. Then Z is a closed subscheme of $\mathbb{A}^1_A$, and g factors as

$$X \xrightarrow{\bar{g}} Z \longrightarrow \mathbb{A}^1_A.$$

The homomorphism $C \to B$ is injective, so $\bar{g}$ is dominant. Since $X \to \operatorname{Spec}(A)$ is proper and $Z \to \operatorname{Spec}(A)$ is separated, the morphism $\bar{g}$ is proper. It is therefore closed. Its image is both dense and closed in Z, hence $\bar{g}$ is surjective.

The scheme Z is separated and of finite type over A. Since it is the surjective image of the proper A-scheme X, the preceding exercise shows that Z is proper over A. But $Z \to \operatorname{Spec}(A)$ is affine, hence finite by the proposition above. Therefore C is a finitely generated A-module. In particular, the class of t in C, and hence b, is integral over A.

Now assume that X is reduced and of finite type over a field k. Let

$$X_1, \ldots, X_r$$

be its irreducible components, endowed with their reduced induced scheme structures. They are proper integral k-schemes. Restriction gives an injective homomorphism

$$\Gamma(\mathcal{O}_X) \longrightarrow \prod_{i=1}^{r} \Gamma(\mathcal{O}_{X_i}).$$

Indeed, on an affine open subset this is the injectivity of a reduced ring into the product of its quotients by its minimal prime ideals.

For each i, the ring $\Gamma(\mathcal{O}_{X_i})$ embeds into the function field $k(X_i)$. By the first part of the theorem, all its elements are algebraic over k. The algebraic closure k_i of k inside the finitely generated field extension $k(X_i)/k$ is a finite extension of k. Hence

$$\Gamma(\mathcal{O}_{X_i}) \subset k_i$$

is finite-dimensional over k. The product of the k_i is finite-dimensional, and so is its k-subspace $\Gamma(\mathcal{O}_X)$. $\square$

Corollary IV.C.10. *Let X be an integral proper k-scheme of finite type. Then $\Gamma(\mathcal{O}_X)$ is a finite field extension of k. In particular, if k is algebraically closed, then*

$$\Gamma(\mathcal{O}_X) = k.$$

Example IV.C.11. For every field $\Bbbk$ and every $n \geq 1$,

$$\Gamma(\mathcal{O}_{\mathbb{P}^n_{\Bbbk}}) = \Bbbk.$$

This was computed directly in the preceding chapter; it also follows from properness and integrality.

Exercise IV.C.12. Let $\Bbbk$ be algebraically closed, let X be an integral proper $\Bbbk$-scheme of finite type, and let $Y = \operatorname{Spec}(A)$ be an affine $\Bbbk$-scheme. Show that every morphism

$$f : X \longrightarrow Y$$

is constant, in the sense that it factors through a $\Bbbk$-rational point of Y.

Solution. By the theorem on morphisms to an affine scheme, the morphism f corresponds to a homomorphism of $\Bbbk$-algebras

$$A \longrightarrow \Gamma(\mathcal{O}_X).$$

Since X is integral and proper and $\Bbbk$ is algebraically closed,

$$\Gamma(\mathcal{O}_X) = \Bbbk.$$

Thus the homomorphism factors as $A \to \Bbbk$, which is precisely a $\Bbbk$-rational point $\operatorname{Spec}(\Bbbk) \to Y$. Therefore f factors through that point.

Exercise IV.C.13. Let $f : X \to Y$ be proper and let $y \in Y$. Show that the fibre

$$X_y \longrightarrow \operatorname{Spec}(\kappa(y))$$

is proper.

Solution. The fibre morphism is obtained from f by base change along

$$\operatorname{Spec}(\kappa(y)) \longrightarrow Y.$$

Properness is preserved by base change, so X_y is proper over $\kappa(y)$.

Chapter 6

A glimpse towards moduli spaces

The aim of this final chapter is not to develop the general theory of moduli spaces, but to show why the language introduced in this course is useful to prepare the ground for the study of moduli spaces. The first ingredient is just one more way to think of a morphism of schemes

$$X \longrightarrow T.$$

We view this morphism as a family of schemes parametrised by T. Namely, over a point $t \in T$, the element of the family is the fibre X_t.

A *moduli space* is a scheme whose T-points themselves parametrise families of geometric objects parametrised by T. The objects that one wants to parametrise, typically, are clear to the person embarking in the study of their moduli space: these could be compact Riemann surfaces of genus g, or vector bundles of a given rank on the projective plane, etc. Sometimes these objects are to be taken up to some equivalence relation (isomorphism of Riemann surfaces, of vector bundles, etc). So actually one starts by setting the *moduli problem*. This is about defining, for each scheme T (which could be over $\mathbb{Z}$, or over some other base scheme), the geometric objects parametrised by T, just as a set. This typically defines a functor, from schemes to sets. Then the question is: is that functor actually given by a space? Or, slightly more precisely, is that functor represented by a scheme?

I The Hilbert scheme functor

We will discuss briefly the idea of moduli space in one of the simplest situations, namely the Hilbert scheme.

I.A Families and the functor of points

Let us fix a ground field $\Bbbk$. For a $\Bbbk$-scheme M, its functor of points is

$$\begin{aligned} h_M : (\mathrm{Sch})_{\Bbbk}^{\mathrm{op}} &\to (\mathrm{Set}) \\ T &\mapsto h_M(T) = \mathrm{Hom}_{(\mathrm{Sch})_{\Bbbk}}(T, M). \end{aligned}$$

Rational $\Bbbk$-points correspond to $T = \mathrm{Spec}(\Bbbk)$. Allowing arbitrary test schemes records more information. For instance, morphisms from

$$\mathrm{Spec}\big(\Bbbk[\varepsilon]/(\varepsilon^2)\big)$$

encode infinitesimal tangent directions, as we have seen. Thus non-reduced test schemes are not an artificial complication: they are precisely what allows a moduli problem to remember first-order deformations.

Imagine now that we start from a functor $\mathcal{F}$. The idea is that this functor is supposed to assign to each scheme T the families of our favourite geometric objects parametrised by T. Thus we start with a functor

$$\mathcal{F} : (\mathrm{Sch})_{\Bbbk}^{\mathrm{op}} \to (\mathrm{Set}).$$

We say that a scheme M represents $\mathcal{F}$ if $\mathcal{F} \simeq h_M$. This means that there are natural bijections

$$\mathrm{Hom}_{(\mathrm{Sch})_{\Bbbk}}(T, M) \simeq \mathcal{F}(T)$$

for every $\Bbbk$-scheme T. This way, if $\mathcal{F}(T)$ is the set of families of objects parametrised by T, then a morphism $T \to M$ is one family of objects parametrised by T. Beware that, under this formalism, a morphism $f : T \to T'$ pulls a T'-family back to a T-family through $\mathcal{F}(f) : \mathcal{F}(T') \to \mathcal{F}(T)$. If the original family is represented by $\alpha' : T' \to M$, then the pull-back family is represented by $\alpha' \circ f : T \to M$.

It is also important to consider $T = M$. Under the representing bijection, the identity morphism of M gives a distinguished tautological element $\mathcal{U} \in \mathcal{F}(M)$. This element has the property that any family comes by pull-back of it. More precisely, for any $\Bbbk$-scheme T, and for any $\mathcal{V} \in \mathcal{F}(T)$ we have a corresponding *moduli map* $\alpha_{\mathcal{V}} \in \mathrm{Hom}_{(\mathrm{Sch})_{\Bbbk}}(T, M)$, given by $\mathcal{V}$ under the bijection $\mathrm{Hom}_{(\mathrm{Sch})_{\Bbbk}}(T, M) \simeq \mathcal{F}(T)$. Then $\mathcal{F}(\alpha_{\mathcal{V}}) : \mathcal{F}(M) \to \mathcal{F}(T)$ sends $\mathcal{U}$ precisely to $\mathcal{V}$. So the slogan is: a T-family determines a moduli map $T \to M$ to the moduli space, in such a way that the family is the pull-back of the tautological family, which is therefore at the source of all families. Of course, a $\Bbbk$-rational point of M is just a single geometric object, and all geometric objects (over $\Bbbk$) individually are in bijection with rational $\Bbbk$-points. If one such object is defined over a field extension κ of $\Bbbk$, then it will be a κ-rational point of M.

I.B The Hilbert scheme functor of n points of the plane

For a systematic treatment of Hilbert schemes of points on surfaces and their matrix description, see [Nak99, Lectures 1–2]. See also [EH00, Chapter VI] for the general moduli philosophy.

Let T be a $\Bbbk$-scheme and write

$$\mathbb{A}^2_T = \mathbb{A}^2_{\Bbbk} \times_{\mathrm{Spec}(\Bbbk)} T.$$

A family of length-n subschemes of the affine plane parametrised by T is a closed subscheme

$$Z \subset \mathbb{A}^2_T$$

such that the projection $p : Z \to T$ is finite and, locally on T, its coordinate algebra admits a basis of n elements over the coordinate ring of the base. More precisely, on an affine open subset $U = \mathrm{Spec}(R) \subset T$, the family is given by an ideal

$$I \subset R[x, y]$$

such that $R[x, y]/I$ is a free R-module of rank n, after possibly replacing U by a smaller affine open cover. This condition is often expressed by saying that $p_*\mathcal{O}_Z$ is locally free of rank n. It implies that every fibre

$$Z_t = Z \times_T \mathrm{Spec}(\kappa(t))$$

has coordinate ring of dimension n over $\kappa(t)$, hence length n. So we let $\mathcal{F}(T)$ be the set of these families. Again, if $T = \mathrm{Spec}(R)$, then

$$\mathcal{F}(\mathrm{Spec}(R)) = \{I \subset R[x, y] \mid R[x, y]/I \text{ is free of rank } n.\}$$

Remark I.B.1. It is tempting to require only that every fibre have length n. Over reduced bases, and under suitable finiteness hypotheses, this often captures the same geometric idea. Over an arbitrary non-reduced base it is not sufficient: infinitesimal torsion may be invisible on all residue-field fibres. The local-basis condition is the down-to-earth form of the flatness requirement appearing in the general Hilbert functor. For projective families, its numerical shadow is the constancy of the Hilbert polynomial; for a zero-dimensional fibre of length n, that polynomial is the constant polynomial n.

Definition I.B.2. The *Hilbert functor of n points on the affine plane* assigns to a $\Bbbk$-scheme T the set of closed subschemes $Z \subset \mathbb{A}^2_T$ satisfying the preceding condition. Pull-back along $T' \to T$ is given by

$$Z_{T'} = Z \times_T T' \subset \mathbb{A}^2_{T'}.$$

Theorem I.B.3 (Representability, stated without proof). *There exists an irreducible quasi-projective $\Bbbk$-scheme*

$$\mathrm{Hilb}^n(\mathbb{A}^2_{\Bbbk})$$

of dimension $2n$ representing the Hilbert functor of n points on $\mathbb{A}^2_{\Bbbk}$. It is smooth; the Barth–Nakajima tangent calculation below explains this last assertion. Thus morphisms

$$T \longrightarrow \mathrm{Hilb}^n(\mathbb{A}^2_{\Bbbk})$$

are naturally the same as families of length-n subschemes of $\mathbb{A}^2_{\Bbbk}$ parametrised by T.

For $T = \mathrm{Spec}(\Bbbk)$, a point of the Hilbert scheme is simply an ideal $I \subset \Bbbk[x, y]$ such that

$$\dim_{\Bbbk} \Bbbk[x, y]/I = n.$$

The open subset corresponding to n distinct reduced points is the unordered configuration space of n distinct points. Its boundary records collisions, and the resulting subschemes are usually non-reduced.

Example I.B.4. For $n = 1$, every colength-one ideal is a maximal ideal $(x - a, y - b)$, so

$$\mathrm{Hilb}^1(\mathbb{A}^2_{\Bbbk}) \simeq \mathbb{A}^2_{\Bbbk}.$$

For $n = 2$, the ideal

$$(y, x^2)$$

defines a non-reduced length-two subscheme supported at the origin. It may be viewed as two points which have collided while retaining an infinitesimal direction.

I.C Commuting matrices and a cyclic vector

Let $I \subset \Bbbk[x, y]$ have colength n and put

$$V = \Bbbk[x, y]/I.$$

Multiplication by x and y gives commuting endomorphisms

$$X, Y \in \mathrm{End}_{\Bbbk}(V), \qquad [X, Y] = 0.$$

The class $v = 1 \bmod I$ is cyclic:

$$V = \Bbbk[X, Y]v.$$

Conversely, let V be an n-dimensional vector space, let $X, Y \in \mathrm{End}(V)$ commute, and let $v \in V$ be cyclic. Then

$$\Bbbk[x, y] \longrightarrow V, \qquad f \longmapsto f(X, Y)v$$

is surjective, and its kernel is an ideal of colength n.

Proposition I.C.1 (Barth–Nakajima description). *The* $\Bbbk$*-points of* $\mathrm{Hilb}^n(\mathbb{A}^2_{\Bbbk})$ *are in bijection with isomorphism classes of triples* (X, Y, v) *where*

$$X, Y \in \mathrm{End}(\Bbbk^n), \qquad [X, Y] = 0,$$

and $v \in \Bbbk^n$ *is cyclic. Two triples are identified when they differ by a simultaneous change of basis by an element of* $\mathrm{GL}_n(\Bbbk)$*.*

Proof. The two constructions just described are inverse. Starting from I, the map $f \mapsto f(X, Y)v$ is the quotient map $\Bbbk[x, y] \to \Bbbk[x, y]/I$, so its kernel is I. Starting from a cyclic triple, the induced identification $\Bbbk[x, y]/I \simeq V$ carries multiplication by x and y back to X and Y, and carries 1 to v.

It remains to account for the choice of a basis of V. Two identifications $V \simeq \Bbbk^n$ differ by an element of $\mathrm{GL}_n(\Bbbk)$, which simultaneously conjugates X and Y and sends v to the new coordinate vector. Conversely, such a change of basis does not change the kernel of $f \mapsto f(X, Y)v$. This proves the asserted bijection. □

In families, the vector space V is replaced by a rank-n vector bundle $\mathcal{E}$ on T, the matrices by commuting endomorphisms $X, Y \in \mathrm{End}(\mathcal{E})$, and the cyclic vector by a section $v \in \Gamma(T, \mathcal{E})$ which generates every fibre under X and Y. This realises the Hilbert scheme as the quotient of the cyclic locus

$$\{(X, Y, v) \mid [X, Y] = 0,\ \Bbbk[X, Y]v = \Bbbk^n\}$$

by GL_n. This is the simplest doorway to framed quiver representations and Nakajima quiver varieties.

Here is what is hidden in the word "quotient". By Cayley–Hamilton, cyclicity can be tested using the finitely many vectors

$$X^a Y^b v, \qquad 0 \le a, b < n.$$

The condition that they span is the non-vanishing of suitable minors, hence defines an open locus. The GL_n-action on this locus is free: an endomorphism commuting with X and Y and fixing v fixes every $f(X, Y)v$, hence is the identity. A family with a rank-n bundle $\mathcal{E}$ becomes a family of matrices after passing to its frame bundle; changing the frame is precisely the GL_n-action. The geometric quotient therefore descends framed matrix data to families of ideals. The existence of this quasi-projective quotient is the non-elementary invariant-theoretic step in the Barth–Nakajima construction.

Example I.C.2. For $I = (y, x^2)$, use the basis $(1, x)$ of $\Bbbk[x, y]/I$. Multiplication gives

$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad Y = 0, \qquad v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The nilpotent Jordan block records the infinitesimal direction of the collision.

I.D Dual numbers and the Barth–Nakajima tangent calculation

Put $A = \Bbbk[x, y]$ and $D = \Bbbk[\varepsilon]/(\varepsilon^2)$. A tangent vector to $\mathrm{Hilb}^n(\mathbb{A}^2_{\Bbbk})$ at the point $[I]$ is a D-valued point specialising to $[I]$. In terms of ideals, it is an ideal

$$\widetilde{I} \subset A \otimes_{\Bbbk} D = A[\varepsilon]/(\varepsilon^2)$$

such that $(A \otimes D)/\widetilde{I}$ is free of rank n over D and $\widetilde{I} \bmod \varepsilon = I$.

Proposition I.D.1 (First-order deformations of an ideal). *There is a natural isomorphism*

$$T_{[I]} \mathrm{Hilb}^n(\mathbb{A}^2_{\Bbbk}) \simeq \mathrm{Hom}_A(I, A/I).$$

More explicitly, if $f + \varepsilon g \in \widetilde{I}$ lifts $f \in I$, the corresponding homomorphism is

$$\varphi_{\widetilde{I}}(f) = g \bmod I.$$

Conversely, $\varphi \in \operatorname{Hom}_A(I, A/I)$ determines

$$\widetilde{I}_\varphi = \{f + \varepsilon g \mid f \in I,\ g \bmod I = \varphi(f)\}.$$

Proof. The local-freeness condition permits every $f \in I$ to be lifted to an element $f + \varepsilon g$ of $\widetilde{I}$. If two such lifts are chosen, then $\varepsilon(g - g') \in \widetilde{I}$. In the free D-module $Q = (A \otimes D)/\widetilde{I}$, the kernel of multiplication by ε is εQ. Reducing modulo ε therefore shows that $g - g' \in I$. Thus $g \bmod I$ is well defined.

Addition of lifts proves additivity. Multiplying a lift by $a \in A$ shows that

$$\varphi_{\widetilde{I}}(af) = a\varphi_{\widetilde{I}}(f),$$

so $\varphi_{\widetilde{I}}$ is A-linear. Conversely, A-linearity makes $\widetilde{I}_\varphi$ an ideal. Choosing a $\Bbbk$-linear section of $A \to A/I$ gives unique normal forms modulo $\widetilde{I}_\varphi$ and identifies the quotient, as a D-module, with

$$D \otimes_{\Bbbk} (A/I).$$

It is therefore free of rank n. The two constructions are inverse. □

We can now see the number $2n$ directly from the commuting-matrix description. Let (X, Y, v) be the cyclic triple corresponding to I and put $V = A/I$. A first-order deformation has the form

$$(X + \varepsilon P,\ Y + \varepsilon Q,\ v + \varepsilon w).$$

The commutation relation modulo ε^2 is

$$[P, Y] + [X, Q] = 0.$$

An infinitesimal change of basis $1 + \varepsilon a$ changes (P, Q, w) by

$$\big([a, X], [a, Y], av\big).$$

Consequently the tangent space is the middle cohomology of

$$\operatorname{End}(V) \xrightarrow{d_0} \operatorname{End}(V) \oplus \operatorname{End}(V) \oplus V \xrightarrow{d_1} \operatorname{End}(V),$$

where

$$d_0(a) = ([a, X], [a, Y], av), \qquad d_1(P, Q, w) = [P, Y] + [X, Q].$$

Proposition I.D.2 (Barth–Nakajima tangent calculation). *For every colength-n ideal $I \subset \Bbbk[x, y]$,*

$$\dim_{\Bbbk} \operatorname{Hom}_A(I, A/I) = 2n.$$

Thus every geometric tangent space of $\operatorname{Hilb}^n(\mathbb{A}^2_{\Bbbk})$ has dimension $2n$.

Proof. Cyclicity makes d_0 injective: if a commutes with X and Y and $av = 0$, then a vanishes on every vector $f(X, Y)v$, hence on all of V.

Use the non-degenerate trace pairing on $\operatorname{End}(V)$. The orthogonal complement of $\operatorname{im}(d_1)$ consists of the endomorphisms commuting with both X and Y. Evaluation at the cyclic vector gives an isomorphism

$$\{c \in \operatorname{End}(V) \mid [c, X] = [c, Y] = 0\} \longrightarrow V, \qquad c \longmapsto cv.$$

It is injective by cyclicity; it is surjective because every $u \in V$ has the form $u = f(X,Y)v$, and the endomorphism $f(X,Y)$ commutes with X and Y. Hence the common centraliser has dimension n, and

$$\operatorname{rank}(d_1) = n^2 - n.$$

It follows that

$$\begin{aligned}\dim\bigl(\ker(d_1)/\operatorname{im}(d_0)\bigr) &= (2n^2+n) - (n^2-n) - n^2 \\ &= 2n.\end{aligned}$$

The preceding proposition identifies this quotient with $\operatorname{Hom}_A(I, A/I)$. Since the representing scheme is irreducible of dimension $2n$, its geometric tangent spaces all have the expected dimension. By the regularity and tangent-space criterion of Theorem VI.C.1, applied after extending the ground field to an algebraic closure, the Hilbert scheme is regular there and hence smooth over $\Bbbk$. ☐

Example I.D.3 (The double point). For $I = (y, x^2)$, the quotient A/I has basis $(1, x)$. The ideal I is generated by y and x^2, with syzygy $(x^2, -y)$. This relation becomes zero in A/I, so the images of both generators under a homomorphism $I \to A/I$ may be chosen freely. Hence

$$T_{[I]} \operatorname{Hilb}^2(\mathbb{A}^2_{\Bbbk}) \simeq (A/I)^{\oplus 2}$$

has dimension 4, just as it does at a pair of distinct points.

I.E Exercises

Exercise I.E.1. Assume that $\Bbbk$ is algebraically closed.

i) Show that every ideal $I \subset \Bbbk[x]$ such that $\dim_{\Bbbk} \Bbbk[x]/I = n$ is generated by a unique monic polynomial

$$f(x) = x^n + a_1 x^{n-1} + \cdots + a_n.$$

ii) Deduce a bijection on $\Bbbk$-points

$$\operatorname{Hilb}^n(\mathbb{A}^1_{\Bbbk})(\Bbbk) \simeq \mathbb{A}^n_{\Bbbk}(\Bbbk).$$

iii) Explain how the factorisation

$$f(x) = \prod_i (x - \lambda_i)^{m_i}$$

records the support and multiplicities of the corresponding subscheme.

iv) State the corresponding functorial result for a base $T = \operatorname{Spec}(R)$ using monic polynomials in $R[x]$.

Solution. The ring $\Bbbk[x]$ is a principal ideal domain, so $I = (f)$ for a unique monic polynomial f. The quotient has dimension $\deg(f)$ with basis $1, x, \ldots, x^{\deg(f)-1}$, hence $\deg(f) = n$. The coefficients $(a_1, \ldots, a_n)$ give the required point of $\mathbb{A}^n_{\Bbbk}$. Over an algebraically closed field, the Chinese remainder theorem identifies the factors $(x - \lambda_i)^{m_i}$ with length-m_i subschemes supported at λ_i. Functorially, a monic polynomial

$$x^n + a_1 x^{n-1} + \cdots + a_n \in R[x]$$

defines a quotient free over R with basis $1, x, \ldots, x^{n-1}$; its coefficients define a morphism $T \to \mathbb{A}^n_{\Bbbk}$. This yields $\operatorname{Hilb}^n(\mathbb{A}^1_{\Bbbk}) \simeq \mathbb{A}^n_{\Bbbk}$.

Exercise I.E.2. Let (X, Y, v) be a commuting cyclic triple on an n-dimensional vector space V. Show directly that

$$I = \{f \in \Bbbk[x, y] \mid f(X, Y)v = 0\}$$

is an ideal and that $\Bbbk[x, y]/I \simeq V$.

Solution. If $f \in I$ and $g \in \Bbbk[x, y]$, then commutativity gives

$$(gf)(X, Y)v = g(X, Y)f(X, Y)v = 0,$$

so $gf \in I$. The map $\Bbbk[x, y] \to V$, $f \mapsto f(X, Y)v$, is surjective because v is cyclic, and its kernel is I. The first isomorphism theorem gives $\Bbbk[x, y]/I \simeq V$.

Exercise I.E.3. Assume that $\Bbbk$ is algebraically closed. Describe the triples associated with

i) two distinct points (a_1, b_1) and (a_2, b_2) of $\mathbb{A}^2_{\Bbbk}$;

ii) the double point defined by (y, x^2).

Solution. For two distinct points, take

$$X = \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix}, \qquad Y = \begin{pmatrix} b_1 & 0 \\ 0 & b_2 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The vector v is cyclic because the two joint eigenvalues are distinct. For (y, x^2), use the basis $(1, x)$ and obtain

$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad Y = 0, \qquad v = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

II What about more general moduli problems?

II.A Hilbert schemes of points on curves, surfaces and threefolds

The geometry of Hilbert schemes of points changes dramatically with the dimension of the ambient variety.

- On the affine line, every colength-n ideal is principal, and

  $$\mathrm{Hilb}^n(\mathbb{A}^1_{\Bbbk}) \simeq \mathbb{A}^n_{\Bbbk}.$$

  Equivalently, a length-n subscheme is determined by one monic polynomial of degree n, with repeated roots recording non-reduced points.

- On a smooth surface, the Hilbert scheme of n points is smooth and irreducible of dimension $2n$; this is the world studied in [Nak99].

- In dimension three, the geometry is much wilder. The scheme $\mathrm{Hilb}^n(\mathbb{A}^3_{\Bbbk})$ is reducible for n sufficiently large. Classical constructions of Iarrobino already produce additional components for length 78; see [Iar72]. Thus the pleasant surface theorem is genuinely special to dimension two.

Remark II.A.1 (Partitions enter geometry). Over $\mathbb{C}$, the torus $(\mathbb{C}^\times)^2$ acts on $\mathbb{A}^2$ by scaling the two coordinates and hence acts on every $\mathrm{Hilb}^n(\mathbb{A}^2)$. Its fixed points are the monomial ideals of colength n. The monomials outside such an ideal form a Young diagram, so the fixed points are indexed by partitions $\lambda \vdash n$. A compatible cell decomposition then gives

$$\chi\big(\mathrm{Hilb}^n(\mathbb{C}^2)\big) = p(n),$$

where $p(n)$ is the number of partitions of n. Euler's identity packages all these Euler characteristics into

$$\sum_{n\geq 0} \chi\big(\mathrm{Hilb}^n(\mathbb{C}^2)\big) q^n = \sum_{n\geq 0} p(n) q^n = \prod_{m\geq 1} \frac{1}{1-q^m}.$$

This elementary-looking product is closely related to the Dedekind eta function. Refined versions of the same generating series lead from Hilbert schemes to representation theory, modular forms and mathematical physics.

II.B Beyond Hilbert schemes of points

Can one construct schemes parametrising algebraic curves, vector bundles, coherent sheaves, or morphisms between schemes? Often the answer is yes, although the resulting moduli spaces may be singular, reducible, or even need to be replaced by more general objects such as algebraic stacks. The language developed in this course – valued points, families, fibres, nilpotents and dimension – is the natural starting point for these constructions.

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